# NEET UG CHEMISTRY

## MASTER TEXTBOOK

Complete Master Theory, Formulas & Experimental Compendium

Strictly Aligned with NEET UG 2027 NMC Syllabus

**Satya Prakash**

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**Authored by**

**Satya Prakash**

**Strictly Calibrated to the Revised NMC Syllabus & Rationalized NCERT Textbooks**

Target Examination: National Eligibility cum Entrance Test (NEET UG 2027)

Published for Independent Medical Aspirants, Educators & Academic Study Circles

Comprehensive Academic Edition · Global Print-on-Demand

**NEET UG CHEMISTRY MASTER TEXTBOOK**

*Complete Master Theory, Formulas & Experimental Compendium strictly aligned with NEET UG 2027 NMC Syllabus*

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**Source Repository & Official Errata:**
https://github.com/spdedsec/neet-study-materials

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**First Edition:** October 2026
**Publishing Imprint:** Independent Academic Publishing / Satya Prakash Editorial Imprint
**Target Examination Cycle:** NEET UG 2027 (Undergraduate Medical Admissions Entrance Examination)
**Publication Format:** Print on Demand Professional Two-Sided Edition
**ISBN-13 (Paperback):** 979-8-1626-6110-8
**ISBN-13 (Hardcover):** 979-8-1626-6110-8 (Reference Library Edition)

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**Typesetting & Typography:** Typeset in LaTeX $2_\varepsilon$ with the Tectonic XeTeX engine using Latin Modern typefaces and custom academic publishing geometry. Chemical nomenclature, SI units, and IUPAC rules strictly observed. Produced and bound under rigorous academic publishing standards.

Printed in the United States of America / United Kingdom / India.

*To my parents, whose boundless love and quiet sacrifices gave me the courage to pursue knowledge;*

*To the chemistry teachers who revealed that reactions are not random rules to be memorized,*
*but the logical dance of electrons seeking thermodynamic stability;*

*And to every medical aspirant battling balancing equations, acid-base equilibria, and organic mechanisms—*
*may this book be your clearest guide, turning what felt overwhelming into your greatest competitive strength.*

# Acknowledgements

This comprehensive chemistry textbook reflects years of academic analysis, teaching, and systematic alignment with national medical entrance examinations. I wish to express my profound gratitude to all who contributed to its realization.

I thank my family for their unending patience, love, and constant moral support during the intensive writing and typesetting process.

I am deeply grateful to the chemistry educators and academic colleagues whose rigorous insights helped shape the structural framework of this book. Their insistence on clear mechanistic arrow-pushing, thermodynamic sign discipline, and crystal field theory clarity has elevated the quality of every discussion.

I owe special thanks to the students whose candid feedback and test error sheets provided direct insight into the most persistent conceptual pitfalls in competitive chemistry. The "NEET Trap" callouts throughout these pages are a direct response to those common stumbling blocks.

Finally, I express my sincere appreciation to the LaTeX $2_\varepsilon$ open-source community for creating typographical tools that allow complex chemical equations, tabular reaction matrices, and color-coded callouts to be typeset with uncompromised aesthetic and mathematical precision.

**Satya Prakash**
*October 2026*

# Author's Preface: Pedagogical Architecture & Analytical Methodology

Mastering chemistry for the National Eligibility cum Entrance Test (NEET UG) requires a decisive transition from passive memorization to analytical problem-solving and mechanistic reasoning.

In competitive medical entrance preparation, candidates often encounter two persistent pedagogical pitfalls. First, many treat chemistry as an encyclopedic catalog of isolated reactions, memorizing reagents and formula sheets without internalizing electron density distributions, stereochemical principles, or thermodynamic driving forces. Consequently, when examination items present novel multi-step transformations, stereochemical configurations, or Assertion-Reasoning pairings, memorized patterns fail.

Second, substantial study time is frequently misallocated to legacy coaching material covering topics eliminated under the rationalized curriculum issued by the National Medical Commission (NMC). Nine former units—Solid State, States of Matter, Surface Chemistry, Hydrogen, the *s*-Block Elements, Metallurgy, Polymers, Chemistry in Everyday Life, and Environmental Chemistry—have been formally excised from the syllabus. This curricular pruning is not a reduction in examination rigor; rather, it allows the examination to test substantially deeper conceptual comprehension across the retained foundational domains: Chemical Bonding, Coordination Chemistry, Reaction Intermediates, Carbonyl Chemistry, Thermodynamics, and Ionic Equilibria.

This master textbook is constructed to provide an uncompromisingly rigorous, self-contained, and syllabus-calibrated pedagogical framework for NEET UG 2027:

1. **First-Principles Physical Chemistry:** Every governing formulation—from de Broglie wave mechanics and Schrödinger orbital definitions to the integrated rate laws, Kohlrausch conductivity, and the Nernst equation—is derived algebraically with rigorous thermodynamic sign discipline and dimensional consistency.
2. **Mechanistic Organic Transformations:** Organic chemistry is presented through the physical movement of electron pairs. Part V provides

a dedicated handbook of curved-arrow reaction mechanisms, arenium ion intermediates, transition states, and carbocation rearrangements spanning all canonical reaction classes.

3. **Inorganic Chemistry Grounded in Electronic Structure:** Rather than relying on disconnected factual recall, inorganic behavior across the *p*-block, *d*-block, and coordination complexes is deduced systematically from effective nuclear charge ($Z_{\text{eff}}$), shielding parameters, crystal field splitting energy ($\Delta_o$ and $\Delta_t$), and stereochemical geometries.
4. **Laboratory Skills & Analytical Protocols:** In strict accordance with the updated practical chemistry mandate, Part VI provides comprehensive protocols for systematic cation group separation (Groups 0–VI), confirmatory tests for acidic radicals, and quantitative redox titrimetry.
5. **Integrated Tactical Callout Architecture:** Five distinct callout environments—`conceptbox`, `formulabox`, `usecasebox`, `trapbox`, and `ncertbox`—demarcate fundamental principles, valid boundary conditions, and the high-frequency pitfalls where aspirants routinely concede negative marks.

Candidates are strongly advised to engage actively with paper and pen: verify every mathematical derivation, trace each curved arrow through its transition state, and balance redox equations systematically by the ion-electron method. Conceptual clarity achieved through active derivation is the most durable asset in competitive examination environments.

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# Syllabus Blueprint & NMC Rationalization Guide

This section details the official NEET UG 2027 Chemistry curriculum prescribed by the National Medical Commission (NMC), highlighting the examination scheme, unit distribution, and critical syllabus exclusions.

## Official Chemistry Examination Architecture

| Component | Compulsory Qs | Marks | Pace Budget | Question Typology |
|---|---|---|---|---|
| **Chemistry** | 45 (All Compulsory) | 180 | 40–45 min | Single Choice MCQs, Reaction Sequences, |
| **Exam Total** | 180 Compulsory | 720 | 180 min (3h) | Assertion-Reason, Statement I/II, Matrix Match |

Table 1: Reformed NEET UG Chemistry Examination Scheme (+4 for correct response, −1 for incorrect response, 0 for unattempted. Optional Section B discontinued by NTA).

## Official 20-Unit Chemistry Syllabus Structure

1. **Unit 1: Some Basic Concepts in Chemistry** (Matter and laws of chemical combination, mole concept, molar mass, stoichiometry and stoichiometric calculations).
2. **Unit 2: Atomic Structure** (Bohr model and limitations, dual nature, de Broglie relation, Heisenberg uncertainty, quantum mechanical model, orbitals, quantum numbers, Hund's rule, Pauli exclusion, Aufbau principle).
3. **Unit 3: Chemical Bonding and Molecular Structure** (Ionic bonding, lattice energy, Born-Haber cycle, covalent bond, VSEPR theory, hybridization, dipole moment, MOT for homonuclear diatomic molecules, hydrogen bonding).

4. **Unit 4: Chemical Thermodynamics** (First law, enthalpy, Hess's law, bond dissociation enthalpy, spontaneous processes, entropy, Gibbs free energy criterion for spontaneity).
5. **Unit 5: Solutions** (Types of solutions, Raoult's law, colligative properties: lowering of vapor pressure, elevation of boiling point, depression of freezing point, osmotic pressure, van 't Hoff factor).
6. **Unit 6: Equilibrium** (Law of mass action, $K_p$ and $K_c$, Le Chatelier's principle, ionic equilibrium: ionization of acids/bases, pH scale, common ion effect, buffer solutions, solubility product $K_{sp}$).
7. **Unit 7: Redox Reactions and Electrochemistry** (Oxidation states, balancing redox, conductance in electrolytic solutions, Kohlrausch's law, electrochemical cells, EMF, Nernst equation, Gibbs energy and EMF).
8. **Unit 8: Chemical Kinetics** (Rate of reaction, rate law, order and molecularity, integrated rate equations for zero and first order, half-life, Arrhenius equation, activation energy).
9. **Unit 9: Classification of Elements and Periodicity** (Modern periodic law, periodic trends: atomic/ionic radii, ionization enthalpy, electron gain enthalpy, electronegativity, valence).
10. **Unit 10: $p$-Block Elements** (General electronic configuration and periodic trends across Groups 13 through 18, comparative chemistry, anomalous properties of first element).
11. **Unit 11: $d$- and $f$-Block Elements** (Transition elements: electronic configuration, oxidation states, catalytic properties, magnetic properties, colored ions, lanthanoids and actinoids, lanthanoid contraction).
12. **Unit 12: Coordination Compounds** (Werner's theory, IUPAC nomenclature, isomerism, Valence Bond Theory, Crystal Field Theory, spectrochemical series, stability and biological significance).
13. **Unit 13: Purification and Characterization of Organic Compounds** (Qualitative and quantitative analysis: Lassaigne's test, Dumas, Kjeldahl, Carius methods).
14. **Unit 14: Some Basic Principles of Organic Chemistry** (IUPAC nomenclature, inductive, electromeric, resonance and hyperconjugation effects, reactive intermediates: carbocations, carbanions, free radicals).
15. **Unit 15: Hydrocarbons** (Alkanes: conformation of ethane; Alkenes: geometrical isomerism, Markovnikov/anti-Markovnikov addition; Alkynes: acidity, addition; Arenes: aromaticity, EAS mechanisms).
16. **Unit 16: Organic Compounds Containing Halogens** (Haloalkanes and haloarenes: $S_N1$ and $S_N2$ mechanisms, environmental effects of polyhalogen compounds).
17. **Unit 17: Organic Compounds Containing Oxygen** (Alcohols, phenols and ethers: Lucas test, Kolbe reaction, Reimer-Tiemann, Williamson synthesis; Aldehydes, ketones and carboxylic acids: nucleophilic addition, aldol condensation, Cannizzaro, Hell-Volhard-Zelinsky).

18. **Unit 18: Organic Compounds Containing Nitrogen** (Amines: basicity, Hinsberg test, carbylamine test; Diazonium salts: synthetic applications).
19. **Unit 19: Biomolecules** (Carbohydrates: aldoses, ketoses, glucose, fructose, glycosidic bond; Proteins: peptide bond, primary/secondary/tertiary structures, denaturation; Nucleic acids: DNA, RNA).
20. **Unit 20: Principles Related to Practical Chemistry** (Qualitative salt analysis: cations and anions; Volumetric titrations: oxalic acid vs $KMnO_4$, Mohr's salt vs $KMnO_4$).

**Zero-Tolerance NMC Exclusions: The 9 Deleted Chemistry Chapters**

Under revised NMC regulations, the following **nine legacy chapters are completely deleted** from the NEET examination:

1. **Solid State** (Unit cells, packing efficiency, point defects).
2. **States of Matter (Gaseous & Liquid Bulk States)** (Ideal gas equations remain in Unit 1; van der Waals equation and critical phenomena are eliminated).
3. **Surface Chemistry** (Adsorption, colloids, micelles, catalysts).
4. **General Principles of Metallurgy** (Extraction of metals, Ellingham diagrams).
5. **Hydrogen** (Hydrides, heavy water, hydrogen peroxide).
6. **s-Block Elements** (Alkali and alkaline earth metals).
7. **Polymers** (Classification, polymerization reactions, plastics).
8. **Chemistry in Everyday Life** (Drugs, medicines, soaps, detergents).
9. **Environmental Chemistry** (Pollution, acid rain, ozone depletion).

**Study Directive:** Do not invest study hours in these eliminated topics. All cognitive energy must focus on the rigorously tested Class 11 and Class 12 units above.

## Empirical Question Weightage Matrix

| Chemistry Domain | Typical Qs | Marks Weightage |
|---|---|---|
| Physical Chemistry (Mole, Atomic, Thermo, Equilibrium, Solutions, Electro, Kinetics) | 15–17 | 60–68 Marks |
| Inorganic Chemistry (Periodicity, Bonding, Coordination, $p$-Block, $d$-&$f$-Block) | 13–15 | 52–60 Marks |
| Organic Chemistry (GOC, Hydrocarbons, Haloalkanes, Oxygen, Nitrogen, Biomolecules) | 14–16 | 56–64 Marks |
| Practical Chemistry (Qualitative Salt Analysis & Volumetric Redox Titrations) | 2–3 | 8–12 Marks |
| **Total Chemistry Paper** | **45 Qs (Compulsory)** | **180 Marks** |

Table 2: Empirical Question Distribution in Recent NEET UG Papers.

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PART I

# Physical Chemistry Foundations

Chapter 1

# Foundations of Physical Chemistry: Stoichiometry, Atomic Structure & Thermodynamics

Physical chemistry provides the quantitative foundation for all chemical systems. This chapter begins with mole concept stoichiometry and empirical laws of chemical combination, progresses into wave mechanics and quantum numbers, and establishes the first and second laws of thermodynamics with explicit enthalpy, entropy, and Gibbs free energy derivations.

## 1.1 NMC 2027 Regulatory Calibration & Scope Demarcation

**Zero-Tolerance NMC Syllabus Alert: Deleted Physical Chemistry Chapters**

Under the revised National Medical Commission (NMC) NEET UG 2027 regulatory curriculum, the following legacy chapters have been **strictly deleted and removed from the examination**:

1. **Solid State:** All unit cells, crystal systems, Bragg's diffraction, packing efficiency derivations, point defects (Schottky and Frenkel defects), electrical and magnetic properties of solids are **completely eliminated**.
2. **States of Matter (Gaseous & Liquid Bulk States):** Macroscopic gas laws, van der Waals equation of state, critical phenomena ($P_c, V_c, T_c$), compressibility factor ($Z$) curves, liquefaction of gases, and bulk liquid properties (surface tension, viscosity) are **completely eliminated**. (Note: Ideal gas stoichiometry at STP is retained in Unit 1; molecular kinetic theory is covered in Physics).

3. **Surface Chemistry:** Adsorption isotherms (Freundlich and Langmuir), colloidal states, lyophilic and lyophobic sols, micelles and CMC, Tyndall effect, electrophoresis, coagulation and Hardy-Schulze rule, and heterogeneous catalysis are **strictly eliminated** from theory.

**Study Directive:** Do not invest study hours in these eliminated topics. All cognitive focus must remain on the rigorously tested Class 11 and Class 12 physical chemistry units detailed below.

## 1.2 Some Basic Concepts in Chemistry: Mole Concept & Stoichiometry (Unit 1)

### 1.2.1 Laws of Chemical Combination

Chemical transformations obey five fundamental empirical laws that form the bedrock of quantitative chemistry:

1. **Law of Conservation of Mass (Lavoisier, 1789):** In any chemical reaction, the total mass of the reactants equals the total mass of the products: $\sum m_{\text{reactants}} = \sum m_{\text{products}}$.
2. **Law of Definite Proportions (Proust, 1799):** A given chemical compound always contains exactly the same proportion of elements by mass, regardless of its source or method of preparation.
3. **Law of Multiple Proportions (Dalton, 1803):** If two elements combine to form more than one compound, the masses of one element that combine with a fixed mass of the other element are in the ratio of small whole numbers (e.g., in CO and $CO_2$, fixed 12 g of carbon combines with 16 g and 32 g of oxygen, ratio 1 : 2).
4. **Gay-Lussac's Law of Gaseous Volumes (1808):** When gases react together, they do so in volumes which bear a simple whole-number ratio to one another and to the volumes of gaseous products, provided all volumes are measured at identical temperature and pressure.
5. **Avogadro's Hypothesis (1811):** Equal volumes of all gases under identical conditions of temperature and pressure contain an equal number of molecules: $V \propto n$ (at constant $T, P$).

### 1.2.2 The Mole Concept & Molar Quantities

The mole is the SI base unit for the amount of substance. Exactly one mole contains $6.02214076 \times 10^{23}$ elementary entities (Avogadro's constant, $N_A$).

**Master Mole Formulations**

For any pure chemical substance of mass $w$ (in g), molar mass $M$ (in $\mathrm{g\,mol^{-1}}$), and elementary particle count $N$:

$$n = \frac{w}{M} = \frac{N}{N_A} = \frac{V_{\text{STP}}\ (\text{in Litres})}{V_m} \tag{1.1}$$

where the standard molar volume $V_m$ depends on the adopted thermodynamic convention:

- **Traditional NCERT Standard** ($T = 273.15\,\mathrm{K}, P = 1\,\mathrm{atm} = 101.325\,\mathrm{kPa}$)**:** $V_m = 22.414\,\mathrm{L\,mol^{-1}} \approx 22.4\,\mathrm{L}$.
- **IUPAC Standard** ($T = 273.15\,\mathrm{K}, P = 1\,\mathrm{bar} = 100.0\,\mathrm{kPa}$)**:** $V_m = 22.711\,\mathrm{L\,mol^{-1}} \approx 22.7\,\mathrm{L}$.

**Vapour Density (VD) Relationship for Ideal Gases:**

$$\text{VD} = \frac{\rho_{\text{gas}}}{\rho_{\text{H}_2}} = \frac{M_{\text{gas}}}{M_{\text{H}_2}} = \frac{M_{\text{gas}}}{2.016} \implies M \approx 2 \times \text{VD} \tag{1.2}$$

## 1.2.3 Empirical and Molecular Formulas

**Empirical vs Molecular Formula Determination**

- **Empirical Formula (EF):** The simplest whole-number ratio of atoms of each element present in a compound.
- **Molecular Formula (MF):** The actual number of atoms of each element present in a molecule of the compound:

$$\text{Molecular Formula} = (\text{Empirical Formula})_n, \qquad n = \frac{\text{Molar Mass}}{\text{Empirical Formula Mass}} \tag{1.3}$$

- **Algorithmic Determination:**
  1. Convert given mass percentages directly to mass in 100 g of sample.
  2. Divide each mass by the respective atomic mass to compute relative moles: $n_i = w_i / A_i$.
  3. Divide each $n_i$ by the smallest mole value among them: $r_i = n_i / n_{\text{min}}$.
  4. If the resulting ratios are non-integers, multiply all ratios by the smallest integer that converts them into whole numbers.

### 1.2.4 Stoichiometry, Limiting Reagents & Reaction Yields

For a general balanced stoichiometric chemical equation:

$$aA + bB \longrightarrow cC + dD \tag{1.4}$$

The stoichiometric coefficients $a, b, c, d$ represent the exact molar combining ratios:

$$\frac{n_A \text{ (consumed)}}{a} = \frac{n_B \text{ (consumed)}}{b} = \frac{n_C \text{ (produced)}}{c} = \frac{n_D \text{ (produced)}}{d} \tag{1.5}$$

**Limiting Reagent (LR) Diagnostic Algorithm**

When arbitrary initial quantities $n_{A,0}$ and $n_{B,0}$ are supplied:

1. Compute the mole-to-coefficient ratio for each reactant:

$$\mathcal{R}_A = \frac{n_{A,0}}{a}, \qquad \mathcal{R}_B = \frac{n_{B,0}}{b} \tag{1.6}$$

2. If $\mathcal{R}_A < \mathcal{R}_B$, reactant $A$ is the **Limiting Reagent** (completely consumed). Reactant $B$ is in excess.
3. All product yields and reactant consumptions are strictly determined by the limiting reagent:

$$n_C = c\left(\frac{n_{A,0}}{a}\right), \qquad n_{B,\text{remaining}} = n_{B,0} - b\left(\frac{n_{A,0}}{a}\right) \tag{1.7}$$

4. **Percentage Yield & Percentage Purity:**

$$\text{Percentage Yield} = \frac{\text{Actual Yield}}{\text{Theoretical Yield}} \times 100\% \tag{1.8}$$

$$\text{Percentage Purity} = \frac{\text{Mass of pure reactant}}{\text{Total mass of impure sample}} \times 100\% \tag{1.9}$$

### 1.2.5 Solution Concentration Expressions & Interconversions

The quantitative composition of solutions is formulated across six distinct mathematical regimes:

| Concentration Unit | Sym. | Mathematical Definition | Temp. Dependence |
|---|---|---|---|
| **Molarity** | $M$ | $M = \frac{n_{\text{solute}}}{V_{\text{sol}}\ (\text{L})} = \frac{w_B \times 1000}{M_B \times V_{\text{sol}}\ (\text{mL})}$ | **Dependent** ($V \propto T \implies M \downarrow$ as $T \uparrow$) |
| **Molality** | $m$ | $m = \frac{n_{\text{solute}}}{W_{\text{solvent}}\ (\text{kg})} = \frac{w_B \times 1000}{M_B \times W_A\ (\text{g})}$ | **Independent** (mass invariant) |
| **Normality** | $N$ | $N = \frac{\text{Gram Eq.}}{V_{\text{sol}}\ (\text{L})} = M \times n\text{-factor}$ | **Dependent** ($V \propto T$) |
| **Mole Fraction** | $x_B$ | $x_B = \frac{n_B}{n_A + n_B}, \quad x_A + x_B = 1$ | **Independent** |
| **Mass Percentage** | $\%(w/w)$ | $\%(w/w) = \frac{w_{\text{solute}}}{w_{\text{solution}}} \times 100$ | **Independent** |
| **Volume Percentage** | $\%(v/v)$ | $\%(v/v) = \frac{V_{\text{solute}}}{V_{\text{solution}}} \times 100$ | **Dependent** |
| **Parts Per Million** | ppm | $\text{ppm} = \frac{w_{\text{solute}}}{w_{\text{solution}}} \times 10^6$ | **Independent** |

Table 1.1: Comprehensive Classification of Solution Concentration Regimes.

**Master Interconversion Formulations ($d$ = density in $\text{g mL}^{-1}$)**

$$M = \frac{10 \times \%(w/w) \times d}{M_B} \tag{1.10}$$

$$m = \frac{1000 \times M}{1000\, d - M\, M_B} \tag{1.11}$$

$$m = \frac{x_B \times 1000}{(1 - x_B) M_A} = \frac{x_B \times 1000}{x_A M_A} \tag{1.12}$$

$$x_B = \frac{m\, M_A}{1000 + m\, M_A} \tag{1.13}$$

**Dilution and Mixing Formulations:**

$$M_1 V_1 = M_2 V_2, \qquad M_{\text{mix}} = \frac{\sum M_i V_i}{\sum V_i}, \qquad N_1 V_1 = N_2 V_2 \quad \text{(at exact redox/acid-ba} \tag{1.14}$$

**NCERT Benchmark: Valence Factor ($n$-factor) Rules**

- **Acids:** $n$-factor = basicity (number of replaceable $H^+$ ions): HCl $\implies$ 1; $H_2SO_4$ $\implies$ 2; $H_3PO_4$ $\implies$ 3; $H_3PO_3$ $\implies$ 2 (one P – H bond is non-ionizable!); $H_3PO_2$ $\implies$ 1 (two P – H bonds are non-ionizable!).
- **Bases:** $n$-factor = acidity (number of replaceable $OH^-$ ions): NaOH $\implies$ 1; $Ca(OH)_2$ $\implies$ 2; $Al(OH)_3$ $\implies$ 3.
- **Salts:** $n$-factor = total positive or negative charge: NaCl $\implies$ 1;

$CaCl_2 \implies 2$; $Al_2(SO_4)_3 \implies 6$.

- **Redox Agents:** $n$-factor = total change in oxidation state per mole of reagent.

**Common Pitfall: STP Molar Volume & Equivalent Mass of Oxyacids**

- **STP Standard Conditions Shift:** In modern IUPAC / NCERT convention, standard temperature and pressure (STP) is defined at $T = 273.15\,\text{K}$ and $P = 1\,\text{bar} = 10^5\,\text{Pa}$. Under these modern conditions, the molar volume of an ideal gas is $V_\text{m} = 22.7\,\text{L}\cdot\text{mol}^{-1}$ (or $22.7\,\text{dm}^3\cdot\text{mol}^{-1}$). The legacy convention ($1\,\text{atm} = 1.01325 \times 10^5\,\text{Pa}$) yielded $22.4\,\text{L}\cdot\text{mol}^{-1}$. Always inspect whether numerical question options in NEET assume legacy (22.4 L) or modern IUPAC (22.7 L).
- **Polyprotic Phosphorus Oxyacids:** Never equate molecular mass divided by 3 to equivalent mass blindly! For orthophosphoric acid ($H_3PO_4$), basicity is $3 \implies E = M/3$. For phosphorous acid ($H_3PO_3$), basicity is $2 \implies E = M/2$ because one hydrogen is directly bonded to phosphorus (P – H) and does not ionize in aqueous solution. For hypophosphorous acid ($H_3PO_2$), basicity is $1 \implies E = M/1$ since two hydrogens are non-ionizable P – H bonds.

## 1.3 Atomic Structure & Wave Mechanics (Unit 2)

### 1.3.1 Dual Nature of Light & The Photoelectric Effect

Einstein formulated the photoelectric effect by treating light as discrete localized wave packets (photons) of energy $E = h\nu$. When a photon strikes a metallic surface:

$$E_\text{photon} = \Phi + K_\text{max} \implies h\nu = h\nu_0 + \frac{1}{2}m_e v_\text{max}^2 = h\nu_0 + eV_0 \tag{1.15}$$

where $\Phi = h\nu_0$ is the work function of the metal, $\nu_0$ is the threshold frequency, and $V_0$ is the stopping potential.

### 1.3.2 Bohr's Model of the Hydrogen Atom: Complete Mathematical Derivations

Niels Bohr synthesized classical electrostatic planetary mechanics with Planckian angular momentum quantization for hydrogen-like species of nuclear

charge $+Ze$ and single orbital electron $-e$:

1. **Postulate 1 (Electrostatic Centripetal Equilibrium):**

$$\frac{m_e v_n^2}{r_n} = \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r_n^2} \implies m_e v_n^2 r_n = \frac{Ze^2}{4\pi\varepsilon_0} \tag{1.16}$$

2. **Postulate 2 (Quantization of Orbital Angular Momentum):**

$$L_n = m_e v_n r_n = \frac{nh}{2\pi} \quad (n = 1, 2, 3, \dots) \tag{1.17}$$

### Derivation of Quantized Orbit Radius ($r_n$)

Dividing Equation 1.17 by $m_e r_n$ yields the electron orbital velocity:

$$v_n = \frac{nh}{2\pi m_e r_n} \tag{1.18}$$

Substituting this expression for $v_n$ into Equation 1.16:

$$m_e \left(\frac{nh}{2\pi m_e r_n}\right)^2 r_n = \frac{Ze^2}{4\pi\varepsilon_0} \implies \frac{n^2h^2}{4\pi^2 m_e r_n} = \frac{Ze^2}{4\pi\varepsilon_0} \tag{1.19}$$

Solving explicitly for $r_n$:

$$r_n = \frac{\varepsilon_0 h^2}{\pi m_e e^2}\frac{n^2}{Z} = a_0\frac{n^2}{Z} \approx 0.529\,\text{Å} \times \frac{n^2}{Z} = 52.9\,\text{pm} \times \frac{n^2}{Z} \tag{1.20}$$

where $a_0 = 0.529177\,\text{Å}$ is the first Bohr radius of hydrogen ($n = 1, Z = 1$).

### Derivation of Quantized Velocity ($v_n$)

Substituting $r_n$ into Equation 1.17:

$$v_n = \frac{nh}{2\pi m_e \left(\frac{\varepsilon_0 h^2 n^2}{\pi m_e e^2 Z}\right)} = \frac{e^2}{2\varepsilon_0 h}\frac{Z}{n} = v_0\frac{Z}{n} \approx 2.188 \times 10^6 \frac{Z}{n}\,\text{m s}^{-1} = \frac{c}{137}\frac{Z}{n} \tag{1.21}$$

### Derivation of Quantized Energies ($K_n, U_n, E_n$)

$$K_n = \frac{1}{2}m_e v_n^2 = \frac{1}{2}\left(\frac{Ze^2}{4\pi\varepsilon_0 r_n}\right) = \frac{Ze^2}{8\pi\varepsilon_0 r_n} \tag{1.22}$$

$$U_n = -\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r_n} = -2K_n \quad \text{(Virial Theorem: } 2\langle K\rangle + \langle U\rangle = 0) \tag{1.23}$$

$$E_n = K_n + U_n = -K_n = \frac{1}{2}U_n = -\frac{Ze^2}{8\pi\varepsilon_0 r_n} \tag{1.24}$$

Substituting the derived value of $r_n$:

$$E_n = -\frac{m_e e^4}{8\varepsilon_0^2 h^2}\frac{Z^2}{n^2} = -R_H hc\frac{Z^2}{n^2} = -2.18\times10^{-18}\,\text{J}\times\frac{Z^2}{n^2} = -13.6\frac{Z^2}{n^2}\,\text{eV} \tag{1.25}$$

### Spectral Transitions & The Rydberg Equation

When an electron drops from an initial upper energy level $n_2$ to a lower level $n_1$ ($n_2 > n_1$):

$$\Delta E = E_{n_2} - E_{n_1} = h\nu = \frac{hc}{\lambda} \implies \bar{\nu} = \frac{1}{\lambda} = R_H Z^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \tag{1.26}$$

where $R_H = \frac{m_e e^4}{8\varepsilon_0^2 h^3 c} \approx 1.09677\times10^7\,\text{m}^{-1}$ ($1/R_H \approx 912\,\text{Å}$).

| Spectral Series | $n_1$ | $n_2$ | Spectral Region | Wavelength Range |
|---|---|---|---|---|
| **Lyman** | 1 | $2,3,4,\ldots,\infty$ | Ultraviolet (UV) | $912\,\text{Å} \le \lambda \le 1216\,\text{Å}$ |
| **Balmer** | 2 | $3,4,5,\ldots,\infty$ | Visible ($H_\alpha$ to $H_\delta$) + Near UV | $3646\,\text{Å} \le \lambda \le 6563\,\text{Å}$ |
| **Paschen** | 3 | $4,5,6,\ldots,\infty$ | Near Infrared (IR) | $8204\,\text{Å} \le \lambda \le 18751\,\text{Å}$ |
| **Brackett** | 4 | $5,6,7,\ldots,\infty$ | Infrared (IR) | $14580\,\text{Å} \le \lambda \le 40512\,\text{Å}$ |
| **Pfund** | 5 | $6,7,8,\ldots,\infty$ | Far Infrared (Far IR) | $22790\,\text{Å} \le \lambda \le 74580\,\text{Å}$ |

Table 1.2: Hydrogen Atom Emission Spectral Series.

## 1.3.3 Wave-Particle Duality: The de Broglie Relation

Louis de Broglie (1924) postulated that matter exhibits intrinsic wave characteristics. Equating Einstein's mass-energy equivalence ($E = mc^2 = pc$) with Planck's photon energy ($E = h\nu = hc/\lambda$):

$$p = \frac{h}{\lambda} \implies \lambda = \frac{h}{p} = \frac{h}{mv} \tag{1.27}$$

For a charged particle of mass $m$ and charge $q$ accelerated from rest through an electrostatic potential difference $V$:

$$K = qV = \frac{p^2}{2m} \implies p = \sqrt{2mqV} \implies \lambda = \frac{h}{\sqrt{2mqV}} \tag{1.28}$$

**Specialized de Broglie Wavelength Formulas:**

- **Accelerated Electron ($m = m_e, q = e$):** $\lambda_e = \frac{1.227}{\sqrt{V}}\,\text{nm} = \frac{12.27}{\sqrt{V}}\,\text{Å}$.
- **Thermal Neutron / Gas Molecule ($K = \frac{3}{2}k_B T$):** $\lambda_{\text{thermal}} = \frac{h}{\sqrt{3mk_B T}}$.

### 1.3.4 Heisenberg's Uncertainty Principle

Werner Heisenberg (1927) proved that it is physically impossible to simultaneously determine both the exact position and exact momentum of a microscopic quantum entity:

$$\Delta x \cdot \Delta p_x \geq \frac{\hbar}{2} = \frac{h}{4\pi} \implies \Delta x \cdot (m\Delta v_x) \geq \frac{h}{4\pi} \implies \Delta x \cdot \Delta v_x \geq \frac{h}{4\pi m} \quad (1.29)$$

**Alternative Conjugate Variables Formulations:**

$$\Delta E \cdot \Delta t \geq \frac{h}{4\pi}, \qquad \Delta\theta \cdot \Delta L \geq \frac{h}{4\pi} \quad (1.30)$$

Because Planck's constant $h = 6.626 \times 10^{-34}\,\mathrm{J\,s}$ is infinitesimally small, the uncertainty product is completely imperceptible for macroscopic bodies ($m \sim 1\,\mathrm{kg}$), but dominates the atomic regime ($m_e \sim 9.1 \times 10^{-31}\,\mathrm{kg}$).

### 1.3.5 Schrödinger Wave Equation & Quantum Mechanical Formalism

Erwin Schrödinger (1926) formulated the stationary-state wave equation for a quantum particle of mass $m$ moving in a potential field $V(x, y, z)$:

$$\nabla^2\psi + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0 \iff \hat{H}\psi = E\psi \quad (1.31)$$

where $\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V$ is the Hamiltonian operator, and $\psi(r, \theta, \phi)$ is the eigenfunction.

**Physical Interpretation of Wave Function $\psi$ and Probability Density $|\psi|^2$**

- **Wave Function Amplitude ($\psi$):** $\psi$ possesses no direct physical meaning; it can be positive, negative, or complex.
- **Born's Probability Density ($|\psi|^2 = \psi^*\psi$):** Represents the probability of finding the electron per unit volume in an infinitesimal volume element $d\tau = dx\,dy\,dz$:

$$dP = |\psi|^2 d\tau, \qquad \int_{\text{all space}} |\psi|^2 d\tau = 1 \quad \text{(Normalization Condition)} \quad (1.32)$$

- **Radial vs Angular Separation:** In spherical coordinates $(r, \theta, \phi)$:

$$\psi(r, \theta, \phi) = R_{n,l}(r) \times Y_{l,m_l}(\theta, \phi) \quad (1.33)$$

$R_{n,l}(r)$ (radial function) dictates orbital size and radial probability; $Y_{l,m_l}(\theta, \phi)$ (angular function) dictates orbital 3D spatial geometry and orientation.

- **Radial Probability Distribution Function ($4\pi r^2 R^2(r)$):** Represents the absolute probability of locating the electron in a spherical shell of radius $r$ and thickness $dr$:

$$P(r)dr = 4\pi r^2 R^2(r)dr \tag{1.34}$$

At $r = 0$, $4\pi r^2 R^2 = 0$ for all orbitals (electron cannot reside at the mathematical point of the nucleus).

**Master Nodal Formulations**

Nodes are spatial domains where the probability density $|\psi|^2$ vanishes identically:

$$\text{Radial Nodes (Spherical Nodes)} = n - l - 1 \tag{1.35}$$

$$\text{Angular Nodes (Nodal Planes / Cones)} = l \tag{1.36}$$

$$\text{Total Nodes} = (n - l - 1) + l = n - 1 \tag{1.37}$$

**Examples:**

- $1s$: Radial $= 1 - 0 - 1 = 0$, Angular $= 0$, Total $= 0$.
- $2s$: Radial $= 2 - 0 - 1 = 1$, Angular $= 0$, Total $= 1$.
- $2p$: Radial $= 2 - 1 - 1 = 0$, Angular $= 1$ (the planar surface passing through nucleus), Total $= 1$.
- $3d$: Radial $= 3 - 2 - 1 = 0$, Angular $= 2$ (two nodal planes/cones), Total $= 2$.
- $4d$: Radial $= 4 - 2 - 1 = 1$, Angular $= 2$, Total $= 3$.

### 1.3.6 Quantum Numbers & Rules of Electron Configuration

An atomic electron is completely described by four quantum numbers:

1. **Principal Quantum Number ($n \in \{1, 2, 3, \dots\}$):** Determines main energy shell, average distance from nucleus, maximum electrons $= 2n^2$, maximum orbitals $= n^2$.
2. **Azimuthal Quantum Number ($l \in \{0, 1, 2, \dots, n-1\}$):** Determines subshell shape ($s : 0, p : 1, d : 2, f : 3$), and orbital angular momentum:

$$L_{\text{orbital}} = \sqrt{l(l+1)}\hbar = \sqrt{l(l+1)}\frac{h}{2\pi} \quad (\text{For any } s\text{-orbital}, l = 0 \implies L = 0!) \tag{1.38}$$

3. **Magnetic Quantum Number ($m_l \in \{-l, \ldots, 0, \ldots, +l\}$):** Determines 3D spatial orientation of the orbital in an external magnetic field ($2l+1$ degenerate orientations).
4. **Spin Quantum Number ($m_s \in \{+\frac{1}{2}, -\frac{1}{2}\}$):** Intrinsic angular momentum orientation. Spin-only magnetic moment:

$$\mu_s = \sqrt{n_u(n_u+2)}\,\text{BM} \quad (n_u = \text{number of unpaired electrons, BM} = \text{Bohr Magne} \tag{1.39}$$

**Fundamental Rules of Electronic Architecture**

- **Aufbau Principle:** In the ground state of an atom, electrons occupy the lowest available atomic energy orbitals first. Energy follows the $(n+l)$ **rule**:
  1. Lower $(n+l)$ orbital fills before higher $(n+l)$ (e.g., $4s : 4+0=4$ fills before $3d : 3+2=5$).
  2. If two orbitals share identical $(n+l)$, the orbital with lower $n$ fills first (e.g., $3d : 3+2=5$ fills before $4p : 4+1=5$).
- **Pauli's Exclusion Principle:** No two electrons in an atom can have the identical set of all four quantum numbers. Consequently, an individual orbital accommodates at most two electrons with antiparallel spins ($\uparrow\downarrow$).
- **Hund's Rule of Maximum Multiplicity:** Pairing of electrons in degenerate orbitals $(p, d, f)$ cannot occur until each orbital of the subshell is singly occupied with parallel spins.
- **Extra Stability of Half-Filled and Fully-Filled Subshells:**
  - **Symmetrical Electron Distribution:** Spherically symmetric charge minimizes interelectronic repulsions.
  - **Exchange Energy:** Quantum mechanical exchange of electrons with parallel spins releases stabilization energy. For $n$ electrons of parallel spin:

$$\text{Number of possible exchanges} = \frac{n(n-1)}{2} \tag{1.40}$$

For Chromium ($[\text{Ar}]3d^5 4s^1$): $3d^5$ allows $\frac{5\times4}{2} = 10$ exchanges, vs $[\text{Ar}]3d^4 4s^2$ which allows only $\frac{4\times3}{2} = 6$ exchanges. Copper ($[\text{Ar}]3d^{10}4s^1$) similarly gains maximum exchange stability over $3d^9 4s^2$.

## 1.4 Chemical Thermodynamics & Thermochemistry (Unit 4)

### 1.4.1 Fundamental Concepts: Systems, Variables & Functions

- **Thermodynamic System:** That part of the universe selected for thermodynamic study.
  - **Open System:** Exchanges both matter and energy with surroundings (e.g., open beaker of boiling water).
  - **Closed System:** Exchanges energy but not matter with surroundings (e.g., sealed metal canister).
  - **Isolated System:** Exchanges neither matter nor energy with surroundings (e.g., perfectly insulated Dewar flask).
- **State Functions vs Path Functions:**
  - **State Functions:** Properties whose values depend solely upon the current thermodynamic state of the system, independent of the path traversed ($U, H, S, G, P, V, T$). Cyclic integral $\oint dX = 0$.
  - **Path Functions:** Quantities whose numerical values depend explicitly upon the transition path traversed between states ($q$ = heat, $w$ = work). Cyclic integral $\oint dq \neq 0$.
- **Intensive vs Extensive Properties:**
  - **Extensive (additive, depends on mass):** Mass ($m$), volume ($V$), internal energy ($U$), enthalpy ($H$), entropy ($S$), Gibbs energy ($G$), heat capacity ($C$).
  - **Intensive (scale-independent, independent of mass):** Temperature ($T$), pressure ($P$), density ($\rho$), molar heat capacity ($C_m$), specific heat ($c$), viscosity ($\eta$), surface tension, refractive index, EMF of a cell ($E_{\text{cell}}$).

### 1.4.2 First Law of Thermodynamics & Expansion Work Derivations

The First Law is the conservation of energy principle applied to thermodynamic systems:

$$\Delta U = q + w \tag{1.41}$$

where $\Delta U = U_2 - U_1$ is the change in internal energy, $q$ is the heat absorbed by the system ($q > 0$ if endothermic, $q < 0$ if exothermic), and $w$ is work done on the system ($w > 0$ for compression, $w < 0$ for expansion under IUPAC sign convention).

#### Derivation of Reversible Isothermal Expansion Work of an Ideal Gas

For an infinitesimal quasi-static reversible expansion, the internal gas pressure $P_{\text{gas}}$ differs from the external opposing pressure $P_{\text{ext}}$ by an infinitesimal

amount $dP$ ($P_{\text{ext}} = P - dP$):

$$\delta w_{\text{rev}} = -P_{\text{ext}} dV = -(P - dP)dV = -PdV + dPdV \approx -PdV \tag{1.42}$$

Integrating from initial state $(P_1, V_1)$ to final state $(P_2, V_2)$ at constant temperature $T$:

$$w_{\text{rev}} = -\int_{V_1}^{V_2} PdV = -\int_{V_1}^{V_2} \frac{nRT}{V} dV = -nRT \ln\left(\frac{V_2}{V_1}\right) = -2.303nRT \log_{10}\left(\frac{V_2}{V_1}\right) \tag{1.43}$$

By Boyle's Law ($P_1V_1 = P_2V_2$ at constant $T$):

$$w_{\text{rev}} = -2.303nRT \log_{10}\left(\frac{P_1}{P_2}\right) \tag{1.44}$$

Since temperature is constant in an isothermal ideal gas expansion ($\Delta U = 0$):

$$q_{\text{rev}} = -w_{\text{rev}} = +2.303nRT \log_{10}\left(\frac{V_2}{V_1}\right) \tag{1.45}$$

#### Irreversible Expansion & Free Expansion

- **Irreversible Expansion against Constant External Pressure ($P_{\text{ext}}$):**

$$w_{\text{irr}} = -\int_{V_1}^{V_2} P_{\text{ext}} dV = -P_{\text{ext}}(V_2 - V_1) = -P_{\text{ext}}\left(\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1}\right) \tag{1.46}$$

  Because $|w_{\text{rev}}| > |w_{\text{irr}}|$, reversible expansion produces the **maximum work output**.

- **Free Expansion into Vacuum ($P_{\text{ext}} = 0$):**

$$w_{\text{free}} = -\int 0 \cdot dV = 0 \tag{1.47}$$

  For an ideal gas in an insulated container undergoing free expansion: $w = 0, q = 0 \implies \Delta U = 0 \implies \Delta T = 0$ (Joule's Law).

### 1.4.3 Enthalpy, Heat Capacities & Mayer's Relation Derivation

Enthalpy is defined as the total heat content:

$$H = U + PV \implies \Delta H = \Delta U + \Delta(PV) \tag{1.48}$$

For a chemical reaction involving gases at constant temperature:

$$\Delta H = \Delta U + \Delta n_g RT \tag{1.49}$$

where $\Delta n_g = \sum n_g(\text{gaseous products}) - \sum n_g(\text{gaseous reactants})$.

**Derivation of Mayer's Relation: $C_p - C_v = R$**

Heat capacities are defined by:

$$C_v = \left(\frac{\partial U}{\partial T}\right)_V = \frac{q_V}{\Delta T}, \qquad C_p = \left(\frac{\partial H}{\partial T}\right)_P = \frac{q_P}{\Delta T} \tag{1.50}$$

For 1 mole of an ideal gas:

$$H = U + PV = U + RT \tag{1.51}$$

Differentiating both sides with respect to temperature $T$:

$$\frac{dH}{dT} = \frac{dU}{dT} + R\frac{d(T)}{dT} \implies C_p = C_v + R \implies C_p - C_v = R \tag{1.52}$$

For $n$ moles: $C_{p,\text{total}} - C_{v,\text{total}} = nR$. **Heat Capacity Ratio ($\gamma$):**

$$\gamma = \frac{C_p}{C_v} = 1 + \frac{R}{C_v} \tag{1.53}$$

- **Monoatomic Gas ($\text{He}, \text{Ne}, \text{Ar}$):** Degrees of freedom $f = 3 \implies C_v = \frac{3}{2}R, C_p = \frac{5}{2}R \implies \gamma = \frac{5}{3} \approx 1.67$.
- **Diatomic Gas ($\text{O}_2, \text{N}_2, \text{CO}$ at moderate $T$):** $f = 5 \implies C_v = \frac{5}{2}R, C_p = \frac{7}{2}R \implies \gamma = \frac{7}{5} = 1.40$.
- **Non-linear Polyatomic Gas ($\text{H}_2\text{O}, \text{NH}_3, \text{CH}_4$):** $f = 6 \implies C_v = 3R, C_p = 4R \implies \gamma = \frac{4}{3} \approx 1.33$.

### 1.4.4 Reversible Adiabatic Expansion of an Ideal Gas

In an adiabatic process, no heat enters or leaves the system ($q = 0$). By the First Law:

$$dU = \delta w \implies nC_v dT = -PdV = -\frac{nRT}{V}dV \tag{1.54}$$

Dividing by $nT$ and separating variables:

$$C_v\frac{dT}{T} = -R\frac{dV}{V} = -(C_p - C_v)\frac{dV}{V} \implies \frac{dT}{T} = -(\gamma - 1)\frac{dV}{V} \tag{1.55}$$

Integrating from state 1 to state 2:

$$\ln\left(\frac{T_2}{T_1}\right) = -(\gamma-1)\ln\left(\frac{V_2}{V_1}\right) = \ln\left[\left(\frac{V_1}{V_2}\right)^{\gamma-1}\right] \implies T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1} = \text{constant} \tag{1.56}$$

Substituting $T = \frac{PV}{nR}$:

$$P_1V_1^\gamma = P_2V_2^\gamma = \text{constant}, \qquad T_1^\gamma P_1^{1-\gamma} = T_2^\gamma P_2^{1-\gamma} = \text{constant} \tag{1.57}$$

**Work Done in Reversible Adiabatic Expansion:**

$$w_{\text{adi}} = \Delta U = nC_v(T_2 - T_1) = \frac{nR(T_2 - T_1)}{\gamma - 1} = \frac{P_2V_2 - P_1V_1}{\gamma - 1} \tag{1.58}$$

Since the gas expands ($V_2 > V_1$), $T_2 < T_1$ (gas cools during adiabatic expansion!). On a $P - V$ diagram, the slope of an adiabat is $\gamma$ times steeper than an isotherm:

$$\left(\frac{dP}{dV}\right)_{\text{adi}} = -\gamma\frac{P}{V} = \gamma\left(\frac{dP}{dV}\right)_{\text{iso}} \tag{1.59}$$

### 1.4.5 Thermochemistry: Heats of Reactions, Hess's Law & Cycles

**Thermochemical Laws & Enthalpy Formulations**

- **Hess's Law of Constant Heat Summation:** If a chemical reaction can be made to take place in a single step or in several successive steps, the overall enthalpy change is identical, independent of the pathway:

$$\Delta_r H^\circ = \Delta H_1 + \Delta H_2 + \Delta H_3 + \dots \tag{1.60}$$

- **Standard Enthalpy of Formation ($\Delta_f H^\circ$):** Enthalpy change when 1 mole of a compound is synthesized from its constituent elements in their standard reference states (298.15 K, 1 bar).

$$\Delta_r H^\circ = \sum \nu_p \Delta_f H^\circ(\text{products}) - \sum \nu_r \Delta_f H^\circ(\text{reactants}) \tag{1.61}$$

- **Standard State Convention Trap:** By definition, $\Delta_f H^\circ = 0$ for all pure elements in their most stable allotropic form at 298 K: C(graphite) = 0 (but C(diamond) = $+1.89\,\text{kJ mol}^{-1}$); S(rhombic) = 0 (monoclinic $\neq 0$); P(white) = 0 (red $\neq 0$); $\text{Br}_{2(\text{l})} = 0$; $\text{I}_{2(\text{s})} = 0$.
- **Bond Dissociation Enthalpy (BE):**

$$\Delta_r H^\circ = \sum \text{BE}(\text{reactants}) - \sum \text{BE}(\text{products}) \quad \text{(all species in gas phase!)} \tag{1.62}$$

- **Kirchhoff's Law (Temperature Dependence of $\Delta_r H$):**

$$\Delta_r H^\circ_{T_2} = \Delta_r H^\circ_{T_1} + \int_{T_1}^{T_2} \Delta C_p dT = \Delta_r H^\circ_{T_1} + \Delta C_p(T_2 - T_1) \tag{1.63}$$

where $\Delta C_p = \sum \nu_p C_{p,\text{products}} - \sum \nu_r C_{p,\text{reactants}}$.

**Born-Haber Cycle for Ionic Crystal Lattice Enthalpy ($\Delta_{\text{lattice}}H^\circ$)**

For crystalline sodium chloride ($NaCl_{(s)}$), the standard enthalpy of formation from elemental sodium and chlorine gas connects directly to lattice enthalpy via five thermochemical steps:

$$Na_{(s)} + \frac{1}{2}Cl_{2(g)} \xrightarrow{\Delta_f H^\circ} NaCl_{(s)} \tag{1.64}$$

1. Sublimation of solid sodium: $Na_{(s)} \rightarrow Na_{(g)}$, $\Delta H^\circ_{\text{sub}} = +108.4\,\text{kJ mol}^{-1}$.
2. Ionization of gaseous sodium: $Na_{(g)} \rightarrow Na^+_{(g)} + e^-$, $\text{IE}_1 = +495.8\,\text{kJ mol}^{-1}$.
3. Dissociation of diatomic chlorine: $\frac{1}{2}Cl_{2(g)} \rightarrow Cl_{(g)}$, $\frac{1}{2}\Delta_{\text{diss}}H^\circ = +\frac{1}{2}(242) = +121.0\,\text{kJ mol}^{-1}$.
4. Electron gain by chlorine: $Cl_{(g)} + e^- \rightarrow Cl^-_{(g)}$, $\Delta_{\text{eg}}H^\circ = -348.6\,\text{kJ mol}^{-1}$.
5. Condensation of gaseous ions into crystal lattice: $Na^+_{(g)} + Cl^-_{(g)} \rightarrow NaCl_{(s)}$, $-\Delta_{\text{lattice}}H^\circ$.

Applying Hess's Law:

$$\Delta_f H^\circ = \Delta H^\circ_{\text{sub}} + \text{IE}_1 + \frac{1}{2}\Delta_{\text{diss}}H^\circ + \Delta_{\text{eg}}H^\circ - \Delta_{\text{lattice}}H^\circ \tag{1.65}$$

$$\Delta_{\text{lattice}}H^\circ = \Delta H^\circ_{\text{sub}} + \text{IE}_1 + \frac{1}{2}\Delta_{\text{diss}}H^\circ + \Delta_{\text{eg}}H^\circ - \Delta_f H^\circ = +788\,\text{kJ mol}^{-1} \tag{1.66}$$

### 1.4.6 Second & Third Laws: Entropy, Free Energy & Spontaneity

- **Entropy ($S$):** Measure of microscopic disorder or thermal energy dispersal. Infinitesimal reversible change:

$$dS = \frac{\delta q_{\text{rev}}}{T} \tag{1.67}$$

- **Entropy Change for an Ideal Gas:**

$$\Delta S = nC_v \ln\left(\frac{T_2}{T_1}\right) + nR\ln\left(\frac{V_2}{V_1}\right) = nC_p \ln\left(\frac{T_2}{T_1}\right) - nR\ln\left(\frac{P_2}{P_1}\right) \tag{1.68}$$

  - **Isothermal Process ($T_1 = T_2$):** $\Delta S = nR\ln(V_2/V_1) = nR\ln(P_1/P_2)$.
  - **Isobaric Process ($P_1 = P_2$):** $\Delta S = nC_p\ln(T_2/T_1)$.
  - **Isochoric Process ($V_1 = V_2$):** $\Delta S = nC_v\ln(T_2/T_1)$.

  - **Reversible Phase Transitions:** $\Delta S_{\text{fusion}} = \frac{\Delta H_{\text{fus}}}{T_f}$, $\Delta S_{\text{vaporization}} = \frac{\Delta H_{\text{vap}}}{T_b}$.
- **Spontaneity Criterion (Second Law of Thermodynamics):** For an isolated system or the universe:

$$\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} \geq 0 \quad (\text{Spontaneous if } > 0; \text{Equilibrium if } = 0 \tag{1.69}$$

- **Third Law of Thermodynamics:** The entropy of a perfectly crystalline pure substance approaches zero as absolute temperature approaches zero: $\lim_{T \to 0} S = 0$.

### 1.4.7 Gibbs Free Energy & Equilibrium Thermodynamics

Gibbs Free Energy ($G = H - TS$) provides the single-parameter spontaneity criterion at constant temperature and pressure:

$$\Delta G = \Delta H - T\Delta S \tag{1.70}$$

- $\Delta G < 0 \implies$ Reaction is spontaneous in the forward direction.
- $\Delta G = 0 \implies$ Dynamic equilibrium is attained.
- $\Delta G > 0 \implies$ Non-spontaneous forward (spontaneous in reverse direction).

| $\Delta H$ | $\Delta S$ | $\Delta G = \Delta H - T\Delta S$ | **Spontaneity Regime & Temperature Dependence** |
|---|---|---|---|
| $-$ | $+$ | Always Negative ($-$) | **Spontaneous at all temperatures** (enthalpy & entropy driven) |
| $+$ | $-$ | Always Positive ($+$) | **Non-spontaneous at all temperatures** (reverse is spontaneous) |
| $-$ | $-$ | Negative at low $T$, Positive at high $T$ | **Spontaneous at low $T$**, non-spontaneous at high $T$ ($\lvert T\Delta S\rvert > \lvert\Delta H\rvert$) |
| $+$ | $+$ | Positive at low $T$, Negative at high $T$ | **Spontaneous at high $T$**, non-spontaneous at low $T$ ($T > \frac{\Delta H}{\Delta S}$) |

Table 1.3: Thermodynamic Spontaneity Temperature Dependency Matrix.

**Master Thermodynamic Equilibrium Links**

For a general reaction with reaction quotient $Q$:

$$\Delta G = \Delta_r G^\circ + RT \ln Q = \Delta_r G^\circ + 2.303 RT \log_{10} Q \tag{1.71}$$

At dynamic chemical equilibrium, $\Delta G = 0$ and $Q = K_{\text{eq}}$:

$$\Delta_r G^\circ = -RT \ln K_{\text{eq}} = -2.303 RT \log_{10} K_{\text{eq}} \iff K_{\text{eq}} = e^{-\Delta_r G^\circ / RT} \tag{1.72}$$

**van 't Hoff Reaction Isobar:**

$$\frac{d \ln K}{dT} = \frac{\Delta_r H^\circ}{RT^2} \implies \log_{10}\left(\frac{K_2}{K_1}\right) = \frac{\Delta_r H^\circ}{2.303 R}\left[\frac{T_2 - T_1}{T_1 T_2}\right] = \frac{\Delta_r H^\circ}{2.303 R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right] \tag{1.73}$$

- **Endothermic Reaction ($\Delta_r H^\circ > 0$):** $K$ increases as temperature increases.
- **Exothermic Reaction ($\Delta_r H^\circ < 0$):** $K$ decreases as temperature increases.

Chapter 2

# Chemical & Ionic Equilibrium

Chemical equilibrium establishes the dynamic balance between forward and reverse reaction rates in closed thermodynamic systems, governed by the law of mass action and Le Chatelier's principle. Ionic equilibrium extends these principles to electrolytic solutions, encompassing acid-base dissociation equilibria, the autoionization of water, buffer solution dynamics via the Henderson-Hasselbalch formulation, quantitative salt hydrolysis, and sparingly soluble salt precipitation criteria governed by the solubility product ($K_{sp}$).

## 2.1 Chemical & Ionic Equilibrium (Unit 6)

### 2.1.1 Chemical Equilibrium: Law of Mass Action & Constant Relations

For a reversible homogeneous gas-phase reaction:

$$aA(g) + bB(g) \rightleftharpoons cC(g) + dD(g) \tag{2.1}$$

The equilibrium constants in terms of molar concentrations ($K_c$) and partial pressures ($K_p$) are:

$$K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}, \qquad K_p = \frac{P_C^c P_D^d}{P_A^a P_B^b} \tag{2.2}$$

**Derivation of $K_p = K_c(RT)^{\Delta n_g}$**

Assuming ideal gas behavior for each component:

$$P_i V = n_i RT \implies P_i = \left(\frac{n_i}{V}\right) RT = [i]RT \tag{2.3}$$

Substituting each partial pressure into the $K_p$ expression:

$$K_p = \frac{([C]RT)^c([D]RT)^d}{([A]RT)^a([B]RT)^b} = \frac{[C]^c[D]^d}{[A]^a[B]^b} \times \frac{(RT)^{c+d}}{(RT)^{a+b}} = K_c(RT)^{(c+d)-(a+b)} \tag{2.4}$$

$$K_p = K_c(RT)^{\Delta n_g} \tag{2.5}$$

where $\Delta n_g = (c+d) - (a+b)$.

- If $\Delta n_g = 0 \implies K_p = K_c$ (dimensionless, pressure independent, e.g., $H_2 + I_2 \rightleftharpoons 2HI$).
- If $\Delta n_g > 0 \implies K_p > K_c$ when $RT > 1$ (units $\text{atm}^{\Delta n_g}$ or $\text{bar}^{\Delta n_g}$, e.g., $PCl_5 \rightleftharpoons PCl_3 + Cl_2$).
- If $\Delta n_g < 0 \implies K_p < K_c$ when $RT > 1$ (e.g., $N_2 + 3H_2 \rightleftharpoons 2NH_3$).

### 2.1.2 Le Chatelier's Principle & Equilibrium Shifts

If a dynamic equilibrium is subjected to a stress in concentration, pressure, volume, temperature, or inert gas, the system shifts in that direction which minimizes or undoes the applied constraint:

- **Concentration:** Adding reactants or removing products shifts forward ($Q < K$).
- **Pressure / Volume:** Increasing pressure (compressing volume) shifts equilibrium toward fewer gaseous moles ($\Delta n_g < 0$).
- **Temperature:** Increasing temperature shifts toward the endothermic direction ($\Delta H > 0$).
- **Inert Gas Addition:**
  - **At Constant Volume:** Total pressure rises, but partial pressures and molar concentrations of reacting gases remain **strictly unchanged** $\implies$ **NO EFFECT** on equilibrium state!
  - **At Constant Pressure:** Volume expands, partial pressures drop $\implies$ equilibrium shifts toward **greater number of gaseous moles** ($\Delta n_g > 0$).
- **Catalyst:** Increases forward and backward rates equally; equilibrium constant $K$ and composition are completely unaffected (shortens time to achieve equilibrium).

**NCERT Focus: Le Chatelier Optimization in Industrial Syntheses**

**Case Study: Synthesis of Ammonia (Haber-Bosch Process)**

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g), \quad \Delta_r H^\circ = -92.4\,\text{kJ}\cdot\text{mol}^{-1}, \quad \Delta n_g = 2-4 = -2$

- **High Pressure ($\approx 200$ atm):** Since $\Delta n_g < 0$, high pressure drives

the equilibrium forward toward product formation, optimizing ammonia yield.

- **Optimum Temperature ($\approx$ 700–775 K):** Because the reaction is exothermic ($\Delta_r H^\circ < 0$), low temperature thermodynamically favors ammonia, but drastically lowers kinetic reaction rates. An optimum temperature with iron oxide catalyst and promoters ($K_2O$ and $Al_2O_3$) is maintained.
- **Inert Gas Injection Trap:** Adding argon or neon at constant volume ($V = \text{const}$) causes zero shift in equilibrium yield. Only addition at constant pressure ($P = \text{const}$) shifts the reaction backward (toward more gaseous moles, $\Delta n_g = -2 < 0 \implies$ leftward shift).

### 2.1.3 Degree of Dissociation from Vapour Density

For a gas dissociating according to $A(g) \rightleftharpoons nB(g)$:

$$\alpha = \frac{D - d}{(n-1)d} = \frac{M_{\text{theoretical}} - M_{\text{observed}}}{(n-1)M_{\text{observed}}} \tag{2.6}$$

where $D$ is the theoretical vapour density of pure reactant, and $d$ is the observed equilibrium mixture vapour density.

### 2.1.4 Ionic Equilibrium: Ostwald Dilution Law & Autoionization of Water

- **Ostwald's Dilution Law for Weak Acid ($HA \rightleftharpoons H^+ + A^-$):**

$$K_a = \frac{C\alpha^2}{1-\alpha} \approx C\alpha^2 \quad (\text{for } \alpha \leq 0.05) \implies \alpha = \sqrt{\frac{K_a}{C}} \tag{2.7}$$

$$[H^+] = C\alpha = \sqrt{K_a C} \implies \text{pH} = \frac{1}{2}\left(\text{p}K_a - \log_{10} C\right) \tag{2.8}$$

- **Self-Ionization of Water ($2H_2O \rightleftharpoons H_3O^+ + OH^-$):**

$$K_w = [H^+][OH^-] = 1.0 \times 10^{-14} \quad (\text{at } 25^\circ\text{C}) \implies \text{pH} + \text{pOH} = \text{p}K_w = 14 \tag{2.9}$$

Because self-ionization is endothermic ($\Delta H > 0$), $K_w$ increases with temperature: at 80°C, $K_w \approx 10^{-13} \implies$ pH of neutral water $= \frac{13}{2} = 6.5$.

### 2.1.5 Salt Hydrolysis: Derivations of $K_h$, $h$, and pH

Salt hydrolysis is the reverse of neutralization, where ions of a salt interact with water to perturb $[H^+]$ and $[OH^-]$:

| Salt Category & Example | Hydrolysis Type | Hydrolysis Constant ($K_h$) | Degree of Hydrolysis ($h$) | Exact pH Formula |
|---|---|---|---|---|
| **Strong Acid + Strong Base** ($NaCl, KNO_3, NaClO_4$) | No Hydrolysis Spectator ions | — | $h = 0$ | pH = 7.00 (Neutral, unaffected by dilution) |
| **Strong Acid + Weak Base** ($NH_4Cl, FeSO_4, AlCl_3$) | Cationic | $K_h = \frac{K_w}{K_b}$ | $h = \sqrt{\frac{K_w}{K_b C}}$ | pH = $7 - \frac{1}{2}pK_b - \frac{1}{2}\log_{10} C$ (Acidic: pH < 7) |
| **Weak Acid + Strong Base** ($CH_3COONa, NaCN, Na_2CO_3$) | Anionic | $K_h = \frac{K_w}{K_a}$ | $h = \sqrt{\frac{K_w}{K_a C}}$ | pH = $7 + \frac{1}{2}pK_a + \frac{1}{2}\log_{10} C$ (Alkaline: pH > 7) |
| **Weak Acid + Weak Base** ($CH_3COONH_4, NH_4CN$) | Both Ions Hydrolyze | $K_h = \frac{K_w}{K_a K_b}$ | $h = \sqrt{\frac{K_w}{K_a K_b}}$ **(Independent of $C$!)** | pH = $\mathbf{7 + \frac{1}{2}pK}$ **(Strictly independent of $C$!)** |

Table 2.1: Master Classification and Analytical Formulations for Salt Hydrolysis.

### 2.1.6 Buffer Solutions & Henderson-Hasselbalch Derivations

A buffer resists pH changes upon addition of small quantities of strong acid or base.

1. **Acidic Buffer (Weak Acid + Salt with Strong Base, e.g.,** $CH_3COOH + CH_3$

$$K_a = \frac{[H^+][A^-]}{[HA]} \implies [H^+] = K_a \frac{[HA]}{[A^-]} \approx K_a \frac{[\text{Acid}]}{[\text{Salt}]} \tag{2.10}$$

Taking $-\log_{10}$ on both sides:

$$\text{pH} = \text{p}K_a + \log_{10}\left(\frac{[\text{Conjugate Base / Salt}]}{[\text{Weak Acid}]}\right) \tag{2.11}$$

2. **Basic Buffer (Weak Base + Salt with Strong Acid, e.g.,** $NH_4OH + NH_4Cl$**):**

$$\text{pOH} = \text{p}K_b + \log_{10}\left(\frac{[\text{Conjugate Acid / Salt}]}{[\text{Weak Base}]}\right), \qquad \text{pH} = 14 - \text{pOH} \tag{2.12}$$

3. **Buffer Capacity ($\phi$):** The number of moles of strong acid or base required to change the pH of 1 L of buffer solution by 1 unit:

$$\phi = \frac{dB}{d(\text{pH})} = 2.303 \frac{[\text{Acid}][\text{Salt}]}{[\text{Acid}] + [\text{Salt}]} \tag{2.13}$$

Buffer capacity is **maximum** when [Acid] = [Salt] $\implies$ pH = p$K_a$ (effective buffer range: p$K_a \pm 1$).

### 2.1.7 Solubility Product ($K_{sp}$) & Common Ion Precipitation

For a sparingly soluble salt $A_xB_y$:

$$A_xB_y(s) \rightleftharpoons xA^{y+}(aq) + yB^{x-}(aq) \tag{2.14}$$

If molar solubility is $S\,\text{mol}\,\text{L}^{-1}$:

$$[A^{y+}] = xS, \qquad [B^{x-}] = yS \tag{2.15}$$

$$K_{sp} = [A^{y+}]^x[B^{x-}]^y = (xS)^x(yS)^y = x^x y^y S^{x+y} \implies S = \left(\frac{K_{sp}}{x^x y^y}\right)^{\frac{1}{x+y}} \tag{2.16}$$

**Solubility Equations by Salt Stoichiometry:**

- $AB$ type ($AgCl, BaSO_4$): $K_{sp} = S^2 \implies S = \sqrt{K_{sp}}$.

- $AB_2$ or $A_2B$ type ($PbCl_2, Ag_2CrO_4, CaF_2$): $K_{sp} = 1^1 \times 2^2 \times S^3 = 4S^3 \implies S = \sqrt[3]{K_{sp}/4}$.
- $AB_3$ or $A_3B$ type ($Al(OH)_3, Fe(OH)_3$): $K_{sp} = 1^1 \times 3^3 \times S^4 = 27S^4 \implies S = \sqrt[4]{K_{sp}/27}$.
- $A_2B_3$ or $A_3B_2$ type ($As_2S_3, Ca_3(PO_4)_2$): $K_{sp} = 2^2 \times 3^3 \times S^5 = 108S^5 \implies S = \sqrt[5]{K_{sp}/108}$.

**Precipitation Criterion:**

- $Q_{sp} < K_{sp} \implies$ Unsaturated solution; no precipitate forms.
- $Q_{sp} = K_{sp} \implies$ Saturated solution at dynamic equilibrium.
- $Q_{sp} > K_{sp} \implies$ Supersaturated solution; **precipitation occurs immediately**.

PART II

# Advanced Physical Chemistry

Chapter 3

# Advanced Physical Chemistry: Solutions, Electrochemistry & Chemical Kinetics

This chapter investigates advanced dynamic physical chemistry systems: homogeneous binary liquid solutions obeying Raoult's law and anomalous colligative properties via the van 't Hoff factor, electrochemical cell energetics governed by the Nernst equation and Kohlrausch conductance, and chemical kinetics exploring differential and integrated rate laws, temperature coefficients, and Arrhenius transition-state collision theory.

## 3.1 Solutions & Colligative Properties (Unit 5)

### 3.1.1 Gas Solubility & Henry's Law

Henry's Law states that the solubility of a gas in a liquid at constant temperature is directly proportional to the partial pressure of the gas above the solution:

$$p = K_H \cdot x \iff x = \frac{p}{K_H} \tag{3.1}$$

where $p$ is the partial pressure of gas, $x$ is the mole fraction of gas dissolved, and $K_H$ is Henry's law constant.

- At a constant pressure $p$, solubility $x$ is **inversely proportional** to $K_H$.
- For a given gas, $K_H$ **increases with increasing temperature** $\implies$ gas solubility in liquids decreases on heating (aquatic life thrives in cold water due to higher dissolved $O_2$).

### 3.1.2 Vapour Pressure & Raoult's Law

For a binary solution of volatile liquids $A$ and $B$, Raoult's law states that the partial vapour pressure of each component is equal to the vapour pressure

of the pure component multiplied by its mole fraction in the liquid phase:

$$P_A = P_A^\circ x_A, \qquad P_B = P_B^\circ x_B \tag{3.2}$$

By Dalton's Law of Partial Pressures, the total vapour pressure is:

$$P_{\text{total}} = P_A + P_B = P_A^\circ x_A + P_B^\circ x_B = P_A^\circ(1-x_B) + P_B^\circ x_B = P_A^\circ + (P_B^\circ - P_A^\circ)x_B \tag{3.3}$$

**Vapour Phase Composition ($y_A, y_B$):**

$$y_A = \frac{P_A}{P_{\text{total}}} = \frac{P_A^\circ x_A}{P_{\text{total}}}, \qquad y_B = \frac{P_B}{P_{\text{total}}} = \frac{P_B^\circ x_B}{P_{\text{total}}} \implies \frac{1}{P_{\text{total}}} = \frac{y_A}{P_A^\circ} + \frac{y_B}{P_B^\circ} \tag{3.4}$$

### 3.1.3 Ideal vs Non-Ideal Solutions & Azeotropes

| Property | Ideal Solution | Positive Dev. | Negative Dev. |
|---|---|---|---|
| **Intermolecular Forces** | $A-B \approx A-A \approx B-B$ | $A-B < A-A, B-B$ (Weaker) | $A-B > A-A, B-B$ (Stronger) |
| **Vapour Pressure** | $P_{\text{tot}} = P_A^\circ x_A + P_B^\circ x_B$ | $P_{\text{tot}} > P_{\text{ideal}}$ | $P_{\text{tot}} < P_{\text{ideal}}$ |
| **Enthalpy of Mixing** | $\Delta H_{\text{mix}} = 0$ | $\Delta H_{\text{mix}} > 0$ (Endo) | $\Delta H_{\text{mix}} < 0$ (Exo) |
| **Volume of Mixing** | $\Delta V_{\text{mix}} = 0$ | $\Delta V_{\text{mix}} > 0$ (Expansion) | $\Delta V_{\text{mix}} < 0$ (Contraction) |
| **Entropy & Free Energy** | $\Delta S > 0, \Delta G < 0$ | $\Delta S > 0, \Delta G < 0$ | $\Delta S > 0, \Delta G < 0$ |
| **Azeotrope Formed** | None (Distillable) | **Min. Boiling Azeotrope** | **Max. Boiling Azeotrope** |
| **High-Yield Examples** | Benzene + Toluene; *n*-hexane + *n*-heptane | Ethanol + Water (95.6%); Acetone + $CS_2$ | $CHCl_3$ + Acetone; $HNO_3 + H_2O$ (68%) |

Table 3.1: Comprehensive Physical and Thermodynamic Profiling of Binary Liquid Mixtures.

### 3.1.4 Colligative Properties & Thermodynamic Derivations

Colligative properties depend exclusively on the ratio of the number of solute particles to solvent particles, irrespective of their chemical nature.

#### 1. Relative Lowering of Vapour Pressure (RLVP)

When a non-volatile solute $B$ is added to volatile solvent $A$, only solvent molecules occupy the vapour phase: $P_{\text{total}} = P_A = P_A^\circ x_A = P_A^\circ(1 - x_B)$.

$$\Delta P = P_A^\circ - P = P_A^\circ x_B \implies \frac{P_A^\circ - P}{P_A^\circ} = x_B = \frac{n_B}{n_A + n_B} \approx \frac{n_B}{n_A} = \frac{w_B M_A}{M_B w_A} \quad \text{(dilute)} \tag{3.5}$$

**Exact Molality Expression for All Concentrations:**

$$\frac{P_A^\circ - P}{P} = \frac{n_B}{n_A} = \frac{m \times M_A}{1000} \implies \text{Notice denominator is } P, \text{ not } P_A^\circ! \quad (3.6)$$

**Common Pitfall: Relative Lowering of Vapour Pressure Denominator**

A frequent numerical trap in NEET involves confusing the theoretical RLVP definition with the exact molality formula:

- **Definition of RLVP:** By Raoult's law, relative lowering with respect to pure solvent vapour pressure is:

$$\frac{P^\circ - P}{P^\circ} = x_B = \frac{n_B}{n_A + n_B}$$

  This equation strictly possesses the pure solvent vapour pressure $P^\circ$ in the denominator.
- **Exact Non-Approximated Molality Conversion:** Inverting and subtracting 1 gives:

$$\frac{P^\circ - P}{P} = \frac{n_B}{n_A} = \frac{w_B \cdot M_A}{M_B \cdot w_A} = \frac{m \cdot M_A}{1000}$$

  Here the denominator is the **solution** vapour pressure $P$, NOT $P^\circ$. Equating $(P^\circ - P)/P^\circ$ directly to $n_B/n_A$ without dilute approximation creates systematic calculation errors in non-dilute questions ($m > 0.1\,\text{m}$).

## 2. Elevation of Boiling Point ($\Delta T_b$)

At boiling point, vapour pressure equals external atmospheric pressure (1 atm). Addition of non-volatile solute lowers vapour pressure; hence the solution must be heated to a higher temperature $T_b > T_b^\circ$:

$$\Delta T_b = T_b - T_b^\circ = K_b \cdot m = \frac{1000 K_b w_B}{M_B w_A} \quad (3.7)$$

**Thermodynamic Derivation of Ebullioscopic Constant ($K_b$):** From the Clausius-Clapeyron relation connecting chemical potential and vapour pressure:

$$\ln\left(\frac{P^\circ}{P}\right) = \frac{\Delta H_{\text{vap}}}{R}\left(\frac{1}{T_b^\circ} - \frac{1}{T_b}\right) \approx \frac{\Delta H_{\text{vap}} \Delta T_b}{R(T_b^\circ)^2} \quad (3.8)$$

Since $\ln(P^\circ/P) = -\ln(1 - x_B) \approx x_B \approx \frac{mM_A}{1000}$:

$$\frac{mM_A}{1000} = \frac{\Delta H_{\text{vap}} \Delta T_b}{R(T_b^\circ)^2} \implies \Delta T_b = \left[\frac{R(T_b^\circ)^2 M_A}{1000 \Delta H_{\text{vap}}}\right] m \quad (3.9)$$

$$K_b = \frac{R(T_b^\circ)^2 M_A}{1000 \Delta H_{\text{vap}}} = \frac{R(T_b^\circ)^2}{1000 l_v} \quad (l_v = \text{latent heat of vaporization per gram}) \tag{3.10}$$

For water: $K_b = 0.512 \approx 0.52\,\text{K kg mol}^{-1}$.

### 3. Depression of Freezing Point ($\Delta T_f$)

Freezing point is the temperature at which the vapour pressure of the liquid solvent equals the vapour pressure of its solid phase. Lowering vapour pressure depresses the freezing point:

$$\Delta T_f = T_f^\circ - T_f = K_f \cdot m = \frac{1000 K_f w_B}{M_B w_A} \tag{3.11}$$

**Thermodynamic Derivation of Cryoscopic Constant ($K_f$):**

$$K_f = \frac{R(T_f^\circ)^2 M_A}{1000 \Delta H_{\text{fusion}}} = \frac{R(T_f^\circ)^2}{1000 l_f} \quad (l_f = \text{latent heat of fusion per gram}) \tag{3.12}$$

For water: $K_f = 1.86\,\text{K kg mol}^{-1}$. Note that $K_f/K_b = 1.86/0.52 \approx 3.58$ (freezing point depression is over 3.5 times more sensitive than boiling point elevation!).

**NEET High-Yield Application: Automotive Antifreeze Formulation**

**Problem Statement:** An automobile radiator requires protection against freezing down to $-15.0\,^\circ\text{C}$ during harsh winter operations. Calculate the mass of ethylene glycol ($C_2H_6O_2$, molar mass $M = 62.07\,\text{g} \cdot \text{mol}^{-1}$) that must be added to 5.00 kg of water ($K_f = 1.86\,\text{K} \cdot \text{kg} \cdot \text{mol}^{-1}$).

1. **Target Freezing Point Depression:**

$$\Delta T_f = T_f^\circ - T_f = 0.00\,^\circ\text{C} - (-15.0\,^\circ\text{C}) = 15.0\,\text{K}$$

2. **Required Molality of Glycol:**

$$m = \frac{\Delta T_f}{K_f} = \frac{15.0\,\text{K}}{1.86\,\text{K} \cdot \text{kg} \cdot \text{mol}^{-1}} = 8.065\,\text{mol} \cdot \text{kg}^{-1}$$

3. **Mass of Ethylene Glycol Required:**

$$w_{\text{glycol}} = m \times W_{\text{water}}\ (\text{in kg}) \times M_{\text{glycol}} = 8.065\,\text{mol} \cdot \text{kg}^{-1} \times 5.00\,\text{kg} \times 62.07\,\text{g} \cdot \text{mol}^{-1} \approx 2$$

**Clinical / Engineering Context:** In addition to lowering the freezing point, the non-volatile ethylene glycol simultaneously elevates the boiling point ($\Delta T_b = K_b \cdot m = 0.52 \times 8.065 \approx 4.19\,\text{K}$), preventing engine coolant boilover during high-temperature engine cycling.

### 4. Osmotic Pressure ($\Pi$)

Osmosis is the spontaneous net flow of solvent molecules through a semi-permeable membrane from pure solvent into solution. Osmotic pressure $\Pi$ is the excess hydrostatic pressure that must be applied to the solution to completely arrest this solvent influx:

$$\Pi = CRT = \left(\frac{n_B}{V}\right) RT = \frac{w_B RT}{M_B V} \implies M_B = \frac{w_B RT}{\Pi V} \tag{3.13}$$

**Why Osmotic Pressure is Superior for Macromolecules (Proteins, DNA, Polymers):**

1. High molecular mass ($M_B \sim 10^4 - 10^6\,\mathrm{g\,mol^{-1}}$) produces immeasurably tiny molalities ($m \sim 10^{-4}\,\mathrm{m}$), causing negligible $\Delta T_b$ ($< 10^{-4}\,\mathrm{K}$), but yielding easily measurable osmotic pressure ($\Pi = 10^{-4} \times 0.0821 \times 300 \approx 2.5 \times 10^{-3}\,\mathrm{atm} \approx 26\,\mathrm{mm\ Hg}$).
2. Measurement occurs at ambient room temperature, preventing thermal denaturation of delicate biological polymers.

## 3.1.5 Abnormal Molar Masses & van 't Hoff Factor ($i$)

When a solute undergoes dissociation or association in solution, colligative properties deviate from theoretical ideal values:

$$i = \frac{\text{Experimental / Observed Colligative Property}}{\text{Calculated / Normal Colligative Property}} = \frac{M_{\text{theoretical / normal}}}{M_{\text{observed / abnormal}}} \tag{3.14}$$

**Derivation of Degree of Dissociation ($\alpha_{\text{diss}}$) and Association ($\alpha_{\text{assoc}}$)**

- **Case 1: Dissociation ($A \rightleftharpoons nB$):** Let 1 mole of electrolyte be dissolved with degree of dissociation $\alpha$:

| | $A$ | $\rightleftharpoons$ | $nB$ |
|---|---|---|---|
| Initial moles: | 1 | | 0 |
| Equilibrium moles: | $1-\alpha$ | | $n\alpha$ |

$$\implies \text{Total moles} = 1 - \alpha + n\alpha = 1 + \tag{3.15}$$

$$i = \frac{1 + (n-1)\alpha}{1} \implies \alpha_{\text{diss}} = \frac{i-1}{n-1} \tag{3.16}$$

- **Case 2: Association / Polymerization ($nA \rightleftharpoons A_n$):** Let 1 mole of solute undergo association to an $n$-mer with degree of association

$\alpha$:

| | $nA$ | $\rightleftharpoons$ | $A_n$ |
|---|---|---|---|
| Initial moles: | 1 | | 0 |
| Equilibrium moles: | $1-\alpha$ | | $\frac{\alpha}{n}$ |

$\implies$ Total moles $= 1-\alpha+\frac{\alpha}{n} = 1-\left(1-\frac{1}{n}\right)\alpha$ (3.17)

$$i = 1-\left(1-\frac{1}{n}\right)\alpha \implies \alpha_{\text{assoc}} = \frac{1-i}{1-\frac{1}{n}} \tag{3.18}$$

For dimerization of acetic acid in benzene ($n=2$): $i = 1-\frac{\alpha}{2} \implies$ if 100% dimerized ($\alpha=1$), $i=0.5$ and observed molar mass is double ($120\,\text{g mol}^{-1}$ vs $60\,\text{g mol}^{-1}$).

# 3.2 Redox Reactions & Electrochemistry (Unit 7)

## 3.2.1 Galvanic Cells & The Daniell Cell

A galvanic cell converts spontaneous chemical redox free energy into electrical work:

$$\text{Zn(s)} + \text{Cu}^{2+}\text{(aq)} \longrightarrow \text{Zn}^{2+}\text{(aq)} + \text{Cu(s)} \tag{3.19}$$

- **Anode (Oxidation, Negative Terminal):** $\text{Zn(s)} \rightarrow \text{Zn}^{2+}\text{(aq)} + 2e^-$, $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76\,\text{V}$.
- **Cathode (Reduction, Positive Terminal):** $\text{Cu}^{2+}\text{(aq)} + 2e^- \rightarrow \text{Cu(s)}$, $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = +0.34\,\text{V}$.
- **Cell Representation:** $\text{Zn(s)} \mid \text{Zn}^{2+}\text{(aq)} \parallel \text{Cu}^{2+}\text{(aq)} \mid \text{Cu(s)}$.
- **Salt Bridge Functions:**
  1. Maintains electrical neutrality in both half-cells by migrating counterions ($\text{K}^+$ and $\text{Cl}^-$ or $\text{NO}_3^-$).
  2. Prevents physical mixing of the two electrolytic solutions.
  3. Minimizes liquid junction potential. Electrolyte chosen ($\text{KCl}, \text{KNO}_3, \text{NH}_4\text{NO}_3$) has nearly identical transport numbers for cation and anion ($v_+ \approx v_-$).

## 3.2.2 Standard Hydrogen Electrode & Electrochemical Series

The Standard Hydrogen Electrode (SHE) is the universal reference standard assigned $E^\circ = 0.000\,\text{V}$ at all temperatures:

$$\text{Pt(s)} \mid \text{H}_2\text{(g, 1 bar)} \mid \text{H}^+\text{(aq, 1 M)}, \qquad 2\text{H}^+ + 2e^- \rightleftharpoons \text{H}_2\text{(g)} \tag{3.20}$$

**Standard Cell EMF ($E^\circ_{\text{cell}}$):**

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} = E^\circ_{\text{right}} - E^\circ_{\text{left}} \quad \text{(using standard reduction potentials)} \tag{3.21}$$

### 3.2.3 Thermodynamic Derivation of the Nernst Equation

The maximum non-expansion electrical work delivered by a galvanic cell at constant $T, P$ equals the decrease in Gibbs free energy:

$$w_{\text{elec}} = q_{\text{charge}} \times E_{\text{cell}} = (nF)E_{\text{cell}} \implies \Delta_r G = -w_{\text{elec}} = -nFE_{\text{cell}} \tag{3.22}$$

Under standard state conditions: $\Delta_r G^\circ = -nFE^\circ_{\text{cell}}$. Substituting these relations into the fundamental thermodynamic equation $\Delta_r G = \Delta_r G^\circ + RT\ln Q$:

$$-nFE_{\text{cell}} = -nFE^\circ_{\text{cell}} + RT\ln Q \tag{3.23}$$

Dividing throughout by $-nF$:

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{RT}{nF}\ln Q = E^\circ_{\text{cell}} - \frac{2.303RT}{nF}\log_{10} Q \tag{3.24}$$

Substituting universal constants $R = 8.314\,\text{J mol}^{-1}\,\text{K}^{-1}$, $F = 96485\,\text{C mol}^{-1}$, and $T = 298.15\,\text{K}$:

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{n}\log_{10} Q \tag{3.25}$$

**Nernst Equation Special Cases & Thermodynamic Properties**

- **Dynamic Equilibrium ($E_{\text{cell}} = 0, Q = K_c$):**

$$0 = E^\circ_{\text{cell}} - \frac{0.0591}{n}\log_{10} K_c \implies \log_{10} K_c = \frac{nE^\circ_{\text{cell}}}{0.0591} \tag{3.26}$$

- **Concentration Cells (Identical Electrodes, Different Concentrations $C_1 < C_2$):** Since standard potentials are identical, $E^\circ_{\text{cell}} = 0$:

$$E_{\text{cell}} = -\frac{0.0591}{n}\log_{10}\left(\frac{C_1}{C_2}\right) = \frac{0.0591}{n}\log_{10}\left(\frac{C_2}{C_1}\right) \quad \text{(Spontaneous if } C_2 > C_1\text{)} \tag{3.27}$$

- **Temperature Coefficient of EMF and Entropy of Cell Reaction:**

$$\left(\frac{\partial \Delta G}{\partial T}\right)_P = -\Delta S \implies \Delta S = nF\left(\frac{\partial E_{\text{cell}}}{\partial T}\right)_P \tag{3.28}$$

$$\Delta H = \Delta G + T\Delta S = -nFE_{\text{cell}} + nFT\left(\frac{\partial E_{\text{cell}}}{\partial T}\right)_P \tag{3.29}$$

### 3.2.4 Electrolytic Conductance & Kohlrausch's Law

- **Conductivity ($\kappa$):** Conductance of a unit volume ($1\,\text{cm}^3$ or $1\,\text{m}^3$) of electrolytic solution:

$$R = \rho\frac{l}{A} \implies G = \frac{1}{R} = \kappa\frac{A}{l} \implies \kappa = \frac{1}{R}\left(\frac{l}{A}\right) = G \cdot G^* \tag{3.30}$$

  where $G^* = l/A$ is the cell constant (units $\text{cm}^{-1}$ or $\text{m}^{-1}$).
- **Molar Conductivity ($\Lambda_m$):** Conductance of volume of solution containing 1 mole of electrolyte:

$$\Lambda_m = \frac{\kappa \times 1000}{M \text{ (in Molarity)}} \quad (\text{in S cm}^2\,\text{mol}^{-1}) = \frac{\kappa}{1000 \times M} \quad (\text{in S m}^2\,\text{mol}^{-1}) \tag{3.31}$$

- **Dilution Effects:**
  - **Conductivity ($\kappa$) Decreases with Dilution:** Dilution reduces the number of current-carrying ions per unit volume.
  - **Molar Conductivity ($\Lambda_m$) Increases with Dilution:** Volume containing 1 mole expands, overcompensating the decrease in $\kappa$.
- **Strong vs Weak Electrolytes:**
  - **Strong Electrolytes:** Obey Debye-Hückel-Onsager equation: $\Lambda_m = \Lambda_m^\circ - A\sqrt{C}$. Plot of $\Lambda_m$ vs $\sqrt{C}$ is linear and can be extrapolated to $C = 0$ to obtain limiting molar conductivity $\Lambda_m^\circ$.
  - **Weak Electrolytes ($CH_3COOH$):** $\Lambda_m$ increases steeply near infinite dilution; cannot be obtained by graphical extrapolation.

**Kohlrausch's Law of Independent Migration of Ions**

At infinite dilution, each ion migrates independently of its co-ion and makes a definite contribution to the total molar conductivity:

$$\Lambda_m^\circ = \nu_+\lambda_+^\circ + \nu_-\lambda_-^\circ \tag{3.32}$$

**High-Yield Applications of Kohlrausch's Law:**

1. **Limiting Molar Conductivity of Weak Electrolytes ($CH_3COOH$):**

$$\Lambda_m^\circ(CH_3COOH) = \Lambda_m^\circ(CH_3COONa) + \Lambda_m^\circ(HCl) - \Lambda_m^\circ(NaCl) \tag{3.33}$$

2. **Degree of Dissociation of Weak Electrolyte ($\alpha$):**

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ} \implies K_a = \frac{C\alpha^2}{1-\alpha} = \frac{C\left(\frac{\Lambda_m}{\Lambda_m^\circ}\right)^2}{1-\frac{\Lambda_m}{\Lambda_m^\circ}} = \frac{C\Lambda_m^2}{\Lambda_m^\circ(\Lambda_m^\circ - \Lambda_m)} \tag{3.34}$$

3. **Solubility and $K_{sp}$ of Sparingly Soluble Salts ($AgCl, BaSO_4$):**

$$\Lambda_m^\circ = \frac{1000\kappa_{\text{salt}}}{S} \implies S = \frac{1000\kappa_{\text{salt}}}{\Lambda_m^\circ} \implies K_{sp} = S^2 \tag{3.35}$$

where $\kappa_{\text{salt}} = \kappa_{\text{solution}} - \kappa_{\text{water}}$.

### 3.2.5 Electrolysis & Faraday's Laws

- **Faraday's First Law:** Mass of substance deposited or liberated at any electrode is directly proportional to the total electrical charge passed:

$$w = Z \cdot Q = Z \cdot I \cdot t = \left(\frac{E}{96500}\right) I \cdot t = \left(\frac{M}{nF}\right) I \cdot t \tag{3.36}$$

where $Z = \frac{E}{F}$ is the electrochemical equivalent, and $E = \frac{M}{n}$ is equivalent mass.

- **Faraday's Second Law:** When the identical quantity of electricity is passed through different electrolytes connected in series, masses deposited are proportional to their chemical equivalent masses:

$$\frac{w_1}{w_2} = \frac{E_1}{E_2} \tag{3.37}$$

- **Preferential Discharge Traps:**
  - **Molten** NaCl**:** Cathode: $Na^+ + e^- \rightarrow Na(l)$; Anode: $2Cl^- \rightarrow Cl_2(g) + 2e^-$.
  - **Aqueous** NaCl**:** At cathode, $H_2O$ reduction ($E^\circ = -0.83\,V$) occurs preferentially over $Na^+$ ($E^\circ = -2.71\,V$) $\implies$ $H_2(g)$ evolved. At anode, oxidation of $Cl^-$ to $Cl_2(g)$ occurs preferentially over $O_2$ evolution due to high **oxygen overpotential** on Pt.
  - **Aqueous** $CuSO_4$ **(Pt electrodes):** Cathode: $Cu^{2+} + 2e^- \rightarrow Cu(s)$; Anode: $2H_2O \rightarrow O_2(g) + 4H^+ + 4e^-$.
  - **Aqueous** $CuSO_4$ **(Cu electrodes):** Cathode: Cu deposition; Anode: Cu dissolution ($Cu \rightarrow Cu^{2+} + 2e^-$); concentration remains constant.

### 3.2.6 Commercial Batteries & Mechanism of Corrosion

**Batteries & Corrosion Reaction Architectures**

- **Lead-Acid Storage Battery (Secondary Reversible Cell):**

– **Discharging Half-Reactions:**

$$\text{Anode: } \mathrm{Pb(s) + SO_4^{2-}(aq) \longrightarrow PbSO_4(s) + 2}e^- \quad (3.38)$$

$$\text{Cathode: } \mathrm{PbO_2(s) + SO_4^{2-}(aq) + 4H^+(aq) + 2}e^- \longrightarrow \mathrm{PbSO_4(s) + 2H_2O(l)} \quad (3.39)$$

$$\text{Net Discharge: } \mathrm{Pb(s) + PbO_2(s) + 2H_2SO_4(aq) \longrightarrow 2PbSO_4(s) + 2H_2O(l)} \quad (3.40)$$

Density of $\mathrm{H_2SO_4}$ drops from $1.30\,\mathrm{g\,cm^{-3}}$ to $< 1.20\,\mathrm{g\,cm^{-3}}$. Recharging reverses all signs.

- **Hydrogen-Oxygen Fuel Cell:**

$$\text{Anode: } \mathrm{2H_2 + 4OH^- \rightarrow 4H_2O + 4}e^-, \qquad \text{Cathode: } \mathrm{O_2 + 2H_2O + 4}e^- \rightarrow \mathrm{4OH^-} \quad (3.41)$$

Theoretical thermodynamic efficiency: $\eta = \frac{\Delta G^\circ}{\Delta H^\circ} \times 100\% \approx 70\% - 75\%$ (far exceeds thermal engines).

- **Electrochemical Mechanism of Rusting of Iron:**

$$\text{Anodic site: } \mathrm{Fe(s) \longrightarrow Fe^{2+}(aq) + 2}e^- \quad (E^\circ = -0.44\,\mathrm{V}) \quad (3.42)$$

$$\text{Cathodic site: } \mathrm{O_2(g) + 4H^+(aq) + 4}e^- \longrightarrow \mathrm{2H_2O(l)} \quad (E^\circ = +1.23\,\mathrm{V}) \quad (3.43)$$

$$\text{Rust Formation: } \mathrm{4Fe^{2+} + O_2 + 4H_2O + 2}x\mathrm{H_2O \longrightarrow 2Fe_2O_3 \cdot }x\mathrm{H_2O(s) + 8H^+} \quad (3.44)$$

Prevented by galvanization (sacrificial zinc coating: $E^\circ_{\mathrm{Zn^{2+}/Zn}} = -0.76\,\mathrm{V} < E^\circ_{\mathrm{Fe^{2+}/Fe}} = -0.44\,\mathrm{V}$).

## 3.3 Chemical Kinetics (Unit 8)

### 3.3.1 Reaction Rate, Rate Laws & Order vs Molecularity

For a general reaction:

$$aA + bB \longrightarrow cC + dD \quad (3.45)$$

The instantaneous stoichiometric rate of reaction is:

$$r = -\frac{1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = +\frac{1}{c}\frac{d[C]}{dt} = +\frac{1}{d}\frac{d[D]}{dt} \quad (3.46)$$

**Experimental Differential Rate Law:**

$$r = k[A]^x[B]^y \tag{3.47}$$

where $k$ is the specific rate constant, $x$ is the order with respect to $A$, $y$ is the order with respect to $B$, and $n = x + y$ is the overall reaction order.

| Feature | Order of Reaction | Molecularity |
|---|---|---|
| **Definition** | Sum of powers of reactant concentrations in rate law. | Reacting species colliding simultaneously in elementary step. |
| **Determination** | Strictly experimental parameter; cannot be deduced from equation. | Theoretical property deduced from elementary mechanism. |
| **Allowed Values** | Can be zero, integer, negative, or fractional $(0, 1, 2, 1/2)$. | Positive non-zero integer $(1, 2, 3)$. Cannot be zero/fractional! |
| **Upper Bound** | Can exceed 3. | Rarely exceeds 3 due to low probability of simultaneous collisions. |
| **Complex Reactions** | Applicable to overall complex reactions. | Meaningless for overall complex reaction; applies to elementary steps. |

Table 3.2: Systematic Comparison: Order of Reaction vs Molecularity.

**Units of Rate Constant ($k$):**

$$\text{Units of } k = \left(\text{mol}\,\text{L}^{-1}\right)^{1-n}\text{s}^{-1} = (\text{atm})^{1-n}\,\text{s}^{-1} \tag{3.48}$$

- Zero-order ($n = 0$): $\text{mol}\,\text{L}^{-1}\,\text{s}^{-1}$.
- First-order ($n = 1$): $\text{s}^{-1}$ (completely independent of concentration units!).
- Second-order ($n = 2$): $\text{L}\,\text{mol}^{-1}\,\text{s}^{-1}$.

### 3.3.2 Integrated Rate Equations: Full Calculus Derivations

#### 1. Zero-Order Kinetics

For $A \to$ Products, rate is independent of $[A]$:

$$-\frac{d[A]}{dt} = k[A]^0 = k \implies d[A] = -k dt \tag{3.49}$$

Integrating from $t = 0$ ($[A] = [A]_0$) to time $t$ ($[A] = [A]_t$):

$$\int_{[A]_0}^{[A]_t} d[A] = -k \int_0^t dt \implies [A]_t - [A]_0 = -kt \implies [A]_t = [A]_0 - kt \tag{3.50}$$

**Half-Life ($t_{1/2}$) and Completion Time ($t_{100\%}$):** At $t = t_{1/2}$, $[A]_t = \frac{[A]_0}{2}$:

$$\frac{[A]_0}{2} = [A]_0 - kt_{1/2} \implies t_{1/2} = \frac{[A]_0}{2k} \quad \text{(Half-life is directly proportional to initial con} \tag{3.51}$$

At $t = t_{100\%}$, $[A]_t = 0$:

$$0 = [A]_0 - kt_{100\%} \implies t_{100\%} = \frac{[A]_0}{k} = 2t_{1/2} \tag{3.52}$$

Zero-order reactions proceed to complete termination in a finite time. **Examples:** Photochemical synthesis of HCl over water ($H_2 + Cl_2 \xrightarrow{h\nu} 2HCl$); decomposition of $NH_3$ on hot Pt/W surface at high pressure.

### 2. First-Order Kinetics

For $A \rightarrow$ Products:

$$-\frac{d[A]}{dt} = k[A] \implies \frac{d[A]}{[A]} = -kdt \tag{3.53}$$

Integrating from $t = 0$ ($[A] = [A]_0$) to time $t$ ($[A] = [A]_t$):

$$\int_{[A]_0}^{[A]_t} \frac{d[A]}{[A]} = -k\int_0^t dt \implies \ln\left(\frac{[A]_t}{[A]_0}\right) = -kt \implies [A]_t = [A]_0 e^{-kt} \tag{3.54}$$

$$k = \frac{1}{t}\ln\left(\frac{[A]_0}{[A]_t}\right) = \frac{2.303}{t}\log_{10}\left(\frac{[A]_0}{[A]_t}\right) = \frac{2.303}{t}\log_{10}\left(\frac{a}{a-x}\right) \tag{3.55}$$

**Half-Life ($t_{1/2}$):** At $t = t_{1/2}$, $[A]_t = \frac{[A]_0}{2}$:

$$k = \frac{2.303}{t_{1/2}}\log_{10}(2) \implies t_{1/2} = \frac{\ln 2}{k} = \frac{0.69315}{k} \approx \frac{0.693}{k} \tag{3.56}$$

The half-life of a first-order reaction is **strictly independent of initial reactant concentration**.

**NCERT Paradigm: Fractional Lifetimes of First-Order Reactions**

For any first-order process ($[A]_t = [A]_0 e^{-kt}$), each half-life interval reduces the remaining reactant concentration by exactly 50%, regardless of initial amount. Rapid shortcuts for NEET:

$$t_{75\%} = \frac{2.303}{k}\log_{10}\left(\frac{100}{25}\right) = \frac{2.303}{k}\log_{10}(4) = 2 \times \frac{0.693}{k} = 2t_{1/2}$$

$$t_{87.5\%} = \frac{2.303}{k}\log_{10}\left(\frac{100}{12.5}\right) = \frac{2.303}{k}\log_{10}(8) = 3t_{1/2}$$

$$t_{93.75\%} = 4t_{1/2} \quad \text{(remaining fraction } = 6.25\%)$$

$$t_{99\%} = \frac{2.303}{k}\log_{10}(100) = \frac{2 \times 2.303}{k} \approx \frac{4.606}{k} \approx 6.64t_{1/2}$$

$$t_{99.9\%} = \frac{2.303}{k}\log_{10}\left(\frac{100}{0.1}\right) = \frac{3 \times 2.303}{k} \approx \frac{6.909}{k} \approx 10t_{1/2}$$

**NCERT Benchmark Conclusion:** The time required for 99.9% completion of a first-order reaction is approximately 10 times its half-life ($t_{99.9\%} \approx 10t_{1/2}$), while $t_{99\%} \approx 2t_{90\%}$.

**Gas Phase First-Order Decomposition ($A(g) \rightarrow B(g)+C(g)$):** If initial pressure of $A$ is $P_i$ and total pressure at time $t$ is $P_t$:

$$P_t = (P_i - p) + p + p = P_i + p \implies p = P_t - P_i \implies P_A = P_i - p = 2P_i - P_t \tag{3.57}$$

$$k = \frac{2.303}{t} \log_{10} \left( \frac{P_i}{2P_i - P_t} \right) \tag{3.58}$$

### 3. Pseudo-First-Order Kinetics

Reactions of higher order that behave kinetically as first-order when one reactant is present in large stoichiometric excess:

- **Acid-Catalyzed Ester Hydrolysis:**

$$CH_3COOC_2H_5 + H_2O \xrightarrow{H^+} CH_3COOH + C_2H_5OH \tag{3.59}$$

Because water is solvent ($[H_2O] \approx 55.5\,M \approx$ constant):

$$r = k'[\text{ester}][H_2O] = k[\text{ester}], \qquad k = k'[H_2O] \tag{3.60}$$

Tracked by titrating aliquots with standard NaOH: $k = \frac{2.303}{t} \log_{10} \left( \frac{V_\infty - V_0}{V_\infty - V_t} \right)$.

- **Inversion of Cane Sugar (Polarimetry):**

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+} C_6H_{12}O_6 \text{ (D-glucose)} + C_6H_{12}O_6 \text{ (D-fructose)} \tag{3.61}$$

The optical rotation changes from dextro ($+66.5^\circ$) to levo ($-39.7^\circ$). Kinetics tracked via polarimeter angle of rotation:

$$k = \frac{2.303}{t} \log_{10} \left( \frac{r_0 - r_\infty}{r_t - r_\infty} \right) \tag{3.62}$$

## 3.3.3 Temperature Dependence: The Arrhenius Equation

Svante Arrhenius (1889) derived the exponential relationship between reaction rate constant and absolute temperature:

$$k = Ae^{-E_a/RT} \tag{3.63}$$

where $A$ is the pre-exponential factor (frequency factor), $E_a$ is the activation energy, and $e^{-E_a/RT}$ represents the Boltzmann fraction of molecular collisions having kinetic energy $\geq E_a$.

$$\ln k = \ln A - \frac{E_a}{RT} \iff \log_{10} k = \log_{10} A - \frac{E_a}{2.303RT} \tag{3.64}$$

An Arrhenius plot of $\log_{10} k$ versus $\frac{1}{T}$ produces a straight line:

$$\text{Slope} = -\frac{E_a}{2.303R}, \qquad y\text{-intercept} = \log_{10} A \tag{3.65}$$

**Two-Temperature Comparative Formulation:**

$$\log_{10}\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right] = \frac{E_a}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right] \tag{3.66}$$

### 3.3.4 Collision Theory & Catalysis Kinetics

According to Max Trautz and William Lewis:

$$k = P \cdot Z_{AB} \cdot e^{-E_a/RT} \tag{3.67}$$

where $Z_{AB}$ is the collision frequency ($10^{25} - 10^{28}\,\text{collisions}\,\text{s}^{-1}\,\text{cm}^{-3}$), and $P$ is the steric or orientation factor representing the fraction of collisions occurring with proper spatial orientation.

**Mechanism of Catalytic Acceleration:** A catalyst accelerates the rate of a chemical transformation without undergoing permanent stoichiometric consumption. It achieves this kinetic enhancement not by shifting molecular energy distributions, but by providing an alternative mechanistic pathway involving an intermediate activated complex characterized by a significantly lower activation barrier $E_a'$ ($E_a' < E_a$), as depicted in Figure 3.1.

According to the Arrhenius relation, the relative catalytic rate enhancement factor is given by:

$$\frac{k_{\text{cat}}}{k_{\text{uncat}}} = \frac{Ae^{-E_a'/RT}}{Ae^{-E_a/RT}} = e^{(E_a - E_a')/RT} = e^{\Delta E_a/RT} \tag{3.68}$$

Because the reduction $\Delta E_a$ resides in the numerator of an exponential exponent, even a modest lowering of the activation energy (e.g., by $15\,\text{kJ}\,\text{mol}^{-1}$) accelerates reaction velocity by several orders of magnitude at physiological temperature.

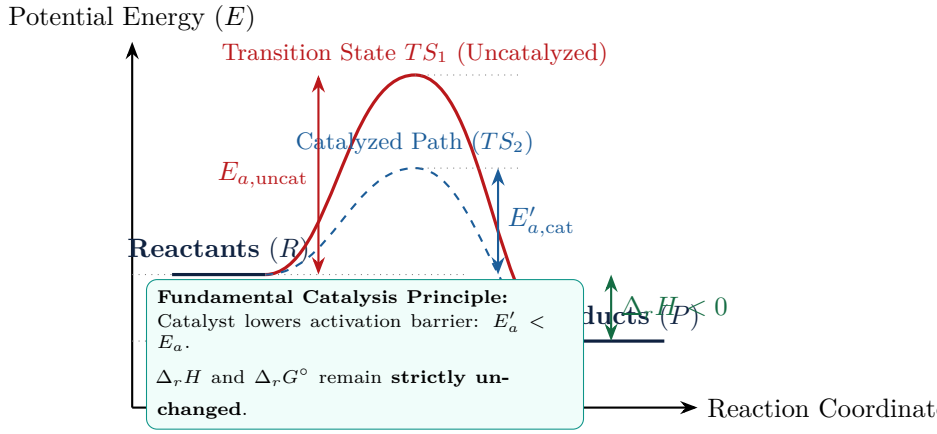



Figure 3.1: Potential Energy Diagram for an Exothermic Reaction Illustrating the Lowering of Activation Energy by a Catalyst.

**NTA Assertion-Reason Traps: Thermodynamic Invariants of Catalysis**

Candidates frequently lose marks on Assertion-Reason questions involving catalytic effects:

- **Equal Barrier Reduction:** A catalyst reduces activation energy for **both forward and reverse reactions by the exact same amount**: $\Delta E_{a,f} = \Delta E_{a,b}$.
- **No Change in Reaction Enthalpy:** Because $\Delta_r H = E_{a,f} - E_{a,b}$, the reaction enthalpy is **strictly unaltered** by the presence of a catalyst.
- **No Change in Equilibrium Constant:** A catalyst accelerates the approach to equilibrium but does **NOT** alter the equilibrium constant $K_c$ or shift equilibrium concentrations.
- **No Change in Spontaneity:** Standard Gibbs free energy $\Delta_r G^\circ$ is completely independent of catalytic mechanisms. A catalyst cannot drive a non-spontaneous ($\Delta G > 0$) process.

PART III

# Inorganic Chemistry & Coordination Compounds

Chapter 4

# Inorganic Chemistry: Periodic Trends, Chemical Bonding, Coordination & Block Elements

Inorganic chemistry requires structural understanding rather than rote memorization. In this comprehensive compendium, we analyze modern periodic trends, molecular orbital theory (MOT) and VSEPR geometries, coordination compounds with crystal field theory (CFT) splitting, and comparative trends across the p-block and transition d/f-block elements.

**A Personal Note from Satya: The Mental Shift from Math to Molecular Geometry**

**Welcome to Inorganic Chemistry! Let's talk honestly about why students fear this subject.**
Most students love Physical Chemistry because it feels like Physics: you plug numbers into formulas ($PV = nRT$, $\Delta G^\circ = -nFE^\circ$), crunch the math, and get an answer. Then you arrive at Inorganic Chemistry, open a traditional coaching module, see 400 pages of reactions and exceptions, and your brain freezes in panic.
Here is the secret: *Inorganic Chemistry is not a collection of random facts; it is the physical architecture of electron clouds.*

- Every periodic anomaly (why Gallium is slightly smaller than Aluminum, why Oxygen has a lower ionization energy than Nitrogen) is rooted in two bedrock concepts: **effective nuclear charge ($Z_{\text{eff}}$)** and **electron-electron repulsion in half-filled subshells**.
- Coordination Chemistry becomes pure geometry when you visualize how incoming negative ligands split the five degenerate $d$-orbitals into $t_{2g}$ and $e_g$ energy levels. Once you see that diagram, high-spin, low-spin, magnetic moments, and colors become obvi-

ous.

**Take Strategic 5-Minute Brain Breaks:** When memorizing chemical properties and group trends, your working memory fills up quickly. Study in 25-minute focused sprints. When the timer rings, stand up, step away from your desk, drink water, and close your eyes for 5 minutes. Do not pick up your phone. Giving your brain zero visual input for five minutes allows the hippocampus to file the atomic radii trends into long-term storage without cognitive interference.

## 4.1 NMC 2027 Regulatory Calibration & Scope Demarcation

**Zero-Tolerance NMC Syllabus Alert: Deleted Inorganic Chemistry Chapters**

Under the revised National Medical Commission (NMC) NEET UG 2027 regulatory curriculum, the following legacy chapters have been **strictly deleted and removed from the examination**:

1. **Hydrogen:** Position in periodic table, isotopes, preparation and properties of dihydrogen, hydrides (ionic, covalent, interstitial), physical and chemical properties of water, hard/soft water, heavy water ($D_2O$), and hydrogen peroxide ($H_2O_2$) are **completely eliminated**.
2. ***s*-Block Elements (Alkali & Alkaline Earth Metals):** Group 1 and Group 2 chemistry, electronic configurations, diagonal relationships, flame coloration, and industrial compounds ($Na_2CO_3$, NaCl, NaOH, $NaHCO_3$, CaO, $CaCO_3$, $CaSO_4 \cdot \frac{1}{2}H_2O$, cement) are **completely eliminated**.
3. **General Principles & Processes of Isolation of Elements (Metallurgy):** Ores, mineral dressing, roasting, calcination, smelting, Ellingham diagrams, blast furnace iron extraction, copper reverberatory smelting, Hall-Héroult aluminum extraction, and refining techniques (Mond process, van Arkel method, zone refining) are **completely eliminated**.
4. **Environmental Chemistry:** Tropospheric/stratospheric pollution, classical and photochemical smog, ozone depletion, and green chemistry are **strictly eliminated**.

**Study Directive:** Do not invest study hours in these eliminated topics. All cognitive focus must remain on the rigorously tested Class 11 and Class 12 inorganic chemistry compendium units detailed below.

# 4.2 Classification of Elements & Periodicity in Properties (Unit 9)

## 4.2.1 Modern Periodic Law & Electronic Architecture

Moseley's Law (1913) demonstrated that the square root of the characteristic X-ray frequency ($\sqrt{\nu}$) emitted by an element is directly proportional to its atomic number $Z$:

$$\sqrt{\nu} = a(Z - b) \tag{4.1}$$

Thus, the physical and chemical properties of the elements are periodic functions of their **atomic numbers**, not their atomic masses. The long form of the periodic table consists of 7 horizontal periods and 18 vertical groups, partitioned into $s, p, d$, and $f$ blocks based on the differentiating electron orbital.

## 4.2.2 Effective Nuclear Charge ($Z_{\text{eff}}$) & Slater's Rules

Electrons in outer shells are electrostatically shielded from the full nuclear charge $+Ze$ by intervening inner-shell electron density:

$$Z_{\text{eff}} = Z - \sigma \tag{4.2}$$

where $\sigma$ is the screening or shielding constant.

- **Shielding Efficiency Hierarchy:** $s > p > d > f$. Spherical $s$-orbitals penetrate closest to the nucleus, providing maximal screening. Diffuse $d$- and $f$-orbitals have poor radial penetration and provide **extraordinarily poor shielding**.
- **Slater's Rules Formulation for $ns, np$ Valence Electrons:**
  1. Electrons in shells higher than the considered electron contribute 0.00.
  2. Other electrons in the identical $(ns, np)$ group contribute 0.35 each (or 0.30 if $1s$).
  3. Each electron in the $(n-1)$ shell contributes 0.85.
  4. Each electron in the $(n-2)$ or lower inner shells contributes 1.00.
- **For $nd, nf$ Valence Electrons:**
  1. Other electrons in the identical $(nd, nf)$ group contribute 0.35 each.
  2. All electrons in all lower underlying shells contribute 1.00 each.

**High-Yield Problem Archetype: Step-by-Step Slater's $Z_{\text{eff}}$ Calculation**

**Archetype Problem:** Calculate the effective nuclear charge ($Z_{\text{eff}}$) experienced by: (a) a $4s$ valence electron in Zinc ($Z = 30$), and (b) a

$3d$ electron in Zinc ($Z = 30$).

1. **Electronic Configuration Grouping:**

$$(1s)^2\ (2s, 2p)^8\ (3s, 3p)^8\ (3d)^{10}\ (4s)^2$$

2. **For the $4s$ Valence Electron ($n = 4$):**
   - 1 other electron in same group ($4s$): $1 \times 0.35 = 0.35$.
   - $(n-1)$ shell (both $3s, 3p$ and $3d$, total 18 electrons): $18 \times 0.85 = 15.30$.
   - $(n-2)$ and lower shells ($1s, 2s, 2p$, total 10 electrons): $10 \times 1.00 = 10.00$.
   - $\sigma_{4s} = 0.35 + 15.30 + 10.00 = 25.65 \implies Z_{\text{eff}} = 30 - 25.65 = \mathbf{4.35}$.
3. **For a $3d$ Electron ($n = 3$):**
   - $4s$ electrons lie in a higher shell $\implies$ contribute 0.00.
   - 9 other electrons in the same ($3d$) group: $9 \times 0.35 = 3.15$.
   - All 18 electrons in lower shells: $18 \times 1.00 = 18.00$.
   - $\sigma_{3d} = 3.15 + 18.00 = 21.15 \implies Z_{\text{eff}} = 30 - 21.15 = \mathbf{8.85}$.
4. **Conclusion:** $3d$ electrons experience more than double the effective nuclear attraction (8.85 vs 4.35) compared to $4s$ electrons. Hence, upon ionization, the $4s$ electrons are expelled first: $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ yields $[\text{Ar}]3d^{10}$, not $[\text{Ar}]3d^8 4s^2$!

### 4.2.3 Atomic and Ionic Radii Trends & Contractions

- **Radius Hierarchy:** For the same element: $r_{\text{cation}} < r_{\text{atom}} < r_{\text{anion}}$. Loss of electrons increases $Z_{\text{eff}} = Z/e$, pulling electrons closer. Gain of electrons increases interelectronic repulsion, expanding the electron cloud.
- **Isoelectronic Series Size Rule:** For ions sharing the identical total electron count:

$$r_{\text{isoelectronic}} \propto \frac{1}{Z} \implies \text{C}^{4-} > \text{N}^{3-} > \text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+} > \text{Si}^{4+} \quad (10e^- \tag{4.3}$$

- **$3d$-Transition Metal Contraction (d-Block Contraction):** Filling of the ten $3d$ orbitals across the first transition series precedes Gallium ($Z = 31$: $[\text{Ar}]3d^{10}4s^2 4p^1$). Because $3d$ electrons shield poorly, $Z_{\text{eff}}$ on the valence $4p$ electron is abnormally high. Consequently:

$$r(\text{Ga}) = 135\,\text{pm} \leq r(\text{Al}) = 143\,\text{pm} \tag{4.4}$$

- **Lanthanoid Contraction ($4f$-Contraction):** Filling of the fourteen diffuse $4f$ orbitals between Lanthanum ($Z = 57$) and Hafnium ($Z = 72$)

results in extremely poor screening of the increasing nuclear charge (+14 units). **Consequences:**

1. Nearly identical covalent and ionic radii for $4d$ and $5d$ congeneric pairs:

$$r(\mathrm{Zr}) \approx r(\mathrm{Hf}) \approx 159{-}160\,\mathrm{pm}, \qquad r(\mathrm{Nb}) \approx r(\mathrm{Ta}) \approx 146\,\mathrm{pm}, \qquad r(\mathrm{Mo}) \approx r(\mathrm{W} \tag{4.5}$$

2. High density of $5d$ elements (roughly double their $4d$ counterparts; e.g., Osmium $\rho \approx 22.6\,\mathrm{g\,cm^{-3}}$).
3. Regular decrease in basicity of lanthanoid hydroxides: $\mathrm{La(OH)_3} > \mathrm{Ce(OH)_3} > \cdot$ (covalent character increases as cation size contracts).

### 4.2.4 Ionization Enthalpy ($\Delta_i H$) & High-Yield Anomalies

Ionization Enthalpy is the minimum energy required to remove the most loosely bound electron from an isolated gaseous atom in its ground state:

$$\mathrm{X(g)} \xrightarrow{\mathrm{IE_1}} \mathrm{X^+(g)} + e^-, \qquad \mathrm{X^+(g)} \xrightarrow{\mathrm{IE_2}} \mathrm{X^{2+}(g)} + e^- \tag{4.6}$$

Successive values always increase: $\mathrm{IE_1} < \mathrm{IE_2} < \mathrm{IE_3} < \ldots$ because removing an electron from an increasingly positive ion encounters greater electrostatic attraction.

### 4.2.5 Electron Gain Enthalpy ($\Delta_{\mathrm{eg}} H$) Trends & Traps

Electron gain enthalpy is the enthalpy change accompanying the addition of an electron to an isolated gaseous atom to form a uninegative gaseous anion:

$$\mathrm{X(g)} + e^- \longrightarrow \mathrm{X^-(g)}, \qquad \Delta_{\mathrm{eg}} H \tag{4.7}$$

Most elements release energy upon gaining an electron ($\Delta_{\mathrm{eg}} H < 0$). However, second electron gain enthalpies are **strictly positive (endothermic)** for all elements (e.g., $\mathrm{O^-(g)} + e^- \rightarrow \mathrm{O^{2-}(g)}$, $\Delta_{\mathrm{eg}} H_2 = +780\,\mathrm{kJ\,mol^{-1}}$) due to overpowering electrostatic repulsion between the incoming electron and the existing negative anion.

**Cardinal Periodic Trap: Fluorine vs Chlorine & Oxygen vs Sulfur**

- **Halogen Group 17 Order:**

$$|\Delta_{\mathrm{eg}} H| : \mathrm{Cl} > \mathrm{F} > \mathrm{Br} > \mathrm{I} \implies \Delta_{\mathrm{eg}} H : \mathrm{Cl}(-349) < \mathrm{F}(-328) < \mathrm{Br}(-325) < \mathrm{I}(- \tag{4.8}$$

**Reason:** Fluorine has an extraordinarily compact $2p$ valence shell ($n = 2$). The incoming electron encounters intense interelectronic

| Element Pair | Observed Trend | | Theoretical Origin & NCERT Rationale |
|---|---|---|---|
| **Be vs B** | $IE_1$(Be) $899\,kJ\,mol^{-1}$ | = | Be ($2s^2$) loses electron from fully-filled subshell. B ($2s^2 2p^1$) loses less penetrating $2p$ electron of higher energy. |
| | > $IE_1$(B) $801\,kJ\,mol^{-1}$ | = | |
| **N vs O** | $IE_1$(N) $1402\,kJ\,mol^{-1}$ | = | N has extra-stable half-filled $2p^3$ subshell. O ($2p^4$) has paired electrons in $2p_x$; mutual repulsion destabilizes the electron. |
| | > $IE_1$(O) $1314\,kJ\,mol^{-1}$ | = | |
| **Mg vs Al** | $IE_1$(Mg) $737\,kJ\,mol^{-1}$ | = | Analogous to Be/B: Mg has fully-filled $3s^2$; Al has solitary $3p^1$ outside neon core. |
| | > $IE_1$(Al) $577\,kJ\,mol^{-1}$ | = | |
| **P vs S** | $IE_1$(P) $1012\,kJ\,mol^{-1}$ | = | Analogous to N/O: P has half-filled $3p^3$; S ($3p^4$) experiences spin-pairing repulsion. |
| | > $IE_1$(S) $1000\,kJ\,mol^{-1}$ | = | |
| **Tl vs In** | $IE_1$(Tl) $589\,kJ\,mol^{-1}$ | = | $4f^{14}$ Lanthanoid contraction in Tl provides poor shielding, raising $Z_{\text{eff}}$ on outer electrons. |
| | > $IE_1$(In) $558\,kJ\,mol^{-1}$ | = | |
| **Pb vs Sn** | $IE_1$(Pb) $715\,kJ\,mol^{-1}$ | = | $4f^{14}$ Lanthanoid contraction raises $Z_{\text{eff}}$ in Lead, arresting downward decrease. |
| | > $IE_1$(Sn) $708\,kJ\,mol^{-1}$ | = | |

Table 4.1: Systematic Catalogue of High-Yield Ionization Enthalpy Anomalies.

repulsion from the 7 tightly packed valence electrons. In Chlorine, the larger, more diffuse $3p$ shell accommodates the incoming electron with substantially less Coulombic repulsion.

- **Chalcogen Group 16 Order:**

$$|\Delta_{\text{eg}}H| : \text{S} > \text{Se} > \text{Te} > \text{Po} > \text{O} \implies \text{Oxygen has the } \textbf{LEAST negative} \text{ valu} \tag{4.9}$$

Oxygen ($\Delta_{\text{eg}}H = -141\,\text{kJ mol}^{-1}$) suffers severe electron-electron repulsion in its tiny $2p$ shell, whereas Sulfur ($\Delta_{\text{eg}}H = -200\,\text{kJ mol}^{-1}$) accommodates the electron far more readily.

- **Positive / Near-Zero $\Delta_{\text{eg}}H$ Values:**
  - **Noble Gases (He, Ne, Ar, Kr, Xe):** Strongly positive $\Delta_{\text{eg}}H > 0$ because the incoming electron must enter a higher unpopulated principal shell.
  - **Alkaline Earth Metals (Be, Mg):** $\Delta_{\text{eg}}H \approx 0$ due to fully-filled stable $ns^2$ subshells.
  - **Nitrogen (N):** $\Delta_{\text{eg}}H \approx +31\,\text{kJ mol}^{-1}$ due to extra-stable half-filled $2p^3$ configuration.

### 4.2.6 Electronegativity & The Inert Pair Effect

- **Electronegativity ($\chi$):** Qualitative tendency of an atom in a covalent molecule to attract shared bonding electrons toward itself.
  - **Pauling Scale:** Based on bond dissociation energy discrepancies:

$$\Delta = E_{A-B} - \sqrt{E_{A-A} \times E_{B-B}} \implies |\chi_A - \chi_B| = 0.208\sqrt{\Delta\ (\text{in kcal.mol}^{-1})} = 0 \tag{4.10}$$

Pauling assigned Fluorine the benchmark value $\chi(\text{F}) = 4.0$. Absolute hierarchy:

$$\text{F}(4.0) > \text{O}(3.5) > \text{N}(3.0) \approx \text{Cl}(3.0) > \text{Br}(2.8) > \text{I}(2.5) \approx \text{C}(2.5) \approx \text{S}(2.5) > \text{H}( \tag{4.11}$$

  - **Mulliken Scale:** Arithmetic mean of ionization enthalpy and electron affinity:

$$\chi_{\text{Mulliken}} = \frac{\text{IE} + \text{EA}}{2}\ (\text{in eV}), \qquad \chi_{\text{Pauling}} \approx \frac{\chi_{\text{Mulliken}}}{2.8} = \frac{\text{IE} + \text{EA}}{5.6} \tag{4.12}$$

  - **Hybridization Influence:** Electronegativity increases with increasing $\%s$-character:

$$sp\ (50\%\ s) > sp^2\ (33.3\%\ s) > sp^3\ (25\%\ s) \tag{4.13}$$

- **The Inert Pair Effect:** The reluctance of the valence $ns^2$ electron pair to participate in chemical bonding down Groups 13, 14, and 15.
  **Physical Origin:**
  1. Intervening $(n-1)d^{10}$ and $(n-2)f^{14}$ electrons shield the nucleus poorly, subjecting the spherical $ns$ orbital to an abnormally high effective nuclear charge $Z_{\text{eff}}$.
  2. Relativistic contraction pulls $ns$ electrons closer to the nucleus, significantly increasing the energy required to unpair and promote an $s$-electron to the $p$-subshell.
  3. Bond energies decrease down the group due to larger orbital size; the energy released by forming two additional covalent bonds is insufficient to compensate for the promotion energy of the $ns^2$ electrons.

  **Consequences on Oxidation State Stabilities:**
  - **Group 13:** Stability of +1 state increases: $B^{3+} > Al^{3+} > Ga^{3+} > In^{3+} > Tl^{3+}$, whereas $Tl^{+} > In^{+} > Ga^{+} > Al^{+}$. TlCl is stable; $Tl^{3+}$ is an oxidant ($Tl^{3+} + 2e^{-} \rightarrow Tl^{+}$).
  - **Group 14:** Stability of +2 state increases: C(+4) > Si(+4) > Ge(+4) > Sn(+4) > Pb(+4), whereas $Pb^{2+} > Sn^{2+} > Ge^{2+}$. Thus, $Sn^{2+}$ is a reducing agent, while $PbO_2$ is a strong oxidant.
  - **Group 15:** Stability of +3 state increases: $Bi^{3+} > Sb^{3+} > As^{3+}$. $Bi_2O_5$ and $BiF_5$ are powerful oxidants because Bi(+5) readily reduces to +3.

# 4.3 Chemical Bonding & Molecular Structure (Unit 3)

## 4.3.1 Ionic Bonding, Lattice Energy & Fajan's Rules

Ionic bonding arises from omnidirectional electrostatic attraction between oppositely charged ions:

$$E_{\text{coulomb}} = -\frac{1}{4\pi\varepsilon_0}\frac{z_+ z_- e^2}{r_0} \tag{4.14}$$

Lattice energy ($\Delta_{\text{lattice}}H^\circ$) is the energy required to completely separate 1 mole of solid ionic crystal into its constituent gaseous ions at infinite separation.

**Fajan's Rules: Covalent Character in Ionic Bonds**

No chemical bond is 100% ionic. When a cation approaches an anion, the cation pulls the valence electron cloud of the anion toward itself while repelling its positive nucleus, polarizing the anion. High

polarization introduces **covalent character**.

1. **Small Cation Size:** Smaller cations possess higher positive charge density and greater polarizing power ($\phi = \text{charge/radius}$):

   $$\text{Covalent character: LiCl} > \text{NaCl} > \text{KCl} > \text{RbCl} > \text{CsCl} \implies \text{Melting Point: M} \tag{4.15}$$

2. **Large Anion Size:** Larger anions have diffuse electron clouds that are easily distorted (high polarizability):

   $$\text{Covalent character: LiI} > \text{LiBr} > \text{LiCl} > \text{LiF} \tag{4.16}$$

3. **High Ionic Charges:** High positive charge on cation and high negative charge on anion dramatically maximize polarization:

   $$\text{Covalent character: } \text{SnCl}_4(\text{Sn}^{4+}) > \text{SnCl}_2(\text{Sn}^{2+}), \qquad \text{AlCl}_3(\text{Al}^{3+}) > \text{MgCl}_2(\text{M} \tag{4.17}$$

4. **Pseudo-Inert Gas Electron Configuration ($ns^2np^6nd^{10}$):** A cation with 18 valence electrons ($\text{Cu}^+, \text{Ag}^+, \text{Zn}^{2+}, \text{Cd}^{2+}$) exerts far greater polarizing power than a noble-gas configuration cation ($ns^2np^6$, 8 electrons; e.g., $\text{Na}^+, \text{K}^+, \text{Ca}^{2+}$) of comparable size and charge, because $d$-electrons shield the nuclear charge very poorly:

   $$\text{CuCl } (18e^-) \text{ is far more covalent and insoluble than NaCl } (8e^-) \quad [r(\text{Cu}^+) \approx r \tag{4.18}$$

### 4.3.2 Lewis Structures & Formal Charge

**Formal Charge Calculation**

Formal charge represents the hypothetical electric charge assigned to an individual atom in a Lewis structure assuming equal sharing of bonding electrons:

$$\text{FC} = V - L - \frac{1}{2}S \tag{4.19}$$

where $V$ is the number of valence electrons of the isolated neutral atom, $L$ is the number of non-bonding lone-pair electrons, and $S$ is the number of shared bonding electrons. **Example: Ozone ($O_3$):**

- Central Oxygen atom (1 double bond, 1 single coordinate bond, 1 lone pair):

  $$\text{FC} = 6 - 2 - \frac{1}{2}(6) = +1 \tag{4.20}$$

- Double-bonded terminal Oxygen (2 lone pairs, 4 shared electrons):

$$\text{FC} = 6 - 4 - \frac{1}{2}(4) = 0 \tag{4.21}$$

- Single-bonded terminal Oxygen (3 lone pairs, 2 shared electrons):

$$\text{FC} = 6 - 6 - \frac{1}{2}(2) = -1 \tag{4.22}$$

### 4.3.3 Dipole Moments & Vector Summation

The electric dipole moment $\vec{\mu}$ measures bond and molecular polarity:

$$\mu = q \times d \quad [1\,\text{Debye (D)} = 3.33564 \times 10^{-30}\,\text{C m}] \tag{4.23}$$

Molecular dipole moment is the vector sum of all individual bond moments and lone-pair moments:

$$\mu_{\text{net}} = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2\cos\theta} \tag{4.24}$$

**The $NH_3$ vs $NF_3$ Dipole Moment Paradox**

Both Ammonia ($NH_3$) and Nitrogen Trifluoride ($NF_3$) exhibit identical trigonal pyramidal geometry ($SN = 4$, 1 lone pair). However:

$$\mu(NH_3) = 1.47\,\text{D} \gg \mu(NF_3) = 0.23\,\text{D} \tag{4.25}$$

**Rigorous Vector Explanation:**

- In $NH_3$, Nitrogen ($\chi = 3.0$) is more electronegative than Hydrogen ($\chi = 2.1$). The three individual N – H bond dipoles point upward toward Nitrogen, **reinforcing and adding constructively** to the upward orbital dipole of the localized lone pair.
- In $NF_3$, Fluorine ($\chi = 4.0$) is far more electronegative than Nitrogen ($\chi = 3.0$). The three individual N – F bond dipoles point downward away from Nitrogen, **opposing and destructively cancelling** the upward lone-pair dipole.

### 4.3.4 VSEPR Theory: Master 3D Geometry Framework

Valence Shell Electron Pair Repulsion (VSEPR) theory dictates that electron pairs surrounding a central atom repel one another and adopt spatial orientations that minimize electrostatic repulsion.

$$\text{Repulsion Hierarchy: } lp - lp > lp - bp > bp - bp \tag{4.26}$$

Steric Number ($SN$) is computed by:

$$SN = (\text{Number of } \sigma\text{-bonds}) + (\text{Number of localized lone pairs on central atom}) \tag{4.27}$$

**Bent's Rule & Bond Length Asymmetry in $PCl_5$**

- **Bent's Rule Formulation:** More electronegative substituents prefer hybrid orbitals with **less $s$-character** (more $p$-character). Lone pairs and electropositive substituents prefer hybrid orbitals with **more $s$-character**.
- **Application to Trigonal Bipyramidal Geometry ($sp^3d = sp^2 + pd$):**
  - Equatorial hybrid orbitals ($sp^2$) possess 33.3% $s$-character.
  - Axial hybrid orbitals ($pd$) possess 0% $s$-character ($50\%p + 50\%d_{z^2}$).
- **Asymmetry in Gaseous $PCl_5$:**
  - The two axial $P-Cl$ bonds experience intense $90^\circ$ repulsion from three equatorial bonds.
  - Consequently, **axial bonds are longer and weaker** (214 pm) than equatorial bonds (202 pm).
  - Upon gentle heating, $PCl_5$ easily dissociates its two axial chlorines: $PCl_5(g) \xrightarrow{\Delta} PCl_3(g) + Cl_2(g)$.
  - In the crystalline solid phase, $PCl_5$ ionizes into ionic tetrahedral and octahedral entities: $\mathbf{2}PCl_5(s) \longrightarrow [PCl_4]^+[PCl_6]^-$. Solid $PBr_5$ exists as $[PBr_4]^+Br^-$.
- **Drago's Rule:** In hydrides of Group 15 and 16 elements from the 3rd period onwards ($PH_3, AsH_3, SbH_3, H_2S, H_2Se, H_2Te$), the central atom does **NOT** undergo hybridization. Bonding occurs via nearly pure $p$-orbitals at $\approx 90^\circ$ angles ($PH_3 : 93.5^\circ, H_2S : 92^\circ$). The lone pair resides in an inert, spherically symmetric $s$-orbital with negligible Lewis basicity.

**High-Yield Problem Archetype: Systematic VSEPR Steric Number Prediction**

**Predicting Hybridization and 3D Geometry for Problematic NEET Species:** Formula for Steric Number of central atom $A$ in $AB_n^q$:

$$SN = \frac{1}{2}\left[V + M - C + A\right]$$

where $V$ = valence $e^-$ of $A$, $M$ = monovalent atoms attached ($H, F, Cl, Br, I$), $C$ = cationic charge, $A$ = anionic charge.

| $SN$ | $\sigma$-bonds | Lone Pairs | **Electron Geometry** | **Molecular Geometry** | **High-Yield Archetypes & Angles** |
|---|---|---|---|---|---|
| 2 | 2 | 0 | Linear | Linear | $BeCl_2, CO_2$, HCN (180°) |
| 3 | 3 | 0 | Trigonal Planar | Trigonal Planar | $BF_3, BCl_3, SO_3$, (120°) |
| 3 | 2 | 1 | Trigonal Planar | Bent / Angular | $SO_2, O_3, NO_2^-$, S (< 120°, ≈ 119°) |
| 4 | 4 | 0 | Tetrahedral | Tetrahedral | $CH_4, CCl_4, NH_4^+$ (109°28′) |
| 4 | 3 | 1 | Tetrahedral | Trigonal Pyramidal | $NH_3, PCl_3, H_3O^+$ (107°) |
| 4 | 2 | 2 | Tetrahedral | Bent / V-shaped | $H_2O, H_2S, OF_2$, S (104.5°) |
| 5 | 5 | 0 | Trigonal Bipyramidal | Trigonal Bipyramidal | $PCl_5, PF_5, AsF_5$ (eq 120°, ax 90°) |
| 5 | 4 | 1 | Trigonal Bipyramidal | Seesaw | $SF_4, SeF_4, TeCl_4$ (173°, 102°) |
| 5 | 3 | 2 | Trigonal Bipyramidal | T-shaped | $ClF_3, BrF_3, IF_3$ (87.5°) |
| 5 | 2 | 3 | Trigonal Bipyramidal | Linear | $XeF_2, I_3^-, ICl_2^-$ (180°) |
| 6 | 6 | 0 | Octahedral | Octahedral | $SF_6, [PF_6]^-$, [AlF (90°) |
| 6 | 5 | 1 | Octahedral | Square Pyramidal | $BrF_5, IF_5$, XeOF (86°) |
| 6 | 4 | 2 | Octahedral | Square Planar | $XeF_4, [ICl_4]^-$, [B (90°) |
| 7 | 7 | 0 | Pentagonal Bipyramidal | Pentagonal Bipyramidal | $IF_7$ (eq 72°, ax 90°) |
| 7 | 6 | 1 | Pentagonal Bipyramidal | Distorted Octahedral | $XeF_6$ (fluxional lone pair |

1. **Triiodide Ion ($I_3^-$):** Central I has $V = 7$, two attached monovalent iodines ($M = 2$), anionic charge $A = 1$:

$$SN = \frac{1}{2}[7 + 2 - 0 + 1] = 5 \implies sp^3d \text{ hybridization}$$

With 2 bond pairs and $5 - 2 = 3$ lone pairs: all three lone pairs occupy equatorial positions at 120° $\implies$ **Linear Geometry** (bond angle 180°).

2. **Xenon Tetrafluoride ($XeF_4$):** Central Xe has $V = 8$, $M = 4$:

$$SN = \frac{1}{2}[8 + 4] = 6 \implies sp^3d^2 \text{ hybridization}$$

With 4 bond pairs and $6 - 4 = 2$ lone pairs: lone pairs occupy opposite trans-axial positions at 180° $\implies$ **Square Planar Geometry** (90°).

3. **Sulfur Tetrafluoride ($SF_4$):** Central S has $V = 6$, $M = 4$:

$$SN = \frac{1}{2}[6 + 4] = 5 \implies sp^3d \text{ hybridization}$$

With 4 bond pairs and 1 lone pair: lone pair occupies an equatorial position $\implies$ **Seesaw Geometry**.

4. **Chlorine Trifluoride ($ClF_3$):** Central Cl has $V = 7$, $M = 3$:

$$SN = \frac{1}{2}[7 + 3] = 5 \implies sp^3d \text{ hybridization}$$

With 3 bond pairs and 2 lone pairs: both lone pairs occupy equatorial positions $\implies$ **T-shaped Geometry**.

### 4.3.5 Molecular Orbital Theory (MOT): Energy Levels, Bond Orders & Magnetism

Molecular Orbital Theory constructs molecular orbitals by Linear Combination of Atomic Orbitals (LCAO):

$$\psi_{\text{MO}} = c_A\psi_A \pm c_B\psi_B \implies \psi_b = \frac{1}{\sqrt{2}}(\psi_A + \psi_B), \qquad \psi_a = \frac{1}{\sqrt{2}}(\psi_A - \psi_B) \tag{4.28}$$

**1. Diatomic Molecules with $\leq 14$ Valence Electrons (with *sp*-Mixing: $B_2, C_2, N_2$)**

Because the energy separation between $2s$ and $2p_z$ atomic orbitals is small in B, C, N, quantum mechanical mixing occurs between $\sigma 2s$ and $\sigma 2p_z$. This raises the energy of $\sigma 2p_z$ above the degenerate $\pi 2p_x, \pi 2p_y$ orbitals:

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z \tag{4.29}$$

**2. Diatomic Molecules with $> 14$ Valence Electrons (without *sp*-Mixing: $O_2, F_2$)**

In Oxygen and Fluorine, higher nuclear charge depresses the $2s$ level, creating a large $\Delta E(2s - 2p_z)$ gap ($> 15\,\text{eV}$) that precludes *sp*-mixing. Normal energy ordering applies:

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z \tag{4.30}$$

**Bond Order & Magnetic Properties Formulation**

$$\text{Bond Order (BO)} = \frac{N_b - N_a}{2} \tag{4.31}$$

where $N_b$ is the number of electrons in bonding MOs, and $N_a$ is the number of electrons in antibonding MOs.

- If all electrons are paired $\implies$ **Diamagnetic**.
- If one or more unpaired electrons exist $\implies$ **Paramagnetic** ($\mu_s = \sqrt{n(n+2)}\,\text{BM}$).

**NCERT High-Yield Benchmark: Oxygen Species Comparison**

- **Bond Order Sequence:** $O_2^+(2.5) > O_2(2.0) > O_2^-(1.5) > O_2^{2-}(1.0)$.
- **Bond Length Sequence (Inversely proportional to Bond Order):**

$$O_2^+(112\,\text{pm}) < O_2(121\,\text{pm}) < O_2^-(128\,\text{pm}) < O_2^{2-}(149\,\text{pm}) \tag{4.32}$$

- **Bond Dissociation Energy Sequence:** $O_2^+ > O_2 > O_2^- > O_2^{2-}$.

## 4.3.6 Hydrogen Bonding: Intermolecular vs Intramolecular

Hydrogen bonding is an electrostatic dipole-dipole attraction occurring between a covalently bound Hydrogen atom (attached to highly electronegative

| Species | Total $e^-$ | Valence MO Configuration | $N_b/N_a$ | BO |
|---|---|---|---|---|
| $B_2$ | 10 | $(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p_x)^1(\pi 2p_y)^1$ | 6/4 | 1.0 |
| $C_2$ | 12 | $(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p_x)^2(\pi 2p_y)^2$ | 8/4 | 2.0 |
| $N_2$ | 14 | $(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p_x)^2(\pi 2p_y)^2(\sigma 2p_z)^2$ | 10/4 | 3.0 |
| $N_2^+$ | 13 | $(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p_x)^2(\pi 2p_y)^2(\sigma 2p_z)^1$ | 9/4 | 2.5 |
| $O_2^+$ | 15 | $(\sigma 2s)^2(\sigma^* 2s)^2(\sigma 2p_z)^2(\pi 2p)^4(\pi^* 2p_x)^1$ | 10/5 | 2.5 |
| $O_2$ | 16 | $(\sigma 2s)^2(\sigma^* 2s)^2(\sigma 2p_z)^2(\pi 2p)^4(\pi^* 2p_x)^1(\pi^* 2p_y)^1$ | 10/6 | 2.0 |
| $O_2^-$ | 17 | $(\sigma 2s)^2(\sigma^* 2s)^2(\sigma 2p_z)^2(\pi 2p)^4(\pi^* 2p_x)^2(\pi^* 2p_y)^1$ | 10/7 | 1.5 |

F, O, N) and another electronegative atom with a localized lone pair.

- **Intermolecular Hydrogen Bonding:** Occurs between distinct molecules (e.g., $H_2O$, HF, $NH_3$, $C_2H_5OH$, *p*-nitrophenol). **Effects:** Elevated boiling points, higher viscosity, lower volatility, and high water solubility. Ice has an open cage-like tetrahedral structure making it less dense than liquid water at 4°C.
- **Intramolecular Hydrogen Bonding (Chelation):** Occurs within the same molecule, forming a 5- or 6-membered ring (e.g., *o*-nitrophenol, salicylaldehyde, salicylic acid). **Effects:** Prevents intermolecular association; results in **significantly lower boiling points**, higher volatility, and **steam volatility** (enabling clean separation of *o*-nitrophenol from *p*-nitrophenol via steam distillation).

## 4.4 Coordination Chemistry (Unit 12)

### 4.4.1 Werner's Coordination Theory

Alfred Werner (1893) postulated that transition metals exhibit two distinct types of valency:

1. **Primary Valency (Ionizable):** Corresponds to the oxidation state of the metal ion. It is non-directional and satisfied strictly by negative anions.
2. **Secondary Valency (Non-Ionizable):** Corresponds to the coordination number of the metal ion (4 or 6). It is strictly directional in 3D space (establishing spatial geometry) and satisfied by neutral molecules or negative ligands.

| Cobalt Ammine Complex | Modern IUPAC Formulation | Moles AgCl ↓ | Total Ions | Ele |
|---|---|---|---|---|
| $CoCl_3 \cdot 6NH_3$ | $[Co(NH_3)_6]Cl_3$ | 3 | 4 (1 : 3 electrolyte) | Hig |
| $CoCl_3 \cdot 5NH_3$ | $[Co(NH_3)_5Cl]Cl_2$ | 2 | 3 (1 : 2 electrolyte) | Mod |
| $CoCl_3 \cdot 4NH_3$ | $[Co(NH_3)_4Cl_2]Cl$ | 1 | 2 (1 : 1 electrolyte) | L |
| $CoCl_3 \cdot 3NH_3$ | $[Co(NH_3)_3Cl_3]$ | 0 | 0 (Neutral molecule) | |

Table 4.4: Werner Formulation, Precipitation Stoichiometry, and Solution Conductivity.

### 4.4.2 Nomenclature & Isomerism in Coordination Entities

### IUPAC Nomenclature Principles

1. Cation is named first in both positive and negative coordination entities.
2. Ligands are listed in strict alphabetical order before the central metal name, regardless of charge.
3. Multiplicative prefixes: *di-, tri-, tetra-* for simple ligands; *bis-, tris-, tetrakis-* for polydentate or ligands containing prefixes (e.g., *ethylenediamine*).
4. Anionic ligands end in *-ido* ($Cl^- \rightarrow$ chlorido, $CN^- \rightarrow$ cyanido, $OH^- \rightarrow$ hydroxido).
5. If the complex sphere is anionic, the metal name ends in ***-ate*** (Fe $\rightarrow$ ferrate, Cu $\rightarrow$ cuprate, Ag $\rightarrow$ argentate).
6. Oxidation state of the metal is designated by Roman numerals in parentheses: e.g., $K_4[Fe(CN)_6] \implies$ Potassium hexacyanidoferrate(II).

### Structural Isomerism

- **Ionization Isomerism:** Exchange of ionizable counter-ion with coordinated ligand:

  $[Co(NH_3)_5SO_4]Br$ (gives $AgBr\downarrow$ with $AgNO_3$) vs $[Co(NH_3)_5Br]SO_4$ (gives Ba (4.33)

- **Solvate / Hydrate Isomerism:** Water acts either as a coordinated ligand or crystal lattice water:

  $[Cr(H_2O)_6]Cl_3$ (Violet, 3 $Cl^-$) vs $[Cr(H_2O)_5Cl]Cl_2{\cdot}H_2O$ (Grey-green, 2 $Cl^-$) (4.34)

- **Linkage Isomerism:** Arises with ambidentate ligands having two distinct coordinating donor atoms:

  $[Co(NH_3)_5(NO_2)]^{2+}$ (Nitro, yellow-brown, via N) vs $[Co(NH_3)_5(ONO)]^{2+}$ (Nit (4.35)

- **Coordination Isomerism:** Interchange of ligands between cationic and anionic complex entities:

  $$[Co(NH_3)_6][Cr(CN)_6] \quad \text{vs} \quad [Cr(NH_3)_6][Co(CN)_6] \tag{4.36}$$

### Stereoisomerism

1. **Geometrical Isomerism (GI):**
   - **Square Planar ($CN = 4$):** $[MA_2B_2]$ exhibits *cis* and *trans* (e.g., *cis*-platin $[Pt(NH_3)_2Cl_2]$ antitumor drug vs inactive *trans*). $[MABCD]$ yields 3 distinct geometrical isomers. Tetrahedral complexes **NEVER** show GI because all 4 positions are mutually adjacent (109.5°).

- **Octahedral ($CN = 6$):**
  - $[MA_4B_2]$: *cis* (90°) and *trans* (180°).
  - $[MA_3B_3]$: **Facial (*fac*)** where three identical ligands occupy corners of the same octahedral triangular face, and **Meridional (*mer*)** where ligands occupy a meridian plane surrounding the metal.

2. **Optical Isomerism (Chirality):** Molecules lacking an improper axis of rotation ($S_n$, including plane of symmetry $\sigma$ and inversion centre $i$) exist as non-superimposable mirror image enantiomers ($d$ and $l$).
   - $[M(AA)_3]$ (e.g., $[Co(en)_3]^{3+}$) is chiral and resolves into $\Delta$ and $\Lambda$ optical enantiomers.
   - $[M(AA)_2B_2]$ (e.g., $[Co(en)_2Cl_2]^{+}$):
     - ***cis*-isomer:** Asymmetric, lacks plane of symmetry $\implies$ **Optically Active** (2 enantiomers).
     - ***trans*-isomer:** Possesses a plane of symmetry ($\sigma_h$) $\implies$ **Optically Inactive**.
   - Square planar complexes have a horizontal plane of symmetry and **NEVER** show optical isomerism.

### 4.4.3 Valence Bond Theory (VBT) & Magnetic States

Linus Pauling applied hybridization of metal $d, s, p$ orbitals to explain coordinate bonding.

- **Inner Orbital Complexes ($d^2sp^3$):** Strong-field ligands force electron pairing in $(n-1)d$ orbitals, providing two empty inner $d$-orbitals (low-spin / spin-paired, e.g., $[Fe(CN)_6]^{4-}$: $3d^6$ paired, diamagnetic, $\mu = 0$).
- **Outer Orbital Complexes ($sp^3d^2$):** Weak-field ligands cannot force pairing; vacant outer $nd$ orbitals are hybridized (high-spin / spin-free, e.g., $[Fe(H_2O)_6]^{2+}$: $3d^6$ with 4 unpaired electrons, paramagnetic, $\mu = \sqrt{4(6)} = 4.90\,\text{BM}$).
- **Limitations of VBT:** Fails to predict absorption spectra or color; cannot quantitatively explain temperature-dependent magnetic susceptibilities; cannot explain why identical $d^8$ configuration forms tetrahedral paramagnetic $[NiCl_4]^{2-}$ but square planar diamagnetic $[Ni(CN)_4]^{2-}$.

### 4.4.4 Crystal Field Theory (CFT): Splitting Diagrams & CFSE

Crystal Field Theory treats the metal-ligand interaction as an electrostatic perturbation arising from point negative charges (anions) or point dipoles (neutral molecules like $NH_3, H_2O$) surrounding the central metal cation. In an isolated gaseous transition metal atom or ion, all five $d$-orbitals ($d_{xy}, d_{yz}, d_{zx}, d_{x^2-y^2}, d_{z^2}$) possess identical energy (five-fold degenerate).

When ligands approach the metal ion, two sequential perturbations occur:

1. A hypothetical spherically symmetric negative field raises the energy of all five $d$-orbitals equally due to overall electron-electron repulsion.
2. An asymmetric ligand field splits this degeneracy according to spatial orientation and ligand coordinate geometry (Figure 4.1).

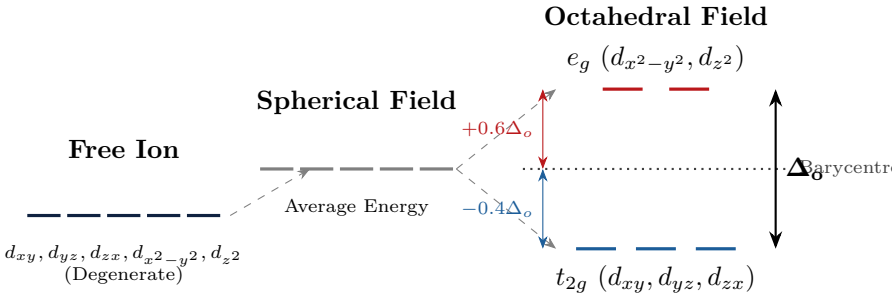



Figure 4.1: Crystal Field Splitting of Five $d$-Orbitals in an Octahedral Complex ($\Delta_o$).

## 1. Octahedral Crystal Field Splitting ($\Delta_o$)

In an octahedral $[ML_6]$ complex, six ligands approach the central metal ion along the Cartesian $\pm x, \pm y, \pm z$ axes:

- The $e_g$ orbitals $(d_{x^2-y^2}, d_{z^2})$ have lobes pointing directly along the Cartesian axes where the ligands lie $\implies$ experience maximal electrostatic repulsion $\implies$ destabilized and raised by $+\mathbf{0.6\Delta_o}$ $(+\frac{3}{5}\Delta_o)$.
- The $t_{2g}$ orbitals $(d_{xy}, d_{yz}, d_{zx})$ have lobes directed in between the Cartesian axes $\implies$ experience minimal direct repulsion $\implies$ stabilized and lowered by $-\mathbf{0.4\Delta_o}$ $(-\frac{2}{5}\Delta_o)$.
- The barycentre (weighted mean energy level) remains invariant: $3(-0.4\Delta_o) + 2(+0.6\Delta_o) = 0$.

## 2. Tetrahedral Crystal Field Splitting ($\Delta_t$)

In a tetrahedral $[ML_4]$ complex, four ligands approach alternate corners of a circumscribed cube containing the metal at its center:

- The $t_2$ orbitals $(d_{xy}, d_{yz}, d_{zx})$ point directly toward the edges of the cube closer to the approaching ligands $\implies$ raised by $+\mathbf{0.4\Delta_t}$.
- The $e$ orbitals $(d_{x^2-y^2}, d_{z^2})$ point toward the faces of the cube further from the ligands $\implies$ lowered by $-\mathbf{0.6\Delta_t}$.

- **Splitting Magnitude Relation:** Because there are only 4 ligands (instead of 6) and none point directly at the $d$-orbitals:

$$\boldsymbol{\Delta_t = \frac{4}{9}\Delta_o} \tag{4.37}$$

Because $\Delta_t$ is less than half of $\Delta_o$, the crystal field splitting energy in tetrahedral geometry is **always smaller than the pairing energy** ($\Delta_t < P$). Consequently, **tetrahedral complexes are universally HIGH-SPIN** (spin-free); low-spin tetrahedral complexes do not exist under standard conditions.

**Spectrochemical Series & CFSE Calculations**

Ligands arranged in order of increasing crystal field splitting magnitude $\Delta_o$:

$$\begin{aligned} &\mathrm{I^- < Br^- < S^{2-} < SCN^- < Cl^- < N_3^- < F^- < OH^- < C_2O_4^{2-}} \\ &< \mathrm{H_2O < NCS^- < edta^{4-} < NH_3 < en < NO_2^- < CN^- < CO} \end{aligned} \tag{4.38}$$

**High-Spin vs Low-Spin Criterion for $d^4, d^5, d^6, d^7$ in Octahedral Fields:**

- **Weak Field Ligands ($\Delta_o < P$):** Electron promotes to $e_g$ before pairing in $t_{2g}$ $\implies$ **High Spin**.
- **Strong Field Ligands ($\Delta_o > P$):** Electrons pair completely in $t_{2g}$ before occupying $e_g$ $\implies$ **Low Spin**.

**Octahedral CFSE Formula:**

$$\mathrm{CFSE} = \left(-0.4\, n_{t_{2g}} + 0.6\, n_{e_g}\right)\Delta_o + mP \tag{4.39}$$

where $m$ is the number of extra electron pairs formed relative to the free gaseous ion.

**High-Yield Problem Archetype: Octahedral CFSE & Spin-Only Magnetic Moments**

**Archetype Problem:** Compare the electronic configuration, CFSE, and magnetic moment ($\mu_s$) for: (a) $[CoF_6]^{3-}$ vs $[Co(en)_3]^{3+}$, and (b) $[Fe(CN)_6]^{3-}$ vs $[Fe(H_2O)_6]^{3+}$.

1. **Cobalt(III) Complexes ($3d^6$ system):**
   - $[CoF_6]^{3-}$**:** $F^-$ is a weak-field ligand ($\Delta_o < P$) $\implies$ **High Spin**.

$$\begin{aligned} \text{Configuration} &= t_{2g}^4 e_g^2, \quad \text{Unpaired } e^- = 4 \\ \mathrm{CFSE} &= (-0.4 \times 4 + 0.6 \times 2)\Delta_o = -0.4\Delta_o \\ \mu_s &= \sqrt{4(4+2)} = \sqrt{24} \approx 4.90\,\mathrm{BM} \quad \text{(Paramagnetic)} \end{aligned}$$

| | High Spin (Weak Field: $\Delta_o < P$) | | | Low Spin (Strong Field | |
|---|---|---|---|---|---|
| $d^n$ | Config | Unpaired | CFSE | Config | Unpaire |
| $d^1$ | $t_{2g}^1 e_g^0$ | 1 | $-0.4\Delta_o$ | $t_{2g}^1 e_g^0$ | 1 |
| $d^2$ | $t_{2g}^2 e_g^0$ | 2 | $-0.8\Delta_o$ | $t_{2g}^2 e_g^0$ | 2 |
| $d^3$ | $t_{2g}^3 e_g^0$ | 3 | $-1.2\Delta_o$ | $t_{2g}^3 e_g^0$ | 3 |
| $d^4$ | $t_{2g}^3 e_g^1$ | 4 | $-0.6\Delta_o$ | $t_{2g}^4 e_g^0$ | 2 |
| $d^5$ | $t_{2g}^3 e_g^2$ | 5 | $\mathbf{0.0\Delta_o}$ | $t_{2g}^5 e_g^0$ | 1 |
| $d^6$ | $t_{2g}^4 e_g^2$ | 4 | $-0.4\Delta_o$ | $t_{2g}^6 e_g^0$ | 0 |
| $d^7$ | $t_{2g}^5 e_g^2$ | 3 | $-0.8\Delta_o$ | $t_{2g}^6 e_g^1$ | 1 |
| $d^8$ | $t_{2g}^6 e_g^2$ | 2 | $-1.2\Delta_o$ | $t_{2g}^6 e_g^2$ | 2 |
| $d^9$ | $t_{2g}^6 e_g^3$ | 1 | $-0.6\Delta_o$ | $t_{2g}^6 e_g^3$ | 1 |
| $d^{10}$ | $t_{2g}^6 e_g^4$ | 0 | $0.0\Delta_o$ | $t_{2g}^6 e_g^4$ | 0 |

Table 4.5: Systematic Crystal Field Stabilization Energies (CFSE) for Octahedral Complexes.

- $[Co(en)_3]^{3-}$**:** *en* is a strong-field chelating ligand ($\Delta_o > P$) $\implies$ **Low Spin**.

$$\text{Configuration} = t_{2g}^6 e_g^0, \quad \text{Unpaired } e^- = 0$$
$$\text{CFSE} = (-0.4 \times 6)\Delta_o + 2P = -2.4\Delta_o + 2P$$
$$\mu_s = 0\,\text{BM} \quad \text{(Diamagnetic)}$$

2. **Iron(III) Complexes ($3d^5$ system):**
   - $[Fe(H_2O)_6]^{3+}$**:** $H_2O$ is weak-field $\implies$ $t_{2g}^3 e_g^2$ $\implies$ CFSE $= 0$, $n = 5$ $\implies$ $\mu_s = 5.92\,\text{BM}$.
   - $[Fe(CN)_6]^{3-}$**:** $CN^-$ is strong-field $\implies$ $t_{2g}^5 e_g^0$ $\implies$ CFSE $= -2.0\Delta_o + 2P$, $n = 1$ $\implies$ $\mu_s = 1.73\,\text{BM}$.

### 4.4.5 Jahn-Teller Distortion Basics

The Jahn-Teller theorem states: *Any non-linear molecular system in a degenerate electronic state is unstable and will undergo geometrical distortion that lowers its symmetry and removes the electronic degeneracy.*

- **Significant / Strong Jahn-Teller Distortion:** Occurs when the $e_g$ **subshell is asymmetrically occupied** because $e_g$ orbitals point di-

rectly along the ligand bond axes:

$$\text{High-spin } d^4\ (t_{2g}^3 e_g^1 : \text{Cr}^{2+}, \text{Mn}^{3+}), \quad \text{Low-spin } d^7\ (t_{2g}^6 e_g^1 : \text{Co}^{2+}), \quad \mathbf{d^9}\ (\mathbf{t_{2g}^6 e_g^3} : \text{Cu}^{2+}) \tag{4.40}$$

- **Tetragonal Elongation ($z$-out distortion):** In $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$, putting the odd electron in $d_{z^2}$ shields the axial ligands, causing the two axial $\text{Cu} - \text{O}$ bonds (240 pm) to elongate significantly relative to the four equatorial bonds (195 pm).

### 4.4.6 Synergic Bonding in Metal Carbonyls

**Synergic Bonding Paradigm in Homoleptic Metal Carbonyls**

In metal carbonyls (e.g., $\text{Ni}(\text{CO})_4, \text{Fe}(\text{CO})_5, \text{Cr}(\text{CO})_6$), the metal-carbon bond comprises a self-reinforcing dual interaction:

1. **$\sigma$-Bond Formation ($M \leftarrow \text{CO}$):** Carbon donates its non-bonding lone pair into an empty hybrid orbital of the transition metal.
2. **$\pi$-Backdonation ($M \rightarrow \text{CO}$):** Filled non-bonding $d$-orbitals of the metal overlap with empty $\pi^*$ antibonding molecular orbitals of Carbon Monoxide.

**Consequences of Synergism:**

- Strengthens the $M - \text{C}$ bond (increases bond order, shortens bond length).
- Populates the $\pi^*$ orbital of CO, **weakening the $\text{C} - \text{O}$ bond** (lowering $\text{C} - \text{O}$ bond order and reducing $\text{C} - \text{O}$ vibrational stretching frequency $\nu_{\text{CO}}$ in IR spectroscopy).
- Greater negative charge on metal increases backdonation $\implies$ weakest $\text{C} - \text{O}$ bond:

$$\text{IR frequency: } [\text{Mn}(\text{CO})_6]^+ > [\text{Cr}(\text{CO})_6] > [\text{V}(\text{CO})_6]^- \tag{4.41}$$

## 4.5 Comparative Chemistry of $p$-Block Elements (Groups 13 to 18)

### 4.5.1 Group 13: Boron Family

- **Anomalous Boron:** Non-metal, high melting point (2453 K), maximum covalency of 4 (absence of $d$-orbitals).
- **Diborane ($\text{B}_2\text{H}_6$):** Electron-deficient dimer with 12 valence electrons. **Bonding Structure:** Contains four terminal two-centre-two-electron

($2c$–$2e$) B – H bonds (119 pm), and two bridging three-centre-two-electron ($3c$–$2e$) B – H – B **banana bonds** (133 pm). **Synthesis & Reactions:**

$$4BF_3 + 3LiAlH_4 \xrightarrow{\text{ether}} 2B_2H_6 + 3LiF + 3AlF_3 \tag{4.42}$$

$$B_2H_6 + 3O_2 \longrightarrow B_2O_3 + 3H_2O \quad (\Delta H = -1976\,\text{kJ mol}^{-1}, \text{green-edged flame}) \tag{4.43}$$

$$3B_2H_6 + 6NH_3 \xrightarrow{450\,\text{K}} 2B_3N_3H_6\ (\textbf{Borazine / Inorganic Benzene}) + 12H_2 \tag{4.44}$$

- **Orthoboric Acid ($H_3BO_3$):** Layer structure with planar $BO_3^{3-}$ units held by H-bonds. Acts as a **monobasic weak Lewis acid** (accepts $OH^-$ from water rather than donating $H^+$):

$$B(OH)_3 + 2H_2O \rightleftharpoons [B(OH)_4]^- + H_3O^+ \quad (pK_a = 9.25) \tag{4.45}$$

  Cannot be directly titrated with NaOH, but addition of *cis*-polyhydroxy compounds (glycerol, mannitol) forms stable chelate complexes, releasing proton quantitatively.
- **Lewis Acidity of Boron Trihalides:**

$$BI_3 > BBr_3 > BCl_3 > BF_3 \tag{4.46}$$

  **Reason:** In $BF_3$, vacant $2p$ orbital of Boron overlaps efficiently with filled $2p$ orbital of Fluorine via $p\pi$–$p\pi$ backbonding, neutralizing electron deficiency. In $BCl_3$ ($2p$–$3p$) and $BI_3$ ($2p$–$5p$), overlap is ineffective due to large size discrepancy.

### 4.5.2 Group 14: Carbon Family

- **Allotropes of Carbon:**
  1. **Diamond:** $sp^3$ hybridized, tetrahedral 3D giant covalent network, C – C bond length 154 pm, electrical insulator, hardest natural substance.
  2. **Graphite:** $sp^2$ hybridized, planar hexagonal layers (C – C = 141.5 pm) separated by 340 pm held by weak van der Waals forces; fourth $\pi$-electron is delocalized over the sheet $\implies$ **electrical conductor** and solid lubricant. **Thermodynamically most stable allotrope** ($\Delta_f H^\circ = 0$).
  3. **Fullerenes ($C_{60}$ Buckminsterfullerene):** Football-shaped cage of 60 carbon atoms comprising 20 six-membered rings and 12 five-membered rings.
- **Silicones:** Organosilicon polymers with repeating $[R_2SiO]_n$ units synthesized by hydrolysis of dialkyldichlorosilanes ($R_2SiCl_2 \xrightarrow{H_2O} R_2Si(OH)_2 \xrightarrow{\text{polymerizatio}}$ silicone). Highly water-repellent, thermally stable, electrical insulators.

- **Silicates:** Basic structural unit is the tetrahedral $[SiO_4]^{4-}$ entity:
  - **Orthosilicates:** Discrete isolated $[SiO_4]^{4-}$ (e.g., Zircon $ZrSiO_4$, Phenacite $Be_2SiO_4$).
  - **Pyrosilicates:** Share 1 oxygen atom: $[Si_2O_7]^{6-}$ (e.g., Thortveitite $Sc_2Si_2O_7$).
  - **Cyclic Silicates:** Share 2 oxygens per tetrahedron: $[Si_3O_9]^{6-}$ or $[Si_6O_{18}]^{12-}$ (e.g., Beryl $Be_3Al_2Si_6O_{18}$).
  - **Chain Silicates (Pyroxenes):** Share 2 oxygens forming linear chains: $[SiO_3]_n^{2n-}$.
  - **Sheet Silicates (Phyllosilicates):** Share 3 oxygens forming 2D sheets: $[Si_2O_5]_n^{2n-}$ (e.g., Talc, Mica).
  - **3D Framework Silicates (Tectosilicates):** Share all 4 oxygens: $SiO_2$ (Quartz, Feldspar, Zeolites).
- **Zeolites:** Microporous aluminosilicates with open 3D honeycomb channel structures. **ZSM-5** acts as a shape-selective catalyst converting alcohols directly into gasoline.

### 4.5.3 Group 15: Nitrogen Family

- **Dinitrogen ($N_2$):** Chemically inert at ambient conditions due to extraordinarily high bond dissociation energy of the $N \equiv N$ triple bond ($941.4\,kJ\,mol^{-1}$).
- **Hydrides of Group 15 ($NH_3, PH_3, AsH_3, SbH_3, BiH_3$):**
  - **Boiling Point:** $BiH_3 > SbH_3 > \mathbf{NH_3} > AsH_3 > PH_3$ ($NH_3$ abnormally high due to H-bonding; $PH_3$ is lowest).
  - **Basicity:** $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$ (nitrogen has smallest size and highest lone-pair charge density).
  - **Thermal Stability:** $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$ ($M-H$ bond length increases).
  - **Reducing Power:** $BiH_3 > SbH_3 > AsH_3 > PH_3 > NH_3$ ($BiH_3$ is the strongest reducing agent).
  - **Bond Angle:** $NH_3(107.8°) > PH_3(93.5°) > AsH_3(91.8°) > SbH_3(91.3°)$ (Drago's rule).
- **Nitric Acid ($HNO_3$ - Ostwald Process):**

$$4NH_3+5O_2 \xrightarrow[Pt/Rh,500\,K]{9\,bar} 4NO+6H_2O, \quad 2NO+O_2 \rightarrow 2NO_2, \quad 3NO_2+H_2O \rightarrow 2HNO_3+N \tag{4.47}$$

**Reaction with Metals (NEET Benchmark):**

$$\text{Copper with Dilute } HNO_3 : 3Cu + 8HNO_3(dil) \longrightarrow 3Cu(NO_3)_2 + \mathbf{2}NO \uparrow + 4H_2O \tag{4.48}$$

$$\text{Copper with Conc. } HNO_3 : Cu + 4HNO_3(conc) \longrightarrow Cu(NO_3)_2 + \mathbf{2}NO_2 \uparrow + 2H_2O \tag{4.49}$$

$$\text{Zinc with Dilute } HNO_3 : 4Zn + 10HNO_3(dil) \longrightarrow 4Zn(NO_3)_2 + N_2O \uparrow + 5H_2O \tag{4.50}$$

$$\text{Zinc with Conc. } HNO_3 : Zn + 4HNO_3(conc) \longrightarrow Zn(NO_3)_2 + \mathbf{2}NO_2 \uparrow + 2H_2O \tag{4.51}$$

Passive metals (Fe, Cr, Al) do not dissolve in concentrated $HNO_3$ due to formation of an impervious surface oxide film.

- **Oxoacids of Phosphorus:**
  - **Hypophosphorous acid ($H_3PO_2$):** Monobasic (1 $P-OH$), contains 2 $P-H$ bonds $\implies$ **powerful reducing agent** (reduces $AgNO_3 \to Ag\downarrow$).
  - **Phosphorous acid ($H_3PO_3$):** Dibasic (2 $P-OH$), contains 1 $P-H$ bond $\implies$ reducing agent. Disproportionates on heating: $4H_3PO_3 \xrightarrow{\Delta} 3H_3PO_4$
  - **Orthophosphoric acid ($H_3PO_4$):** Tribasic (3 $P-OH$), zero $P-H$ bonds $\implies$ non-reducing.
  - **Pyrophosphoric acid ($H_4P_2O_7$):** Tetrabasic (4 $P-OH$), contains 1 $P-O-P$ bridge.

### 4.5.4 Group 16: Oxygen Family

- **Ozone ($O_3$):** Prepared by passing silent electric discharge through pure dry dioxygen ($3O_2 \rightleftharpoons 2O_3, \Delta H^\circ = +142\,\text{kJ mol}^{-1}$). Quantitative estimation by adding neutral buffered KI (pH 9.2, borate buffer) and titrating liberated $I_2$ against standard sodium thiosulfate:

$$2I^- + H_2O + O_3 \longrightarrow 2OH^- + I_2 + O_2 \tag{4.52}$$

  Depletion by nitric oxide: $NO + O_3 \to NO_2 + O_2$.
- **Allotropes of Sulfur:**
  - **$\alpha$-Rhombic Sulfur:** Stable below 369 K (95.6°C), yellow, puckered crown $S_8$ ring.
  - **$\beta$-Monoclinic Sulfur:** Stable above 369 K, needle-shaped, crown $S_8$ ring.
  - **Transition Temperature (369 K):** Both $\alpha$ and $\beta$ allotropes coexist in dynamic equilibrium.
  - **Vapour Phase Paramagnetism:** Above 1000 K, sulfur exists predominantly as diatomic $S_2$ molecules which are **paramagnetic** due to two unpaired electrons in antibonding $\pi^*$ orbitals (identical to $O_2$).

- **Sulfuric Acid ($H_2SO_4$ - Contact Process):**

$$2SO_2+O_2 \xrightarrow[V_2O_5,720\,K]{2\,bar} 2SO_3 \quad (\Delta H^\circ = -196.6\,kJ\,mol^{-1}), \quad SO_3+H_2SO_4 \rightarrow H_2S_2O_7 \text{ (Ol} \tag{4.53}$$

- **Peroxo Oxoacids of Sulfur:**
  - **Caro's Acid (Peroxomonosulfuric, $H_2SO_5$):** Contains one $-O-O-$ peroxo bond. Oxidation state of $S$ is $+6$ (NOT $+8$!).
  - **Marshall's Acid (Peroxodisulfuric, $H_2S_2O_8$):** Contains one $-O-O-$ peroxo bridge connecting two sulfur atoms. Oxidation state of $S$ is $+6$.

### 4.5.5 Group 17: Halogen Family

- **Bond Dissociation Enthalpy Anomaly:**

$$Cl_2(242.6) > Br_2(192.8) > F_2(158.8) > I_2(151.1)\,kJ\,mol^{-1} \tag{4.54}$$

  Fluorine has an abnormally low bond dissociation energy due to intense electrostatic repulsion between non-bonding lone pairs on the tiny, closely spaced fluorine atoms.
- **Oxidizing Power Hierarchy:** $F_2 > Cl_2 > Br_2 > I_2$. Although $\Delta_{eg}H$ is more negative for Chlorine, Fluorine is the most powerful oxidizing agent because of:
  1. Low bond dissociation enthalpy of $F_2$.
  2. Extraordinarily high hydration enthalpy of the small $F^-$ ion ($\Delta_{hyd}H = -515\,kJ\,mol^{-1}$).
- **Hydrogen Halides ($HF, HCl, HBr, HI$):**
  - **Boiling Point:** $HF(293\,K) > HI(238\,K) > HBr(206\,K) > HCl(189\,K)$ (HF is liquid due to intermolecular H-bonding).
  - **Acidic Strength:** $HF < HCl < HBr < HI$ ($H-X$ bond dissociation enthalpy decreases).
  - **Reducing Power:** $HF < HCl < HBr < HI$ (HI is the strongest reducing agent).
- **Oxoacids of Halogens:**
  - **Acidic Strength Order:** $HClO_4 > HClO_3 > HClO_2 > HClO$ (higher oxidation state of halogen increases conjugate base resonance stabilization via $d\pi - p\pi$ bonding).
  - **Oxidizing Power Order:** $HClO > HClO_2 > HClO_3 > HClO_4$ (hypohalous acids are the most powerful oxidants due to ready oxygen atom liberation).

### 4.5.6 Group 18: Noble Gases

Neil Bartlett (1962) observed that $PtF_6$ oxidizes molecular dioxygen to $[O_2]^+[PtF_6]^-$. Because the first ionization enthalpy of Xenon ($1170\,kJ\,mol^{-1}$) is nearly identical to that of $O_2$ ($1175\,kJ\,mol^{-1}$), he reacted Xenon with $PtF_6$ to synthesize the first noble gas compound: $Xe^+[PtF_6]^-$.

**Xenon Fluorides: Preparation, Structures & Hydrolysis Pathways**

$$Xe + F_2 \xrightarrow[1\,bar]{673\,K} XeF_2 \quad (2:1 \text{ ratio}) \qquad [\text{Linear, } SN = 5, 3 \text{ lone pairs equato} \tag{4.55}$$

$$Xe + 2F_2 \xrightarrow[7\,bar]{873\,K} XeF_4 \quad (1:5 \text{ ratio}) \qquad [\text{Square Planar, } SN = 6, 2 \text{ lone pairs a} \tag{4.56}$$

$$Xe + 3F_2 \xrightarrow[60-70\,bar]{573\,K} XeF_6 \quad (1:20 \text{ ratio}) \qquad [\text{Distorted Octahedral, } SN = 7, 1 \text{ lone p} \tag{4.57}$$

**Hydrolysis Pathways (Cardinal NEET Questions):**

$$2XeF_2 + 2H_2O \longrightarrow 2Xe + 4HF + O_2 \uparrow \quad \text{(Redox hydrolysis)} \tag{4.58}$$

$$6XeF_4 + 12H_2O \longrightarrow 4Xe + 2XeO_3 + 24HF + 3O_2 \uparrow \quad \text{(Disproportionation: } Xe^{IV} \tag{4.59}$$

$$XeF_6 + 3H_2O \longrightarrow XeO_3 \text{ (Explosive solid, pyramidal)} + 6HF \quad \text{(Complete non} \tag{4.60}$$

$$XeF_6 + H_2O \longrightarrow XeOF_4 \text{ (Liquid, square pyramidal)} + 2HF \quad \text{(Partial hydrol} \tag{4.61}$$

$$XeF_6 + 2H_2O \longrightarrow XeO_2F_2 \text{ (Seesaw shape)} + 4HF \quad \text{(Partial hydrolysis)} \tag{4.62}$$

**Fluoride Ion Donor and Acceptor Reactions:**

$$XeF_2 + PF_5 \longrightarrow [XeF]^+[PF_6]^-, \qquad XeF_6 + MF \longrightarrow M^+[XeF_7]^- \quad (M = Na, K, Rb, \tag{4.63}$$

**Direct NCERT Benchmark: $p$-Block Cardinal Diagnostic Extracts**

- **Basicity vs Reducing Power in Group 15 Hydrides:** Basicity decreases ($NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$), whereas reducing power increases ($BiH_3 > SbH_3 > AsH_3 > PH_3 > NH_3$). $BiH_3$ is the strongest reducing hydride; $NH_3$ is a mild reducing agent.
- **Phosphorus Oxoacid Reducing Power:** Arises exclusively from the presence of directly bonded $P-H$ bonds:

  - $H_3PO_2$ (Hypophosphorous / Phosphinic): Has 2 $P-H$ bonds $\implies$ most powerful reducing oxoacid.
  - $H_3PO_3$ (Phosphorous / Phosphonic): Has 1 $P-H$ bond $\implies$ moderate reducing agent.
  - $H_3PO_4$ (Orthophosphoric): Has 0 $P-H$ bonds $\implies$ strictly non-reducing.
- **Paramagnetism of Diatomic Sulfur ($S_2$):** In the vapour phase at elevated temperatures ($> 1000\,K$), sulfur exists as $S_2$ molecules which possess two unpaired electrons in degenerate $\pi^*$ antibonding orbitals, exactly analogous to $O_2$.
- **Commercial Preparation of Dinitrogen from Azides:** Ultra-pure nitrogen gas is produced by thermal decomposition of sodium or barium azide:

$$2NaN_3 \xrightarrow{\Delta} 2Na + 3N_2\uparrow, \qquad Ba(N_3)_2 \xrightarrow{\Delta} Ba + 3N_2\uparrow$$

# 4.6 Comparative Chemistry of $d$- and $f$-Block Elements (Unit 11)

## 4.6.1 General Trends of $3d$ Transition Series

A transition element is defined as an element having an incompletely filled $d$-subshell in its ground state or in any of its common oxidation states. Zinc, Cadmium, and Mercury ($d^{10}$) are **non-transition elements** because they have completely filled $d$-orbitals in both ground and ionic states.

- **Atomic Radii across $3d$ Series:** Decreases initially (Sc to Cr), remains nearly constant (Mn to Ni) because increasing nuclear charge is balanced by $d-d$ shielding, and slightly increases at the end (Cu, Zn) due to interelectronic repulsion in filled $d^{10}$.
- **High Enthalpies of Atomization:** Reaches maximum at middle of series (V, Cr, Mo, W) due to maximum unpaired electrons participating in strong interatomic metallic bonding.
- **Variable Oxidation States:** Arises because $(n-1)d$ and $ns$ electrons have comparable energies. Sc exhibits only +3; Mn exhibits the widest range from +2 to +7.
- **Standard Electrode Potentials ($E^\circ(M^{2+}/M)$):**

$$M(s) \xrightarrow{\Delta_{sub}H} M(g) \xrightarrow{IE_1+IE_2} M^{2+}(g) \xrightarrow{\Delta_{hyd}H} M^{2+}(aq) \tag{4.64}$$

  Cu has an anomalous **positive standard reduction potential ($+0.34\,V$)** because its high enthalpy of atomization and high ionization enthalpy are

not compensated by the hydration enthalpy of $Cu^{2+}$. Thus, Copper does not liberate $H_2$ from non-oxidizing acids.

- **Disproportionation of Copper(I):**

$$2Cu^{+}(aq) \longrightarrow Cu^{2+}(aq) + Cu(s) \quad (E^{\circ}_{cell} = +0.36\,V) \tag{4.65}$$

$Cu^{2+}$ is far more stable in aqueous solution than $Cu^{+}$ because the much higher hydration enthalpy of $Cu^{2+}$ ($\Delta_{hyd}H \approx -2121\,kJ\,mol^{-1}$) overcompensates for the second ionization enthalpy of Copper.

- **Spin-Only Magnetic Moment Formula:**

$$\mu_s = \sqrt{n(n+2)}\,BM \tag{4.66}$$

  - $n = 1 \implies \mu = \sqrt{3} = 1.73\,BM$ ($Ti^{3+}, Cu^{2+}$).
  - $n = 2 \implies \mu = \sqrt{8} = 2.83\,BM$ ($Ti^{2+}, Ni^{2+}$).
  - $n = 3 \implies \mu = \sqrt{15} = 3.87\,BM$ ($V^{2+}, Cr^{3+}$).
  - $n = 4 \implies \mu = \sqrt{24} = 4.90\,BM$ ($Cr^{2+}, Fe^{2+}$).
  - $n = 5 \implies \mu = \sqrt{35} = 5.92\,BM$ ($Mn^{2+}, Fe^{3+}$).

### 4.6.2 Master Transition Metal Compounds: $K_2Cr_2O_7$ & $KMnO_4$

#### 1. Potassium Dichromate ($K_2Cr_2O_7$)

**Industrial Preparation from Chromite Ore ($FeCr_2O_4$):**

$$4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \longrightarrow 8Na_2CrO_4\ (\text{Yellow}) + 2Fe_2O_3 + 8CO_2 \tag{4.67}$$

$$2Na_2CrO_4 + 2H^{+} \longrightarrow Na_2Cr_2O_7\ (\text{Orange}) + 2Na^{+} + H_2O \tag{4.68}$$

$$Na_2Cr_2O_7 + 2KCl \longrightarrow K_2Cr_2O_7\ (\text{Orange crystals}) + 2NaCl \tag{4.69}$$

$K_2Cr_2O_7$ is preferred over $Na_2Cr_2O_7$ as a primary volumetric standard because $Na_2Cr_2O_7$ is deliquescent. **pH-Dependent Chromate-Dichromate Interconversion:**

$$\underset{\text{Yellow}}{2CrO_4^{2-}} + 2H^{+} \rightleftharpoons \underset{\text{Orange}}{Cr_2O_7^{2-}} + H_2O \quad (\text{Chromate in alkali; Dichromate in acid; Oxidatio} \tag{4.70}$$

**Structures:**

- Chromate ($CrO_4^{2-}$): Tetrahedral ($T_d$), four equivalent $Cr-O$ bonds (166 pm) due to resonance.

- Dichromate ($Cr_2O_7^{2-}$): Two tetrahedra sharing one oxygen corner; bridging $Cr-O-Cr$ bond angle is 126°; bridging $Cr-O$ bond length is 179 pm, while six terminal $Cr-O$ bonds are shorter (163 pm).

**Oxidizing Reactions in Acidic Medium ($n$-factor = 6):**

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O \quad (E^\circ = +1.33\,V) \tag{4.71}$$

$$6Fe^{2+} \longrightarrow 6Fe^{3+} + 6e^- \tag{4.72}$$

$$6I^- \longrightarrow 3I_2 + 6e^- \tag{4.73}$$

$$3H_2S \longrightarrow 3S\downarrow + 6H^+ + 6e^- \tag{4.74}$$

$$3Sn^{2+} \longrightarrow 3Sn^{4+} + 6e^- \tag{4.75}$$

**Chromyl Chloride Test:** Confirmation of chloride ions:

$$K_2Cr_2O_7 + 4NaCl + 6H_2SO_4(conc) \xrightarrow{\Delta} 2CrO_2Cl_2\uparrow \text{ (Deep red vapors)} + 2KHSO_4 + 4NaHSO_4 \tag{4.76}$$

Pass into NaOH → yellow $Na_2CrO_4$ $\xrightarrow{Pb(CH_3COO)_2}$ bright yellow precipitate of lead chromate ($PbCrO_4\downarrow$).

## 2. Potassium Permanganate ($KMnO_4$)

**Industrial Preparation from Pyrolusite Ore ($MnO_2$):**

$$2MnO_2 + 4KOH + O_2 \xrightarrow{\text{fuse}} 2K_2MnO_4 \text{ (Dark green)} + 2H_2O \tag{4.77}$$

$$3MnO_4^{2-} + 4H^+ \longrightarrow 2MnO_4^- \text{ (Purple)} + MnO_2 + 2H_2O \quad \text{(Disproportionation)} \tag{4.78}$$

**Manganate vs Permanganate Comparison:**

- Manganate ($MnO_4^{2-}$): Green, tetrahedral, Mn(+6), $3d^1 \implies$ **Paramagnetic** (1 unpaired electron).
- Permanganate ($MnO_4^-$): Deep purple, tetrahedral, Mn(+7), $3d^0 \implies$ **Diamagnetic**. **Origin of Color:** Deep purple color is **NOT** due to $d-d$ transition (it is $d^0$!). Color arises from **Ligand-to-Metal Charge Transfer (LMCT)** where oxygen $2p$ electron density temporarily shifts into vacant manganese $3d$ orbitals.

**Permanganate Redox Media & Valence Factors ($n$-factor)**

1. **In Acidic Medium ($n$-factor = 5):**

$$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O \quad (E^\circ = +1.51\,V) \tag{4.79}$$

$$5Fe^{2+} \longrightarrow 5Fe^{3+} + 5e^- \tag{4.80}$$

$$5C_2O_4^{2-} \longrightarrow 10CO_2 \uparrow + 10e^- \quad \text{(titration heated to } 60-70^\circ C\text{)} \tag{4.81}$$

$$10I^- \longrightarrow 5I_2 + 10e^- \tag{4.82}$$

$$5NO_2^- + 5H_2O \longrightarrow 5NO_3^- + 10H^+ + 10e^- \tag{4.83}$$

$$5SO_3^{2-} + 5H_2O \longrightarrow 5SO_4^{2-} + 10H^+ + 10e^- \tag{4.84}$$

2. **In Neutral / Faintly Alkaline Medium (Baeyer's Reagent, $n$-factor $= 3$):**

$$MnO_4^- + 2H_2O + 3e^- \longrightarrow MnO_2 + 4OH^- \quad (E^\circ = +1.23\,V) \tag{4.85}$$

**Cardinal NEET Reaction Differences:**

- Iodide is oxidized to **Iodate** ($IO_3^-$), NOT $I_2$:

$$I^- + 2MnO_4^- + H_2O \longrightarrow IO_3^- + 2MnO_2 + 2OH^- \tag{4.86}$$

- Thiosulfate is oxidized to **Sulfate** ($SO_4^{2-}$):

$$8MnO_4^- + 3S_2O_3^{2-} + H_2O \longrightarrow 8MnO_2 + 6SO_4^{2-} + 2OH^- \tag{4.87}$$

3. **In Strongly Alkaline Medium ($n$-factor $= 1$):**

$$MnO_4^- + e^- \longrightarrow MnO_4^{2-} \tag{4.88}$$

**High-Yield Problem Archetype: Permanganometric Volumetric Stoichiometry**

**Archetype Problem:** Determine the volume of 0.02 M $KMnO_4$ required to completely oxidize: (a) 25.0 mL of 0.1 M Mohr's salt ($FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$), and (b) 25.0 mL of 0.1 M Oxalic acid ($H_2C_2O_4 \cdot 2H_2O$) in acidic medium.

1. **Equivalence Principle:** At the exact equivalence endpoint:

$$N_1V_1 = N_2V_2 \iff (M_1 \times n_1)V_1 = (M_2 \times n_2)V_2$$

For $KMnO_4$ in acidic medium, $n_1 = 5$ ($Mn^{VII} \rightarrow Mn^{II}$).

2. **Case (a): Titration with Mohr's Salt:** Mohr's salt acts as a single-electron reducing agent: $Fe^{2+} \rightarrow Fe^{3+} + e^- \implies n_2 = 1$.

$(0.02\times5)\times V_1 = (0.1\times1)\times25.0 \implies 0.10\times V_1 = 2.50 \implies V_1 = \mathbf{25.0}$ mL

3. **Case (b): Titration with Oxalic Acid:** Oxalate ion undergoes a 2-electron oxidation: $C_2O_4^{2-} \rightarrow 2CO_2 + 2e^- \implies n_2 = 2$.

$(0.02\times5)\times V_1 = (0.1\times2)\times25.0 \implies 0.10\times V_1 = 5.00 \implies V_1 = \mathbf{50.0}$ mL

4. **Practical Laboratory Condition:** The oxalic acid titration must be conducted at $60-70^\circ$C because the initial reduction is slow. The reaction is **autocatalyzed** by the produced $Mn^{2+}$ ions.

### 4.6.3 The $f$-Block Elements: Lanthanoids & Actinoids

**1. Lanthanoids ($4f$ Series: $Ce_{58}$ to $Lu_{71}$)**

- **Electronic Configuration:** $[Xe]4f^{1-14}5d^{0-1}6s^2$.
- **Oxidation States:** Dominant and most stable state is +3.
  - +2 exhibited by $Eu^{2+}$ ($[Xe]4f^7$, half-filled) and $Yb^{2+}$ ($[Xe]4f^{14}$, fully-filled) $\implies$ act as **strong reducing agents** reverting to +3.
  - +4 exhibited by $Ce^{4+}$ ($[Xe]4f^0$, empty shell) and $Tb^{4+}$ ($[Xe]4f^7$) $\implies$ $Ce^{4+}$ acts as a **powerful analytical oxidizing agent** ($E^\circ_{Ce^{4+}/Ce^{3+}} = +1.74\,V$).
- **Alloy: Mischmetal:** Consists of Lanthanoid metals ($\sim 95\%$), Iron ($\sim 5\%$), and traces of S, C, Ca, Al. Used in cigarette lighter flints and tracer bullets.

**NCERT Hallmark: Lanthanoid Contraction & Chemical Consequences**

The imperfect shielding of one $4f$ electron by another causes a progressive increase in effective nuclear charge ($Z_{eff}$), resulting in a steady decrease in atomic and ionic radii from $Ce^{3+}$ (103 pm) to $Lu^{3+}$ (86 pm).
**Three Major Chemical Consequences for NEET:**

1. **Covalent Character and Basicity of Hydroxides:** Size of $M^{3+}$ decreases from $La^{3+}$ to $Lu^{3+}$. By Fajan's rules, covalent character of the $M-OH$ bond increases; therefore, basic strength strictly decreases:

   $La(OH)_3 > Ce(OH)_3 > \cdots > Lu(OH)_3$ ($La(OH)_3$ is most basic, $Lu(OH)_3$ is least b

2. **Pairing of 4d and 5d Transition Radii:** Due to the intervention of the 14 lanthanoid elements, elements of the 4d and 5d series show almost identical atomic and ionic radii:

   $r(Zr) \approx r(Hf) \approx 160\,pm$; $r(Nb) \approx r(Ta) \approx 146\,pm$; $r(Mo) \approx r(W)$

   Consequently, Zr and Hf exhibit identical chemical properties and occur together in nature, posing difficult separation challenges.
3. **Chromatographic Separation:** Because radii vary by only $\sim$ 17 pm across 14 elements, chemical separation is difficult, but their

slight differences in complex formation constants and hydrated ionic radii permit successful separation via ion-exchange chromatography.

### 2. Actinoids ($5f$ Series: $\mathrm{Th}_{90}$ to $\mathrm{Lr}_{103}$)

- **Electronic Configuration:** $[\mathrm{Rn}]5f^{1-14}6d^{0-2}7s^2$.
- **Oxidation States:** Much wider spectrum of oxidation states ($+3, +4, +5, +6, +7$) than Lanthanoids because the energy difference between $5f, 6d$, and $7s$ subshells is exceedingly small.
- **Actinoid Contraction:** Steady decrease in ionic radii across the actinoid series. The contraction is **significantly greater** from element to element than the Lanthanoid contraction because $5f$ electrons exert even poorer shielding than $4f$ electrons.
- **Nuclear Instability:** All actinoids possess unstable nuclei; elements beyond Uranium ($Z > 92$) are synthetic **transuranic elements**.

PART IV

# Organic Chemistry: Principles, Pathways & Biomolecules

Chapter 5

# Organic Chemistry: Electronic Effects, Functional Groups & Biomolecules

This chapter spans the core of NEET Organic Chemistry. We develop the foundational electronic effects (inductive, resonance, hyperconjugation) and reactive intermediates, proceed through aliphatic and aromatic hydrocarbons, haloalkanes, oxygen-containing functional groups (alcohols, phenols, ethers, carbonyls, carboxylic acids), nitrogen derivatives (amines, diazonium salts), and conclude with biomolecules (carbohydrates, proteins, nucleic acids).

**A Personal Note from Satya: Stop Memorizing 200 Reactions — Follow the Electrons!**

**Take a deep breath: Organic Chemistry is about to become your highest-scoring subject.**
Aspirants make Organic Chemistry a living nightmare by trying to memorize 200 named reactions like isolated words in a foreign language. They stare at Aldol, Cannizzaro, Reimer-Tiemann, and Gabriel Phthalimide and wonder how anyone can hold this in their head.
Here is the secret of physical organic chemistry that unlocks everything:

***Every single reaction in organic chemistry is just an electron-rich center attacking an electron-poor center.***

That is literally it!

- A nucleophile (lone pair, negative charge, or $\pi$-bond) looks for an electrophile (partial positive charge $\delta^+$, carbocation, or carbonyl carbon).
- When you look at an aldehyde, don't memorize what reagent adds to it. Look at the C = O double bond: the oxygen is electronegative ($\delta^-$), pulling electron density away from the carbonyl carbon

($\delta^+$). The carbon is electrophilic! Naturally, any nucleophile (Grignard, cyanide, hydride, enolate) will attack that carbon.

Once you train your eyes to spot the electron flow with curved arrows, you will never panic in the exam hall. Even if NTA presents an unfamiliar substrate, you can deduce the product step-by-step from scratch. Trust the electronic principles, draw every mechanism out with your pen, and let's conquer this subject together!

## 5.1 NMC Syllabus Demarcation & Zero-Tolerance Deletions

**Curriculum Alignment & Pedagogical Scope**

The revised National Medical Commission (NMC) NEET UG 2027 Chemistry syllabus demands a rigorous, mechanistic understanding of organic transformations rather than superficial rote memorization. The curriculum is built upon three integrated pillars:

1. **General Organic Chemistry (GOC) & Reaction Intermediates:** Quantitative and qualitative evaluation of electron displacement effects (inductive, electromeric, resonance, hyperconjugation, and aromaticity) and the structural thermodynamics of transient intermediates (carbocations, carbanions, free radicals, carbenes, and arynes).
2. **Systematic Functional Group Reaction Pathways:** Exhaustive mechanism sequences spanning aliphatic/aromatic hydrocarbons, haloalkanes/haloarenes, oxygenated functions (alcohols, phenols, ethers, aldehydes, ketones, carboxylic acids), and nitrogenous compounds (amines, diazonium salts).
3. **Biomolecules of Life:** Structural stereochemistry of carbohydrates, amino acids, peptide linkages, primary through quaternary protein architecture, nucleic acid Watson-Crick geometry, vitamins, and hormones.

**Zero-Tolerance NMC Deleted Topics Alert: Do Not Study!**

To optimize preparation efficiency and adhere strictly to the regulatory mandate issued by the NMC for NEET UG 2024–2027, the following entire domains are **permanently excluded** from the examination:

- **Polymers (Entire Chapter Deleted):** Classification of polymers; addition and condensation polymerization; natural and syn-

thetic rubber, vulcanization; polythene (LDPE/HDPE), Teflon, PAN, Nylon-6, Nylon-6,6, Terylene/Dacron, Bakelite, melamine-formaldehyde, Buna-S, Buna-N; biodegradable polymers (PHBV, Nylon-2-Nylon-6).

- **Chemistry in Everyday Life (Entire Chapter Deleted):** Drugs and pharmacodynamics; analgesics (narcotic and non-narcotic); tranquilizers, antiseptics (Dettol, Bithionol), disinfectants, antibiotics, antacids, antihistamines; artificial sweeteners (saccharin, aspartame, alitame, sucralose); food preservatives; soaps and synthetic detergents (anionic, cationic, non-ionic).

Candidates must **not** allocate study time to these deleted units. Any occurrence of these keywords in this text serves solely as an exclusion boundary warning.

## 5.2 General Organic Chemistry (GOC) & Electronic Effects

### 5.2.1 Inductive Effect ($\pm I$)

**Physical Origin & Characteristics of Inductive Effect**

The inductive effect is a permanent polarization of a $\sigma$-bond arising from electronegativity differences between bonded atoms. The more electronegative atom pulls the bonding $\sigma$-electron pair toward itself, establishing a static dipole:

$$-\mathrm{C}^{\delta\delta\delta+} - \mathrm{C}^{\delta\delta+} - \mathrm{C}^{\delta+} - \mathrm{X}^{\delta-} \quad (\text{where } \chi_X > \chi_C) \tag{5.1}$$

**Key Physical Attributes:**

- **Permanent Distortion:** It involves shifting of $\sigma$-electrons without complete bond cleavage or free electron transfer.
- **Distance Attenuation:** The magnitude of the fractional charge decreases rapidly with distance and becomes virtually negligible beyond the third $\sigma$-carbon atom ($C_3$).
- **Hydrogen Reference Standard:** The electronegativity of hydrogen is taken as the arbitrary zero benchmark: $I_{\mathrm{H}} = 0$.

## Authoritative $-I$ and $+I$ Relative Strength Series

**1. Electron-Withdrawing ($-I$) Series (Decreasing Order of Pull):**

$$-\mathrm{NF_3^+} > -\mathrm{NR_3^+} > -\mathrm{NH_3^+} > -\mathrm{NO_2} > -\mathrm{SO_3H} > -\mathrm{CN} > -\mathrm{COOH} > -\mathrm{F} > -\mathrm{Cl} > -\mathrm{Br} > -\mathrm{I} > -\mathrm{OR} > -\mathrm{OH} > -\mathrm{C} \equiv \mathrm{CR} > -\mathrm{NH_2} > -\mathrm{C_6H_5} > -\mathrm{CH} = \mathrm{CH_2} > \tag{5.2}$$

**2. Electron-Releasing ($+I$) Series (Decreasing Order of Push):**

$$-\mathrm{O^-} > -\mathrm{COO^-} > -\mathrm{C(CH_3)_3} > -\mathrm{CH(CH_3)_2} > -\mathrm{CH_2CH_3} > -\mathrm{CH_3} > -\mathrm{D} > -\mathrm{H} \tag{5.3}$$

**Physical Rationales for Key Sequences:**

- **Positive Formal Charges:** Positively charged ammonium and nitro centers exhibit dramatically higher $-I$ than neutral halogens because full ionic charges create immense electrostatic attraction.
- **Hybridization Dependency:** Higher $\%s$-character pulls electrons closer to the carbon nucleus, raising electronegativity:

  $$sp \text{ carbon } (50\%s, \chi \approx 3.29) > sp^2 \text{ carbon } (33.3\%s, \chi \approx 2.75) > sp^3 \text{ carbon } (25\%s, \chi \tag{5.4}$$

  Hence, alkynyl ($-\mathrm{C} \equiv \mathrm{CR}$) exerts a stronger $-I$ effect than vinyl ($-\mathrm{CH} = \mathrm{CH_2}$) and phenyl ($-\mathrm{C_6H_5}$).
- **Isotope Effect:** Tritium ($-\mathrm{T}$) and Deuterium ($-\mathrm{D}$) have shorter and stronger $\mathrm{C-D}$ bonds than $\mathrm{C-H}$ due to lower zero-point vibrational energy, causing less average nuclear distance and greater electron release: $+I(-\mathrm{T}) > +I(-\mathrm{D}) > +I(-\mathrm{H})$.

## Quantitative Applications: Acidic Strength of Carboxylic Acids

The acidic strength ($K_a$, where $pK_a = -\log K_a$) of carboxylic acids is determined by the stability of the conjugate carboxylate anion ($\mathrm{R-COO^-}$):

$$\text{Acidic Strength} \propto \text{Stability of Conjugate Base } (\mathrm{R-COO^-}) \propto \frac{-I \text{ effect}}{+I \text{ effect}} \tag{5.5}$$

**1. Influence of Halogen Identity (Electronegativity):**

$$\mathrm{FCH_2COOH}\ (pK_a = 2.66) > \mathrm{ClCH_2COOH}\ (2.86) > \mathrm{BrCH_2COOH}\ (2.90) > \mathrm{ICH_2COOH}\ (3.17) > \mathrm{CH_3COOH}\ (4.76) \tag{5.6}$$

**2. Number of Electron-Withdrawing Groups:**

$$\mathrm{Cl_3C-COOH}\ (pK_a = 0.65) > \mathrm{Cl_2CH-COOH}\ (1.29) > \mathrm{ClCH_2-COOH}\ (2.86) > \mathrm{CH_3} \tag{5.7}$$

**3. Distance Dependence ($\alpha, \beta, \gamma$-Positions):**

$$\mathrm{CH_3-CH_2-CH(Cl)-COOH}\ (\alpha\text{-isomer}, pK_a = 2.84) > \mathrm{CH_3-CH(Cl)-CH_2-} > \mathrm{CH_2(Cl)-CH_2-CH_2} \tag{5.8}$$

**Direct NCERT Line: Formic Acid vs Benzoic Acid vs Acetic Acid**

The electron-donating $+I$ effect of the methyl group destabilizes the acetate anion relative to the formate anion:

$$\mathrm{HCOOH}\ (pK_a = 3.75) > \mathrm{C_6H_5COOH}\ (pK_a = 4.20) > \mathrm{CH_3COOH}\ (pK_a = 4.76) \tag{5.9}$$

**Crucial Insight:** Even though phenyl is electron-withdrawing by $-I$ relative to $sp^3$ alkyl, its $+M$ resonance into the carbonyl group weakens acidity compared to formic acid. Thus, formic acid is stronger than benzoic acid!

## 5.2.2 Resonance / Mesomeric Effect ($\pm M$)

**The Quantum Mechanical Basis of Resonance**

Resonance describes the delocalization of $\pi$-electrons or non-bonding lone pairs across parallel, unhybridized $p$-orbitals in planar conjugated frameworks. **Mandatory Conditions for Conjugation:**

1. **Planarity:** All participating atomic nuclei must lie in a single plane (or near-planar geometry) to permit effective sideways overlap of adjacent $p$-orbitals.
2. **Conjugated Topologies:** Conjugation occurs across alternating systems:
   - $\pi - \sigma - \pi$ (e.g., 1,3-butadiene, benzene)
   - $\pi - \sigma - p_{\text{vacant}}$ (e.g., allyl carbocation, tropylium cation)
   - $\pi - \sigma -$ lone pair / non-bonding (e.g., vinyl chloride, phenoxide, aniline)
   - $\pi - \sigma -$ unpaired electron ($\bullet$) (e.g., allyl free radical)
   - Lone pair $-\sigma - p_{\text{vacant}}$ (e.g., $\mathrm{CH_3-\overset{+}{C}H-\ddot{O}CH_3}$)

**Hierarchical Rules for Evaluating Canonical Structure Stability**

When ranking the thermodynamic contribution of contributing resonance structures (canonical forms) to the true resonance hybrid, apply

the following hierarchy strictly in order:

1. **Complete Octet Rule (Paramount Criterion):** Structures in which all second-period atoms ($C, N, O, F$) possess complete valence octets contribute overwhelmingly more than structures with incomplete octets, **even if** a formal positive charge resides on an electronegative atom ($O$ or $N$):

$$\mathrm{CH_3-\overset{+}{O}=CH_2}\ \text{(all octets complete, Major)} \longleftrightarrow \mathrm{CH_3-\ddot{O}-\overset{+}{C}H_2}\ \text{(carbon has } 6e^-, \tag{5.10}$$

2. **Maximum Number of Covalent Bonds:** Greater number of covalent bonds maximizes attractive nuclear-electronic interactions, conferring greater stability.
3. **Charge Separation vs Uncharged Species:** Neutral structures without formal charge separation are intrinsically more stable than dipolar (zwitterionic) structures.
4. **Electronegativity & Charge Concordance:** Among charge-separated structures:
   - Negative formal charges reside most stably on the most electronegative atoms ($\mathrm{O^- > N^- > C^-}$).
   - Positive formal charges reside most stably on the least electronegative atoms ($\mathrm{C^+ > N^+ > O^+}$).
5. **Proximity of Like/Unlike Charges:** Canonical forms with like charges on adjacent atoms are violently destabilized (electrostatic repulsion); unlike charges close together confer electrostatic stabilization.

**Resonance Energy ($E_{\text{res}}$) Formulation**

Resonance energy is defined as the difference in energy between the actual resonance hybrid and the most stable theoretical canonical Lewis structure:

$$E_{\text{res}} = E_{\text{canonical (most stable)}} - E_{\text{actual hybrid}} \tag{5.11}$$

Experimentally, resonance energy is determined from standard enthalpies of hydrogenation ($\Delta_{\text{hydro}}H^\circ$):

$$\begin{aligned}
\text{Cyclohexene} + \mathrm{H_2} \rightarrow \text{Cyclohexane} \quad \Delta H^\circ &= -119.7\ \text{kJ/r} \\
\text{Theoretical 1,3,5-Cyclohexatriene (no resonance)} \implies 3 \times (-119.7) &= -359.1\ \text{kJ/mol} \\
\text{Experimental Benzene} + 3\mathrm{H_2} \rightarrow \text{Cyclohexane} \quad \Delta H^\circ &= -208.5\ \text{kJ/r} \\
E_{\text{res}}(\text{Benzene}) = |-359.1 - (-208.5)| &= 150.6\ \text{kJ/m}
\end{aligned} \tag{5.12}$$

**NEET Trap: Halogen Paradox in EAS ($+M$ vs $-I$)**

Halogens ($-$F, $-$Cl, $-$Br, $-$I) attached to benzene present a classical dual nature:

- **Reactivity (Net Deactivating):** Halogens are strongly electronegative ($\chi_F = 4.0, \chi_{Cl} = 3.16$), creating a powerful $-I$ effect that drains $\sigma$-electron density from the ring. The $-I$ effect dominates over the $+M$ resonance effect for overall rate, making haloarenes **less reactive** than benzene toward electrophiles.
- **Regioselectivity (Ortho/Para Directing):** During attack by an incoming electrophile ($E^+$), the halogen donates its $p$-lone pair via $+M$ resonance to stabilize the positively charged arenium ion specifically when $E^+$ attacks the **ortho** or **para** positions (giving a complete-octet canonical structure [$Cl^+ = C_6H_5E$]). At the meta position, the halogen lone pair cannot stabilize the carbocation.

**Conclusion:** Halogens are uniquely **deactivating ortho/para-directors** in EAS.

### 5.2.3 Hyperconjugation (Baker-Nathan Effect)

**The $\sigma \to \pi^*$ and $\sigma \to p$ No-Bond Resonance**

Hyperconjugation involves the delocalization of bonding electrons in a C $-$ H (or C $-$ C) $\sigma$-bond into an adjacent vacant or partially filled non-bonding $p$-orbital or an antibonding $\pi^*$ molecular orbital:

$$\sigma_{C-H} \longrightarrow p_{vacant} \quad \text{(in carbocations)} \quad \text{or} \quad \sigma_{C-H} \longrightarrow \pi^* \quad \text{(in alkenes)} \tag{5.13}$$

**Structural Characteristics:**

1. **Number of Contributing Structures:** For an alkene, carbocation, or free radical with $n$ $\alpha$-hydrogens (hydrogens on the $sp^3$ carbon directly attached to the $sp^2$ system), the total number of hyperconjugative canonical forms is:

   $$\text{Total Hyperconjugative Structures} = n+1 \quad (\text{where } n = \text{number of } \alpha\text{-H}) \tag{5.14}$$

2. **Bond Shortening & Bond Lengthening:** The single bond between the $sp^3$ and $sp^2$ carbons acquires partial double bond character (e.g., the C $-$ C single bond in propene is 1.49 Å, significantly shorter than standard 1.54 Å in ethane). The $C_\alpha-H$ bond is lengthened and weakened.
3. **Reverse Hyperconjugation:** Observed when electronegative halogen atoms reside at the $\alpha$-carbon (e.g., $-CX_3$ in trifluoromethyl-

benzene $C_6H_5 - CF_3$). Electron density flows from the $\pi$-system into the vacant $\sigma^*_{C-F}$ antibonding orbital, causing meta-direction and ring deactivation.

**Alkene Stability & Heat of Hydrogenation ($\Delta_{hydro}H^\circ$)**

Alkene thermodynamic stability is directly proportional to the number of hyperconjugative $\alpha$-hydrogens:

$$\text{Alkene Stability} \propto (\text{Number of } \alpha\text{-H}) \propto \frac{1}{\text{Heat of Hydrogenation } (\Delta_{hydro}H^\circ)} \tag{5.15}$$

**Authoritative Stability Sequence:**

$$\begin{aligned}(CH_3)_2C=C(CH_3)_2\ (12\alpha\text{-H}) &> (CH_3)_2C=CHCH_3\ (9\alpha\text{-H})\\ &> CH_2=C(CH_3)_2\ (6\alpha\text{-H, gem-dimethyl})\\ &> \textit{trans}{-}CH_3CH=CHCH_3\ (6\alpha\text{-H})\\ &> \textit{cis}{-}CH_3CH=CHCH_3\ (6\alpha\text{-H})\\ &> CH_3CH_2CH=CH_2\ (2\alpha\text{-H})\\ &> CH_2=CH_2\ (0\alpha\text{-H})\end{aligned} \tag{5.16}$$

**The *trans* vs *cis* Isomer Trap:** *trans*-2-butene is more stable than *cis*-2-butene because the two bulky methyl groups are oriented at 180°, eliminating the steric van der Waals repulsion present in the eclipsed-like geometry of the *cis*-isomer. The *gem*-dimethyl alkene ($CH_2=C(CH_3)_2$) is thermodynamically more stable than both *cis*- and *trans*-2-butene due to superior electrostatic distribution across the two methyl groups attached to the same $sp^2$ carbon.

## 5.2.4 Electromeric Effect ($\pm E$)

**Temporary Reagent-Induced $\pi$-Polarization**

The electromeric effect is a purely **temporary** electronic displacement operating exclusively in molecules containing multiple bonds ($C=C, C\equiv C, C=O$) upon the approach of an attacking polar reagent. Upon removal of the reagent, the system instantly reverts to its unpolarized ground state.

- **Positive Electromeric Effect ($+E$):** The $\pi$-electron pair shifts

toward the atom to which the attacking reagent attaches:

$$\mathrm{CH_2=CH_2 + H^+ \xrightarrow{+E} CH_2^+ - CH_2 - H} \quad \text{(Electrophilic attack on alkene)} \tag{5.17}$$

- **Negative Electromeric Effect ($-E$):** The $\pi$-electron pair shifts away from the atom to which the attacking reagent attaches:

$$\mathrm{>C=O + CN^- \xrightarrow{-E} >C(CN) - O^-} \quad \text{(Nucleophilic attack on carbonyl)} \tag{5.18}$$

### 5.2.5 Aromaticity & Hückel's Rule

**The Four Cardinal Criteria for Aromaticity**

To be classified as **aromatic**, a chemical entity must simultaneously satisfy all four structural and electronic requirements formulated by Erich Hückel:

1. **Cyclic Structure:** The system must possess a continuous closed loop of bonded atoms.
2. **Planar Geometry:** All ring vertices must be strictly planar (or possess $< 0.1$ Å deviation) to maintain parallel $p$-orbital orientation.
3. **Complete Unbroken Conjugation:** Every atom in the cyclic ring must possess an unhybridized $p$-orbital (i.e., fully $sp^2$ or $sp$ hybridized; zero tetrahedral $sp^3$ carbons in the ring loop).
4. **Hückel's $(4n+2)\pi$ Electron Rule:** The number of delocalized $\pi$-electrons must equal $(4n+2)$, where $n$ is a non-negative integer ($n = 0, 1, 2, 3, \cdots \implies 2, 6, 10, 14, 18 \ldots$ $\pi$-electrons).

## Aromatic vs Anti-Aromatic vs Non-Aromatic Classification

| Property | Aromatic | Anti-Aromatic | Non-Aromatic |
|---|---|---|---|
| **Cyclic & Planar?** | Yes | Yes | Lacks cyclicity or planarity |
| **Conjugation** | Unbroken throughout ring | Unbroken throughout ring | Interrupted by $sp^3$ carbon |
| **$\pi$-Electron Count** | $(4n+2)\pi$ electrons | $4n\pi$ electrons ($n = 1, 2, \dots$) | Arbitrary count |
| **Thermodynamic Stability** | Exceptionally High ($E_{\text{res}} \gg 0$) | Exceptionally Low ($E_{\text{strain}} \gg 0$) | Normal open-chain analog |
| **Magnetic Induced Field** | Diatropic (Diamagnetic ring current) | Paratropic (Paramagnetic ring current) | No ring current |
| **Canonical Examples** | Benzene ($6\pi$), Tropylium ($6\pi$) | Cyclobutadiene ($4\pi$), Cyclopentadienyl cation ($4\pi$) | 1,3-Cyclohexadiene, COT (tub-shaped) |

## Annulenes & Non-Planar Ring Escapes

**1. Cyclooctatetraene (COT, [8]-Annulene):** COT possesses $8\pi$-electrons ($4n$ with $n = 2$). If planar, it would be severely anti-aromatic. To escape extreme anti-aromatic electronic destabilization, the flexible 8-membered ring distorts into a non-planar **tub conformation**. Because $p$-orbitals are perpendicular at the edges, conjugation is broken: COT is strictly **non-aromatic**. However, reduction with potassium metal yields the cyclooctatetraenyl dianion ($C_8H_8^{2-}$), which has $10\pi$-electrons ($4n+2, n=2$), becomes strictly planar, and is **aromatic**!
**2. [10]-Annulene:** Possesses $10\pi$-electrons ($4n+2, n=2$). In the planar configuration, severe steric repulsion between the two internal hydrogen atoms forces the ring out of planarity $\implies$ **non-aromatic**.

If the internal hydrogens are replaced with a methylene bridge (1,6-methano[10]annulene), the ring achieves planarity and exhibits robust **aromaticity**.

**Heterocyclic Aromaticity & Lone Pair Involvement**

When determining Hückel $\pi$-electron counts in heterocycles containing $N, O, S$:

- **Pyridine ($C_5H_5N$):** The nitrogen atom is $sp^2$-hybridized. Its unhybridized $p$-orbital contributes $1e^-$ to the $6\pi$ aromatic sextet. The lone pair resides in an $sp^2$ hybrid orbital in the ring plane, perpendicular to the $\pi$-cloud. It does **not** participate in aromaticity $\implies$ Pyridine is basic ($pK_b \approx 8.8$).
- **Pyrrole ($C_4H_4NH$):** The nitrogen atom is $sp^2$-hybridized. Its lone pair resides in an unhybridized $p$-orbital, contributing $2e^-$ to complete the $6\pi$ aromatic sextet. The lone pair is tied up in aromaticity $\implies$ Pyrrole is an exceptionally weak base ($pK_b \approx 13.6$) and protonation on nitrogen destroys aromaticity.
- **Furan ($C_4H_4O$) & Thiophene ($C_4H_4S$):** The heteroatom contributes one lone pair ($2e^-$) to the aromatic sextet ($6\pi$ electrons); the second lone pair resides in an $sp^2$ orbital orthogonal to the $\pi$-system.

## 5.3 Reactive Intermediates & Molecular Rearrangements

### 5.3.1 Carbocations

**Geometry, Electronic Configuration & Stability Hierarchy**

Carbocations are trivalent carbon cations possessing a positive formal charge and a sextet ($6e^-$) of valence electrons:

- **Geometry & Hybridization:** The positively charged carbon is $sp^2$-hybridized with trigonal planar geometry ($120^\circ$ bond angles). The remaining unhybridized $2p_z$ orbital is completely vacant.
- **Stability Factors:** Stabilized by $+M$ resonance delocalization, aromatic stabilization, hyperconjugation ($n_{\alpha-H}$), and $+I$ inductive effects.

$$\begin{aligned}\text{Tropylium } (6\pi\text{-aromatic}) \approx \text{Cyclopropylcarbinyl (dancing resonance)} \\ > (C_6H_5)_3C^+ > (C_6H_5)_2CH^+ > C_6H_5CH_2^+ \approx CH_2 = CH - C \\ > 3^\circ\ (CH_3)_3C^+\ (9\alpha\text{-H}) > 2^\circ\ (CH_3)_2CH^+\ (6\alpha\text{-H}) \\ > 1^\circ\ CH_3CH_2^+\ (3\alpha\text{-H}) > CH_3^+\ (0\alpha\text{-H}) > \text{vinyl } (CH_2 = CH^+) >\end{aligned} \tag{5.19}$$

## The Driving Force & Pathways of Carbocation Rearrangements

A carbocation will spontaneously rearrange if and only if the rearranged carbocation possesses lower Gibbs free energy (greater thermodynamic stability). Rearrangements occur via intramolecular 1,2-migrations of a hydride, alkyl, or aryl group into the adjacent vacant $p$-orbital:

1. **1,2-Hydride Shift ($H^-$):**

$$CH_3 - CH_2 - CH_2^+\ (1^\circ, 2\alpha\text{-H}) \xrightarrow{1,2\text{-}H^-} CH_3 - \overset{+}{C}H - CH_3\ (2^\circ, 6\alpha\text{-H}) \tag{5.20}$$

2. **1,2-Methyl / Alkyl Shift ($R^-$):**

$$(CH_3)_3C - \overset{+}{C}H_2\ (\text{neopentyl}, 1^\circ) \xrightarrow{1,2\text{-}CH_3^-} CH_3 - \overset{+}{C}(CH_3) - CH_2CH_3\ (3^\circ, 8\alpha\text{-H}) \tag{5.21}$$

3. **Ring Expansion (Angle Strain Relief):** Four- and five-membered rings with an exocyclic carbocation undergo instantaneous 1,2-bond migration to expand the ring:
   - **Cyclobutylcarbinyl Cation → Cyclopentyl Cation:** The 90° ring angle expands to 108°, relieving ≈ 108.8 kJ/mol of Baeyer angle strain.
   - **Cyclopentylcarbinyl Cation → Cyclohexyl Cation:** Expansion from 5-membered ring (108°) to strain-free chair cyclohexane (109.5°).

## NEET Trap: Migratory Aptitude in Carbocation Rearrangements

When competing groups are attached to the migrating carbon, the intrinsic migratory aptitude is:

$$H^- > \text{Aryl } (Ar^-) > \text{Alkyl } (3^\circ > 2^\circ > 1^\circ > CH_3^-) \tag{5.22}$$

For substituted aryl groups: electron-donating groups accelerate migration:

$$p\text{-anisyl } (p{-}CH_3O - C_6H_4{-}) > p\text{-tolyl } (p{-}CH_3 - C_6H_4{-}) > \text{phenyl } (C_6H_5{-}) > p\text{-chlorophenyl} > p\text{-nitrophenyl} \tag{5.23}$$

**Crucial Rule:** Never perform a less-stable to less-stable rearrangement unless it immediately opens a direct pathway to an overwhelmingly more stable intermediate (e.g., resonance-delocalized benzylic or allylic system).

## 5.3.2 Carbanions

**Structure, Pyramidal Inversion & Stability Hierarchy**

Carbanions possess a trivalent carbon bearing a formal negative charge and eight valence electrons (3 bonding pairs + 1 lone pair):

- **Hybridization & Inversion:** Unconjugated carbanions are $sp^3$-hybridized with trigonal pyramidal geometry. The lone pair undergoes rapid umbrella inversion (flipping barrier $\approx$ 25 kJ/mol), making isolated chiral carbanions rapidly racemize.
- **Conjugated Carbanions ($sp^2$ Planarity):** If the carbanion is adjacent to a $\pi$-system or an electron-withdrawing carbonyl group ($-C=O$), it rehybridizes to $sp^2$ to accommodate the lone pair in an unhybridized $p$-orbital for maximum resonance delocalization.
- **Stability Sequence:**

$$\text{Cyclopentadienyl Anion } (6\pi) > (C_6H_5)_3C^- > (C_6H_5)_2CH^- > C_6H_5CH_2^- \approx CH_2 = CH - CH_2^- > HC \equiv C^- ( > CH_2 = CH^- \ (sp^2) \approx C_6H_5^- > CH_3^- > 1^\circ > 2^\circ \tag{5.24}$$

**Active Methylene Compounds & Enolate Chemistry**

Methylene groups ($-CH_2-$) flanked between two strong electron-withdrawing groups (e.g., diethyl malonate, ethyl acetoacetate, acetylacetone) possess exceptionally acidic hydrogens ($pK_a \approx 9 - 13$):

$$CH_3 - CO - CH_2 - CO - CH_3 + EtO^- \rightleftharpoons [CH_3 - CO - \bar{C}H - CO - CH_3 \longleftrightarrow 2\ E \tag{5.25}$$

The resulting enolate anion is an **ambident nucleophile** that can attack electrophiles via carbon (*C*-alkylation) or oxygen (*O*-alkylation), driving fundamental $C-C$ bond-forming reactions (Aldol, Claisen, Michael additions).

### 5.3.3 Free Radicals

**Structure, Generation & Allylic Halogenation via NBS**

Carbon free radicals are neutral trivalent species possessing seven valence electrons (one unpaired electron):

- **Structure:** Planar $sp^2$ hybridized with the odd electron occupying the unhybridized $2p$ orbital, or shallow rapidly inverting pyramid.
- **Stability:** Stabilized by hyperconjugation and resonance delocalization:

$$(C_6H_5)_3C^{\bullet} > (C_6H_5)_2\dot{C}H > C_6H_5\dot{C}H_2 \approx CH_2 = CH - \dot{C}H_2 > 3^{\circ} > 2^{\circ} > 1^{\circ} > \dot{C}H_3 > \quad (5.26)$$

- **N-Bromosuccinimide (NBS) Allylic/Benzylic Halogenation:** NBS selectively brominates the allylic or benzylic position without adding across the $C=C$ double bond. **Mechanism:** Trace HBr reacts with NBS to maintain a constant, extremely low concentration of molecular bromine ($Br_2$):

$$Br^{\bullet} + CH_3 - CH = CH_2 \longrightarrow HBr + [\dot{C}H_2 - CH = CH_2 \longleftrightarrow CH_2 = CH - \dot{C}H_2] \quad (R \quad (5.27)$$

$$NBS + HBr \longrightarrow \text{Succinimide} + Br_2 \quad (\text{Maintains } [Br_2] \approx 10^{-6} \text{ M}) \quad (5.28)$$

$$\dot{C}H_2 - CH = CH_2 + Br_2 \longrightarrow BrCH_2 - CH = CH_2 + Br^{\bullet} \quad (5.29)$$

The low $[Br_2]$ prevents electrophilic addition across the double bond, ensuring exclusive allylic radical substitution!

## 5.3.4 Carbenes, Nitrenes & Arynes

**Singlet vs Triplet Carbenes ($:CR_2$)**

| Feature | Singlet Carbene ($^1:CR_2$) | Triplet Carbene ($^3:CR_2$) |
|---|---|---|
| **Electron Spin State** | Spin-paired ($\uparrow\downarrow$, $S = 0, 2S+1 = 1$) | Parallel spins ($\uparrow\uparrow$, $S = 1, 2S+1 = 3$) |
| **Hybridization & Geometry** | $sp^2$, Bent ($102 - 108°$) | $sp$ linear ($180°$) or bent $sp^2$ ($130 - 150°$) |
| **Orbital Configuration** | Lone pair in $sp^2$, vacant $p$-orbital | One electron in $sp^2$, one in $p$-orbital |
| **Relative Stability** | Thermodynamically less stable (except halocarbenes $:CCl_2$) | Ground state for $:CH_2$ (Hund's rule) |
| **Addition to Alkenes** | **Stereospecific syn-addition** (concerted, geometry retained: cis → cis) | **Non-stereospecific** (diradical intermediate, spin flip allows rotation → cis + trans) |

**Generation of Dichlorocarbene & The Benzyne Mechanism**

**1. Dichlorocarbene ($:CCl_2$) via $\alpha$-Elimination:** Chloroform treated with strong alkali undergoes two-step $\alpha$-elimination:

$$CHCl_3 + OH^- \overset{\text{fast}}{\rightleftharpoons} :\bar{C}Cl_3 + H_2O \xrightarrow[\text{RDS}]{\text{loss of } Cl^-} :CCl_2 \quad \text{(Electrophilic singlet carbene)} \tag{5.30}$$

$:CCl_2$ serves as the key electrophilic intermediate in the **Reimer-Tiemann reaction** and the **Carbylamine reaction**.

**2. Benzyne Intermediate (Elimination-Addition):** Aryl halides lacking electron-withdrawing groups react with extremely strong bases (e.g., $KNH_2$/liq.$NH_3$) via elimination of HX to yield **benzyne** ($C_6H_4$):

- **Bonding in Benzyne:** The formal triple bond is formed by sideways overlap of two $sp^2$ hybrid orbitals in the plane of the ring.

This bond does not disturb the $6\pi$ aromatic sextet, but overlap is extremely weak due to poor orbital alignment $\implies$ severe ring strain and immense reactivity.

- **Cine-Substitution:** Addition of $NH_3$ across the strained triple bond occurs with equal probability at either carbon, producing isotopic scrambling (50% direct substitution + 50% cine-substitution).

## 5.4 Hydrocarbons: Aliphatic & Aromatic Pathways

### 5.4.1 Alkanes

**Free-Radical Halogenation Kinetics & Hammond's Postulate**

Free radical halogenation follows a three-stage chain mechanism (Initiation, Propagation, Termination):

$$\text{Initiation: } X_2 \xrightarrow{h\nu \text{ or } \Delta} 2X^\bullet \tag{5.31}$$

$$\text{Propagation: } R-H + X^\bullet \xrightarrow{k_1} R^\bullet + HX \quad \text{(RDS for bromination)} \tag{5.32}$$

$$R^\bullet + X_2 \xrightarrow{k_2} R-X + X^\bullet \tag{5.33}$$

**Relative Reactivity Rates at 298 K:**

- **Chlorination (Reactivity over Selectivity):** $1^\circ : 2^\circ : 3^\circ = 1.0 : 3.8 : 5.0$. Chlorination is highly exothermic; the transition state is early (reactant-like per Hammond's postulate), meaning $C-H$ bond cleavage has barely progressed, yielding low regioselectivity.
- **Bromination (Selectivity over Reactivity):** $1^\circ : 2^\circ : 3^\circ = 1.0 : 82 : 1600$. Bromination is endothermic; the transition state is late (product-like), with extensive radical character on carbon, leading to massive selectivity for the $3^\circ$ alkyl halide ($> 99\%$).

**Conformational Analysis of Ethane and n-Butane**

- **Ethane:** Infinite conformations between **staggered** (dihedral angle $\theta = 60^\circ$, minimum energy) and **eclipsed** ($\theta = 0^\circ$, maximum energy). The energy barrier is 12.5 kJ/mol (3.0 kcal/mol), caused by **torsional strain** (electron cloud repulsion between adjacent $C-H$ bonds).

- **n-Butane (Rotation about $C_2 - C_3$):**

  $$\text{Anti } (\theta = 180^\circ) > \text{Gauche } (\theta = 60^\circ) > \text{Partially Eclipsed } (\theta = 120^\circ) > \text{Fully E} \tag{5.34}$$

  The anti conformation is the global minimum. Gauche is destabilized by 3.8 kJ/mol due to van der Waals steric repulsion between the two bulky methyl groups. The fully eclipsed form is destabilized by 19.0 kJ/mol due to combined torsional and steric strain.
- **Gauche Exception (Intramolecular H-Bonding):** In ethylene glycol ($HO - CH_2 - CH_2 - OH$) and 2-fluoroethanol ($F - CH_2 - CH_2 - OH$), the **gauche** conformation is more stable than anti due to stabilizing intramolecular hydrogen bonding.

### 5.4.2 Alkenes

**Electrophilic Addition of Halogens: Cyclic Halonium Ion**

Electrophilic addition of $Br_2$ (in nonpolar solvent $CCl_4$) to alkenes proceeds via a cyclic three-membered **bromonium ion**:

1. The $\pi$-cloud of the alkene polarizes the $Br - Br$ bond, displacing the distal bromide ion to form a bridged cyclic bromonium ion intermediate.
2. The nucleophilic $Br^-$ attacks the cyclic intermediate from the **backside** (anti-attack) with complete inversion at the attacked carbon.
3. **Stereospecificity:**
   - *trans*-Alkene + $Br_2$ (anti-addition) $\longrightarrow$ ***meso*-dibromide**.
   - *cis*-Alkene + $Br_2$ (anti-addition) $\longrightarrow$ **racemic mixture ($\pm$) of enantiomers**.
   - *(Mnemonic: CAR – Cis + Anti → Racemic; TAM – Trans + Anti → Meso).*

**Markovnikov Addition vs Anti-Markovnikov Peroxide Effect**

**1. Markovnikov's Rule:** In electrophilic addition of unsymmetrical $HX$ to unsymmetrical alkenes, the electrophilic proton ($H^+$) attaches to the $sp^2$ carbon carrying the greater number of hydrogen atoms to generate the more stable carbocation intermediate (which may rearrange!).
**2. Anti-Markovnikov (Kharasch / Peroxide Effect):** Operates exclusively when **HBr** is added in the presence of organic peroxides

(R – O – O – R):

$$\text{Initiation: } \mathrm{ROOR} \xrightarrow{\Delta} 2\mathrm{RO}^{\bullet}; \quad \mathrm{RO}^{\bullet} + \mathrm{HBr} \rightarrow \mathrm{ROH} + \mathrm{Br}^{\bullet} \tag{5.35}$$

$$\text{Propagation 1: } \mathrm{CH_3 - CH = CH_2} + \mathrm{Br}^{\bullet} \rightarrow \mathrm{CH_3 - \dot{C}H - CH_2Br} \quad (\Delta H^{\circ} = -42 \text{ kJ/mol} \tag{5.36}$$

$$\text{Propagation 2: } \mathrm{CH_3 - \dot{C}H - CH_2Br} + \mathrm{HBr} \rightarrow \mathrm{CH_3 - CH_2 - CH_2Br} + \mathrm{Br}^{\bullet} \quad (\Delta H^{\circ} = - \tag{5.37}$$

**Thermochemical Reason: Why Peroxide Effect Operates ONLY for HBr**

NEET frequently tests why the peroxide effect fails completely for HCl and HI:

- **Failure with HCl:** The H – Cl bond is exceptionally strong (431 kJ/mol). Step 2 ($\mathrm{R}^{\bullet} + \mathrm{HCl} \rightarrow \mathrm{R - H} + \mathrm{Cl}^{\bullet}$) is strongly **endothermic** ($\Delta H^{\circ} \approx +27$ kJ/mol). The chain reaction cannot propagate.
- **Failure with HI:** The H – I bond is weak (297 kJ/mol), making $\mathrm{I}^{\bullet}$ generation easy. However, iodine radicals are large and unreactive toward alkenes: Step 1 (Alkene + $\mathrm{I}^{\bullet} \rightarrow$ Radical) is **endothermic** ($\Delta H^{\circ} \approx +54$ kJ/mol). Instead of adding to the double bond, iodine radicals dimerize instantaneously: $2\mathrm{I}^{\bullet} \rightarrow \mathrm{I_2}$.

**Conclusion:** Only for HBr are **both propagation steps exothermic**, permitting chain reaction!

**Comparative Summary: The Three Alkene Hydration Pathways**

| Parameter | Acid-Catalyzed Hydration | Hydroboration-Oxidation (HBO) | Oxymercuration-Demercuration (OMDM) |
|---|---|---|---|
| **Reagents** | $H_2O/H_2SO_4$ | 1. $BH_3 \cdot THF$, 2. $H_2O_2, OH^-$ | 1. $Hg(OAc)_2, H_2O$, 2. $NaBH_4$ |
| **Regiochemistry** | Markovnikov | **Anti-Markovnikov** | **Markovnikov** |
| **Stereochemistry** | Non-stereospecific | **Syn-addition** | Overall non-stereospecific (anti in step 1) |
| **Intermediate** | Free Carbocation | 4-center cyclic transition state | Cyclic mercurinium ion |
| **Rearrangement?** | **Yes (frequent)** | **Zero rearrangement** | **Zero rearrangement** |
| **Propene Product** | Propan-2-ol | Propan-1-ol | Propan-2-ol |
| **3,3-Dimethylbut-1-ene** | 2,3-dimethylbutan-2-ol (rearranged) | 3,3-dimethylbutan-1-ol | 3,3-dimethylbutan-2-ol (clean Markovnikov) |

**Direct NCERT Extraction: Ozonolysis & Structure Elucidation**

Ozonolysis cleaves the C = C double bond into two carbonyl fragments:

- **Reductive Ozonolysis ($O_3$ followed by $Zn/H_2O$ or $(CH_3)_2S$):** Aldehydes and ketones are preserved. Zinc destroys hydrogen peroxide ($H_2O_2 + Zn \rightarrow ZnO + H_2O$), preventing oxidation of aldehydes to carboxylic acids.
- **Oxidative Ozonolysis ($O_3$ followed by $H_2O_2$ or $H_2O$):** All aldehydes are oxidized to carboxylic acids ($-CHO \rightarrow -COOH$); terminal $= CH_2$ becomes $CO_2 + H_2O$; ketones remain unchanged.

## 5.4.3 Alkynes

**Terminal Alkyne Acidity & Selective Hydrogenation**

**1. Acidity of Terminal Alkynes ($-C \equiv CH$):** The *sp*-hybridized carbon has 50%*s*-character, placing valence electrons closest to the carbon nucleus. Terminal alkynes are acidic ($pK_a \approx 25$):

- React with sodamide: $R - C \equiv CH + NaNH_2 \rightarrow R - C \equiv C^-Na^+ + NH_3 \uparrow$.
- Tollens' Test: Forms a **white precipitate** of silver acetylide: $R - C \equiv CH + [Ag(NH_3)_2]^+OH^- \rightarrow R - C \equiv C - Ag \downarrow + 2NH_3 + H_2O$.
- Ammoniacal $Cu_2Cl_2$: Forms a **red precipitate** of cuprous acetylide: $R - C \equiv CH + [Cu(NH_3)_2]^+OH^- \rightarrow R - C \equiv C - Cu \downarrow + 2NH_3 + H_2O$.
- *Note: Internal alkynes ($R - C \equiv C - R'$) give negative tests with both reagents!*

**2. Stereoselective Reduction to Alkenes:**

- **Lindlar's Catalyst ($H_2/Pd - CaCO_3$ poisoned with quinoline / lead acetate):** Syn-addition of hydrogen yielding ***cis*-alkene**.
- **Birch Reduction (Na/liq. $NH_3$ at $-33°C$):** Anti-addition proceeding via radical-anion intermediate with trans-geometry to minimize electron-electron repulsion, yielding ***trans*-alkene**.

**3. Kucherov Hydration ($HgSO_4$/dil.$H_2SO_4$ at 333 K):** Electrophilic addition of water across the triple bond yields an enol, which immediately undergoes keto-enol tautomerism:

$$HC \equiv CH + H_2O \xrightarrow{Hg^{2+}/H^+} [CH_2 = CH - OH] \rightleftharpoons CH_3 - CHO \quad \text{(Acetaldehyde, or}$$
$$CH_3 - C \equiv CH + H_2O \xrightarrow{Hg^{2+}/H^+} [CH_3 - C(OH) = CH_2] \rightleftharpoons CH_3 - CO - CH_3 \quad \text{(Aceton} \tag{5.38}$$

## 5.4.4 Aromatic Hydrocarbons: EAS Mechanics & Substituent Effects

**Fundamental 3-Step EAS Mechanism & Arenium Ion ($\sigma$-Complex)**

Electrophilic aromatic substitution ($S_EAr$) preserves the massive resonance stabilization of the benzene ring (150.6 kJ/mol):

1. **Generation of Active Electrophile ($E^+$):** Lewis or Brønsted acid generates $E^+$ ($NO_2^+, Cl^+, SO_3, R^+, RCO^+$).

2. **Formation of Arenium Ion ($\sigma$-Complex / Wheland Intermediate) [RDS]:** $C_6H_6 + E^+ \xrightarrow{\text{slow}} [C_6H_6E]^+$. The attacked carbon rehybridizes from $sp^2$ to $sp^3$, disrupting aromaticity. Positive charge delocalizes over ortho and para positions.
3. **Rapid Deprotonation & Rearomatization:** Fast abstraction of the proton from the $sp^3$ carbon restores the planar $sp^2$ aromatic sextet.

## Directing and Activating Effects of Ring Substituents

| Group Type | Examples | Electronic Mechanism | Directing Influence |
|---|---|---|---|
| **Strongly Activating** | $-OH$, $-O^-$, $-NH_2$, $-NR_2$ | Dominant $+M$ resonance | Ortho / Para |
| **Moderately Activating** | $-OCH_3$, $-NHCOCH_3$, $-OCOR$ | $+M$ moderated by cross-conjugation | Ortho / Para |
| **Weakly Activating** | $-CH_3$, $-CH_2CH_3$, $-C_6H_5$ | Hyperconjugation & $+I$ | Ortho / Para |
| **Deactivating (Halogens)** | $-F$, $-Cl$, $-Br$, $-I$ | Strong $-I$ dominates rate; $+M$ directs | **Ortho / Para** |
| **Moderately Deactivating** | $-CHO$, $-COR$, $-COOH$, $-COOR$ | $-M$ resonance & $-I$ | Meta |
| **Strongly Deactivating** | $-NO_2$, $-CF_3$, $-CCl_3$, $-CN$, $-SO_3H$, $-NR_3^+$ | Powerful $-M$ / $-I$ withdraws electron density | Meta |

**Friedel-Crafts Reactions: Critical Limitations & Exceptions**

**1. Carbocation Rearrangement in Alkylation:** Reaction of benzene with 1-chloropropane ($CH_3CH_2CH_2Cl$) and $AlCl_3$ yields **isopropylbenzene (cumene)** as major product ($65-70\%$) due to 1,2-hydride shift of the primary propyl carbocation to the more stable secondary carbocation!

$$\text{Benzene} + CH_3CH_2CH_2Cl \xrightarrow{AlCl_3} C_6H_5-CH(CH_3)_2\ (\text{Major}) + C_6H_5-CH_2CH_2CH \tag{5.39}$$

**Solution:** To synthesize linear alkylbenzenes, carry out **Friedel-Crafts Acylation** with propanoyl chloride to yield ethyl phenyl ketone (no rearrangement because acylium ion $R-C\equiv O^+$ is resonance-stabilized), followed by **Clemmensen reduction** ($Zn-Hg/HCl$) or **Wolff-Kishner reduction**.

**2. Polyalkylation vs Monoacylation:** Alkyl groups activate the ring toward further substitution (polyalkylation occurs). Acyl groups ($-COR$) are strongly deactivating, ensuring clean monoacylation.

**3. Complete Failure with Aniline and Phenol:** Aniline ($-NH_2$) and Phenol ($-OH$) fail Friedel-Crafts reactions because their lone pairs form an insoluble Lewis acid-base adduct with $AlCl_3$ (e.g., $Ph-NH_2\cdot AlCl_3$), placing a positive charge adjacent to the ring and rendering it deactivated toward electrophiles!

## 5.5 Haloalkanes & Haloarenes

### 5.5.1 Nucleophilic Aliphatic Substitution ($S_N1$ vs $S_N2$ vs $S_Ni$)

**Comprehensive Mechanistic Comparison: $S_N1$ vs $S_N2$**

| **Parameter** | $S_N1$ **(Substitution Nucleophilic Unimolecular)** | $S_N2$ **(Substitution Nucleophilic Bimolecular)** |
|---|---|---|
| **Number of Steps** | Two discrete elementary steps | Single concerted step |
| **Rate Law & Kinetics** | Rate $= k[\mathrm{R-X}]$ (First order) | Rate $= k[\mathrm{R-X}][\mathrm{Nu^-}]$ (Second order) |
| **Substrate Reactivity** | $3^\circ > 2^\circ \gg 1^\circ > \mathrm{CH_3}$ (governed by carbocation stability) | $\mathrm{CH_3} > 1^\circ > 2^\circ \gg 3^\circ$ (governed by steric hindrance) |
| **Transition State** | Involves carbocation intermediate | Pentacoordinate trigonal bipyramidal TS |
| **Stereochemical Outcome** | Partial racemization with net inversion ($50-70\%$ inversion due to front-side ion pair shielding) | **100% Complete Walden Inversion** (backside attack) |
| **Nucleophile Strength** | Weak, neutral nucleophiles ($\mathrm{H_2O}, \mathrm{ROH}$) suffice | Requires strong, anionic nucleophiles ($\mathrm{OH^-}, \mathrm{RO^-}, \mathrm{CN^-}$) |
| **Solvent Preference** | **Polar Protic** ($\mathrm{H_2O}$, EtOH, AcOH): solvates carbocation and stabilizes leaving group anion | **Polar Aprotic** (DMSO, DMF, acetone): solvates cations, leaving naked, highly reactive anions |
| **Rearrangements** | Carbocation rearrangements (shifts, expansions) common | **Zero rearrangement possible** |
| **Leaving Group Order** | $\mathrm{I^-} > \mathrm{Br^-} > \mathrm{Cl^-} \gg \mathrm{F^-}$ | $\mathrm{I^-} > \mathrm{Br^-} > \mathrm{Cl^-} \gg \mathrm{F^-}$ |

**The $S_Ni$ Mechanism: Darzens Process with Thionyl Chloride**

Reaction of alcohols with $SOCl_2$ proceeds with different stereochemical outcomes depending on the solvent:

- **In Dioxane / Diethyl Ether ($S_Ni$, Internal Return):** Reaction forms an alkyl chlorosulfite ester ($R-O-SOCl$). In nonpolar solvent, it dissociates into an intimate ion pair [$R^+$ $SO_2Cl^-$]. Decomposition releases $SO_2 \uparrow$, and the chloride is delivered to the carbocation from the **same face** from which the oxygen departed. Outcome: **Retention of configuration**.
- **In Pyridine ($S_N2$ Inversion):** Pyridine reacts with generated HCl and converts the chlorosulfite into an active pyridinium intermediate. Free chloride ion attacks from the **backside**. Outcome: **Complete Walden inversion**.

## 5.5.2 Elimination Reactions ($E1$ vs $E2$ vs $E1cB$)

**Saytzeff (Zaitsev) vs Hofmann Regioselectivity**

**1. Saytzeff Rule (Thermodynamic Control):** Dehydrohalogenation yields preferentially the more stable, more highly substituted alkene (greater number of hyperconjugative $\alpha$-hydrogens).

- **Conditions for Saytzeff Selectivity:** Small, unhindered bases ($EtO^-, MeO^-, OH^-$) acting on substrates with good leaving groups ($I^-, Br^-, Cl^-$).

$$CH_3-CH_2-CH(Br)-CH_3 \xrightarrow{\text{alc. KOH}} CH_3-CH=CH-CH_3 \text{ (Saytzeff, 81\%)} + CH_3-CH_2-CH=CH_2 \text{ (Hofmann,} \tag{5.40}$$

**2. Hofmann Rule (Kinetic Control):** Elimination yields preferentially the less substituted, terminal alkene.

- **Conditions for Hofmann Selectivity:**
  1. **Bulky Sterically Hindered Bases:** Potassium tert-butoxide ($t-BuO^-K^+$), diisopropylamide. The base cannot access the more hindered internal $\beta$-hydrogen and abstracts the accessible terminal $\beta$-hydrogen.
  2. **Poor Leaving Groups:** Fluoride ($-F$) or quaternary ammonium ($-NR_3^+$) and sulfonium ($-SR_2^+$) salts. Due to extreme electronegativity, the $\beta$-hydrogen becomes acidic, developing significant carbanion character in the transition state. The less substituted primary carbanion is more stable than secondary, directing elimination to the terminal alkene.

**3.** $E2$ **Stereochemistry:** Elimination requires strict **anti-periplanar** geometry: the $\beta-$H, the two carbons, and the leaving group $X$ must lie in a single plane with dihedral angle $\theta = 180°$.

### 5.5.3 Nucleophilic Aromatic Substitution ($S_N$Ar)

**Addition-Elimination via Meisenheimer Complex**

Unactivated haloarenes are virtually inert toward nucleophiles at ambient conditions because the C − X bond possesses partial double bond character (1.69 Å in chlorobenzene vs 1.77 Å in chloroethane) due to $+M$ resonance, and the $sp^2$ carbon holds electrons tightly. **Activation by Nitro Groups ($S_N$Ar):** Introducing strong electron-withdrawing nitro ($-NO_2$) groups at **ortho** and **para** positions dramatically enhances substitution rate:

$$\text{Chlorobenzene (623 K, 300 atm)} \gg p\text{-nitro (443 K)} \gg 2,4\text{-dinitro (368 K)} \gg 2,4,6\text{-trin} \tag{5.41}$$

**Mechanism:** Nucleophilic attack on the ipso carbon forms a resonance-stabilized carbanionic intermediate (**Meisenheimer complex**), where negative charge delocalizes into the ortho and para nitro groups. Loss of halide in Step 2 restores aromaticity. **Leaving Group Anomaly:** Unlike aliphatic substitutions, the rate order in $S_N$Ar is:

$$\text{Ar} - \text{F} \gg \text{Ar} - \text{Cl} > \text{Ar} - \text{Br} > \text{Ar} - \text{I} \tag{5.42}$$

The rate-determining step is nucleophilic attack (Step 1). The extreme electronegativity of fluorine stabilizes the Meisenheimer intermediate via $-I$ pull, accelerating the rate!

## 5.6 Alcohols, Phenols & Ethers

### 5.6.1 Alcohols: Acidity, Oxidation & Dehydration

**Acidity & The Lucas Distinction Test**

**1. Acidity Order of Alcohols:**

$$H_2O\ (pK_a = 15.7) > CH_3OH\ (15.5) > 1°\ RCH_2OH\ (16) > 2°\ R_2CHOH\ (16.5) > 3°\ R_3C(\text{OH}) \tag{5.43}$$

Alkoxide anions ($R-O^-$) are destabilized by the $+I$ inductive effect of alkyl groups. Water is a stronger acid than all alcohols except methanol!

**2. The Lucas Test (anhydrous $ZnCl_2$ + conc. HCl):** Distinguishes $1^\circ, 2^\circ, 3^\circ$ alcohols via $S_N1$ conversion to insoluble alkyl chlorides:

$$R-OH + HCl \xrightarrow{ZnCl_2} R-Cl \downarrow \text{(cloudy turbidity)} + H_2O \qquad (5.44)$$

- $3^\circ$ **Alcohols:** Immediate turbidity within 5–10 seconds at room temperature.
- $2^\circ$ **Alcohols:** Turbidity appears within 5 minutes.
- $1^\circ$ **Alcohols:** Solution remains completely clear at room temperature; turbidity appears only on prolonged boiling.

**Oxidation Reagents for Alcohols**

- **Pyridinium Chlorochromate (PCC in $CH_2Cl_2$):** Mild selective oxidant. Oxidizes $1^\circ$ alcohols to **aldehydes** (stops cleanly without over-oxidation to carboxylic acid) and $2^\circ$ alcohols to ketones.
- **Jones Reagent ($CrO_3$/aq. $H_2SO_4$/acetone):** Converts $1^\circ$ alcohols directly to carboxylic acids; $2^\circ$ alcohols to ketones.
- **Heated Copper Catalyst at 573 K (300°C):**
  - $1^\circ$ Alcohol $\xrightarrow{Cu,573K}$ Aldehyde $+H_2\uparrow$ (Dehydrogenation).
  - $2^\circ$ Alcohol $\xrightarrow{Cu,573K}$ Ketone $+H_2\uparrow$ (Dehydrogenation).
  - $3^\circ$ Alcohol $\xrightarrow{Cu,573K}$ **Alkene** $+H_2O$ (**Dehydration**, not dehydrogenation!).

## 5.6.2 Phenols: Acidic Character & Named Reactions

**Phenol Acidity & Substituent Influence**

Phenol ($pK_a \approx 10$) is $10^6$ times more acidic than aliphatic alcohols ($pK_a \approx 16-18$):

- **Resonance Stabilization of Phenoxide:** The phenoxide ion ($C_6H_5O^-$) delocalizes negative charge into the benzene ring over ortho and para positions (5 resonance structures). Alkoxide ions ($RO^-$) have zero resonance stabilization.

- **Substituent Effects on Phenol Acidity:**

  Picric acid ($pK_a = 0.38$) > 2, 4-dinitrophenol (4.0) > *p*-nitrophenol (7.15) > *o*-nitrophenol (7.23) > Phenol (10.0) > *p*-cresol (10.2) (5.45)

  *Note: p-nitrophenol is more acidic than o-nitrophenol because the ortho isomer is stabilized by intramolecular hydrogen bonding, reducing proton release!*

**Kolbe's & Reimer-Tiemann Reactions**

**1. Kolbe's Reaction (Synthesis of Salicylic Acid):** Sodium phenoxide is treated with weak electrophile $CO_2$ at 400 K and $4-7$ atm, followed by acidification:

$$C_6H_5ONa + CO_2 \xrightarrow[4-7\text{ atm}]{400\text{ K}} \text{Sodium Salicylate} \xrightarrow{H^+} \text{Salicylic Acid (\textbf{o}-hydroxybenzoic acid)} \quad (5.46)$$

Salicylic acid acetylates with acetic anhydride to yield **Aspirin** (acetylsalicylic acid).

**2. Reimer-Tiemann Reaction (Synthesis of Salicylaldehyde):** Phenol treated with $CHCl_3$ and aqueous NaOH at 340 K:

$$C_6H_5OH + CHCl_3 + 3NaOH \longrightarrow \text{Salicylaldehyde (\textbf{o}-hydroxybenzaldehyde)} + 3NaCl + 2H_2O \quad (5.47)$$

**Mechanistic Detail:** The electrophile is singlet dichlorocarbene ($:CCl_2$) generated in situ. Attack occurs predominantly at the ortho position due to cyclic sodium chelation in the transition state. If $CCl_4$ is used instead of $CHCl_3$, **salicylic acid** is formed directly!

## 5.6.3 Ethers: Williamson Synthesis & HI Cleavage

**NEET Trap: Substrate Rules for Williamson Ether Synthesis**

Williamson ether synthesis involves nucleophilic displacement of an alkyl halide by an alkoxide:

$$R - O^-Na^+ + R' - X \xrightarrow{S_N2} R - O - R' + NaX \quad (5.48)$$

**The Strict Constraint:** The alkyl halide ($R' - X$) **must be methyl or primary** ($1^\circ$).

- **Correct Preparation of tert-Butyl Methyl Ether:**

$$(CH_3)_3C-O^-Na^+ \text{ (sodium } \textit{tert}\text{-butoxide)} + CH_3-I \xrightarrow{S_N2} (CH_3)_3C-O-CH \tag{5.49}$$

- **Fatal Reversal Trap:** If sodium methoxide is treated with tert-butyl bromide:

$$CH_3-O^-Na^+ + (CH_3)_3C-Br \xrightarrow{E2} CH_2=C(CH_3)_2 \text{ (2-methylpropene)} + CH_3C \tag{5.50}$$

Because tert-butyl bromide is sterically crowded, the strongly basic methoxide ion induces **exclusive $E2$ elimination**!

**Cleavage of Ethers with Hydroiodic Acid (HI)**

When unsymmetrical ether $R-O-R'$ is cleaved with cold, concentrated HI:

1. **Case 1: Both Alkyl Groups are $1^\circ$ or $2^\circ$ ($S_N2$ Pathway):** The protonated ether $[R-OH^+-R']$ is attacked by iodide ion ($I^-$) at the **less sterically hindered carbon**:

$$CH_3-O-CH_2CH_2CH_3 + HI \xrightarrow{S_N2} CH_3I + CH_3CH_2CH_2OH \tag{5.51}$$

2. **Case 2: One Alkyl Group is $3^\circ$ or Resonance-Stabilized ($S_N1$ Pathway):** Heterolysis occurs first to produce the stable $3^\circ$ or benzylic carbocation:

$$(CH_3)_3C-O-CH_3 + HI \xrightarrow{S_N1} (CH_3)_3C-I + CH_3OH \tag{5.52}$$

3. **Case 3: Alkyl Aryl Ethers (Anisole $C_6H_5-O-CH_3$):** The $C_{\text{aromatic}}-O$ bond has partial double bond character due to resonance and cannot be cleaved. Cleavage yields exclusively:

$$C_6H_5-O-CH_3 + HI \longrightarrow C_6H_5OH \text{ (phenol)} + CH_3I \tag{5.53}$$

*Note: Excess hot* HI *converts all aliphatic alcohols to alkyl iodides:* $R-OH + HI \rightarrow R-I + H_2O$. *Phenol never converts to iodobenzene!*

## 5.7 Aldehydes, Ketones & Carboxylic Acids

## 5.7.1 Nucleophilic Addition Mechanisms

**Carbonyl Electrophilicity & Relative Reactivity Order**

The carbonyl carbon is $sp^2$-hybridized with planar geometry (120°). Strong electronegativity of oxygen ($\chi = 3.44$ vs $\chi_C = 2.55$) creates a permanent dipole:

$$> C^{\delta+} = O^{\delta-} \longleftrightarrow > \overset{+}{C} - \overset{-}{O} \tag{5.54}$$

Nucleophiles attack perpendicular to the carbonyl plane (107° Bürgi-Dunitz angle), transforming the planar $sp^2$ carbon into a tetrahedral $sp^3$ alkoxide intermediate. **Reactivity Hierarchy Toward Nucleophiles:**

$$HCHO > CH_3CHO > CH_3COCH_3 > C_6H_5CHO > C_6H_5COCH_3 > C_6H_5COC_6H_5 \tag{5.55}$$

**Governing Factors:**

- **Steric Effect:** Bulky alkyl or aryl groups crowd the transition state, hindering nucleophilic approach.
- **Electronic Effect:** Electron-donating $+I$ alkyl groups reduce partial positive charge on the carbonyl carbon. $+M$ resonance from aromatic rings strongly stabilizes the ground state, lowering electrophilicity.

**Standard Nucleophilic Addition Reactions**

1. **Addition of** HCN **(Cyanohydrin Formation):** Catalyzed by base ($OH^-$ generates active nucleophile $CN^-$). Cyanohydrins hydrolyze in acid to $\alpha$-hydroxy acids:

$$CH_3CHO + HCN \xrightarrow{OH^-} CH_3CH(OH)CN \xrightarrow{H_3O^+} CH_3CH(OH)COOH \text{ (Lactic Acid)} \tag{5.56}$$

2. **Addition of Sodium Bisulfite** ($NaHSO_3$)**:** Aldehydes and methyl ketones form crystalline, water-soluble bisulfite adducts. The adduct regenerates pure carbonyl on treatment with dilute mineral acid or base (used for industrial separation and purification).
3. **Addition of Alcohols (Acetal/Ketal Formation):** Carbonyls react with 1 equivalent of alcohol in dry HCl to form hemiacetals, and with 2 equivalents to form acetals. Acetals are stable to bases, oxidants, and reducing agents, acting as universal carbonyl protecting groups.
4. **Nucleophilic Addition-Elimination with Ammonia Deriva-**

**tives ($NH_2 - G$):**

$$> C = O + H_2N - G \xrightarrow{pH4.5} [> C(OH) - NH - G] \xrightarrow{-H_2O} > C = N - G \tag{5.57}$$

- Hydroxylamine ($NH_2OH$) → **Oxime** ($> C = N - OH$).
- Hydrazine ($NH_2NH_2$) → **Hydrazone** ($> C = N - NH_2$).
- 2,4-Dinitrophenylhydrazine (2,4-DNP / Brady's Reagent) → **Yellow/Orange precipitate** (used to detect aldehydes/ketones).
- Semicarbazide ($NH_2 - CO - NHNH_2$) → **Semicarbazone**. (*Trap: Only the hydrazine $-NH_2$ is nucleophilic; the amide $-CONH_2$ nitrogen lone pair is delocalized into the carbonyl!*)

## 5.7.2 Reactions Involving $\alpha$-Hydrogen: Aldol, Cannizzaro & Haloform

**Step-by-Step Mechanism: Aldol Condensation**

Aldehydes or ketones possessing at least one $\alpha$-hydrogen react in dilute alkali (dil. NaOH):

1. **Step 1: Enolate Generation (Acid-Base Equilibrium):**

$$CH_3 - CHO + OH^- \rightleftharpoons [:\bar{C}H_2 - CHO \longleftrightarrow CH_2 = CH - O^-] + H_2O \tag{5.58}$$

2. **Step 2: Nucleophilic Attack on Second Carbonyl Molecule (RDS):**

$$CH_3 - CHO + :\bar{C}H_2 - CHO \longrightarrow CH_3 - CH(O^-) - CH_2 - CHO \tag{5.59}$$

3. **Step 3: Protonation to Aldol ($\beta$-Hydroxyaldehyde):**

$$CH_3 - CH(O^-) - CH_2 - CHO + H_2O \longrightarrow CH_3 - CH(OH) - CH_2 - CHO \text{ (3-hy} \tag{5.60}$$

4. **Step 4: Dehydration on Heating ($E1cB$ Mechanism):** Deprotonation of the acidic $\alpha$-carbon creates an enolate, followed by expulsion of hydroxide ion ($OH^-$) to yield a conjugated $\alpha,\beta$-unsaturated aldehyde:

$$CH_3 - CH(OH) - CH_2 - CHO \xrightarrow{\Delta, -H_2O} CH_3 - CH = CH - CHO \text{ (But-2-enal} \tag{5.61}$$

**Step-by-Step Mechanism: Cannizzaro Reaction**

Aldehydes **lacking $\alpha$-hydrogen** ($HCHO, C_6H_5CHO, (CH_3)_3CCHO$) disproportionate in concentrated alkali (50% NaOH):

1. **Step 1: Reversible Addition of Hydroxide:**

$$C_6H_5CHO + OH^- \rightleftharpoons C_6H_5 - CH(OH) - O^- \quad \text{(Monoanion)} \quad (5.62)$$

2. **Step 2: Rate-Determining Hydride Shift ($H^-$):** The alkoxide reforms the carbonyl double bond, transferring a hydride ion directly to a second aldehyde molecule:

$$C_6H_5 - CH(O^-) - OH + C_6H_5CHO \xrightarrow{\text{slow, RDS}} C_6H_5COOH + C_6H_5CH_2O^- \quad (5.63)$$

3. **Step 3: Fast Acid-Base Proton Exchange:**

$$C_6H_5COOH + C_6H_5CH_2O^- \xrightarrow{\text{fast}} C_6H_5COO^-Na^+ \text{ (Sodium Benzoate)} + C_6H_5CH_2OH \quad (5.64)$$

**Cross-Cannizzaro Reaction:** When benzaldehyde reacts with formaldehyde in concentrated alkali, formaldehyde is oxidized exclusively to sodium formate (HCOONa) while benzaldehyde is reduced to benzyl alcohol:

$$C_6H_5CHO + HCHO + NaOH \longrightarrow C_6H_5CH_2OH + HCOONa \quad (5.65)$$

*Cause: Formaldehyde is sterically uncrowded and far more electrophilic than benzaldehyde, undergoing nucleophilic attack by $OH^-$ first!*

**The Haloform (Iodoform) Reaction Test**

**Structural Requirement:** Reagent $I_2$/NaOH (or NaOI) oxidizes and cleaves compounds containing:

1. $CH_3 - C = O$ (Methyl ketone group, e.g., acetaldehyde, acetone, acetophenone).
2. $CH_3 - CH(OH) -$ (Methyl carbinol group oxidizable in situ to methyl ketone, e.g., ethanol, propan-2-ol, butan-2-ol).

$$R - CO - CH_3 + 3I_2 + 4NaOH \longrightarrow CHI_3 \downarrow \text{ (Yellow precipitate, m.p. } 119^\circ C) + R - COON \quad (5.66)$$

**Classical NEET Distinction Traps:**

- **Ethanol vs Methanol:** Ethanol gives positive iodoform test; methanol gives negative test.
- **Pentan-2-one vs Pentan-3-one:** Pentan-2-one is a methyl ketone (positive); pentan-3-one is not (negative).
- **Acetaldehyde vs Benzaldehyde:** Acetaldehyde gives yellow iod-

oform; benzaldehyde does not.

### 5.7.3 Distinction Tests & Reductions

**Distinction Tests for Carbonyl Functions**

- **Tollens' Test (Silver Mirror Test):** Tollens' reagent ($[Ag(NH_3)_2]^+OH^-$) is reduced by **all aldehydes** (aliphatic and aromatic) and $\alpha$-hydroxy ketones to a shining silver mirror (Ag $\downarrow$):

$$R-CHO+2[Ag(NH_3)_2]^+ + 3OH^- \longrightarrow R-COO^- + 2Ag\downarrow + 4NH_3 + 2H_2O \tag{5.67}$$

- **Fehling's Test:** Alkaline cupric tartrate complex ($Cu^{2+}$) is reduced to a **red precipitate of cuprous oxide** ($Cu_2O$ $\downarrow$) by **aliphatic aldehydes only**:

$$R-CHO + 2Cu^{2+} + 5OH^- \longrightarrow R-COO^- + Cu_2O\downarrow(\text{red}) + 3H_2O \tag{5.68}$$

*NEET Trap: Benzaldehyde and all aromatic aldehydes FAIL Fehling's test because aromatic resonance stabilizes the carbonyl, making it an insufficiently strong reducing agent to reduce the tartrate-chelated $Cu^{2+}$ ion!*

**Clemmensen vs Wolff-Kishner Carbonyl Reduction**

- **Clemmensen Reduction:** Uses amalgamated zinc and concentrated hydrochloric acid ($Zn-Hg$/conc. HCl):

$$>C=O + 4[H] \xrightarrow{Zn-Hg/\text{conc. HCl}} > CH_2 + H_2O \tag{5.69}$$

*Constraint: Cannot be used for substrates containing acid-sensitive groups ($-OH, -OR, -C=C-$), which undergo elimination or addition in strong acid.*

- **Wolff-Kishner Reduction:** Carbonyl converted to hydrazone, then heated with KOH in ethylene glycol at $453-473$ K:

$$>C=O + NH_2NH_2 \xrightarrow{-H_2O} >C=N-NH_2 \xrightarrow[\text{glycol},\Delta]{KOH} > CH_2 + N_2\uparrow \tag{5.70}$$

*Constraint: Cannot be used for substrates containing base-sensitive groups (alkyl halides, esters), which undergo nucleophilic displacement or saponification.*

## 5.7.4 Carboxylic Acids & Derivatives

### Resonance in Carboxylic Acids & The Ortho Effect

**1. Resonance of Carboxylate Anion:** Deprotonation of carboxylic acid yields the carboxylate anion ($R-COO^-$), which is stabilized by two **completely equivalent** resonance structures sharing negative charge equally between two electronegative oxygen atoms ($C-O$ bond lengths identical at 1.26 Å). This makes carboxylic acids vastly more acidic than phenols.

**2. The Ortho Effect in Benzoic Acids:** Any substituent at the **ortho position** of benzoic acid—regardless of whether it is electron-withdrawing ($-NO_2$) or electron-releasing ($-CH_3, -OCH_3$)—increases the acidic strength compared to benzoic acid:

$$o\text{-toluic acid } (pK_a = 3.91) > \text{benzoic acid } (pK_a = 4.20) > m\text{-toluic acid } (4.27) > p\text{-tolui} \tag{5.71}$$

**Physical Cause:** Steric crowding by the ortho substituent twists the $-COOH$ group out of the plane of the benzene ring. This **steric inhibition of resonance (SIR)** eliminates cross-conjugation between the ring and the carbonyl, allowing unhindered resonance delocalization of negative charge across the two carboxylate oxygens!

### Hell-Volhard-Zelinsky (HVZ) Reaction & Decarboxylation

**1. HVZ Reaction ($\alpha$-Halogenation):** Carboxylic acids possessing $\alpha$-hydrogen react with chlorine or bromine in the presence of catalytic red phosphorus:

$$R-CH_2-COOH + X_2 \xrightarrow{\text{red P}} R-CH(X)-COX \xrightarrow{H_2O} R-CH(X)-COOH\ (\alpha\text{-haloca} \tag{5.72}$$

*Note: Formic acid (*$HCOOH$*) and benzoic acid (*$C_6H_5COOH$*) fail the HVZ reaction due to absence of $\alpha$-hydrogens!*

**2. Soda-Lime Decarboxylation:** Sodium carboxylates heated with soda lime ($NaOH + CaO$ in 3 : 1 mass ratio) undergo loss of carbon dioxide:

$$R-COONa + NaOH \xrightarrow[\Delta]{CaO} R-H + Na_2CO_3 \tag{5.73}$$

$CaO$ acts as a desiccant, keeping $NaOH$ dry and raising the reaction fusion temperature.

# 5.8 Nitrogen-Containing Compounds: Amines & Diazonium Chemistry

## 5.8.1 Basicity of Amines: Gas Phase vs Aqueous Phase

**Thermodynamic Factors Governing Amine Basicity**

The basicity of amines is measured by $K_b$ or $pK_b$ ($pK_b = -\log K_b$; lower $pK_b$ indicates stronger base). Basicity reflects the thermodynamic stability of the conjugate ammonium cation ($RNH_3^+$). **1. Gas Phase (Isolated Molecules):** In the gas phase, solvent molecules are absent. Basicity is governed strictly by the electron-donating $+I$ inductive effect of alkyl groups, which stabilizes the positive charge on nitrogen:

$$\text{Gas Phase Basicity: } 3^\circ\ R_3N > 2^\circ\ R_2NH > 1^\circ\ RNH_2 > NH_3 \quad (5.74)$$

**2. Aqueous Solution (Interplay of Three Opposing Factors):** In water, three thermodynamic effects operate concurrently:

1. **Inductive Effect ($+I$):** Stabilizes cation in order $3^\circ > 2^\circ > 1^\circ > NH_3$.
2. **Hydration Energy (H-Bonding):** The conjugate ammonium ion forms hydrogen bonds with water molecules. Greater number of hydrogens attached to positive nitrogen yields higher hydration energy: $RNH_3^+$ (3 H-bonds) > $R_2NH_2^+$ (2 H-bonds) > $R_3NH^+$ (1 H-bond). Hydration stabilizes in order: $NH_4^+ > 1^\circ > 2^\circ > 3^\circ$.
3. **Steric Hindrance to Protonation:** Bulky alkyl groups shield the nitrogen lone pair, impeding proton attack in order: $1^\circ > 2^\circ > 3^\circ$.

**Authoritative Aqueous Basicity Orders (NEET Core Benchmark)**

The combined net effect produces the following experimentally established basicity sequences in aqueous medium:

1. **Methyl-Substituted Amines ($2^\circ > 1^\circ > 3^\circ > NH_3$, Mnemonic: "213"):**

$$(CH_3)_2NH\ (pK_b = 3.27) > CH_3NH_2\ (3.38) > (CH_3)_3N\ (4.22) > NH_3\ (4.75) \quad (5.75)$$

2. **Ethyl-Substituted Amines ($2^\circ > 3^\circ > 1^\circ > NH_3$, Mnemonic: "231"):**

$$(C_2H_5)_2NH\ (pK_b = 3.00) > (C_2H_5)_3N\ (3.25) > C_2H_5NH_2\ (3.29) > NH_3\ (4.75) \quad (5.76)$$

3. **Arylamines (Aniline):** Aniline ($C_6H_5NH_2$, $pK_b = 9.38$) is vastly weaker than ammonia ($pK_b = 4.75$) because its nitrogen lone pair is delocalized over the ortho and para positions of the aromatic ring (5 canonical forms). The anilinium cation ($C_6H_5NH_3^+$) has only 2 resonance structures, making protonation thermodynamically disfavored!

## 5.8.2 Gabriel Phthalimide & Hoffmann Bromamide Mechanisms

**Gabriel Phthalimide Synthesis: Pure Primary Aliphatic Amines**

Gabriel synthesis produces pure 1° aliphatic amines free from secondary or tertiary contamination:

1. **Potassium Salt Formation:** Phthalimide is deprotonated by ethanolic KOH to form potassium phthalimide (the imide hydrogen is acidic due to two flanking carbonyls).
2. **Alkylation ($S_N2$):** Nucleophilic attack of the phthalimide anion on a primary alkyl halide ($R-X$) yields $N$-alkylphthalimide.
3. **Hydrazinolysis / Alkaline Hydrolysis:** Alkaline hydrolysis with NaOH yields pure primary amine and sodium phthalate:

$$\text{Phthalimide} \xrightarrow{\text{KOH}} \text{Potassium Phthalimide} \xrightarrow{\text{R}-\text{X}} N\text{-Alkylphthalimide} \xrightarrow{\text{NaOH(aq)}} R-NH_2 + \text{Sodium Phthalate} \tag{5.77}$$

**NEET Trap: Why Aromatic Amines CANNOT be Prepared by Gabriel Synthesis**

Aniline ($C_6H_5NH_2$) cannot be prepared by Gabriel phthalimide synthesis because the required aryl halide ($C_6H_5-X$) does not undergo nucleophilic aromatic substitution ($S_N2$) with the phthalimide anion under standard conditions due to partial double bond character of the $C-X$ bond and steric/electronic repulsion of the aromatic ring!

**Step-by-Step Mechanism: Hoffmann Bromamide Degradation**

Primary acid amides react with bromine and aqueous alkali ($Br_2 + 4KOH$) to yield primary amines with **one less carbon atom**:

$$R-CONH_2 + Br_2 + 4KOH \longrightarrow R-NH_2 + K_2CO_3 + 2KBr + 2H_2O \tag{5.78}$$

**Elementary Mechanistic Steps:**

1. **Formation of *N*-Bromoamide:** $\mathrm{RCONH_2} + \mathrm{OH^-} \rightarrow \mathrm{RCON\bar{H}} \xrightarrow{\mathrm{Br_2}} \mathrm{R-CONHBr} + \mathrm{Br^-}$.
2. **Formation of Bromoamide Anion:** $\mathrm{R-CONHBr} + \mathrm{OH^-} \rightleftharpoons [\mathrm{R-CON-Br}]^- + \mathrm{H_2O}$.
3. **Rate-Determining Concerted 1,2-Shift (Isocyanate Formation):** Migration of alkyl/aryl group $R$ with its electron pair from carbonyl carbon to electron-deficient nitrogen with simultaneous loss of bromide ion:

$$[\mathrm{R-CO-\bar{N}-Br}] \xrightarrow{\text{slow, RDS}} \mathrm{R-N=C=O\ (Alkyl\ Isocyanate)} + \mathrm{Br^-} \tag{5.79}$$

*Stereochemical Note: The migrating group R migrates with **complete retention of configuration**! If chiral, optical activity is preserved.*

4. **Alkaline Hydrolysis of Isocyanate:**

$$\mathrm{R-N=C=O} + 2\mathrm{OH^-} \longrightarrow \mathrm{R-NH_2} + \mathrm{CO_3^{2-}} \tag{5.80}$$

### 5.8.3 Distinction Tests for Amines

**Hinsberg, Carbylamine & Nitrous Acid Tests**

1. **Hinsberg Test (Benzenesulfonyl Chloride $\mathrm{C_6H_5SO_2Cl}$):**
   - 1° **Amines:** React to form *N*-alkylbenzenesulfonamide ($\mathrm{C_6H_5SO_2NHR}$). The remaining hydrogen on nitrogen is strongly acidic due to the electron-withdrawing $-\mathrm{SO_2}-$ group $\Longrightarrow$ **Dissolves completely in aqueous** KOH.
   - 2° **Amines:** React to form *N*,*N*-dialkylbenzenesulfonamide ($\mathrm{C_6H_5SO_2NR_2}$). Lacks acidic hydrogen on nitrogen $\Longrightarrow$ **Insoluble in aqueous** KOH.
   - 3° **Amines:** Possess no hydrogen on nitrogen $\Longrightarrow$ **Does not react at all**.
2. **Carbylamine Test (Isocyanide Test):** Exclusive to **primary amines** (both aliphatic and aromatic). Heated with chloroform and alcoholic KOH:

$$\mathrm{R-NH_2} + \mathrm{CHCl_3} + 3\mathrm{KOH} \xrightarrow{\Delta} \mathrm{R-N\equiv C\ (Carbylamine,\ foul\ odor)} + 3\mathrm{KCl} + 3\mathrm{H_2O} \tag{5.81}$$

*Secondary and tertiary amines fail the carbylamine test!*

3. **Reaction with Nitrous Acid ($\mathrm{HNO_2} = \mathrm{NaNO_2} + \mathrm{HCl}$):**

- 1° **Aliphatic Amines:** Yield unstable aliphatic diazonium salts that decompose quantitatively to alcohols with quantitative evolution of nitrogen gas: $RNH_2 + HNO_2 \rightarrow ROH + N_2 \uparrow + H_2O$.
- 1° **Aromatic Amines (Aniline):** Form stable benzenediazonium chloride at ice-cold temperature ($0 - 5°C$).
- 2° **Amines (Aliphatic & Aromatic):** React to form yellow oily *N*-nitrosamines ($R_2N - NO$), which give **Liebermann's nitroso test**.
- 3° **Aliphatic Amines:** Form water-soluble trialkylammonium nitrite salts ($[R_3NH]^+NO_2^-$).

## 5.8.4 Benzenediazonium Chemistry & Synthetic Transformations

**Sandmeyer, Gattermann, Balz-Schiemann & Coupling Reactions**

Benzenediazonium chloride ($C_6H_5N_2^+Cl^-$) is prepared by diazotization of aniline with $NaNO_2 + 2HCl$ at $273 - 278$ K ($0 - 5°C$). It undergoes two major classes of reactions: **1. Replacement Reactions (Loss of $N_2 \uparrow$):**

- **Sandmeyer Reaction:** $Cu_2Cl_2/HCl \rightarrow C_6H_5Cl$; $Cu_2Br_2/HBr \rightarrow C_6H_5Br$; $CuCN/KCN \rightarrow C_6H_5CN$ (higher yields via cuprous salts).
- **Gattermann Reaction:** Cu powder/HCl $\rightarrow C_6H_5Cl$; Cu powder/HBr $\rightarrow C_6H_5Br$.
- **Balz-Schiemann Reaction:** $HBF_4 \rightarrow C_6H_5N_2^+BF_4^- \xrightarrow{\Delta} C_6H_5F + BF_3 + N_2 \uparrow$ (best method for fluorobenzene).
- **Replacement by Iodine:** Warm with aqueous KI $\rightarrow C_6H_5I + KCl + N_2 \uparrow$ (no copper catalyst needed).
- **Deamination (Replacement by H):** Reduction with hypophosphorous acid ($H_3PO_2/H_2O$) or ethanol ($CH_3CH_2OH$):

$$C_6H_5N_2^+Cl^- + H_3PO_2 + H_2O \xrightarrow{Cu^+} C_6H_6 + H_3PO_3 + HCl + N_2 \uparrow \quad (5.82)$$

**2. Azo Coupling Reactions (Retention of Diazo Linkage $-N=N-$):** The diazonium ion acts as a weak electrophile, attacking highly activated aromatic rings at the para position:

- **Coupling with Phenol (Alkaline Medium, pH $9-10$):** Forms **orange azo dye** (*p*-hydroxyazobenzene):

$$C_6H_5N_2^+Cl^- + C_6H_5OH \xrightarrow{pH9-10} C_6H_5 - N = N - C_6H_4 - OH \text{ (Orange Dye)} + HCl \quad (5.83)$$

- **Coupling with Aniline (Mildly Acidic Medium, pH $4-5$):** Forms **yellow azo dye** (*p*-aminoazobenzene):

$$C_6H_5N_2^+Cl^- + C_6H_5NH_2 \xrightarrow{pH4-5} C_6H_5 - N = N - C_6H_4 - NH_2 \text{ (Yellow Dye)} + \text{I} \tag{5.84}$$

## 5.9 Biomolecules (NEET Unit 19)

### 5.9.1 Carbohydrates: Classification & Stereochemistry

**Definition & D/L Stereochemical Nomenclature**

Carbohydrates are optically active polyhydroxy aldehydes or polyhydroxy ketones, or substances that produce them upon hydrolysis.

- **D/L Configuration (Rosanoff Convention):** Assigned by comparing the configuration of the highest-numbered chiral carbon (the penultimate asymmetric carbon, $C_5$ in hexoses) in a vertical Fischer projection to **D-glyceraldehyde**:
  - **D-Sugar:** The $-OH$ group at the lowest asymmetric center projects to the **right**.
  - **L-Sugar:** The $-OH$ group projects to the **left**.
- *Crucial Principle: D and L designate configurational stereochemistry only. They have ZERO correlation to experimental dextrorotation $(+)$ or levorotation $(-)$. D-glucose is dextrorotatory, whereas D-fructose is strongly levorotatory!*

**Structure Elucidation of D-(+)-Glucose**

Experimental evidence establishing the open-chain structure of D-glucose:

1. **Molecular Formula:** $C_6H_{12}O_6$.
2. **Straight Chain of 6 Carbons:** Prolonged heating with hydroiodic acid and red phosphorus yields **n-hexane**:

$$C_6H_{12}O_6 \xrightarrow[\Delta]{HI/red\ P} CH_3 - (CH_2)_4 - CH_3 \ (n\text{-hexane}) \tag{5.85}$$

3. **Carbonyl Group ($>C=O$):** Reacts with hydroxylamine ($NH_2OH$) to form an oxime, and with HCN to form a cyanohydrin.
4. **Aldehyde Function ($-CHO$):** Oxidation with mild bromine water ($Br_2/H_2O$) selectively oxidizes glucose to **gluconic acid**

$(HOOC - (CHOH)_4 - CH_2OH)$. (Ketones are not oxidized by bromine water).

5. **Five Hydroxyl Groups ($-OH$):** Acetylation with acetic anhydride ($Ac_2O$) yields **glucose penta-acetate**. Because it is stable, all five $-OH$ groups reside on separate carbon atoms.
6. **Primary Alcohol at $C_6$ ($-CH_2OH$):** Oxidation with concentrated nitric acid ($HNO_3$) oxidizes both the terminal aldehyde and the primary alcohol to dicarboxylic **saccharic acid (glucaric acid)**:

$$CHO - (CHOH)_4 - CH_2OH \xrightarrow{HNO_3} HOOC - (CHOH)_4 - COOH \text{ (Saccharic Acid)} \quad (5.86)$$

## 5.9.2 Cyclic Pyranose Structures, Anomers & Mutarotation

**Failure of Open-Chain Structure & The Cyclic Pyranose Ring**

**1. Evidence Opposing the Simple Open-Chain Aldehyde:**

- Glucose does not react with $NaHSO_3$, does not form 2,4-DNP adduct, and does not restore the pink color of Schiff's reagent.
- Glucose penta-acetate does not react with hydroxylamine (proves absence of a free $-CHO$ group).
- Glucose exists in two distinct crystalline stereoisomers ($\alpha$ and $\beta$) with different physical properties.

**2. Cyclic Hemiacetal Formation:** Intramolecular nucleophilic attack of the $C_5 - OH$ group onto the $C_1$ carbonyl carbon produces a 6-membered **pyranose ring**. The $C_1$ carbon becomes an asymmetric center known as the **anomeric carbon**:

- **$\alpha$-D-Glucopyranose:** In Haworth projection, the anomeric $-OH$ at $C_1$ points **downward** (*trans* to the $-CH_2OH$ group at $C_6$). Melting point 146°C, specific rotation $[\alpha]_D = +112°$.
- **$\beta$-D-Glucopyranose:** The anomeric $-OH$ at $C_1$ points **upward** (*cis* to the $-CH_2OH$ group at $C_6$). In the chair conformation, all bulky groups ($-OH$ and $-CH_2OH$) occupy equatorial positions $\implies$ thermodynamically more stable. Melting point 150°C, specific rotation $[\alpha]_D = +18.7°$.

**3. Mutarotation Mechanism:** When pure crystals of either $\alpha$-D-glucose (+112°) or $\beta$-D-glucose (+18.7°) are dissolved in water, the optical rotation spontaneously drifts over time until reaching an equi-

librium value of +52.7°:

$$\alpha\text{-D-glucose } (+112^\circ) \underset{\approx 36\%}{\overset{H_2O}{\rightleftharpoons}} \text{Open-Chain Form } (0.02\%) \underset{\approx 64\%}{\overset{H_2O}{\rightleftharpoons}} \beta\text{-D-glucose } (+18.7 \tag{5.87}$$

This reversible interconversion between anomers via the open-chain aldehyde is called **mutarotation**.

### 5.9.3 Disaccharides, Glycosidic Linkages & Polysaccharides

**Disaccharide Chemistry: Linkages & Reducing Character**

| Sugar | Monomer Units | Glycosidic Bond | Free Anomeric C? | Reducing? |
|---|---|---|---|---|
| **Sucrose** (Cane sugar) | $\alpha$-D-glucose + $\beta$-D-fructose | $\alpha(\mathbf{1} \rightarrow \mathbf{2})\beta$ | None ($C_1$ of Glc and $C_2$ of Fru tied up) | **Non-Reducing** |
| **Maltose** (Malt sugar) | $\alpha$-D-glucose + $\alpha$-D-glucose | $\alpha(\mathbf{1} \rightarrow \mathbf{4})$ | Yes ($C_1$ of second glucose free) | **Reducing** |
| **Lactose** (Milk sugar) | $\beta$-D-galactose + $\beta$-D-glucose | $\beta(\mathbf{1} \rightarrow \mathbf{4})$ | Yes ($C_1$ of glucose free) | **Reducing** |

**Invert Sugar & The Optical Inversion of Sucrose**

Sucrose is **dextrorotatory** ($[\alpha]_D = +66.5^\circ$). Upon acid hydrolysis or enzymatic cleavage by *invertase*, it yields an equimolar mixture of D-(+)-glucose and D-(-)-fructose:

$$C_{12}H_{22}O_{11} \text{ (Sucrose, } +66.5^\circ) + H_2O \xrightarrow{H^+} \text{D-(+)-glucose } (+52.7^\circ) + \text{D-(-)-fructose } (- \tag{5.88}$$

Because the levorotation of fructose ($-92.4^\circ$) is numerically far greater than the dextrorotation of glucose ($+52.7^\circ$), the net resulting hy-

drolyzed mixture is **levorotatory** ($[\alpha]_{\text{net}} \approx -39.7^\circ$).
**Definition:** This reversal in the sign of optical rotation from dextro (+) to levo (−) is called the **inversion of cane sugar**, and the resulting equimolar mixture is termed **invert sugar** (found naturally in honey).

**Polysaccharides: Starch, Cellulose & Glycogen**

- **Starch (Plant Storage Polysaccharide):** Polymer of $\alpha$-D-glucose consisting of two distinct structural components:
  1. **Amylose ($15-20\%$):** Water-soluble, linear unbranched chain of $200-1000$ $\alpha$-D-glucose units linked by $\alpha(1\to4)$ **glycosidic bonds**. Forms a hollow helical coil that entraps triiodide ions ($I_3^-$) to produce a characteristic deep blue-black color.
  2. **Amylopectin ($80-85\%$):** Water-insoluble, highly branched polymer. Linear chains contain $\alpha(1\to4)$ linkages; branching points occur every $24-30$ units via $\alpha(1\to6)$ **glycosidic bonds**.
- **Cellulose (Plant Cell Wall Structural Polymer):** Unbranched, linear polymer composed exclusively of $\beta$-D-glucose units joined by $\beta(1\to4)$ **glycosidic bonds**. Chains lie parallel, held together by massive intermolecular hydrogen bonds to form rigid microfibrils. Humans cannot digest cellulose due to lack of the enzyme cellulase.
- **Glycogen (Animal Starch):** Stored in mammalian liver and muscle. Structurally similar to amylopectin, but possesses far more frequent branching ($\alpha(1\to6)$ branches every $8-12$ glucose residues).

## 5.9.4 Amino Acids, Zwitterions & The Isoelectric Point ($pI$)

**Structure, Stereochemistry & Essential Amino Acids**

Amino acids contain both an amino group ($-NH_2$) and a carboxyl group ($-COOH$):

$$R-CH(NH_2)-COOH \quad (\alpha\text{-amino acid}) \tag{5.89}$$

- **Chirality Exception:** All 20 standard proteinogenic amino acids are chiral and optically active **except Glycine** ($H_2N-CH_2-COOH$), where $R=H$, which is achiral and optically inactive!
- **L-Configuration:** All natural amino acids in proteins belong to

the L-series (the $-NH_2$ group resides on the left in Fischer projections).

- **Essential Amino Acids (Cannot be synthesized in human body; must be supplied in diet):** Valine, Leucine, Isoleucine, Phenylalanine, Tryptophan, Threonine, Methionine, Lysine (plus Arginine and Histidine, semi-essential in infants). *(Mnemonic: PVT TIM HaLL).*

**Zwitterion Structure & Formulation of Isoelectric Point ($pI$)**

In aqueous solution, the acidic carboxyl group transfers a proton to the basic amino group, generating a dipolar internal salt known as a **zwitterion**:

$$R - CH(NH_2) - COOH \rightleftharpoons R - CH(NH_3^+) - COO^- \quad \text{(Zwitterion: net charge } = 0 \tag{5.90}$$

**1. Behavior in Varying pH Media:**

- In acidic solution (pH $<$ $pI$): Protonation gives cation $R - CH(NH_3^+) - COOH \implies$ migrates toward **cathode** in electric field.
- In basic solution (pH $>$ $pI$): Deprotonation gives anion $R - CH(NH_2) - COO^- \implies$ migrates toward **anode** in electric field.

**2. Isoelectric Point ($pI$):** The exact pH at which the amino acid carries no net electrical charge and does not migrate in an electric field:

$$pI = \frac{pK_{a1}(\alpha\text{–COOH}) + pK_{a2}(\alpha\text{–}NH_3^+)}{2} \quad \text{(for neutral amino acids)} \tag{5.91}$$

*Crucial Property: At its isoelectric point ($pI$), an amino acid exhibits **minimum aqueous solubility**, permitting selective precipitation and separation!*

## 5.9.5 Peptide Bond Resonance & Structural Hierarchy of Proteins

**The Peptide Bond (-CO – NH–) & Planarity**

A peptide bond is an amide linkage formed by condensation of the $-COOH$ group of one amino acid with the $-NH_2$ group of another

with loss of water:

$$\mathrm{R-CH(NH_2)-COOH + H_2N-CH(R')-COOH} \xrightarrow{-\mathrm{H_2O}} \mathrm{R-CH(NH_2)-\mathbf{CO}-\mathbf{NH}-CH(R')-COOH} \tag{5.92}$$

**Resonance Stabilization of the Peptide Bond:** Delocalization of the nitrogen lone pair into the carbonyl group confers $\approx 40\%$ **partial double bond character** to the C – N bond:

$$-\mathrm{C(=O)-NH-} \longleftrightarrow -\mathrm{C(O^-)=\overset{+}{N}H-} \tag{5.93}$$

**Consequences:**

1. The C – N bond length (1.32 Å) is substantially shorter than a normal C – N single bond (1.47 Å).
2. Free rotation about the C – N bond is restricted; the six atoms of the peptide group ($C_\alpha$–CO – NH – $C_\alpha$) are strictly coplanar.
3. The *trans*-conformation is overwhelmingly favored to minimize steric clash between adjacent side chains ($R$).

## The Four Structural Levels of Protein Organization

1. **Primary Structure (1°):** The linear sequence of amino acids linked covalently by peptide bonds. Any alteration in sequence alters biological function (e.g., in sickle-cell anemia, glutamate at position 6 of the $\beta$-chain of hemoglobin is mutated to valine).
2. **Secondary Structure (2°):** Regular folding of the polypeptide backbone stabilized exclusively by **hydrogen bonds** between > C = O and −NH− groups:
   - **$\alpha$-Helix:** Polypeptide coils into a right-handed screw. Stabilized by intramolecular hydrogen bonding between the > C = O of residue $n$ and the −NH− of residue $(n+4)$. Each turn contains 3.6 amino acids with pitch 5.4 Å. Proline acts as a helix breaker.
   - **$\beta$-Pleated Sheet:** Polypeptide chains are stretched side-by-side in zigzag sheets, held by intermolecular hydrogen bonds. Can be parallel or antiparallel (e.g., silk fibroin).
3. **Tertiary Structure (3°):** Overall 3D spatial folding of the polypeptide chain into compact shapes:
   - **Fibrous Proteins:** Linear, insoluble in water, structural role (keratin in hair/nails, collagen in tendons, myosin in muscle).
   - **Globular Proteins:** Spherical, water-soluble, metabolic role (enzymes, hemoglobin, albumin).

- **Stabilizing Forces:** Hydrogen bonds, **disulfide bonds ($-S-S-$ between cysteine residues, covalent)**, ionic/salt bridges ($-COO^- \cdots NH_3^+$), hydrophobic interactions, and van der Waals forces.

4. **Quaternary Structure ($4^\circ$):** Spatial arrangement of two or more independent tertiary polypeptide subunits into an oligomeric protein (e.g., adult Hemoglobin is a tetramer consisting of $2\alpha$ and $2\beta$ subunits).

**Denaturation of Proteins: What Survives and What Is Destroyed?**

When a native protein is exposed to physical changes (heating, radiation) or chemical changes (extreme pH, urea, organic solvents, heavy metals):

- **Destroyed:** Hydrogen bonds, salt bridges, and hydrophobic interactions rupture. The delicate $2^\circ, 3^\circ$, and $4^\circ$ spatial architectures unfold. Globular proteins precipitate, and biological activity is irreversibly lost.
- **Retained Intact: The primary structure ($1^\circ$) remains 100% completely intact**! Covalent peptide bonds ($-CO-NH-$) are not cleaved under normal denaturation conditions.
- **Classic NCERT Examples:** Coagulation of egg white (albumin) upon boiling; curdling of milk caused by lactic acid produced by bacteria.

## 5.9.6 Nucleic Acids: DNA & RNA Architecture

**Chemical Composition: Sugars, Bases & Nucleotides**

Nucleic acids are unbranched biopolymers of nucleotides:

1. **Pentose Sugar:** $\beta$-D-ribose in RNA; $\beta$-D-2-deoxyribose in DNA (lacks oxygen atom at $C_2'$ position).
2. **Nitrogenous Bases:**
   - **Purines (Bicyclic 9-membered):** Adenine (A) and Guanine (G) in both DNA and RNA.
   - **Pyrimidines (Monocyclic 6-membered):** Cytosine (C) and Thymine (T, 5-methyluracil) in DNA; Cytosine (C) and Uracil (U) in RNA.
3. **Nucleoside vs Nucleotide:**
   - Nucleoside = Pentose Sugar + Base (joined by $N$-glycosidic linkage between $C_1'$ of sugar and $N_9$ of purine or $N_1$ of pyrimidine).

- Nucleotide = Nucleoside + Phosphate (esterified at $C_5'$ of pentose sugar).

4. **Phosphodiester Linkage:** Nucleotides polymerize via $3', 5'$-**phosphodiester bonds** between the $3' - OH$ of one nucleotide sugar and the $5' - OH$ of the next nucleotide sugar, establishing directional polarity ($5' \to 3'$).

**Watson-Crick Double Helix Model & Chargaff's Rules**

James Watson and Francis Crick (1953) established the B-DNA double helix structure:

1. **Antiparallel Strands:** Two polynucleotide chains wound around a central axis in a right-handed helix. One strand runs $5' \to 3'$, and the complementary strand runs $3' \to 5'$.
2. **Complementary Base Pairing (Hydrogen Bonds):**

$$\mathbf{A = T} \quad \text{(Adenine pairs with Thymine via 2 Hydrogen Bonds)} \tag{5.94}$$

$$\mathbf{G \equiv C} \quad \text{(Guanine pairs with Cytosine via 3 Hydrogen Bonds)} \tag{5.95}$$

   Because $G \equiv C$ pairs have 3 hydrogen bonds, DNA with high GC content has a higher melting temperature ($T_m$).
3. **Geometric Dimensions of B-DNA:**
   - Helix Diameter: 2.0 nm (20 Å).
   - Pitch of Helix (one complete 360° turn): 3.4 nm (34 Å).
   - Base Pairs per Turn: 10 base pairs.
   - Distance between Adjacent Base Pairs: 0.34 nm (3.4 Å).
4. **Chargaff's Rules (Applies Exclusively to Double-Stranded DNA):**

$$[\text{A}] = [\text{T}], \quad [\text{G}] = [\text{C}] \implies [\text{A+G}] = [\text{T+C}] \quad \left(\frac{\text{Total Purines}}{\text{Total Pyrimidines}} = 1.0\right) \tag{5.96}$$

   *Note: The ratio* $\frac{\text{A+T}}{\text{G+C}}$ *is not necessarily equal to 1; it is variable and species-specific.*

### 5.9.7 Vitamins & Hormones

**Comprehensive Vitamin Classification & Deficiency Pathology**

| Vitamin | Chemical Name | Solubility Class | Deficiency Disease / Symptoms |
|---|---|---|---|
| **Vitamin A** | Retinol | Fat-Soluble | Night blindness (Nyctalopia), Xerophthalmia (corneal hardening) |
| **Vitamin** $B_1$ | Thiamine | Water-Soluble | **Beriberi** (loss of appetite, muscular weakness, neuropathy) |
| **Vitamin** $B_2$ | Riboflavin | Water-Soluble | Cheilosis (fissuring at corners of mouth), glossitis |
| **Vitamin** $B_3$ | Niacin / Nicotinic acid | Water-Soluble | **Pellagra** (dermatitis, |

**Endocrine Hormones: Chemical Classification**

Hormones are intercellular chemical messengers secreted directly into the bloodstream by ductless endocrine glands:

- **Steroid Hormones (Lipid-Soluble, Cholesterol Derivatives):** Secreted by adrenal cortex and gonads: Testosterone, Estrogens (Estradiol), Progesterone, Cortisol, Aldosterone.
- **Peptide / Protein Hormones (Water-Soluble):**
  - **Insulin (51 amino acids):** Secreted by $\beta$-cells of islets of Langerhans in pancreas; promotes glucose uptake and conversion to glycogen; deficiency causes *Diabetes mellitus*.
  - **Glucagon:** Secreted by $\alpha$-cells of pancreas; stimulates glycogenolysis to elevate blood glucose.
  - **Oxytocin & Vasopressin (ADH):** Nonapeptides released by posterior pituitary gland.
- **Amine Hormones (Amino Acid Derivatives):**
  - **Epinephrine (Adrenaline) & Norepinephrine:** Derived from tyrosine; secreted by adrenal medulla ("fight or flight" hormones).
  - **Thyroxine ($T_4$):** Iodinated derivative of tyrosine produced by thyroid gland; regulates basal metabolic rate (BMR); deficiency causes goiter and hypothyroidism.

## 5.10 Master Distinction Matrix & High-Yield Synthesis Archetypes

### 5.10.1 Exhaustive Chemical Distinction Matrix

## High-Yield Laboratory Chemical Tests for Functional Groups

| Test Name | Reagents | Target Species | Positive Observation / Chemistry |
|---|---|---|---|
| **Lucas Test** | Anhyd. $ZnCl_2$ + conc. HCl | 1°, 2°, 3° Alcohols | 3°: Immediate turbidity; 2°: in 5 min; 1°: on boiling |

## 5.10.2 Solved Master Synthesis Archetypes

**Archetype 1: Multi-Step Synthesis with Stereochemical Retention**

**Problem:** Convert Benzene into pure *p*-bromoaniline. Identify why direct bromination of aniline fails, and formulate the complete protected sequence. **Step-by-Step Solution:**

1. **Pitfall of Direct Bromination:** Aniline contains a powerfully activating $-NH_2$ group ($+M$). Bromination with $Br_2/H_2O$ yields instantaneous white precipitate of 2,4,6-tribromoaniline; monosubstitution cannot be controlled.
2. **Step 1 (Nitration):** Benzene $\xrightarrow[323\text{–}333\ K]{\text{conc. } HNO_3+H_2SO_4} C_6H_5NO_2$ (Nitrobenzene).
3. **Step 2 (Reduction):** $C_6H_5NO_2 \xrightarrow{\text{Sn/HCl or Fe/HCl}} C_6H_5NH_2$ (Aniline).
4. **Step 3 (Protection via Acetylation):** $C_6H_5NH_2 + (CH_3CO)_2O \xrightarrow{\text{pyridine}} C_6H_5NHCOCH_3$ (Acetanilide). The lone pair on nitrogen is delocalized into the acetyl carbonyl, moderating ring activation.
5. **Step 4 (Regioselective Bromination):** $C_6H_5NHCOCH_3 + Br_2 \xrightarrow{CH_3COOH}$ *p*-bromoacetanilide (exclusive para product due to steric hindrance at ortho positions).
6. **Step 5 (Deprotection):** *p*-bromoacetanilide $\xrightarrow{H_3O^+ \text{ or } OH^-,\Delta}$ **p-bromoaniline** + $CH_3COOH$.

**Archetype 2: Carbon-Carbon Chain Lengthening & Functional Transformation**

**Problem:** Formulate the synthetic conversion of Ethanol into Propan-1-amine and Propan-2-ol. **Step-by-Step Solution:**

1. **Conversion of Ethanol into Propan-1-amine (One Carbon Ascent):**

$$CH_3CH_2OH \xrightarrow{PCl_5 \text{ or } SOCl_2} CH_3CH_2Cl \quad \text{(Ethyl chloride)} \tag{5.97}$$

$$CH_3CH_2Cl \xrightarrow{\text{ethanolic KCN}} CH_3CH_2CN \quad \text{(Propanenitrile: nucleophilic substitu} \tag{5.98}$$

$$CH_3CH_2CN \xrightarrow{LiAlH_4 \text{ or } Na/EtOH} CH_3CH_2CH_2NH_2 \quad \text{(Propan-1-amine: complete} \tag{5.99}$$

2. **Conversion of Ethanol into Propan-2-ol (Grignard Path-**

**way):**

$$CH_3CH_2OH \xrightarrow{PCC/CH_2Cl_2} CH_3CHO \quad \text{(Acetaldehyde: selective oxidat} \tag{5.100}$$

$$CH_3CHO + CH_3MgBr \xrightarrow{\text{dry ether}} [CH_3 - CH(OMgBr) - CH_3] \tag{5.101}$$

$$[CH_3 - CH(OMgBr) - CH_3] \xrightarrow{H_3O^+} CH_3 - CH(OH) - CH_3 \quad \text{(Propan-2-ol)} \tag{5.102}$$

**Master Summary: Key Takeaways for NEET UG 2027 Organic Chemistry**

- **Always inspect carbocation intermediates** for possible 1,2-hydride, 1,2-methyl, or ring-expansion rearrangements before forming products in $S_N1, E1$, and electrophilic addition reactions.
- **In Williamson ether synthesis**, always employ the alkoxide of the bulky tertiary component and a primary alkyl halide to avoid elimination.
- **In biomolecules**, remember that sucrose is non-reducing and lacks mutarotation; all monosaccharides, maltose, and lactose are reducing.
- **In DNA double helix**, $A = T$ (2 H-bonds) and $G \equiv C$ (3 H-bonds); Chargaff's rules hold strictly for double-stranded DNA.
- **In aromatic synthesis**, protect aniline by acetylation with acetic anhydride before nitration to avoid the anomalous 47% meta-product.

PART V

# Organic Reaction Mechanisms Handbook

Chapter 6

# Organic Reaction Mechanisms Handbook: Step-by-Step Arro Pushing Pathways

To answer multi-step synthesis and Assertion-Reasoning questions with confidence, students must master the underlying reaction mechanisms. This chapter presents a rigorous, curved-arrow mechanistic breakdown of electrophilic aromatic substitution, nucleophilic aliphatic substitution ($S_N1$ vs $S_N2$), elimination pathways ($E1$ vs $E2$), carbonyl additions and condensations, and carboxylic acid derivative substitutions.

## 6.1 Electrophilic Aromatic Substitution ($S_E$Ar / EAS)

**Fundamental 3-Step EAS Pathway & The Arenium Ion ($\sigma$-Complex)**

Electrophilic aromatic substitution is the hallmark reaction of electron-rich aromatic systems. The driving force is the preservation of aromatic resonance energy ($\approx 150.6$ kJ/mol in benzene). The reaction proceeds through three discrete elementary steps:

1. **Step 1: Generation of the Active Electrophile ($E^+$):** A Lewis acid or strong Brønsted acid polarizes or cleaves the reagent precursor to produce a potent cationic or polarized electrophilic species ($E^+$).
2. **Step 2: Formation of the Arenium Ion ($\sigma$-Complex / Wheland Intermediate) [RDS]:** The aromatic $\pi$-electron cloud attacks $E^+$. The attacked carbon rehybridizes from planar $sp^2$ to tetrahedral $sp^3$, temporarily disrupting aromaticity. The resulting carbocation is resonance-stabilized across the ortho and para posi-

tions:

$$C_6H_6 + E^+ \xrightarrow{\text{slow, RDS}} [C_6H_6E]^+ \quad \text{(Arenium Ion: positive charge delocalized over 5 ri} \tag{6.1}$$

3. **Step 3: Fast Deprotonation & Rearomatization:** The conjugate base abstracts the proton from the $sp^3$ carbon. The electron pair returns to the ring sextet, restoring full aromaticity ($sp^2$ rehybridization).

**Kinetics, Energy Profile & Primary Kinetic Isotope Effect**

**1. Rate Law:** Overall rate is governed by the second step (formation of the arenium ion):

$$\text{Rate} = k[\text{Ar} - \text{H}][E^+] \quad \text{(Second-order kinetics: first order in benzene, first order in elec} \tag{6.2}$$

**2. Kinetic Isotope Effect ($k_H/k_D$):**

- **Nitration, Chlorination, Bromination, Alkylation:** $k_H/k_D \approx 1.0$. The $C - H$ bond cleavage occurs in Step 3 (fast step) after the rate-determining transition state. Hence, $C_6H_6$, $C_6D_6$, and $C_6T_6$ react at identical rates.
- **Sulfonation & Iodination:** $k_H/k_D > 1.0$ ($k_H/k_D \approx 2 - 3$ for sulfonation; $\approx 4$ for iodination). Here, Step 3 or intermediate decomposition has a barrier comparable to Step 2; Step 3 is reversible and partially rate-limiting.

### 6.1.1 Nitration of Aromatic Rings

**Nitration Mechanism & Nitronium Ion Generation**

Nitration is carried out using a "nitrating mixture" consisting of concentrated nitric acid ($HNO_3$) and concentrated sulfuric acid ($H_2SO_4$) at $323 - 333$ K:

1. **Generation of Nitronium Ion ($NO_2^+$):** Sulfuric acid ($pK_a \approx -3$) is a stronger acid than nitric acid ($pK_a \approx -1.4$). Nitric acid acts as

a Brønsted-Lowry **base**, accepting a proton on its hydroxyl oxygen:

$$HNO_3 + H_2SO_4 \rightleftharpoons H_2\overset{+}{O} - NO_2 + HSO_4^- \quad \text{(Protonation of nitric acid)} \tag{6.3}$$

$$H_2\overset{+}{O} - NO_2 \rightleftharpoons NO_2^+ + H_2O \quad \text{(Heterolytic loss of water)} \tag{6.4}$$

$$H_2O + H_2SO_4 \rightleftharpoons H_3O^+ + HSO_4^- \quad \text{(Protonation of water by second } H_2 \tag{6.5}$$

$$\text{Net: } HNO_3 + 2H_2SO_4 \rightleftharpoons NO_2^+ + H_3O^+ + 2HSO_4^- \quad (NO_2^+ \text{ is linear, } sp\text{-hybridiz} \tag{6.6}$$

2. **Electrophilic Attack (RDS):** $C_6H_6 + NO_2^+ \xrightarrow{\text{slow}} [C_6H_6NO_2]^+$ ($\sigma$-complex).
3. **Rearomatization:** $[C_6H_6NO_2]^+ + HSO_4^- \xrightarrow{\text{fast}} C_6H_5NO_2 + H_2SO_4$.

**NCERT Critical Extraction: Nitration of Aniline & Protecting Groups**

Direct nitration of aniline with conc. $HNO_3/H_2SO_4$ gives an anomalous product distribution:

$$\text{Aniline} \xrightarrow[288\ \text{K}]{\text{conc. } HNO_3+H_2SO_4} p\text{-nitroaniline (51\%)} + \textbf{m-nitroaniline (47\%)} + o\text{-nitroa} \tag{6.7}$$

**Mechanistic Cause:** In strongly acidic medium, the highly activating $-NH_2$ group ($+M$) is protonated to form the anilinium ion ($-NH_3^+$), which is a powerful $-I$ deactivating group that directs meta. **Synthetic Solution (Acetylation Protection):** Aniline is first protected with acetic anhydride ($Ac_2O$) in pyridine to form acetanilide ($Ph-NHCOCH_3$). The lone pair on nitrogen is delocalized into the carbonyl group by resonance, reducing its basicity (preventing protonation) and modulating activating strength. Nitration then gives exclusively **para-nitroacetanilide**, followed by acid/base hydrolysis to yield pure **para-nitroaniline**.

### 6.1.2 Halogenation of Benzene

**Halogenation Pathways & Lewis Acid Catalysis**

- **Chlorination / Bromination:** Benzene reacts with $Cl_2$ or $Br_2$ in the presence of anhydrous Lewis acids ($FeCl_3, AlCl_3, FeBr_3$):

$$Cl-Cl+FeCl_3 \rightleftharpoons [Cl^{\delta+}\cdots Cl\cdots {FeCl_3}^{\delta-}] \rightarrow Cl^+ + [FeCl_4]^- \quad (6.8)$$

  Electrophilic $Cl^+$ attacks the ring to form the chlorocyclohexadienyl cation, followed by $[FeCl_4]^-$ deprotonation yielding chlorobenzene and regenerating $FeCl_3 + HCl$.
- **Fluorination:** Direct fluorination is violently exothermic ($\Delta H \ll 0$) leading to explosive rupture of $C-C$ bonds. Fluorobenzene is prepared via the Balz-Schiemann reaction: $Ar-NH_2 \xrightarrow{NaNO_2/HCl} Ar-N_2^+Cl^- \xrightarrow{HBF_4} Ar-N_2^+BF_4^- \xrightarrow{\Delta} Ar-F+BF_3+N_2\uparrow$.
- **Iodination:** Reversible due to formation of reducing agent HI:

$$C_6H_6 + I_2 \rightleftharpoons C_6H_5I + HI \quad (6.9)$$

  To drive equilibrium forward, an in situ oxidizing agent ($HNO_3, HIO_3$, or $HgO$) is added to oxidize HI to $I_2$: $5HI + HIO_3 \rightarrow 3I_2 + 3H_2O$.

### 6.1.3 Friedel-Crafts Alkylation & Acylation

**Friedel-Crafts Alkylation: Mechanism & Classical Carbocation Rearrangements**

**1. Mechanism:**

$$R-Cl + AlCl_3 \rightleftharpoons R^+[AlCl_4]^- \quad \text{(Carbocation intermediate generated)} \quad (6.10)$$

$$C_6H_6 + R^+ \xrightarrow{\text{slow, RDS}} [C_6H_6R]^+ \xrightarrow{[AlCl_4]^-} C_6H_5R + HCl + AlCl_3 \quad (6.11)$$

**2. Carbocation Rearrangement Trap:** Whenever 1° alkyl halides with $\geq 3$ carbons are used, the primary carbocation rearranges via 1,2-hydride or 1,2-methide shift to a more stable 2° or 3° carbocation prior to electrophilic attack:

$$C_6H_6 + CH_3CH_2CH_2Cl \xrightarrow{\text{anhyd. } AlCl_3} \textbf{Cumene (isopropylbenzene, 65–70\%)} + n\text{-propy} \quad (6.12)$$

*Mechanism of shift:* $CH_3 - CH_2 - \overset{+}{C}H_2 \xrightarrow{\text{1,2-hydride shift}} CH_3 - \overset{+}{C}H - CH_3$ (2° carbocation is stabilized by 6 $\alpha$-H vs 2 $\alpha$-H).

**Four Cardinal NEET Limitations of Friedel-Crafts Alkylation**

1. **Polyalkylation Problem:** The introduced alkyl group ($-R$) is electron-donating ($+I, +H$), making the monoalkylated product more reactive toward further EAS than benzene itself. Hence, ortho/para dialkyl and polyalkyl benzenes form readily.
2. **Carbocation Rearrangement:** Straight unbranched alkyl benzenes cannot be synthesized cleanly in high yield by direct alkylation.
3. **Deactivated Ring Immunity:** Rings carrying moderate to strong electron-withdrawing groups ($-M/-I$ like $-NO_2, -CF_3, -SO_3H, -COOH, -COOR, -CN, -CHO$) do **not** undergo Friedel-Crafts reactions. Nitrobenzene is routinely used as an inert solvent for Friedel-Crafts reactions!
4. **Aniline / Phenol Lewis Acid Failure:** Aniline and phenol do not undergo Friedel-Crafts alkylation/acylation cleanly because the Lewis acid ($AlCl_3$) forms a strong coordinate complex with the lone pair on $-NH_2$ or $-OH$:

$$C_6H_5NH_2 + AlCl_3 \rightarrow C_6H_5 - \overset{+}{N}H_2 - \overset{-}{A}lCl_3 \tag{6.13}$$

This converts the powerfully activating $-NH_2$ into a strongly deactivating $-I$ group ($-\overset{+}{N}H_2 - AlCl_3^-$), completely terminating EAS.

**Friedel-Crafts Acylation: Acylium Ion Intermediate & Remedy for Polyalkylation**

Acylation avoids both the rearrangement and polyalkylation defects of alkylation:

1. **Generation of the Acylium Ion:**

$$R - CO - Cl + AlCl_3 \rightleftharpoons [R - \overset{+}{C} = O \longleftrightarrow R - C \equiv \overset{+}{O}] + [AlCl_4]^- \tag{6.14}$$

The acylium ion is exceptionally stable because all atoms possess a complete octet in the $R - C \equiv O^+$ canonical form. Consequently, it **never undergoes carbocation rearrangement**!

2. **Attack & Single Substitution:** $C_6H_6 + RCO^+ \xrightarrow{\text{RDS}}$

$[C_6H_6COR]^+ \rightarrow C_6H_5COR + HCl$.

3. **No Polyacylation:** The introduced acyl group ($-COR$) is strongly electron-withdrawing ($-M, -I$), deactivating the product ring against further electrophilic attack.
4. **Synthesis of Unbranched Alkylbenzenes:** The acyl product is subsequently reduced to an unbranched alkyl chain via either:
   - **Clemmensen Reduction:** $Ar - CO - R \xrightarrow{Zn-Hg/\text{conc. } HCl} Ar - CH_2 - R$ (for acid-stable substrates).
   - **Wolff-Kishner Reduction:** $Ar - CO - R \xrightarrow[KOH,\text{ethylene glycol, } \Delta]{NH_2NH_2} Ar - CH_2 - R + N_2 \uparrow$ (for base-stable substrates).

## 6.1.4 Orientation & Reactivity: Arenium Ion Resonance Analysis

**Arenium Canonical Analysis: Ortho/Para vs Meta Directing Effects**

The directing effect is governed entirely by the relative thermodynamic stability of the arenium intermediate for ortho, meta, and para electrophilic attack:

- **Ortho/Para Directing Groups ($+M > -I$, e.g., $-OH, -OR, -NH_2, -NHCOR$ or $+I/+H$, e.g., $-CH_3$):** Electrophilic attack at the ortho or para position places the positive charge directly on the ring carbon attached to the substituent. If the substituent has a lone pair ($+M$), it donates electrons into the ring, forming an especially stable **fourth resonance structure** where every atom has a complete octet.
- **Meta Directing Groups ($-M/-I$, e.g., $-NO_2, -CN, -CHO, -COOH, -SO_3H$):** Ortho and para attack generates a canonical form where the positive charge is placed directly adjacent to the positively polarized substituent carbon ($C^+ - \overset{\delta+}{C} = O$ or $C^+ - \overset{+}{N}O_2^-$), resulting in catastrophic electrostatic repulsion. Meta attack never places the positive charge directly on the ipso carbon, thereby minimizing destabilization. Hence, meta attack has a lower activation barrier than ortho/para attack.

**The Halogen Paradox in EAS: Deactivating yet Ortho/Para Directing**

Halogens ($-$F, $-$Cl, $-$Br, $-$I) are unique in organic chemistry:

- **Reactivity (Deactivating):** The inductive effect ($-I$) of the highly electronegative halogen strongly withdraws $\sigma$-electron density from the entire $\pi$-system, reducing overall ring electron density. Hence, halobenzenes react **slower** than benzene in all EAS reactions.
- **Orientation (Ortho/Para):** When attack occurs at the ortho or para position, the halogen lone pair can be donated via resonance ($+M$) to form a fourth canonical contributor ($= \overset{+}{\text{X}}$) with complete valence octets. This stabilization is absent for meta attack.
- **Conclusion:** Inductive effect controls reactivity (deactivating), while resonance effect controls orientation ($o/p$-directing).

# 6.2 Nucleophilic Aliphatic Substitution ($S_N1$ vs $S_N2$)

## 6.2.1 The $S_N1$ Reaction Pathway

**Two-Step Unimolecular Substitution ($S_N1$)**

The $S_N1$ mechanism operates through a stepwise dissociation-addition mechanism:

1. **Step 1: Heterolysis to form Carbocation Intermediate [RDS]:**

$$\mathrm{R-X} \xrightarrow[\text{slow, RDS}]{k_1} \mathrm{R^+ + X^-} \quad \text{(Planar, trigonal } sp^2 \text{ hybridized carbocation intermediat} \tag{6.15}$$

The rate depends solely on the concentration of the alkyl halide.

2. **Step 2: Nucleophilic Attack [Fast]:**

$$\mathrm{R^+ + Nu^-} \xrightarrow[\text{fast}]{k_2} \mathrm{R-Nu} \tag{6.16}$$

The nucleophile can attack either the front face or back face of the unhindered planar carbocation.

**$S_N1$ Governing Equations, Stereochemistry & Solvation**

**Rate Law:** $\text{Rate} = k[\mathrm{R-X}] \quad (\text{Order} = 1, \text{Molecularity} = 1)$

**Substrate Hierarchy:** $\text{Benzylic} \approx \text{Allylic} \approx 3^\circ > 2^\circ \gg 1^\circ > \mathrm{CH_3X}$

**Solvent Acceleration:** Favored by polar protic solvents ($\mathrm{H_2O}, \mathrm{MeOH}, \mathrm{EtOH}, \mathrm{AcOH}$) with high dielectric constant ($\epsilon$), which solvate and stabilize both the carbocation and leaving group anion via hydrogen bonding.

**Stereochemical Outcome (Racemization with Partial Inversion):** The leaving group $\mathrm{X^-}$ does not diffuse away instantaneously; it forms an **intimate ion pair** ($\mathrm{R^+ \cdots X^-}$) that partially shields the front face. Consequently, backside attack occurs more frequently than frontside attack, leading to **net partial racemization with predominant inversion** (typically $\approx 50-70\%$ inversion $+30-50\%$ retention, corresponding to $60-100\%$ racemic mixture with $0-40\%$ net optical inversion).

**Carbocation Rearrangements in $S_N1$**

Because a free carbocation is formed in Step 1, $S_N1$ reactions are prone to Wagner-Meerwein rearrangements:

- **1,2-Hydride Shift:** 2-chloro-3-methylbutane solvolyzed in ethanol yields predominantly 2-ethoxy-2-methylbutane (3° product via hydride shift from 2° carbocation).
- **1,2-Methyl Shift:** Neopentyl bromide solvolyzed in water yields 2-methylbutan-2-ol (3° carbocation via methide shift).
- **Ring Expansion:** Cyclobutylmethyl halide yields cyclopentanol derivatives to relieve 90° ring angle strain.

## 6.2.2 The $S_N2$ Reaction Pathway

**One-Step Bimolecular Concerted Substitution ($S_N2$)**

The $S_N2$ mechanism occurs in a single, concerted step without any reactive intermediate:

1. The nucleophile approaches the electrophilic carbon from the **backside** (180° relative to the C − X leaving bond).
2. The reaction passes through a single, pentacoordinate, trigonal bipyramidal transition state:

$$\mathrm{Nu^- + R-CH_2-X \xrightarrow{slow} \left[Nu^{\delta-}\cdots CH_2R\cdots X^{\delta-}\right]^{\ddagger} \xrightarrow{fast} Nu-CH_2-R + X^-} \tag{6.17}$$

   The central carbon is approximately $sp^2$ hybridized with three equatorial bonds, while the nucleophile and leaving group share a collinear $p$-orbital axis at 180°.

**$S_N2$ Rate Law, Steric Effects & Walden Inversion**

**Rate Law:** $\mathrm{Rate} = k[\mathrm{R-X}][\mathrm{Nu^-}]$ (Order = 2, Molecularity = 2)

**Substrate Reactivity:** $\mathrm{CH_3X} > 1° > 2° \ggg 3°$ (Steric hindrance is dominant

**Relative Rates:** $\mathrm{CH_3Br}$ (1200) > EtBr (40) > *i*-PrBr (1) > *t*-BuBr (0.0

**Stereochemical Outcome (100% Walden Inversion):** Backside attack forces the three equatorial substituents to flip through a planar transition state, resembling an umbrella blown inside out by a gust of wind. This results in complete inversion of configuration at the stereocenter ($R \rightarrow S$ or $S \rightarrow R$, depending on Cahn-Ingold-Prelog priorities).

**The Neopentyl Halide $S_N2$ Trap & Solvent Effects**

- **Neopentyl Halide Inertness:** Neopentyl bromide, $(CH_3)_3C-CH_2-Br$, is a primary alkyl halide (1°). However, its $S_N2$ rate is $10^{-5}$ times slower than ethyl bromide! The three bulky methyl groups on the $\beta$-carbon completely block backside access to the electrophilic $\alpha$-carbon.
- **Polar Aprotic Solvents Accelerate $S_N2$:** Solvents such as DMSO (dimethyl sulfoxide), DMF ($N,N$-dimethylformamide), acetone, and acetonitrile coordinate strongly to cations via their electronegative oxygen atoms, but cannot hydrogen-bond to nucleophilic anions. The anions remain "naked" and desolvated, boosting their nucleophilicity by up to $10^5$-fold.

### 6.2.3 Master Diagnostic: $S_N1$ vs $S_N2$ vs $S_N$i

| Parameter | $S_N1$ Mechanism | $S_N2$ Mechanism |
|---|---|---|
| **Number of Steps** | Two steps (carbocation intermediate). | One concerted step (pentacoordinate transition state). |
| **Kinetic Rate Law** | Rate $= k[R-X]$ (1st order, unimolecular). | Rate $= k[R-X][Nu^-]$ (2nd order, bimolecular). |
| **Substrate Preference** | $3° > 2° \gg 1°$; benzylic $\approx$ allylic $> 3°$. | $CH_3X > 1° > 2° \ggg 3°$ (neopentyl extremely slow). |
| **Nucleophile** | Weak nucleophile is sufficient ($H_2O$, ROH). | Strong, unhindered nucleophile required ($OH^-$, $RO^-$, $CN^-$, $I^-$). |
| **Solvent Requirement** | Polar protic solvent ($H_2O$, MeOH, EtOH) stabilizes ions. | Polar aprotic solvent (DMSO, DMF, acetone) desolvates $Nu^-$. |
| **Stereochemistry** | Racemization with slight excess of inversion. | 100% Walden Inversion of configuration. |
| **Rearrangements** | Readily observed (hydride/alkyl shifts, ring expansion). | Completely absent (no carbocation intermediate). |
| **Leaving Group Order** | $I^- > Br^- > Cl^- \gg F^-$ (governed by C – X bond energy). | $I^- > Br^- > Cl^- \gg F^-$ (governed by leaving group stability). |

**$S_N$i Mechanism (Substitution Nucleophilic Internal): Thionyl Chloride Reaction**

When an alcohol reacts with thionyl chloride ($SOCl_2$):

- **In Ether (No Pyridine):** $R-OH + SOCl_2 \xrightarrow{\text{ether}} R-Cl + SO_2 \uparrow + HCl \uparrow$. Proceeds via an internal chlorosulfite intermediate ($R-O-SOCl$) with intramolecular delivery of chlorine from the same face $\rightarrow$ **Complete Retention of Configuration ($S_N$i).**
- **In Pyridine Medium:** Pyridine neutralizes HCl to form pyridinium chloride ($C_5H_5NH^+Cl^-$), releasing free chloride ions ($Cl^-$). These chloride ions attack the chlorosulfite from the backside $\rightarrow$

**Complete Inversion of Configuration ($S_N2$).**

# 6.3 Elimination Reactions ($E1$ vs $E2$) & Dehydration of Alcohols

## 6.3.1 $E1$ Elimination Mechanism

**Unimolecular Elimination ($E1$): Steps & Thermodynamic Drive**

$E1$ elimination proceeds through ionization to a planar carbocation, identical to Step 1 of $S_N1$:

1. **Step 1 (RDS):** $\mathrm{R-X} \xrightarrow{\text{slow, RDS}} \mathrm{R^+ + X^-}$.
2. **Step 2 (Fast):** A weak base ($\mathrm{H_2O}, \mathrm{EtOH}$) abstracts a $\beta$-hydrogen from the carbocation:

$$\mathrm{R-CH_2-\overset{+}{C}H-CH_3 + B: \xrightarrow{\text{fast}} R-CH=CH-CH_3 + BH^+} \tag{6.18}$$

3. **Thermodynamic Competition ($S_N1$ vs $E1$):** Because Step 1 is shared, $S_N1$ and $E1$ compete directly. Elimination yields 2 product molecules from 1 reactant ($\Delta S^\circ > 0$). In the Gibbs equation $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$, higher temperatures increase the magnitude of $-T\Delta S^\circ$, favoring $E1$ **elimination over $S_N1$ substitution at elevated temperatures**.
4. **Regiochemistry:** Governed by **Saytzeff's Rule** (the more substituted, thermodynamically stable alkene with more hyperconjugative $\alpha$-hydrogens is the major product; $trans > cis$).

## 6.3.2 $E2$ Elimination: Anti-Periplanar Geometry & Regioselectivity

**Bimolecular Elimination ($E2$): Stereoelectronic Rigidity**

$E2$ elimination occurs in a concerted manner upon treatment with a strong base (such as alcoholic KOH or $\mathrm{NaOCH_2CH_3}$):

$$\text{Rate} = k[\mathrm{R-X}][\text{Base}] \quad (\text{Order} = 2, \text{Molecularity} = 2) \tag{6.19}$$

**Substrate Order:** $3^\circ > 2^\circ > 1^\circ$ (developing double bond in transition state is stabilized by alkyl groups).

**Stereoelectronic Requirement (Anti-Periplanar Geometry):** The $\mathrm{C_\beta - H}$ bond and $\mathrm{C_\alpha - X}$ bond must lie in the same plane at a dihedral angle of $180^\circ$ (**anti-coplanar**). This orientation allows the electron pair from the breaking $\mathrm{C-H}$ bond to smoothly overlap into

the developing $\pi$-bond while minimizing electron repulsion between the incoming base and departing halide.

**Saytzeff (Zaitsev) vs Hofmann Regioselectivity Matrix**

**1. Saytzeff Rule (Thermodynamic Major Product):** With normal unhindered strong bases ($OH^-, OEt^-$) and good leaving groups ($I^-, Br^-, Cl^-$), the transition state resembles the product alkene. The more substituted, hyperconjugated alkene is formed as the major product:

$$CH_3 - CH_2 - CH(Br) - CH_3 \xrightarrow{\text{alc. KOH}, \Delta} \textbf{But-2-ene (81\%, Saytzeff)} + \text{But-1-en} \tag{6.20}$$

**2. Hofmann Rule (Kinetic Major Product):** The less substituted, terminal alkene is obtained as the major product under two distinct scenarios:

- **Scenario A: Poor Leaving Group (e.g., Alkyl Fluorides, Quaternary Ammonium Hydroxides):** Because $F^-$ or $-NR_3^+$ is an extremely poor leaving group, the base begins abstracting the $\beta$-proton before the $C - X$ bond breaks significantly. This develops substantial **carbanionic character** in the transition state. Since primary carbanions are more stable than secondary carbanions ($1^\circ > 2^\circ > 3^\circ$), the base preferentially deprotonates the more acidic, least hindered methyl carbon:

  $$CH_3CH_2CH(F)CH_3 \xrightarrow{EtO^-Na^+} \textbf{But-1-ene (80\%, Hofmann)} + \text{But-2-ene (20\%} \tag{6.21}$$

- **Scenario B: Sterically Bulky Bases (e.g., Potassium tert-butoxide, *t*-BuOK or LDA):** The bulky tert-butoxide anion cannot access the sterically congested internal secondary $\beta$-hydrogen, forcing abstraction of the easily accessible, unhindered terminal $\beta$-hydrogen:

  $$CH_3CH_2CH(Br)CH_3 \xrightarrow{(CH_3)_3CO^-K^+} \textbf{But-1-ene (73\%, Hofmann)} + \text{But-2-en} \tag{6.22}$$

### 6.3.3 Acid-Catalyzed Dehydration of Alcohols

**Dehydration of Alcohols: Step-by-Step Mechanism**

Dehydration of alcohols to alkenes is an acid-catalyzed elimination ($E1$ for 2° and 3° alcohols; $E2$ for 1° alcohols):

1. **Step 1: Protonation of Hydroxyl Group (Fast Equilibrium):**

$$R-OH + H^+ \rightleftharpoons R-\overset{+}{O}H_2 \quad \text{(Converts poor leaving group } -OH \text{ into excellent leaving} \tag{6.23}$$

2. **Step 2: Heterolysis to form Carbocation [RDS]:**

$$R-\overset{+}{O}H_2 \xrightarrow{\text{slow, RDS}} R^+ + H_2O \tag{6.24}$$

3. **Step 3: Carbocation Rearrangement (if applicable):** 1,2-hydride or 1,2-methyl shifts to yield the most stable carbocation.
4. **Step 4: Deprotonation (Fast):** Water or $HSO_4^-$ abstracts $\beta$-H to form the Saytzeff alkene.

**NCERT Class 12 Benchmark: Reaction Conditions & Temperature Trap**

- **Reactivity Hierarchy:** 3° alcohol > 2° alcohol > 1° alcohol (parallels carbocation stability: 3° > 2° > 1°).
- **Ethanol Conditions:**
  - At 443 K(170°C) with conc. $H_2SO_4$: **Ethene** ($CH_2 = CH_2$) forms via intramolecular dehydration.
  - At 413 K(140°C) with excess ethanol: **Diethyl ether** ($C_2H_5OC_2H_5$) forms via bimolecular nucleophilic substitution ($S_N2$).
- 2° **Alcohol (Propan-2-ol):** Requires only 85% $H_3PO_4$ at 440 K.
- 3° **Alcohol (2-Methylpropan-2-ol):** Requires only 20% $H_2SO_4$ at mild temperature of 358 K.

# 6.4 Carbonyl Chemistry Mechanisms

## 6.4.1 Nucleophilic Addition to Aldehydes & Ketones ($Ad_N$)

**Carbonyl Electrophilicity & Trajectory of Nucleophilic Attack**

The carbonyl group ($> C = O$) is strongly polarized due to the high electronegativity of oxygen ($\chi_O = 3.5$ vs $\chi_C = 2.5$). The carbonyl carbon is planar $sp^2$ with an empty $\pi^*$ LUMO.

1. **Burgi-Dunitz Angle:** The nucleophile attacks the electrophilic carbon at an angle of $\approx 107°$ relative to the C = O bond axis, maximizing overlap with $\pi^*$ while minimizing repulsion with oxygen lone pairs.
2. **Reactivity Sequence:**

$$HCHO > CH_3CHO > CH_3COCH_3 > C_6H_5CHO > C_6H_5COCH_3 > C_6H_5COC_6 \tag{6.25}$$

**Governing Factors:**

- **Electronic Effect:** Electron-donating alkyl groups ($+I$) disperse the partial positive charge on carbon ($\delta+$), reducing its electrophilicity. Conjugation with an aromatic ring (resonance) stabilizes the ground state carbonyl, diminishing reactivity.
- **Steric Effect:** As substituents on the carbonyl carbon increase in bulk, transition state crowding during conversion from trigonal planar (120°) to tetrahedral (109.5°) increases, raising the activation energy.

**Key Nucleophilic Addition Mechanisms**

**1. Addition of HCN (Cyanohydrin Formation):**

- Pure HCN is a weak acid ($pK_a = 9.2$) and reacts extremely slowly. The reaction is base-catalyzed:

$$HCN + OH^- \rightleftharpoons CN^- + H_2O \quad \text{(Generates strong nucleophile } CN^-) \tag{6.26}$$

- Step 1 (RDS): Attack of $CN^-$ on carbonyl carbon: $R_2C=O + CN^- \xrightarrow{\text{slow}} R_2C(CN)O^-$.
- Step 2 (Fast): Proton transfer: $R_2C(CN)O^- + HCN \xrightarrow{\text{fast}} R_2C(CN)OH + CN^-$.

**2. Addition of Sodium Bisulfite ($NaHSO_3$):** Sulfur is more nucleophilic than oxygen. The bisulfite nucleophile attacks via sulfur to form a tetrahedral intermediate, followed by proton transfer to give a

crystalline, water-soluble bisulfite adduct:

$$\mathrm{RCHO + NaHSO_3 \rightleftharpoons R - CH(OH)SO_3^- Na^+} \quad \text{(White crystalline precipitate)} \tag{6.27}$$

Used for the separation and purification of aldehydes and aliphatic methyl ketones (sterically hindered ketones do not react). **3. Addition of Ammonia Derivatives ($H_2N - Z$):** Reagents: $Z = -OH$ (hydroxylamine $\rightarrow$ oxime), $-NH_2$ (hydrazine $\rightarrow$ hydrazone), $-NHC_6H_5$ (phenylhydrazine $\rightarrow$ phenylhydrazone), $-NH - C_6H_3(NO_2)_2$ (2,4-DNP $\rightarrow$ orange Brady's precipitate), $-NHCONH_2$ (semicarbazide $\rightarrow$ semicarbazone).

- **Optimum pH Control (pH $\approx$ 4.5):** If medium is too acidic (pH $< 3$), the amine reagent is completely protonated to $H_3\overset{+}{N} - Z$, destroying its nucleophilicity. If medium is too basic (pH $> 6$), acid-catalyzed dehydration of the carbinolamine intermediate is too slow.

## 6.4.2 Aldol Condensation & Cross-Aldol Mechanism

**Aldol Condensation: Step-by-Step Mechanism**

Aldol condensation occurs in aldehydes and ketones possessing at least one $\alpha$-hydrogen when treated with dilute alkali ($NaOH, Ba(OH)_2, Na_2CO_3$):

1. **Step 1: Enolate Generation (Acid-Base Equilibrium):** Hydroxide abstracts an acidic $\alpha$-proton ($pK_a \approx 19 - 20$, acidified by resonance stabilization and $-I$ of carbonyl):

$$\mathrm{CH_3CHO + OH^- \rightleftharpoons [^-CH_2CHO \longleftrightarrow CH_2 = CHO^-] + H_2O} \quad \text{(Enolate carbanion)} \tag{6.28}$$

2. **Step 2: Nucleophilic Addition [RDS]:** The enolate carbanion attacks the electrophilic carbonyl carbon of a second un-ionized aldehyde molecule:

$$\mathrm{CH_3CHO + CH_2 = CHO^- \xrightarrow{\text{slow, RDS}} CH_3 - CH(O^-) - CH_2 - CHO} \quad \text{(Alkoxide inter} \tag{6.29}$$

3. **Step 3: Protonation to Form $\beta$-Hydroxyaldehyde (Aldol):**

$$\mathrm{CH_3 - CH(O^-) - CH_2 - CHO + H_2O \rightleftharpoons CH_3 - CH(OH) - CH_2 - CHO} \quad \text{(3-hydroxyb} \tag{6.30}$$

4. **Step 4: Condensation / Dehydration ($E1cB$ Mechanism):** Upon heating ($\Delta$), the aldol eliminates water. Hydroxide abstracts

the remaining acidic $\alpha$-hydrogen to form an enolate, followed by expulsion of hydroxide ion ($OH^-$, poor leaving group expelled due to conjugation driving force):

$$CH_3 - CH(OH) - CH_2 - CHO \xrightarrow[\Delta]{OH^-} CH_3 - CH = CH - CHO \text{ (Crotonaldehyd} \tag{6.31}$$

The resulting $\alpha,\beta$-unsaturated enal/enone is thermodynamically stabilized by extended conjugation between the C = C double bond and the carbonyl group.

**Cross-Aldol Condensation: Product Complexity & Directed Syntheses**

- **Mixture of Two Aldehydes with $\alpha$-H:** Reacting ethanal ($CH_3CHO$) with propanal ($CH_3CH_2CHO$) gives a complex mixture of **four** condensation products:
  1. Self-condensation of ethanal: But-2-enal.
  2. Self-condensation of propanal: 2-Methylpent-2-enal.
  3. Cross-condensation (ethanal enolate + propanal electrophile): Pent-2-enal.
  4. Cross-condensation (propanal enolate + ethanal electrophile): 2-Methylbut-2-enal.
- **Claisen-Schmidt Condensation (Directed Cross-Aldol):** To obtain a single clean product, one carbonyl component must lack $\alpha$-hydrogens (e.g., benzaldehyde, $C_6H_5CHO$, or formaldehyde, HCHO). The partner with $\alpha$-hydrogens acts exclusively as the nucleophile:

$$PhCHO + CH_3COCH_3 \xrightarrow[293\text{–}303\ K]{\text{dil. NaOH}} Ph - CH = CH - COCH_3 \text{ (Benzalacetone)+H} \tag{6.32}$$

### 6.4.3 Cannizzaro Reaction & Hydride Transfer Mechanism

**Cannizzaro Disproportionation: Step-by-Step Hydride Shift**

Aldehydes that lack $\alpha$-hydrogens (e.g., $HCHO, C_6H_5CHO, (CH_3)_3CCHO$) cannot form an enolate. Upon heating with concentrated alkali (50% NaOH or KOH), they undergo self-oxidation-reduction (disproportionation):

1. **Step 1: Rapid Nucleophilic Addition of Hydroxide:**

$$\mathrm{Ar-CHO + OH^- \rightleftharpoons Ar-CH(OH)O^-} \quad \text{(Monoanion hydrate)} \tag{6.33}$$

2. **Step 2: Second Deprotonation in Strongly Basic Media:**

$$\mathrm{Ar-CH(OH)O^- + OH^- \rightleftharpoons Ar-CH(O^-)_2 + H_2O} \quad \text{(Dianion, extremely electron-rich} \tag{6.34}$$

3. **Step 3: Intermolecular Hydride ($H^-$) Transfer [RDS]:** The electron-rich dianion expels a hydride ion ($H^-$) directly to the carbonyl carbon of a second aldehyde molecule. This direct hydride transfer is the slow, rate-determining step:

$$\mathrm{Ar-CH(O^-)_2 + Ar-CHO \xrightarrow{\text{slow, RDS}} Ar-COO^- + Ar-CH_2O^-} \tag{6.35}$$

4. **Step 4: Rapid Proton Exchange:** Alkoxide ($ArCH_2O^-$) rapidly abstracts a proton from water to yield alcohol ($ArCH_2OH$), leaving the carboxylate salt ($ArCOO^-Na^+$).

**Kinetics & Crossed Cannizzaro Reaction**

**1. Kinetic Order:**

$$\text{At moderate } [OH^-]: \quad \text{Rate} = k[\mathrm{RCHO}]^2[\mathrm{OH^-}] \quad \text{(Third order overall)}$$

$$\text{At high } [OH^-]: \quad \text{Rate} = k[\mathrm{RCHO}]^2[\mathrm{OH^-}]^2 \quad \text{(Fourth order, involving the dianion)}$$

**2. Crossed Cannizzaro (Regioselectivity of Oxidation):** When formaldehyde (HCHO) is mixed with another non-enolizable aldehyde (e.g., benzaldehyde, $C_6H_5CHO$), nucleophilic addition of $OH^-$ occurs preferentially on formaldehyde due to its superior electrophilicity and absence of steric hindrance:

$$\mathrm{C_6H_5CHO + HCHO \xrightarrow{\text{conc. NaOH}}} \textbf{ Benzyl alcohol } (\mathrm{C_6H_5CH_2OH}) + \textbf{Sodium formate } (\mathrm{H} \tag{6.36}$$

*Universal Rule:* In crossed Cannizzaro involving formaldehyde, **formaldehyde is always oxidized to formate**, and the other aldehyde is reduced to the alcohol.

# 6.5 Carboxylic Acid Derivatives & Mechanisms

## 6.5.1 Fischer Esterification Mechanism

**Acid-Catalyzed Esterification: Tetrahedral Intermediate Pathway**

The reaction of a carboxylic acid with an alcohol in the presence of an inorganic acid catalyst (conc. $H_2SO_4$ or dry HCl) is an equilibrium addition-elimination process:

$$\mathrm{R-COOH + R'-OH \overset{H^+}{\rightleftharpoons} R-COOR' + H_2O} \tag{6.37}$$

1. **Step 1: Protonation of Carbonyl Oxygen:** Protonation occurs selectively at the carbonyl oxygen (giving a resonance-stabilized cation $[\mathrm{R-C(OH)_2}]^+$), increasing the electrophilicity of the carbonyl carbon:

$$\mathrm{R-COOH + H^+ \rightleftharpoons R-\overset{+}{C}(OH)_2} \tag{6.38}$$

2. **Step 2: Nucleophilic Attack of Alcohol [RDS]:** The alcohol oxygen lone pair attacks the activated carbonyl carbon:

$$\mathrm{R-\overset{+}{C}(OH)_2 + R'OH \xrightarrow{\text{slow, RDS}} R-C(OH)_2(\overset{+}{O}HR')} \tag{6.39}$$

3. **Step 3: Proton Transfer (Tautomerism):** Proton shifts from the oxonium group to one of the hydroxyl groups:

$$\mathrm{R-C(OH)_2(\overset{+}{O}HR') \rightleftharpoons R-C(OH)(\overset{+}{O}H_2)(OR')} \tag{6.40}$$

   This converts a poor leaving group ($-$OH) into a good leaving group ($H_2O$).

4. **Step 4: Elimination of $H_2O$ & Rearomatization/Deprotonation:** The hydroxyl oxygen lone pair assists in expelling $H_2O$, followed by loss of $H^+$ to regenerate the acid catalyst and yield the ester (RCOOR′).

**Isotopic Tracer Proof of Acyl-Oxygen Bond Cleavage**

When benzoic acid reacts with $^{18}$O-labeled methanol ($CH_3{}^{18}OH$), mass spectrometry confirms that the heavy isotope $^{18}$O resides entirely in

methyl benzoate, and **no** $^{18}O$ is found in the produced water:

$$C_6H_5 - CO - OH + H - {}^{18}OCH_3 \overset{H^+}{\rightleftharpoons} C_6H_5 - CO - {}^{18}OCH_3 + H_2O \tag{6.41}$$

**Deduction:** The reaction proceeds with **Acyl-Oxygen bond fission** ($R - CO$—$OH$ cleavage) in the carboxylic acid, and **Alkyl-Oxygen bond retention** ($R'O$—$H$ cleavage) in the alcohol!

**Steric Hindrance Hierarchy in Esterification**

Because the rate-determining step involves transition from planar $sp^2$ (120°) to a crowded tetrahedral $sp^3$ intermediate (109.5°), esterification rates are governed primarily by steric bulk:

**Reactivity of Carboxylic Acids:** $HCOOH > CH_3COOH > RCH_2COOH > R_2CHC$

**Reactivity of Alcohols:** $CH_3OH >$ Primary (1°) > Secondary (2°) $\gg$ Ter

*Note on 3° Alcohols:* Tertiary alcohols (e.g., tert-butanol) do **not** react via Fischer esterification. Instead, they dehydrate rapidly to alkenes or react via alkyl-oxygen cleavage through an $S_N1$ carbocation intermediate ($R'^+$).

## 6.5.2 Decarboxylation of Carboxylic Acids

**Soda-Lime Decarboxylation: Mechanism & The Carbanion Intermediate**

Heating sodium or potassium salts of carboxylic acids with soda lime (a dry mixture of NaOH and CaO in a 3 : 1 mass ratio) eliminates carbon dioxide as carbonate, yielding an alkane with **one carbon atom less** than the parent carboxylate:

$$R - COONa + NaOH \xrightarrow[\Delta]{CaO} R - H + Na_2CO_3 \tag{6.42}$$

**Step-by-Step Reaction Mechanism:**

1. **Step 1: Dissociation & Attack by Base:** Under high thermal fusion ($\approx$ 600 K), the carboxylate anion interacts with hydroxide ion:

$$R - COO^- + OH^- \rightleftharpoons [R - COO \cdots H \cdots O]^{2-} \tag{6.43}$$

2. **Step 2: Heterolytic Cleavage to Free Carbanion [RDS]:** The $C - C$ bond connecting the alkyl group to the carboxylate carbon

undergoes heterolytic cleavage, expelling carbon dioxide (which instantly neutralizes with base to form carbonate) and generating a reactive **carbanion intermediate** ($R^-$):

$$R - COO^- \xrightarrow{\text{slow, RDS}} \mathbf{R^-}\ (\textbf{Carbanion Intermediate}) + CO_2 \quad (CO_2 + 2OH^- \text{-} \tag{6.44}$$

3. **Step 3: Rapid Proton Capture:** The generated carbanion instantly abstracts a proton from water or amphoteric media to form the alkane:
$$R^- + H_2O \xrightarrow{\text{fast}} R - H + OH^- \tag{6.45}$$

**Rate-Determining Rule:** Because the rate-determining step generates a carbanion ($R^-$), the rate of soda-lime decarboxylation is **directly proportional to the stability of the departing carbanion**:

$$\text{Decarboxylation Rate} \propto \text{Carbanion Stability: } CH_3^- > 1^\circ > 2^\circ > 3^\circ \quad (\text{Reverse of c} \tag{6.46}$$

Electron-withdrawing groups $(-I, -M)$ that disperse the negative charge tremendously accelerate the rate:

$$O_2N - CH_2 - COOH > NC - CH_2 - COOH > Cl_3C - COOH > CH_3COOH \tag{6.47}$$

## Kolbe's Electrolytic Decarboxylation & Free Radical Coupling

When a concentrated aqueous solution of sodium or potassium carboxylate undergoes electrolysis, decarboxylative dimerization occurs at the anode:

$$2R - COO^-K^+ + 2H_2O \xrightarrow{\text{Electrolysis}} \mathbf{R - R} + 2CO_2 \uparrow + H_2 \uparrow + 2KOH \tag{6.48}$$

**Electrochemical Mechanism at Electrodes:**

- **Anode Oxidation (Decarboxylation & Radical Dimerization):**

$$2R - COO^- \xrightarrow{-2e^-} 2[R - COO^\bullet] \quad (\text{Acyloxy free radical}) \tag{6.49}$$

$$2[R - COO^\bullet] \xrightarrow{\text{fast, RDS}} 2R^\bullet + 2CO_2 \uparrow \quad (\text{Homolytic cleavage of C} - \text{C bond}) \tag{6.50}$$

$$2R^\bullet \xrightarrow{\text{fast}} \mathbf{R - R} \quad (\text{Radical-radical coupling}) \tag{6.51}$$

- **Cathode Reduction (Evolution of $H_2$ & Alkalinization):**

$$2H_2O + 2e^- \rightarrow \mathbf{H_2} \uparrow + 2OH^- \tag{6.52}$$

**Diagnostic Electrolytic Phenomena:**

1. **pH Drift:** As the reaction proceeds, $H^+$ is evolved as $H_2$ gas while $OH^-$ accumulates in solution. Hence, the **pH of the electrolyte continuously rises** (pH $> 7$, alkaline).
2. **Gas Ratio at Electrodes:** At the anode, $CO_2$ and alkane vapor are evolved. At the cathode, pure $H_2$ gas is evolved. For sodium acetate, the molar ratio of gases is (Anodic gases : Cathodic gas) $= (2CO_2 + 1C_2H_6) : (1H_2) = 3 : 1$.

**Four Cardinal NEET Traps in Decarboxylation Reactions**

1. **The Methane Exclusion in Kolbe Electrolysis:** Methane ($CH_4$) **cannot** be prepared by Kolbe electrolysis because radical coupling requires at least two alkyl radicals ($2CH_3^\bullet \rightarrow CH_3 - CH_3$). Potassium formate (HCOOK) electrolyzed yields $H_2$ and $CO_2$, not alkane!
2. **Radical Disproportionation By-Products:** When sodium propanoate ($CH_3CH_2COONa$) is electrolyzed, the intermediate ethyl radicals ($CH_3CH_2^\bullet$) undergo both coupling (*n*-butane) and disproportionation:

$$CH_3CH_2^\bullet + CH_3CH_2^\bullet \rightarrow CH_2 = CH_2 \text{ (Ethene)} + CH_3 - CH_3 \text{ (Ethane)} \tag{6.53}$$

3. **Dicarboxylic Acid Electrolysis Shortcuts:**
   - Potassium succinate ($KOOC - CH_2 - CH_2 - COOK$) yields **Ethene** ($CH_2 = CH_2$).
   - Potassium maleate or fumarate ($KOOC - CH = CH - COOK$) yields **Ethyne** ($CH \equiv CH$).
4. **Role of CaO in Soda Lime:** CaO is not a reactant. It is used because:
   - It is intensely hygroscopic and keeps solid NaOH completely dry (deliquescence prevention).
   - It prevents melting/fusion of NaOH at high temperatures, avoiding corrosion and cracking of the glass tube.

**Thermal Decarboxylation of $\beta$-Keto & Gem-Dicarboxylic Acids**

Carboxylic acids possessing a carbonyl group at the $\beta$-carbon ($\beta$-keto acids like acetoacetic acid, and gem-dicarboxylic acids like malonic acid) undergo spontaneous decarboxylation upon gentle heating ($\approx 373 - 423$ K) **without any added base or catalyst**:

$$R - CO - CH_2 - COOH \xrightarrow{\Delta} R - CO - CH_3 + CO_2 \uparrow \tag{6.54}$$

**Pericyclic Mechanism via 6-Membered Cyclic Transition State:** The acidic carboxylic proton forms an intramolecular hydrogen bond with the $\beta$-carbonyl oxygen. Heating triggers a concerted 6-electron cyclic electron redistribution:

$$\mathrm{R-CO-CH_2-COOH} \xrightarrow[\text{cyclic 6-membered T.S.}]{\Delta} \mathrm{R-C(OH)=CH_2\ (Enol)} + \mathrm{CO_2}\uparrow \tag{6.55}$$

$$\xrightarrow{\text{tautomerism}} \mathrm{R-CO-CH_3\ (Stable\ Methyl\ Ketone)} \tag{6.56}$$

The enol intermediate undergoes instantaneous, virtually irreversible keto-enol tautomerism to yield the thermodynamically favored **methyl ketone**.

### 6.5.3 Hell-Volhard-Zelinsky (HVZ) Reaction & $\alpha$-Halogenation

**HVZ Mechanism: In Situ Enolization via Acyl Halide**

Carboxylic acids undergo selective $\alpha$-halogenation when treated with bromine or chlorine in the presence of a catalytic amount of red phosphorus:

$$\mathrm{R-CH_2-COOH} \xrightarrow[\text{(ii) } \mathrm{H_2O}]{\text{(i) } \mathrm{X_2/red\ P}} \mathrm{R-CH(X)-COOH} \quad (\mathrm{X=Cl,Br}) \tag{6.57}$$

**Step-by-Step Mechanistic Progression:**

1. **Formation of Phosphorus Trihalide:** $2\mathrm{P} + 3\mathrm{X_2} \rightarrow 2\mathrm{PX_3}$.
2. **Acyl Halide Generation:** Carboxylic acids themselves enolize too slowly for direct $\alpha$-halogenation. The $\mathrm{PX_3}$ converts a small catalytic fraction of carboxylic acid to the corresponding acyl halide:

$$3\mathrm{R-CH_2-COOH} + \mathrm{PX_3} \rightarrow 3\mathrm{R-CH_2-COX} + \mathrm{H_3PO_3} \tag{6.58}$$

3. **Rapid Enolization:** The acyl halide enolizes millions of times faster than the carboxylic acid because the electronegative halogen increases the acidity of the $\alpha$-protons:

$$\mathrm{R-CH_2-COX} \rightleftharpoons \mathrm{R-CH=C(OH)X} \quad \text{(Enol form)} \tag{6.59}$$

4. **Electrophilic Halogenation:** The enol attacks $\mathrm{X_2}$ to produce the $\alpha$-haloacyl halide:

$$\mathrm{R-CH=C(OH)X} + \mathrm{X_2} \rightarrow \mathrm{R-CH(X)-COX} + \mathrm{HX} \tag{6.60}$$

5. **Transacylation Exchange:** The $\alpha$-haloacyl halide reacts with unreacted parent carboxylic acid to yield the $\alpha$-halocarboxylic acid and regenerate the acyl halide, sustaining the catalytic cycle:

$$R-CH(X)-COX + R-CH_2-COOH \rightleftharpoons R-CH(X)-COOH + R-CH_2-COX \tag{6.61}$$

**HVZ Substrate Scope & Synthetic Traps**

- **Requirement for $\alpha$-Hydrogen:** Carboxylic acids devoid of $\alpha$-hydrogens (e.g., formic acid HCOOH, benzoic acid $C_6H_5COOH$, and trimethylacetic acid $(CH_3)_3CCOOH$) **never** undergo the HVZ reaction.
- **Halogen Limitations:** The reaction works efficiently with $Br_2$ and $Cl_2$. Direct fluorination causes extensive fragmentation; direct iodination fails because the reverse reaction with HI dominates.
- **Downstream Synthetic Utility:** The $\alpha$-halogen is readily substituted:

$$R-CH(Br)-COOH \xrightarrow{\text{aq. KOH}} R-CH(OH)-COOH \quad (\alpha\text{-hydroxy acid})$$

$$R-CH(Br)-COOH \xrightarrow{\text{excess } NH_3} R-CH(NH_2)-COOH \quad (\alpha\text{-amino acid})$$

# 6.6 Aromatic & Nitrogen Mechanisms

## 6.6.1 Diazotization Mechanism of Primary Aromatic Amines

**Diazotization Mechanism & Nitrosonium Ion Electrophile**

Primary aromatic amines react with nitrous acid at ice-cold temperature (273 – 278 K) to form stable arenediazonium salts:

$$\mathrm{Ar-NH_2 + NaNO_2 + 2HCl \xrightarrow{273-278\ K} Ar-N_2^+Cl^- + NaCl + 2H_2O} \tag{6.62}$$

**Step-by-Step Reaction Sequence:**

1. **Generation of Nitrosonium Ion ($NO^+$):** Nitrous acid is unstable and prepared in situ:

   $$\mathrm{NaNO_2 + HCl \rightarrow HNO_2 + NaCl}$$

   $$\mathrm{HNO_2 + H^+ \rightleftharpoons H_2\overset{+}{O} - NO \rightleftharpoons NO^+ + H_2O} \quad \text{(Nitrosonium electrophile)}$$

2. **Nucleophilic Attack:** The aniline nitrogen lone pair attacks $NO^+$ to form *N*-nitrosaniline cation:

   $$\mathrm{Ar-NH_2 + NO^+ \rightleftharpoons Ar-\overset{+}{N}H_2-NO \xrightarrow{-H^+} Ar-NH-NO} \quad (N\text{-nitrosamine}) \tag{6.63}$$

3. **Proton Tautomerism & Dehydration:**

   $$\mathrm{Ar-NH-N=O \rightleftharpoons Ar-N=N-OH \xrightarrow{H^+} Ar-N=N-\overset{+}{O}H_2 \xrightarrow{-H_2O} Ar-\overset{+}{N}} \tag{6.64}$$

   The diazonium ion is stabilized by resonance with the aromatic $\pi$-system.

**Thermal Decomposition & Aliphatic vs Aromatic Diazonium Salts**

- **Temperature Sensitivity ($> 278$ K/5°C):** If the temperature rises above 5°C, benzenediazonium chloride decomposes rapidly in aqueous solution:

  $$\mathrm{Ar-N_2^+Cl^- + H_2O \xrightarrow{>278\ K} Ar-OH + N_2\uparrow + HCl} \quad \text{(Forms phenol quantitative} \tag{6.65}$$

- **Aliphatic Primary Amines:** Aliphatic diazonium salts ($R-N_2^+Cl^-$) are exceptionally unstable even at 0°C because the alkyl group cannot disperse the positive charge by resonance. They instantaneously lose molecular $N_2$ to generate high-energy carbocations ($R^+$), yielding complex mixtures of alcohols, alkenes, and

alkyl chlorides. The quantitative evolution of $N_2$ is utilized analytically for amino acid estimation (Van Slyke method).

## 6.6.2 Azo Dye Coupling Mechanism (Phenol vs Aniline)

**Azo Coupling: Mild $S_EAr$ & pH-Directed Chemoselectivity**

Arenediazonium cations are weak electrophiles ($Ar - N_2^+$) that undergo electrophilic aromatic substitution only on highly activated aromatic rings:

1. **Coupling with Phenol (Orange Dye):**
   - **Optimum pH:** Alkaline medium (**pH $\approx 9 - 10$**).
   - **Mechanistic Rationale:** Phenol is a weak activator, but at pH $\approx 9 - 10$ it is deprotonated into the **phenoxide ion** ($Ph - O^-$), which is a vastly superior $+M$ donor. The weak diazonium electrophile attacks predominantly at the unhindered **para-position**:

$$Ph - N_2^+Cl^- + Ph - OH \xrightarrow[\text{pH 9–10}]{OH^-} p\text{-}HO - C_6H_4 - N = N - Ph \ (\textbf{Orange Dye}) + C \tag{6.66}$$

   - **High pH Hazard:** If pH $> 12$, the diazonium ion reacts with hydroxide to form unreactive diazotate ion ($Ar - N = N - O^-$), terminating the coupling reaction.
2. **Coupling with Aniline (Yellow Dye):**
   - **Optimum pH:** Mildly acidic medium (**pH $\approx 4 - 5$**).
   - **Mechanistic Rationale:** Aniline is already strongly activated. If pH $< 3$, aniline is protonated into the non-nucleophilic anilinium ion ($-NH_3^+$). At pH $\approx 4 - 5$, sufficient free aniline exists to couple at the para-position:

$$Ph - N_2^+Cl^- + Ph - NH_2 \xrightarrow[\text{pH 4–5}]{H^+} p\text{-}H_2N - C_6H_4 - N = N - Ph \ (\textbf{Yellow Dye}) + H \tag{6.67}$$

   - If the medium is basic, diazonium attacks the nitrogen lone pair instead of the ring, producing diazoaminobenzene ($Ph - N = N - NH - Ph$).

## 6.6.3 Reimer-Tiemann Reaction Mechanism

**Dichlorocarbene Intermediate & Formylation of Phenol**

The conversion of phenol to salicylaldehyde using chloroform and aqueous sodium hydroxide:

1. **Step 1: Generation of Dichlorocarbene (: $CCl_2$) [$\alpha$-Elimination]:** Hydroxide ion deprotonates the relatively acidic proton of chloroform ($CHCl_3, pK_a \approx 15.5$ due to $-I$ of three chlorines):

$$CHCl_3 + OH^- \rightleftharpoons H_2O + {}^-:CCl_3 \quad \text{(Trichloromethyl carbanion)}$$

$${}^-:CCl_3 \xrightarrow{\text{slow, } \alpha\text{-elimination}} :CCl_2 \text{ (Singlet Dichlorocarbene)} + Cl^-$$

   **Nature of** $:CCl_2$**:** Neutral reactive intermediate with 6 valence electrons and an empty unhybridized $p$-orbital $\rightarrow$ **powerful electrophile**.

2. **Step 2: Electrophilic Attack on Phenoxide:** Phenoxide attacks $:CCl_2$ predominantly at the ortho position, forming a non-aromatic cyclohexadienone carbanion intermediate:

$$C_6H_5O^- + :CCl_2 \rightarrow [o\text{-}C_6H_4(=O) - CCl_2^-] \tag{6.68}$$

3. **Step 3: Intramolecular Proton Shift & Rearomatization:** The aromatic proton from the ring shifts to the $-CCl_2^-$ carbon, restoring aromatic sextet stability and forming the dichloromethylphenoxide intermediate:

$$[o\text{-}C_6H_4(=O) - CCl_2^-] \rightarrow o\text{-}O^- - C_6H_4 - CHCl_2 \tag{6.69}$$

4. **Step 4: Alkaline Hydrolysis & Dehydration:** Hydroxide displaces both chlorine atoms via nucleophilic substitution, producing an unstable gem-diol intermediate that spontaneously loses water upon acidification to afford **salicylaldehyde** ($o$-hydroxybenzaldehyde).

**Reimer-Tiemann Variant Trap: Chloroform vs Carbon Tetrachloride**

- If **$CHCl_3$** is employed: The intermediate is a gem-dichloride ($-CHCl_2$), which hydrolyzes to give **Salicylaldehyde** ($o$-hydroxybenzaldehyde).
- If **$CCl_4$** is employed: Phenoxide attacks $CCl_4$ to generate a trichloromethyl intermediate ($-CCl_3$). Alkaline hydrolysis of $-CCl_3$ introduces three hydroxyl groups, which eliminate water to

yield **Salicylic Acid** (*o*-hydroxybenzoic acid)!

## 6.6.4 Kolbe's (Kolbe-Schmitt) Reaction Mechanism

**Carboxylation of Phenoxide via Carbon Dioxide Electrophile**

Treatment of sodium phenoxide with carbon dioxide gas at 400 K and $4-7$ atm, followed by acidification, produces salicylic acid:

$$C_6H_5OH \xrightarrow{NaOH} C_6H_5O^-Na^+$$

$$C_6H_5O^-Na^+ \xrightarrow[4\text{–}7 \text{ atm, } 400 \text{ K}]{CO_2} \text{Sodium Salicylate} \xrightarrow{H^+} \textbf{Salicylic Acid } (o\text{-hydroxybenzoic a}$$

**Mechanism & Ortho-Directing Sodium Chelate Effect:**

1. **Electrophile:** Carbon dioxide ($O=C=O$) is a relatively weak, neutral electrophile. Un-ionized phenol is too unreactive to react with $CO_2$; conversion to sodium phenoxide ($Ph-O^-$) provides the requisite nucleophilic activation.
2. **6-Membered Cyclic Transition State:** The $Na^+$ counter-ion coordinates simultaneously to the phenoxide oxygen and one of the carbonyl oxygens of $CO_2$. This cyclic coordination holds $CO_2$ directly above the ortho-carbon, directing electrophilic attack selectively to the **ortho position**.
3. **Rearomatization:** Proton transfer from the ring carbon to oxygen regenerates the aromatic ring, followed by acidification to yield salicylic acid.

**Cation Size Effect: Sodium vs Potassium Phenoxide**

- **Sodium Phenoxide ($Na^+$):** Forms a tight 6-membered cyclic chelate due to the small ionic radius of $Na^+$ (102 pm), giving **salicylic acid (ortho)** as the major product ($> 80\%$).
- **Potassium Phenoxide ($K^+$):** $K^+$ has a larger ionic radius (138 pm) and cannot form an effective cyclic chelate. At 473 K, attack shifts predominantly to the less hindered para position, yielding **p-hydroxybenzoic acid** as the major product.
- **Downstream Aspirin Synthesis:**

$$\text{Salicylic acid} + (CH_3CO)_2O \xrightarrow{H^+} \textbf{Acetylsalicylic acid (Aspirin)} + CH_3COOH \tag{6.70}$$

### 6.6.5 Hofmann Bromamide Degradation Mechanism

**Hofmann Bromamide Reaction: Concerted Intramolecular Rearrangement**

Primary acid amides react with bromine in aqueous or ethanolic sodium hydroxide to yield primary amines containing **one carbon less** than the parent amide:

$$\mathrm{R-CONH_2 + Br_2 + 4NaOH \rightarrow R-NH_2 + Na_2CO_3 + 2NaBr + 2H_2O} \tag{6.71}$$

**Detailed Step-by-Step Mechanism:**

1. **Step 1: Deprotonation & Formation of *N*-Bromoamide:** Hydroxide abstracts a proton from the amide nitrogen ($pK_a \approx 15-16$), generating an amide anion that attacks $Br_2$:

$$\mathrm{R-CONH_2 + OH^- \rightleftharpoons R-CON^-H + H_2O \xrightarrow{Br_2} R-CONHBr\ (\textit{N}\text{-bromoamide})} \tag{6.72}$$

2. **Step 2: Second Deprotonation to Bromoamide Anion:** The remaining proton on nitrogen is more acidic due to the $-I$ effect of bromine:

$$\mathrm{R-CONHBr + OH^- \rightleftharpoons [R-CO-\bar{N}-Br] + H_2O} \tag{6.73}$$

3. **Step 3: Concerted 1,2-Alkyl Shift [RDS] → Formation of Isocyanate:** Expulsion of the bromide leaving group is concerted with the simultaneous intramolecular migration of the alkyl group R (with its electron pair) from the carbonyl carbon to the electron-deficient nitrogen atom:

$$\mathrm{[R-CO-\bar{N}-Br] \xrightarrow[\text{slow, RDS}]{-Br^-} \mathbf{R-N=C=O}} \quad \textbf{(Alkyl Isocyanate Interme} \tag{6.74}$$

**Stereochemical Consequence:** The alkyl group migrates with **100% retention of stereochemical configuration** at the migrating chiral center!

4. **Step 4: Hydrolysis of Isocyanate to Carbamic Acid:** Hydroxide attacks the electrophilic carbonyl carbon of the isocyanate, followed by proton transfer to form an unstable carbamic acid:

$$\mathrm{R-N=C=O + OH^- \rightarrow R-NH-COO^-} \tag{6.75}$$

5. **Step 5: Spontaneous Decarboxylation:** The carbamate undergoes spontaneous decarboxylation, releasing $CO_2$ (which is trapped by alkali as $Na_2CO_3$) and yielding the **primary amine ($\mathrm{R-NH_2}$)**.

**Hofmann Bromamide Pitfalls, Stoichiometry & Reagent Trapping**

- **Exact Stoichiometry:** For every 1 mol of amide, exactly **1** mol of $\mathbf{Br_2}$ and **4** mol of **NaOH** are consumed. A frequent NEET trap involves asking for the molar ratio: Amide : $Br_2$ : NaOH = 1 : 1 : 4.
- **Secondary and Tertiary Amides Fail:** Secondary amides ($RCONHR'$) have only one hydrogen on nitrogen and cannot form the necessary *N*-bromoamide anion; tertiary amides have zero hydrogens. Thus, only 1° amides undergo this reaction.
- **Isocyanate Trapping Proof:** If the reaction is conducted in an anhydrous alcoholic medium ($R'OH$) instead of water, the isocyanate intermediate is intercepted by alcohol to form a stable **urethane / carbamate** ($R-NH-COOR'$), proving the existence of the isocyanate intermediate!
- **Phthalimide Reaction:** Phthalimide treated with $Br_2$/NaOH undergoes degradation to yield **anthranilic acid** (*o*-aminobenzoic acid).

## 6.7 Master Diagnostic Matrix: Reaction Mechanisms Overview

| Reaction Name | Key Intermediate | Kinetic Order | Regio / Stereo-chemistry | Critical NEET Exam Pitfall |
|---|---|---|---|---|
| **Nitration of Benzene** | Arenium Ion ($\sigma$-complex) | 2nd order $(1+1)$ | $k_H/k_D \approx 1$ (no isotope effect) | $HNO_3$ acts as a base; aniline gives 47% meta without $-NH_2$ protection. |
| **Friedel-Crafts Alkylation** | $1^\circ, 2^\circ, 3^\circ$ Carbocation | 2nd order | Thermodynamic rearrange-ment | Polyalkylation occurs; deactivated rings and aniline fail. |
| **Friedel-Crafts Acylation** | Acylium Ion ($R-C\equiv\overset{+}{O}$) | 2nd order | No rear-rangement | Deactivates ring against polyacyla-tion; requires $>1$ eq. $AlCl_3$. |
| **Halogenation of Halobenzenes** | Arenium Ion ($+M$ donor) | 2nd order | Ortho/Para orientation | Halogens are deactivating overall ($-I > +M$) but direct ortho/para. |
| $S_N1$ **Substitution** | Planar Carbocation ($sp^2$) | 1st order: $k$[RX] | Net partial racemization + excess inversion | Rearrangements occur; hindered substrate ($3^\circ$) and polar protic solvent required. |
| $S_N2$ **Substitution** | Trigonal Bipyramid T.S. | 2nd order: $k$[RX][Nu] | 100% Walden Inversion | Neopentyl halides are extremely unreactive; polar aprotic solvents accelerate. |
| $E1$ **Elimination** | Planar Carbocation ($sp^2$) | 1st order: $k$[RX] | Saytzeff major (more substituted) | High temperature favors $E1$ over $S_N1$ due to $+T\Delta S^\circ$. |
| $E2$ **Elimination** | Anti-periplanar T.S. | 2nd order: $k$[RX][Base] | Anti-elimination; Saytzeff vs Hofmann | Hofmann product forms with poor leaving group ($F^-$) or bulky base ($t$-BuOK). |
| **Dehydration of Alcohols** | Alkyloxonium $\rightarrow$ Carbocation | 1st order in alcohol | Saytzeff alkene | Ethanol at 443 K $\rightarrow$ ethene; at 413 K $\rightarrow$ diethyl ether ($S_N2$). |
| **Aldol Condensation** | Enolate Carbanion $\rightarrow$ Aldol | 2nd order overall | $\alpha,\beta$-unsaturated enal/enone | Must possess $\alpha$-H; cross-aldol yields 4 products unless one partner lacks $\alpha$-H. |
| **Cannizzaro** | Dianion hydrate | 3rd / 4th | Intermolecular | Requires NO |

PART VI

# Practical Chemistry & Qualitative Analysis Lab Manual

Chapter 7

# Practical Chemistry: Qualitative Salt Analysis & Volumetric Redox Titrations

With the NMC's renewed emphasis on practical laboratory skills, qualitative and quantitative analysis questions appear regularly. This chapter provides a complete laboratory manual covering systematic group separation of basic radicals (cations), confirmatory tests for acidic radicals (anions), and volumetric redox titrations with $KMnO_4$.

## 7.1 Systematic Qualitative Analysis: Acidic Radicals (Anions)

**Class I: Dilute $H_2SO_4$ Soluble Anions ($CO_3^{2-}, S^{2-}, SO_3^{2-}, NO_2^-$)**

Treated with dilute $H_2SO_4$ to observe volatile evolved gases:

| Anion | Gas | Preliminary Observation | Confirmatory Test & Governing Equation |
|---|---|---|---|
| $CO_3^{2-}$ | $CO_2$ | Brisk effervescence; odorless, colorless gas. | Lime water turns milky ($CaCO_3\downarrow$); clears with excess $CO_2$: $CaCO_3 + H_2O + CO_2 \rightarrow Ca(HCO_3)_2$. |
| $S^{2-}$ | $H_2S$ | Colorless gas with rotten-egg smell. | Turns lead acetate paper black ($PbS\downarrow$); purple/violet with nitroprusside: $S^{2-} + [Fe(CN)_5NO]^{2-} \rightarrow [Fe(CN)_5NOS]^{4-}$. |
| $SO_3^{2-}$ | $SO_2$ | Colorless gas with suffocating odor of sulfur. | Acidified $K_2Cr_2O_7$ turns green: $K_2Cr_2O_7 + H_2SO_4 + 3SO_2 \rightarrow Cr_2(SO_4)_3 + K_2SO_4 + H_2O$. |
| $NO_2^-$ | $NO_2$ | Pungent reddish-brown fumes. | Starch-iodide turns blue: $2NO_2^- + 2I^- + 4H^+ \rightarrow 2NO + I_2 + 2H_2O$; brown ring with dil $H_2SO_4 + FeSO_4$. |

### Class II: Concentrated $H_2SO_4$ Anions ($Cl^-, Br^-, I^-, NO_3^-, CH_3COO^-$)

Treated with concentrated $H_2SO_4$ and gently heated:

| Anion | Observation | Confirmatory Reactions & Identification |
|---|---|---|
| $Cl^-$ | Colorless pungent fumes (HCl); white fumes with $NH_4OH$ glass rod. | **Chromyl Chloride Test:** Salt $+K_2Cr_2O_7$ + conc. $H_2SO_4 \xrightarrow{\Delta} CrO_2Cl_2$ (red vapors) $\xrightarrow{NaOH} Na_2CrO_4 \xrightarrow{Pb^{2+}} PbCrO_4\downarrow$ (Yellow). |
| $Br^-$ | Reddish brown vapors ($Br_2$) intensifying with $MnO_2$. | **Organic Layer Test:** Salt $+Cl_2$-water + $CHCl_3 \xrightarrow{shake}$ orange-brown layer ($2Br^- + Cl_2 \rightarrow 2Cl^- + Br_2$). |
| $I^-$ | Deep violet vapors ($I_2$) turning starch paper blue. | **Layer Test:** Salt $+Cl_2$-water + $CHCl_3 \rightarrow$ violet layer. $AgNO_3$ gives bright yellow AgI $\downarrow$ insoluble in conc $NH_4OH$. |
| $NO_3^-$ | Dense brown fumes of $NO_2$ with Cu turnings. | **Brown Ring Test:** Salt $+FeSO_4 \xrightarrow{\text{trickle conc. } H_2SO_4} [Fe(H_2O)_5(NO)]^{2+}$ (Brown Ring at junction). |
| $CH_3COO^-$ | Characteristic vinegar odor. | **Ester Test:** Sweet fruity smell ($CH_3COOC_2H_5$). $FeCl_3$ **Test:** Blood-red coloration ($[Fe_3(OH)_2(CH_3COO)_6]^+$); boiling yields brown basic acetate. |

### Class III: Independent Group Anions ($SO_4^{2-}, PO_4^{3-}$)

Do not evolve gases with dilute or concentrated acids; identified by direct precipitation:

**Sulfate** ($SO_4^{2-}$): Salt $+ BaCl_2 \xrightarrow{\text{dil. HCl}} BaSO_4\downarrow$ (White precipitate insoluble in

**Phosphate** ($PO_4^{3-}$): Salt + conc. $HNO_3 + (NH_4)_2MoO_4 \xrightarrow{\Delta} (NH_4)_3[P(Mo_3O_{10})_4]\downarrow$

## 7.2 Systematic Qualitative Analysis: Basic Radicals (Cations)

| Group | Cations | Group Reagent | Precipitated As (Color) | Solubility Product ($K_{sp}$) Principle | Confirmatory Test & Governing Equation |
|---|---|---|---|---|---|
| **Group 0** | $NH_4^+$ | Dilute NaOH (gently warm) | Evolved gas $NH_3 \uparrow$ | High volatility of free ammonia | Nessler test: brown precipitate of iodide of Millon base ($H_2N-Hg-O-Hg-I\downarrow$). |
| **Group I** | $Pb^{2+}$ | Dilute HCl | $PbCl_2 \downarrow$ (White) | Lowest $K_{sp}$ among metal chlorides | Precipitate dissolves in hot water, recrystallizes as golden spangles on cooling. $K_2CrO_4 \rightarrow PbCrO_4 \downarrow$ (Yellow). |
| **Group II** | $Pb^{2+}, Cu^{2+}, As^{3+}$ | $H_2S$ gas in presence of dilute HCl | $PbS, CuS$ (Black); $As_2S_3$ (Yellow) | **Common Ion Effect:** $H^+$ from HCl suppresses $H_2S$ ionization, maintaining $[S^{2-}]$ extremely low; only sparingly soluble Group II sulfides exceed $K_{sp}$. | $Cu^{2+}$ + excess $NH_4OH \rightarrow [Cu(NH_3)_4]^{2+}$ (Deep Azure Blue); $K_4[Fe(CN)_6] \rightarrow Cu_2[Fe(CN)_6] \downarrow$ (Chocolate Brown). |
| **Group III** | $Fe^{3+}, Al^{3+}$ | $NH_4OH$ in presence of solid $NH_4Cl$ | $Fe(OH)_3 \downarrow$ (Reddish-brown); $Al(OH)_3 \downarrow$ (White gelatinous) | **Common Ion Effect:** $NH_4^+$ from $NH_4Cl$ suppresses $NH_4OH$ ionization, keeping $[OH^-]$ sufficiently low to precipitate only Group III hydroxides. | $Fe^{3+} + K_4[Fe(CN)_6] \rightarrow Fe_4[Fe(CN)_6]_3 \downarrow$ (Prussian Blue); $Al(OH)_3$ dissolves in excess NaOH forming soluble aluminate $NaAlO_2$. |
| **Group IV** | $Co^{2+}, Ni^{2+}, Mn^{2+}, Zn^{2+}$ | $H_2S$ gas in presence of $NH_4OH + NH_4Cl$ | $CoS, NiS$ (Black); $MnS$ (Buff/flesh); $ZnS$ (Dirty white) | Alkaline medium: $OH^-$ consumes $H^+$, shifting $H_2S \rightleftharpoons 2H^+ + S^{2-}$ forward; $[S^{2-}]$ increases, exceeding higher $K_{sp}$ of Group IV sulfides. | $Ni^{2+} + DMG + NH_4OH \rightarrow [Ni(DMG)_2] \downarrow$ (Rosy-red); $Zn^{2+} + K_4[Fe(CN)_6] \rightarrow Zn_2[Fe(CN)_6] \downarrow$ (Bluish-white). |
| **Group V** | $Ba^{2+}, Sr^{2+}, Ca^{2+}$ | $(NH_4)_2CO_3 + NH_4OH + NH_4Cl$ | $BaCO_3, SrCO_3, CaCO_3$ (White) | Carbonate precipitation in ammoniacal medium; presence of $NH_4Cl$ prevents precipitation of $MgCO_3$ by keeping $[CO_3^{2-}]$ below its $K_{sp}$. | Flame Tests: $Ba^{2+}$ gives Apple Green; $Sr^{2+}$ gives Crimson Red; $Ca^{2+}$ gives Brick Red. $Ba^{2+} + K_2CrO_4 \rightarrow BaCrO_4 \downarrow$ (Yellow). |
| **Group VI** | $Mg^{2+}$ | $Na_2HPO_4 + NH_4OH$ | $Mg(NH_4)PO_4 \downarrow$ (White crystalline) | Direct precipitation of ammonium magnesium phosphate in strongly ammoniacal alkaline medium | Scratching test tube walls with glass rod accelerates crystallization of white $Mg(NH_4)PO_4$. |

Table 7.1: Master Analytical Scheme: Systematic Qualitative Group Separation of Basic Radicals (Cations).

**High-Yield Confirmatory Tests & Diagnostic Colorations**

**1. Nickel Ion ($Ni^{2+}$) with Dimethylglyoxime (DMG):** Rosy-red precipitate stabilized by intramolecular hydrogen bonding:

$$Ni^{2+} + 2\,DMG + 2\,NH_4OH \longrightarrow [Ni(DMG)_2]\downarrow (\text{Rosy Red}) + 2\,NH_4^+ + 2\,H_2O \tag{7.1}$$

**2. Ferric Ion ($Fe^{3+}$) Confirmatory Tests:**

With Potassium Ferrocyanide: $4Fe^{3+} + 3[Fe(CN)_6]^{4-} \longrightarrow Fe_4[Fe(CN)_6]_{\mathbf{3}}\downarrow$ (**Prussian**

With Potassium Thiocyanate: $Fe^{3+} + SCN^- \longrightarrow [Fe(SCN)]^{\mathbf{2+}}$ (**Blood Red coloratio**

**3. Copper Ion ($Cu^{2+}$) Confirmatory Tests:**

With Excess $NH_4OH$ : $Cu^{2+} + 4NH_3 \longrightarrow [Cu(NH_3)_4]^{\mathbf{2+}}$ (**Azure Blue**)

With Potassium Ferrocyanide: $2Cu^{2+} + [Fe(CN)_6]^{4-} \longrightarrow Cu_2[Fe(CN)_6]\downarrow$ (**Chocolate**

**4. Flame Test Colors of Alkaline Earth Metals:**

$Ba^{2+} \Longrightarrow$ Apple Green $\quad Sr^{2+} \Longrightarrow$ Crimson Red $\quad Ca^{2+} \Longrightarrow$ Brick Red (7.2)

## 7.3 Volumetric Quantitative Redox Titrations

**$KMnO_4$ vs Mohr's Salt Titration (Standardization)**

Mohr's salt is a primary standard double salt: $(NH_4)_{\mathbf{2}}SO_4 \cdot FeSO_4 \cdot \mathbf{6}H_2O$ (M.W. $= 392$ g/mol).

Oxidation Half: $5[Fe^{2+} \longrightarrow Fe^{3+} + e^-]$

Reduction Half: $MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$

**Overall Net Redox:** $MnO_4^- + \mathbf{5}Fe^{2+} + \mathbf{8}H^+ \longrightarrow Mn^{2+} + \mathbf{5}Fe^{3+} + \mathbf{4}H_2O$

$$\frac{\mathbf{M_1V_1}}{\mathbf{1}}(KMnO_4) = \frac{\mathbf{M_2V_2}}{\mathbf{5}}(\text{Mohr's Salt}) \implies \mathbf{N_1V_1 = N_2V_2} \tag{7.3}$$

(Valence factor: $n = 5$ for $KMnO_4$, $n = 1$ for $Fe^{2+}$)

**Crucial Practical Precautions:**

- **Acidifying Agent:** Strictly use **dilute** $H_2SO_4$.
  - **Why NOT** HCl? Permanganate oxidizes HCl to chlorine gas ($2MnO_4^- + 16HCl \rightarrow 2Mn^{2+} + 5Cl_2\uparrow + 8H_2O + 6Cl^-$), artificially consuming excess $KMnO_4$!

  - **Why NOT $HNO_3$?** Nitric acid is a strong oxidizing agent itself and competes with $KMnO_4$ in oxidizing $Fe^{2+}$.
- **Indicator:** $KMnO_4$ acts as a **self-indicator**. The endpoint is marked by the appearance of a **permanent faint pink color** (excess single drop of $MnO_4^-$).

### $KMnO_4$ vs Oxalic Acid Titration

Oxalic acid is a primary standard dihydrate: $(COOH)_2 \cdot 2H_2O$ (M.W. $= 126\,g/mol$, Equivalent Weight $= 63\,g/eq$).

$$\text{Oxidation Half: } 5[C_2O_4^{2-} \longrightarrow 2CO_2 \uparrow + 2e^-]$$

$$\text{Reduction Half: } 2[MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O]$$

$$\textbf{Overall Net Redox: } \mathbf{2}MnO_4^- + \mathbf{5}C_2O_4^{2-} + \mathbf{16}H^+ \longrightarrow \mathbf{2}Mn^{2+} + \mathbf{10}CO_2 \uparrow + \mathbf{8}H_2$$

$$\frac{\mathbf{M_1V_1}}{\mathbf{2}}(KMnO_4) = \frac{\mathbf{M_2V_2}}{\mathbf{5}}(\text{Oxalic Acid}) \implies \mathbf{N_1V_1 = N_2V_2} \quad (7.4)$$

(Valence factor: $n = 5$ for $KMnO_4$, $n = 2$ for $C_2O_4^{2-}$)

**Crucial Practical Precautions:**

- **Pre-Heating Requirement:** Oxalic acid solution must be heated to $60 - 70°C$ before starting titration. At room temperature, the reaction is extremely slow.
- **Autocatalysis:** The initial disappearance of pink color is slow, but once the first few drops react, the reaction becomes rapid because product $Mn^{2+}$ **acts as an autocatalyst**!
- **Do NOT boil ($> 70°C$):** High temperature causes thermal decomposition of oxalic acid into $CO_2$, $CO$, and $H_2O$.

# Appendix: Comprehensive Chemistry Formula, Named Reaction & Reagent Matrices

Appendix A

# Physical Chemistry Core Formulas & Thermodynamic Equations

This chapter compiles all mathematical formulations, equilibrium relations, colligative equations, electrochemical relations, and kinetic rate laws for rapid revision before major examinations.

## A.1 Some Basic Concepts in Chemistry: Mole Concept (Unit 1)

### A.1.1 Stoichiometric Mole Formulations

**Fundamental Mole Formulations & Gas Volumes**

$$n = \frac{w\ [\text{mass in g}]}{M\ [\text{molar mass in g mol}^{-1}]} = \frac{N\ [\text{number of particles}]}{N_A} = \tag{A.1}$$

$$N_A = 6.022 \times 10^{23}\,\text{mol}^{-1}, \qquad 1\,\text{amu} = 1\,\text{u} = \frac{1}{N_A}\,\text{g} \approx 1.66 \times 10 \tag{A.2}$$

$$\text{Vapour Density (VD)} = \frac{\text{Density of gas at } T, P}{\text{Density of } H_2 \text{ gas at same } T, P} \implies \text{Molar Mass} = 2 \times \tag{A.3}$$

$$\text{Mass \% of element } X = \frac{(\text{Number of atoms of } X) \times (\text{Atomic mass of } X)}{\text{Molar mass of compound}} \times 100 \tag{A.4}$$

**Limiting Reagent Condition for Reaction $aA + bB \to cC + dD$:**

Compare ratio $\frac{n_A}{a}$ and $\frac{n_B}{b} \implies$ If $\frac{n_A}{a} < \frac{n_B}{b}$, $A$ is the limiting reagent (completely c

(A.5)

## A.1.2 Solution Concentrations & Dilution Formulations

**Molarity, Molality, Normality & Interconversions**

$$M = \frac{n_{\text{solute}}}{V_{\text{solution}}\ (\text{in L})} = \frac{w_B \times 1000}{M_B \times V_{\text{sol}}\ (\text{in mL})}$$ [Molarity: Tempe

(A.6)

$$m = \frac{n_{\text{solute}}}{W_{\text{solvent}}\ (\text{in kg})} = \frac{w_B \times 1000}{M_B \times W_A\ (\text{in g})}$$ [Molality: Tempe

(A.7)

$$N = \frac{\text{Gram equivalents of solute}}{V_{\text{solution}}\ (\text{in L})} = M \times n\text{-factor}$$ [Normality: $n$-fac

(A.8)

$$x_B = \frac{n_B}{n_A + n_B}, \qquad x_A + x_B = 1$$ [Mole Fraction: T

(A.9)

$$\text{ppm} = \frac{\text{Mass of component}}{\text{Total mass of solution}} \times 10^6, \qquad \%(\text{w/w}) = \frac{w_{\text{solute}}}{w_{\text{solution}}} \times 100$$

(A.10)

**High-Yield Interconversion Formulas ($d$ = solution density in $\text{g mL}^{-1}$):**

$$M = \frac{10 \times \%(w/w) \times d}{M_B}, \qquad m = \frac{1000M}{1000d - MM_B}, \qquad m = \frac{x_B \times 1000}{(1 - x_B)M_A}$$

(A.11)

**Dilution and Mixing of Solutions:**

$$M_1V_1 = M_2V_2, \qquad M_{\text{mix}} = \frac{M_1V_1 + M_2V_2 + \dots}{V_1 + V_2 + \dots}, \qquad N_1V_1 = N_2V_2$$ [at equivalence po

(A.12)

**Mole Traps: STP Standards & Temperature Dependence**

- **STP Standard Volume:** In IUPAC revised standard conditions ($T = 273.15\,\text{K}, P = 1\,\text{bar} = 10^5\,\text{Pa}$), molar volume is $22.7\,\text{L}$. Under traditional NCERT standard pressure ($1\,\text{atm} = 1.01325 \times 10^5\,\text{Pa}$), molar volume is $22.4\,\text{L}$. Unless specified as $1\,\text{bar}$, calculate using $22.4\,\text{L}$.
- **Molarity Changes with Temperature:** Because volume expands on heating ($V \propto T$), molarity $M$ and normality $N$ **decrease with increasing temperature**. Molality $m$, mole fraction $x$, and $\%(w/w)$ involve mass only and remain **strictly constant**.

# A.2 Structure of Atom (Unit 2)

## A.2.1 Quantum Mechanics, Wave-Particle Duality & Uncertainty

**de Broglie, Heisenberg Uncertainty & Orbital Angular Momentum**

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2mqV}} \quad [h = 6.626 \times 10^{-34}\,\mathrm{J\,s}] \tag{A.13}$$

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi} \implies \Delta x \cdot (m\Delta v) \geq \frac{h}{4\pi} \implies \Delta x \cdot \Delta v \geq \frac{h}{4\pi m} \quad [\text{Heisenberg Uncertainty} \tag{A.14}$$

$$\Delta E \cdot \Delta t \geq \frac{h}{4\pi} \quad [\text{Energy-Time Uncertain} \tag{A.15}$$

$$L_{\text{orbital}} = \sqrt{l(l+1)}\hbar = \sqrt{l(l+1)}\,\frac{h}{2\pi} \quad [\text{Orbital Angular Momer} \tag{A.16}$$

$$\mu_{\text{spin}} = \sqrt{n(n+2)}\,\text{BM} \quad [\text{BM} = \text{Bohr Magneton}] \quad [n = \text{number of unpaire} \tag{A.17}$$

## A.2.2 Quantum Numbers, Nodes & Electron Filling Principles

**Nodal Formulations for Atomic Orbitals**

$$\text{Radial Nodes (Spherical Nodes)} = n - l - 1 \tag{A.18}$$

$$\text{Angular Nodes (Nodal Planes)} = l \tag{A.19}$$

$$\text{Total Nodes} = (n - l - 1) + l = n - 1 \tag{A.20}$$

$$\text{Maximum Electrons in Shell } n = 2n^2, \qquad \text{Maximum Orbitals in Shell } n = n^2 \tag{A.21}$$

$$\text{Maximum Electrons in Subshell } l = 2(2l+1) = 4l + 2 \tag{A.22}$$

| Quantum Number | Symbol | Allowed Values | Physical Significance / Property |
|---|---|---|---|
| Principal | $n$ | $1, 2, 3, \ldots, \infty$ | Main energy level, shell size, distance from nucleus |
| Azimuthal / Orbital Angular | $l$ | $0, 1, 2, \ldots, (n-1)$ | Subshell shape ($l = 0 : s, 1 : p, 2 : d, 3 : f$) |
| Magnetic | $m_l$ | $-l, \ldots, 0, \ldots, +l$ | Spatial orientation of orbital in magnetic field |
| Spin | $m_s$ | $+\frac{1}{2}, -\frac{1}{2}$ | Intrinsic spin angular momentum orientation |

Table A.1: The Four Quantum Numbers and Boundary Constraints.

**Orbital Filling Rules & Exchange Energy Stability**

- **Aufbau Principle ($(n+l)$ Rule):** Electrons enter lowest energy subshell first. Lower $(n+l)$ has lower energy. If $(n+l)$ identical, lower $n$ has lower energy (e.g. $4s$ has $4+0=4$, $3d$ has $3+2=5 \implies 4s$ fills before $3d$).
- **Pauli Exclusion Principle:** No two electrons in an atom can have the identical set of all four quantum numbers. An orbital holds maximum 2 electrons with opposite spins.
- **Hund's Rule of Maximum Multiplicity:** Pairing does not occur in degenerate orbitals until each holds one electron with parallel spin.
- **Exchange Energy of Half/Fully Filled Subshells:** Number of possible electron exchanges among electrons of parallel spin:

$$K = \frac{n_p(n_p-1)}{2} \implies \text{For } d^5 : K = \frac{5 \times 4}{2} = 10 \text{ exchanges (extraordinarily stab}$$

# A.3 Chemical Thermodynamics & Thermochemistry (Unit 4)

## A.3.1 First Law, Work, Enthalpy & Heat Capacities

**First Law, PV-Work & Enthalpy-Internal Energy Coupling**

$$\Delta U = q + w \quad \text{[IUPAC Sign: Work done ON system is positive; BY system is} \tag{A.23}$$

$$w_{\text{irr}} = -P_{\text{ext}}(V_2 - V_1) = -P_{\text{ext}}\Delta V \tag{A.24}$$

$$w_{\text{rev, isothermal}} = -\int_{V_1}^{V_2} P\,dV = -nRT\ln\left(\frac{V_2}{V_1}\right) = -2.303nRT\log_{10}\left(\frac{P_1}{P_2}\right) \tag{A.25}$$

$$w_{\text{free expansion}} = 0 \quad (P_{\text{ext}} = 0 \text{ into vacuum}) \implies \Delta U = 0, \quad \Delta T = 0 \text{ for ideal gas} \tag{A.26}$$

$$H = U + PV \implies \Delta H = \Delta U + \Delta(PV) = \Delta U + \Delta n_g RT \tag{A.27}$$

$$C_p - C_v = R \implies \Delta H = q_p = nC_p\Delta T, \qquad \Delta U = q_v = nC_v\Delta T \tag{A.28}$$

## A.3.2 Thermochemistry & Hess's Law

**Hess's Law, Reaction Enthalpy & Bond Dissociation Enthalpies**

$$\Delta_r H^\circ = \sum \Delta_f H^\circ(\text{Products}) - \sum \Delta_f H^\circ(\text{Reactants}) \quad \text{[}\Delta_f H^\circ \text{ of pure elements} \tag{A.29}$$

$$\Delta_r H^\circ = \sum BE(\text{Reactants}) - \sum BE(\text{Products}) \quad \text{[All species MUST be in} \tag{A.30}$$

$$\Delta_r H = \sum \Delta_{\text{comb}} H^\circ(\text{Reactants}) - \sum \Delta_{\text{comb}} H^\circ(\text{Products}) \quad \text{[From Enthalpies of Com} \tag{A.31}$$

**Kirchhoff's Law of Temperature Dependence of Enthalpy:**

$$\frac{\Delta H_{T_2} - \Delta H_{T_1}}{T_2 - T_1} = \Delta C_p, \qquad \frac{\Delta U_{T_2} - \Delta U_{T_1}}{T_2 - T_1} = \Delta C_v \quad \left[\Delta C_p = \sum C_{p,\text{prod}} - \sum C_p \right. \tag{A.32}$$

### A.3.3 Entropy, Second Law & Gibbs Free Energy Spontaneity

**Entropy, Third Law & Spontaneity Criteria**

$$\Delta S = \frac{q_{\text{rev}}}{T}, \qquad \Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} \geq 0 \quad \text{[Second Law]} \tag{A.33}$$

$$\Delta S_{\text{system}} = nC_v \ln\left(\frac{T_2}{T_1}\right) + nR \ln\left(\frac{V_2}{V_1}\right) = nC_p \ln\left(\frac{T_2}{T_1}\right) - nR \ln\left(\frac{P_2}{P_1}\right) \tag{A.34}$$

$$\Delta S_{\text{isothermal}} = nR \ln\left(\frac{V_2}{V_1}\right) = 2.303 nR \log_{10}\left(\frac{P_1}{P_2}\right), \qquad \Delta S_{\text{isochoric}} = nC_v \ln\left(\frac{T_2}{T_1}\right) \tag{A.35}$$

$$G = H - TS \implies \Delta G = \Delta H - T\Delta S \quad \text{[Gibbs-Helmholtz Equation at c} \tag{A.36}$$

$$\Delta G^\circ = -RT \ln K = -2.303 RT \log_{10} K \iff K = e^{-\Delta G^\circ / RT} \tag{A.37}$$

**Thermodynamics Traps: Standard States & Bond Enthalpy Reversal**

- **Bond Enthalpy Subtraction Order:** Enthalpy of reaction using formation enthalpies is [Products – Reactants], but using bond enthalpies it is [**Reactants – Products**]!
- **Reference Standard State Formation Enthalpy is Zero:** Only the most stable allotropic form at 298.15 K and 1 bar has $\Delta_f H^\circ = 0$: Carbon is **Graphite** (Diamond has $\Delta_f H^\circ = 1.9\,\text{kJ mol}^{-1}$), Sulfur is **Rhombic**, Phosphorus is **White**.

| $\Delta H$ | $\Delta S$ | $\Delta G = \Delta H - T\Delta S$ | **Spontaneity Criterion** |
|---|---|---|---|
| $-$ | $+$ | Always Negative $(-)$ | Spontaneous at ALL temperatures |
| $+$ | $-$ | Always Positive $(+)$ | Non-spontaneous at ALL temperatures (reverse is spontaneous) |
| $-$ | $-$ | Negative at low $T$ $(T < \frac{\Delta H}{\Delta S})$ | Spontaneous at LOW temperature; non-spontaneous at high $T$ |
| $+$ | $+$ | Negative at high $T$ $(T > \frac{\Delta H}{\Delta S})$ | Spontaneous at HIGH temperature; non-spontaneous at low $T$ |

Table A.2: Spontaneity Matrix Governed by Enthalpy and Entropy Interplay.

## A.4 Equilibrium (Unit 6)

### A.4.1 Chemical Equilibrium, $K_p - K_c$ Coupling & Le Chatelier

**Equilibrium Constants, Reaction Quotient & Degree of Dissociation**

$$K_p = K_c(RT)^{\Delta n_g} \quad \left[R = 0.0821\,\text{L atm mol}^{-1}\,\text{K}^{-1} \text{ or } 0.0831\,\text{L bar mol}^{-1}\,\text{K}^{-1}\right] \tag{A.38}$$

$$Q < K \implies \text{Forward direction}; \quad Q = K \implies \text{Equilibrium}; \quad Q > K \implies \text{Backward} \tag{A.39}$$

$$\alpha = \frac{D - d}{(n-1)d} = \frac{M_{\text{theoretical}} - M_{\text{observed}}}{(n-1)M_{\text{observed}}} \quad [n = \text{number of moles produced per mole di} \tag{A.40}$$

**van 't Hoff Equation (Temperature Dependence of Equilibrium Constant):**

$$\ln\left(\frac{K_2}{K_1}\right) = \frac{\Delta H^\circ}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right) \implies \log_{10}\left(\frac{K_2}{K_1}\right) = \frac{\Delta H^\circ}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right] \tag{A.41}$$

### A.4.2 Ionic Equilibrium: Ostwald Dilution, Water Autoionization & pH

**Ostwald's Dilution Law, Water Ion Product & Acid-Base pH**

$$K_a = \frac{C\alpha^2}{1-\alpha} \approx C\alpha^2 \implies \alpha = \sqrt{\frac{K_a}{C}}, \quad [\mathrm{H}^+] = C\alpha = \sqrt{K_a C} \quad (\text{for } \alpha \leq 0.05) \tag{A.42}$$

$$K_w = [\mathrm{H}^+][\mathrm{OH}^-] = 1.0 \times 10^{-14} \text{ at } 25^\circ\mathrm{C} \implies \mathrm{pH} + \mathrm{pOH} = 14 = \mathrm{p}K_w \tag{A.43}$$

$$\mathrm{pH} = -\log_{10}[\mathrm{H}^+], \qquad [\mathrm{H}^+] = 10^{-\mathrm{pH}} \tag{A.44}$$

### A.4.3 Salt Hydrolysis Formulations

### A.4.4 Buffer Solutions & Henderson-Hasselbalch Equations

**Acidic and Basic Buffer Governing Equations**

$$\mathrm{pH} = \mathrm{p}K_a + \log_{10}\left(\frac{[\text{Salt}]}{[\text{Weak Acid}]}\right) = \mathrm{p}K_a + \log_{10}\left(\frac{[\text{Conjugate Base}]}{[\text{Acid}]}\right) \tag{A.45}$$

$$\mathrm{pOH} = \mathrm{p}K_b + \log_{10}\left(\frac{[\text{Salt}]}{[\text{Weak Base}]}\right) \implies \mathrm{pH} = 14 - \mathrm{pOH} \tag{A.46}$$

$$\text{Buffer Range} = \mathrm{p}K_a \pm 1, \qquad \beta = \frac{dB}{d(\mathrm{pH})} \tag{A.47}$$

### A.4.5 Solubility Product ($K_{sp}$) & Precipitation Criteria

**Solubility Product Equations & Precipitation**

$$A_xB_y(s) \rightleftharpoons xA^{y+}(aq) + yB^{x-}(aq) \implies K_{sp} = [A^{y+}]^x[B^{x-}]^y = (xS)^x(yS)^y = x^x y^y \tag{A.48}$$

**Canonical Stoichiometric Solubility Forms:**

| **Salt Type (Example)** | **Hydrolysis Const. ($K_h$)** | **Degree of Hydrolysis ($h$)** | **pH Formula at 25°C** |
|---|---|---|---|
| Strong Acid + Strong Base (NaCl) | No Hydrolysis | $h = 0$ | pH = 7.00 |
| Weak Acid + Strong Base ($CH_3COONa$) | $K_h = \frac{K_w}{K_a}$ | $h = \sqrt{\frac{K_w}{CK_a}}$ | $\text{pH} = 7 + \frac{1}{2}\text{p}K_a + \frac{1}{2}\log C$ |
| Strong Acid + Weak Base ($NH_4Cl$) | $K_h = \frac{K_w}{K_b}$ | $h = \sqrt{\frac{K_w}{CK_b}}$ | $\text{pH} = 7 - \frac{1}{2}\text{p}K_b - \frac{1}{2}\log C$ |
| Weak Acid + Weak Base ($CH_3COONH_4$) | $K_h = \frac{K_w}{K_a K_b}$ | $h = \sqrt{\frac{K_w}{K_a K_b}}$ | $\text{pH} = 7 + \frac{1}{2}\text{p}K_a - \frac{1}{2}\text{p}K_b$ |

Table A.3: Exact Hydrolysis Parameters and pH Formulations for Canonical Salts.

- 1 : 1 Salt (AgCl): $K_{sp} = S^2 \implies S = \sqrt{K_{sp}}$
- 1 : 2 or 2 : 1 Salt ($PbCl_2, CaF_2, Ag_2CrO_4$): $K_{sp} = 1^1 \times 2^2 \times S^{1+2} = 4S^3 \implies S = \sqrt[3]{\frac{K_{sp}}{4}}$
- 1 : 3 or 3 : 1 Salt ($Al(OH)_3, Fe(OH)_3$): $K_{sp} = 1^1 \times 3^3 \times S^{1+3} = 27S^4 \implies S = \sqrt[4]{\frac{K_{sp}}{27}}$
- 2 : 3 or 3 : 2 Salt ($Ca_3(PO_4)_2, As_2S_3$): $K_{sp} = 2^2 \times 3^3 \times S^{2+3} = 108S^5 \implies S = \sqrt[5]{\frac{K_{sp}}{108}}$

**Precipitation Criterion:**

$$Q_{sp} < K_{sp} \implies \text{Unsaturated (No precipitate)}; \quad Q_{sp} = K_{sp} \implies \text{Saturated}; \quad Q_{s} \tag{A.49}$$

**Ionic Equilibrium Traps: Autoionization of Water & pH Scaling**

- **Low Concentration Acid pH ($10^{-8}$ M HCl):** Never write pH = 8 for an acid! In extremely dilute solutions ($< 10^{-6}$ M), autoionization of water ($[H^+]_{water} = 10^{-7}$ M) cannot be neglected:

  $$[H^+]_{total} = 10^{-8} + 10^{-7} = 1.1 \times 10^{-7}\,\text{M} \implies \text{pH} \approx 6.96 \quad (\text{strictly} < 7)$$

- **Temperature Dependence of $K_w$:** The autoionization of water ($H_2O \rightleftharpoons H^+ + OH^-$) is endothermic ($\Delta H > 0$). Heating water to 60°C increases $K_w \approx 10^{-13}$. Neutral water at 60°C has pH = $\frac{1}{2}pK_w = 6.5$, but remains strictly **neutral** because $[H^+] = [OH^-]$.

## A.5 Solutions & Colligative Properties (Unit 5)

### A.5.1 Gas Solubility & Raoult's Law

**Henry's Law, Raoult's Law & Vapour Pressure**

$$p = K_H \cdot x \implies x = \frac{p}{K_H} \quad \text{[Henry's Law: Higher } K_H \text{ a}$$

(A.50)

$$P_{\text{total}} = P_A + P_B = P_A^\circ x_A + P_B^\circ x_B = P_A^\circ + (P_B^\circ - P_A^\circ)x_B \quad \text{[Raoult's Law for Volatile B}$$

(A.51)

$$y_A = \frac{P_A}{P_{\text{total}}} = \frac{P_A^\circ x_A}{P_{\text{total}}}, \qquad \frac{1}{P_{\text{total}}} = \frac{y_A}{P_A^\circ} + \frac{y_B}{P_B^\circ} \quad [y_A, y_B \text{ are mole fractions in}$$

(A.52)

## A.5.2 Colligative Properties & van 't Hoff Factor

## The Four Colligative Properties & van 't Hoff Factor ($i$)

$$\frac{P_A^\circ - P}{P_A^\circ} = i\,x_B = i\left(\frac{n_B}{n_A + n_B}\right) \approx i\left(\frac{w_B M_A}{M_B w_A}\right) \quad \text{[Relative Lowering of Vapour P} \tag{A.53}$$

$$\Delta T_b = T_b - T_b^\circ = i\,K_b\,m = i\left(\frac{1000 K_b w_B}{M_B w_A}\right) \quad \text{[Elevation of Boiling Point; } K_b \tag{A.54}$$

$$\Delta T_f = T_f^\circ - T_f = i\,K_f\,m = i\left(\frac{1000 K_f w_B}{M_B w_A}\right) \quad \text{[Depression of Freezing Point; } K \tag{A.55}$$

$$\Pi = i\,C\,R\,T = i\left(\frac{n_B}{V}\right) RT = i\left(\frac{w_B RT}{M_B V}\right) \quad \text{[Osmotic Pressure; isotonic: } \Pi_1 \tag{A.56}$$

**van 't Hoff Factor ($i$) Formulations:**

$$i = \frac{\text{Experimental / Observed Colligative Property}}{\text{Calculated / Normal Colligative Property}} = \frac{\text{Normal M}}{\text{Abnormal / Obse}} \tag{A.57}$$

$$\alpha_{\text{dissociation}} = \frac{i-1}{n-1} \implies i = 1 + (n-1)\alpha \quad \text{[e.g. for } K_4[Fe(CN)_6]: n = 5 \implies i = \tag{A.58}$$

$$\alpha_{\text{association}} = \frac{1-i}{1-\frac{1}{n}} \implies i = 1 - \left(1 - \frac{1}{n}\right)\alpha \quad \text{[for dimerization } n = 2 : i = 1 - \frac{\alpha}{2} \tag{A.59}$$

## Solutions Traps: Molality in RLVP & Abnormal Molar Mass

- **RLVP with Molality Shortcut:** For an aqueous solution, relative lowering of vapour pressure can be directly expressed in terms of molality $m$:

$$\frac{P^\circ - P}{P} = \frac{m \times M_{\text{solvent}}}{1000} \implies \text{For water: } \frac{P^\circ - P}{P} = \frac{18m}{1000}$$

  Notice the denominator is $P$ (solution vapour pressure), NOT $P^\circ$!
- **Elevation vs Depression Constant:** For water, $K_b = 0.52\,\text{K kg mol}^{-1}$, whereas $K_f = 1.86\,\text{K kg mol}^{-1}$. Freezing point depression is roughly 3.6 times more sensitive than boiling point elevation for the identical molal concentration.

# A.6 Electrochemistry (Unit 7)

## A.6.1 Galvanic Cells, Nernst Equation & Equilibrium

**EMF, Nernst Equation & Free Energy Formulations**

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} = E^\circ_{\text{right}} - E^\circ_{\text{left}} \quad \text{[Standard Reduction P}$$
(A.60)

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{RT}{nF}\ln Q = E^\circ_{\text{cell}} - \frac{2.303RT}{nF}\log_{10} Q \quad \text{[Nernst Equation]}$$
(A.61)

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{n}\log_{10} Q \quad (\text{at } T = 298.15\,\text{K}) \quad [F = 96485\,\text{C mol}^{-1} \approx$$
(A.62)

$$E^\circ_{\text{cell}} = \frac{0.0591}{n}\log_{10} K_c \iff \log_{10} K_c = \frac{nE^\circ_{\text{cell}}}{0.0591} \quad \text{[At Equilibrium: } E_{\text{cell}}$$
(A.63)

$$\Delta_r G = -nFE_{\text{cell}}, \qquad \Delta_r G^\circ = -nFE^\circ_{\text{cell}} = -2.303RT\log_{10} K_c \quad \text{[Thermodynamic Link}$$
(A.64)

**Concentration Cells (Identical Electrodes, Different Concentrations $C_1 < C_2$):**

$$E^\circ_{\text{cell}} = 0 \implies E_{\text{cell}} = \frac{0.0591}{n}\log_{10}\left(\frac{C_2}{C_1}\right) \quad (\text{Spontaneous if } C_2 > C_1)$$
(A.65)

## A.6.2 Electrolytic Conductance & Kohlrausch's Law

**Conductivity, Molar Conductivity & Kohlrausch Formulations**

$$G = \frac{1}{R} = \frac{A}{\rho L} = \kappa \frac{A}{L} \implies \kappa = G\left(\frac{L}{A}\right) = \frac{G^*}{R} \tag{A.66}$$

$$\Lambda_m = \frac{1000\,\kappa}{M\ [\text{Molarity}]}\ [\text{S cm}^2\,\text{mol}^{-1}], \qquad \Lambda_{\text{eq}} = \frac{1000\,\kappa}{N\ [\text{Normality}]} = \frac{\Lambda_m}{n\text{-factor}} \tag{A.67}$$

$$\Lambda_m = \Lambda_m^\circ - A\sqrt{C} \tag{A.68}$$

$$\Lambda_m^\circ(A_xB_y) = x\lambda_m^\circ(A^{y+}) + y\lambda_m^\circ(B^{x-}) \tag{A.69}$$

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ}, \qquad K_a = \frac{C\alpha^2}{1-\alpha} = \frac{C\Lambda_m^2}{\Lambda_m^\circ(\Lambda_m^\circ - \Lambda_m)}, \qquad S = \frac{1000\,\kappa_{\text{salt}}}{\Lambda_m^\circ} \tag{A.70}$$

## A.6.3 Faraday's Laws of Electrolysis

**Faraday's First & Second Laws of Electrolysis**

$$w = Z\,I\,t = \left(\frac{E}{96500}\right) I\,t = \left(\frac{M}{n \times 96500}\right) I\,t \quad [\text{Faraday's First Law; } Q = It] \tag{A.71}$$

$$\frac{w_1}{w_2} = \frac{E_1}{E_2} \quad \text{(for identical charge } Q \text{ passed)} \qquad [\text{Faraday's Second Law}] \tag{A.72}$$

**Electrochemistry Traps: Intensive $E^\circ$ vs Extensive $\Delta G^\circ$**

- **Electrode Potentials are Intensive Properties:** Never multiply $E^\circ$ by stoichiometric coefficients when balancing electrons! For half-reaction multiplied by 2: $\Delta G^\circ$ doubles, but $E^\circ$ remains **unchanged**. When combining non-standard half-cells, always combine $\Delta G^\circ = -nFE^\circ$, never add potentials directly!
- **Conductivity vs Molar Conductivity on Dilution:** On dilution: Specific conductivity $\kappa$ **decreases** (fewer ions per unit volume), but molar conductivity $\Lambda_m$ **increases** (interionic attractions weaken for strong electrolytes; $\alpha$ increases for weak electrolytes).

# A.7 Chemical Kinetics (Unit 8)

## A.7.1 Rate Laws, Order & Molecularity

**Reaction Rates & General Rate Expressions**

For general reaction: $aA + bB \to cC + dD$:

$$\text{Rate} = -\frac{1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = +\frac{1}{c}\frac{d[C]}{dt} = +\frac{1}{d}\frac{d[D]}{dt} \tag{A.73}$$

$$\text{Differential Rate Law} = k[A]^x[B]^y \implies \text{Overall Order } n = x + y \quad \text{(Order can be} \tag{A.74}$$

$$\text{SI Units of Rate Constant } k = (\text{mol}\,\text{L}^{-1})^{1-n}\,\text{s}^{-1} = \text{mol}^{1-n}\,\text{L}^{n-1}\,\text{s}^{-1} \tag{A.75}$$

## A.7.2 Integrated Rate Equations: Zero, First & $n$-th Order

**First-Order Fractional Lifetimes Shortcuts**

For any first-order process:

$$t_{50\%} = 1 \times t_{1/2} = \frac{0.693}{k}$$
$$t_{75\%} = 2 \times t_{1/2} \quad (75\% \text{ consumed, } 25\% \text{ remaining})$$
$$t_{87.5\%} = 3 \times t_{1/2} \quad (87.5\% \text{ consumed, } 12.5\% \text{ remaining})$$
$$t_{99\%} = 2 \times t_{90\%} \approx 6.64 \times t_{1/2} \approx \frac{4.606}{k}$$
$$t_{99.9\%} = 10 \times t_{1/2} \approx 3 \times t_{90\%} = \frac{6.909}{k}$$

| Reaction Order | Differential Form | Integrated Equation | Half-Life ($t_{1/2}$) | Comple |
|---|---|---|---|---|
| Zero Order ($n = 0$) | $-\frac{d[A]}{dt} = k$ | $[A]_t = [A]_0 - kt$ | $t_{1/2} = \frac{[A]_0}{2k} \propto [A]_0$ | $t_{100\%}$ |
| First Order ($n = 1$) | $-\frac{d[A]}{dt} = k[A]$ | $k = \frac{2.303}{t} \log_{10} \frac{[A]_0}{[A]_t}$ | $t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{k}$ | $t_{100\%} \to$ |
| Second Order ($n = 2$) | $-\frac{d[A]}{dt} = k[A]^2$ | $\frac{1}{[A]_t} - \frac{1}{[A]_0} = kt$ | $t_{1/2} = \frac{1}{k[A]_0} \propto \frac{1}{[A]_0}$ | |
| General $n$-th Order | $-\frac{d[A]}{dt} = k[A]^n$ | $\frac{1}{[A]_t^{n-1}} - \frac{1}{[A]_0^{n-1}} = (n-1)kt$ | $t_{1/2} \propto \frac{1}{[A]_0^{n-1}}$ | |

Table A.4: Systematic Summary of Integrated Rate Formulations and Half-Life Scaling.

## A.7.3 Temperature Dependence: Arrhenius Equation & Catalysis

**Arrhenius Equation, Activation Energy & Collision Theory**

$$k = A\,e^{-E_a/RT} \implies \ln k = \ln A - \frac{E_a}{RT} \tag{A.76}$$

$$\log_{10} k = \log_{10} A - \frac{E_a}{2.303RT} \tag{A.77}$$

$$\log_{10}\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right] \qquad [\text{Two Temperatures } T_1, T_2] \tag{A.78}$$

$$\text{Slope} = -\frac{E_a}{2.303R}, \qquad \text{Intercept} = \log_{10} A \qquad \left[\log_{10} k \text{ vs } \frac{1}{T}\right] \tag{A.79}$$

$$\Delta_r H = E_{a,\text{forward}} - E_{a,\text{backward}} \qquad [\Delta H < 0 \text{ Exothermic}] \tag{A.80}$$

$$\text{Rate} = P \cdot Z_{AB} \cdot e^{-E_a/RT} \qquad [P = \text{Steric Factor}, Z = \text{Freq.}] \tag{A.81}$$

**Kinetics Traps: Molecularity vs Order & Catalyst Action**

- **Molecularity is NEVER Zero or Fractional:** Molecularity is the number of reacting species colliding simultaneously in an elementary step; it is strictly an integer $(1, 2, 3)$. Reactions with molecularity $> 3$ are statistically rare. Order is experimentally determined and can be 0, fractional, or negative.
- **Catalysts Do NOT Alter Equilibrium State:** A catalyst accelerates both forward and reverse reaction rates by the identical factor by providing an alternate reaction pathway with lower activation energy ($E_a' < E_a$). It does **not** alter $\Delta_r H$, $\Delta_r G^\circ$, or the equilibrium constant $K_c$.

## Appendix B

# Inorganic Trends & Organic Named Reactions Compendiu

This quick-reference compendium brings together inorganic electronic configurations, crystal field splitting patterns, and all 35+ prescribed organic named reactions, complete with reagents, conditions, and functional distinction tests.

## B.1 Classification of Elements & Periodicity (Unit 9)

### B.1.1 Effective Nuclear Charge & Atomic Radii Trends

**Slater's Rules, Screening Constant & Radius Scaling**

$$Z_{\text{eff}} = Z - \sigma \qquad [Z = \text{Atomic Number}, \quad \sigma = \text{Screening} \tag{B.1}$$

$$\text{Shielding Power Order} : s > p > d > f \qquad [f\text{- and } d\text{-electrons provide extraordinar} \tag{B.2}$$

$$r_{\text{cation}} < r_{\text{atom}} < r_{\text{anion}} \qquad [r_{\text{isoelectronic}} \propto \frac{1}{Z} \implies \text{Size: } O^{2-} > F^{-} \tag{B.3}$$

| Periodic Property | General Trend Across Period / Down Group | | High-Yield NEET Exceptions & Anomalies |
|---|---|---|---|
| **Ionization Enthalpy (IE)** | Across: Increases | Down: Decreases | Be > B ($2s^2$ fully filled vs $2p^1$); N > O ($2p^3$ half filled vs $2p^4$); Mg > Al; P > S; Tl > In (Lanthanoid contraction). |
| **Electron Gain Enthalpy ($\Delta_{eg}H$)** | Across: More Negative | Down: Less Negative | Cl > F and S > O! (Compact $2p$ shell in F and O creates intense inter-electronic repulsions, lowering incoming electron affinity). |
| **Electronegativity ($\chi$)** | Across: Increases | Down: Decreases | Most electronegative: F(4.0) > O(3.5) > N(3.0) ≈ Cl(3.0) > Br(2.8) > I ≈ C ≈ S(2.5). |

Table B.1: Systematic Summary of Periodic Trends and Anomaly Anchors.

## B.1.2 Ionization Enthalpy, Electron Gain Enthalpy & Electronegativity

**Electronegativity Scales & Dipole Moment Formulations**

$$\chi_{\text{Mulliken}} = \frac{\text{IE} + \text{EA}}{2} \text{ (in eV)}, \qquad \chi_{\text{Pauling}} = \frac{\chi_{\text{Mulliken}}}{2.8} = \frac{\text{IE} + \text{EA}}{5.6} \tag{B.4}$$

$$\mu = q \times d \quad [1\,\text{Debye (D)} = 3.33564 \times 10^{-30}\,\text{C m}], \qquad \%\,\text{Ionic Character} = \tag{B.5}$$

$$\mu_{\text{net}} = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2\cos\theta} \implies \mu_{\text{ortho}} > \mu_{\text{meta}} > \mu_{\text{para}} = 0 \quad \text{(for identica} \tag{B.6}$$

**Periodic Traps: $NH_3$ vs $NF_3$ Dipole Moment & Electron Gain Enthalpy**

- **Dipole Moment Comparison ($\mu_{NH_3} > \mu_{NF_3}$):** In $NH_3$, bond dipoles of three N − H bonds reinforce the lone-pair dipole ($\mu = 1.47\,\text{D}$). In $NF_3$, the highly electronegative fluorines pull electron density away, opposing the lone-pair dipole ($\mu = 0.23\,\text{D}$).
- **Most Negative $\Delta_{eg}H$:** The element with the **most negative electron gain enthalpy** in the entire periodic table is **Chlorine (Cl)**, NOT Fluorine ($\text{Cl} : -349\,\text{kJ mol}^{-1}$ vs $\text{F} : -328\,\text{kJ mol}^{-1}$).

# B.2 Chemical Bonding & Molecular Structure (Unit 3)

## B.2.1 Formal Charge & VSEPR Theory Master Geometry Table

**Formal Charge Equation & Steric Number Rule**

$$\text{Formal Charge (FC)} = V - L - \frac{1}{2}S \qquad [V = \text{Valence } e^-, \tag{B.7}$$

$$\text{Steric Number (SN)} = \text{Bond pairs } (\sigma) + \text{Lone pairs on central atom} \tag{B.8}$$

**Bent's Rule Formulations in Trigonal Bipyramidal ($sp^3d$)**

- **Bent's Postulate:** More electronegative substituents prefer hybrid orbitals having **less $s$-character** (Axial bonds in TBP: pure $pd$ character, longer and weaker).
- **Lone Pair Preference:** Lone pairs and electropositive substituents prefer hybrid orbitals having **more $s$-character** (Equatorial bonds in TBP: $sp^2$ character, shorter and stronger).
- **Application to $PCl_5$ and $ClF_3$:** In gaseous $PCl_5$, axial P − Cl bonds (214 pm) are longer than equatorial bonds (202 pm). In solid state, it dissociates into ionic $[PCl_4]^+[sp^3]$ and $[PCl_6]^-[sp^3d^2]$.

## B.2.2 Molecular Orbital Theory (MOT)

**Molecular Orbital Energy Levels, Bond Order & Magnetic State**

$$\text{Bond Order (BO)} = \frac{1}{2}(N_b - N_a), \qquad \text{Stability} \propto \text{BO} \propto \text{Bond Dissociation Energy} \propto \frac{}{\text{B}} \tag{B.9}$$

**Case 1: Total Electrons $\leq 14$ ($B_2, C_2, N_2$ with $sp$-mixing):**

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z \tag{B.10}$$

| SN | Hybridization | Bond Pairs | Lone Pairs | Molecular Geometry / Shape | High-Yield NEET Examples |
|---|---|---|---|---|---|
| 2 | $sp$ | 2 | 0 | Linear (180°) | $BeCl_2, CO_2, HCN, C_2$ |
| 3 | $sp^2$ | 3 | 0 | Trigonal Planar (120°) | $BF_3, BCl_3, SO_3, NO_3^-$ |
| 3 | $sp^2$ | 2 | 1 | Bent / V-Shaped (< 120°) | $SO_2, O_3, NO_2^-$ |
| 4 | $sp^3$ | 4 | 0 | Tetrahedral (109.5°) | $CH_4, CCl_4, NH_4^+, SO_4^{2-}$ |
| 4 | $sp^3$ | 3 | 1 | Trigonal Pyramidal (107°) | $NH_3, PCl_3, H_3O^+, Cl$ |
| 4 | $sp^3$ | 2 | 2 | Bent / Angular (104.5°) | $H_2O, H_2S, OF_2, SCl_2$ |
| 5 | $sp^3d$ | 5 | 0 | Trigonal Bipyramidal | $PCl_5, PF_5, SbCl_5$ |
| 5 | $sp^3d$ | 4 | 1 | Seesaw / Distorted Tetrahedral | $SF_4, SeF_4$ |
| 5 | $sp^3d$ | 3 | 2 | T-Shaped | $ClF_3, BrF_3, ICl_3$ |
| 5 | $sp^3d$ | 2 | 3 | Linear | $XeF_2, I_3^-, ICl_2^-$ |
| 6 | $sp^3d^2$ | 6 | 0 | Octahedral (90°) | $SF_6, [AlF_6]^{3-}, PF_6^-$ |
| 6 | $sp^3d^2$ | 5 | 1 | Square Pyramidal | $BrF_5, IF_5, XeOF_4$ |
| 6 | $sp^3d^2$ | 4 | 2 | Square Planar | $XeF_4, [ICl_4]^-$ |
| 7 | $sp^3d^3$ | 7 | 0 | Pentagonal Bipyramidal | $IF_7$ |
| 7 | $sp^3d^3$ | 6 | 1 | Distorted Octahedral | $XeF_6$ |

Table B.2: Master VSEPR Geometries, Hybridization and Lone Pair Partitions.

**Case 2: Total Electrons $> 14$ ($O_2, F_2$ without $sp$-mixing):**

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z \tag{B.11}$$

| Species | Total Electrons | Configuration Key | Bond Order | Magnetic B |
|---|---|---|---|---|
| $N_2$ | 14 | $(\pi 2p)^4(\sigma 2p_z)^2$ | 3.0 | Diamagr |
| $N_2^+$ | 13 | $(\pi 2p)^4(\sigma 2p_z)^1$ | 2.5 | Paramagnetic (1 |
| $O_2^{2+}$ | 14 | $(\sigma 2p_z)^2(\pi 2p)^4$ | 3.0 | Diamagr |
| $O_2^+$ | 15 | $(\pi^* 2p)^1$ | 2.5 | Paramagnetic (1 |
| $O_2$ | 16 | $(\pi^* 2p_x)^1(\pi^* 2p_y)^1$ | 2.0 | Paramagnetic (2 |
| $O_2^-$ (Superoxide) | 17 | $(\pi^* 2p)^3$ | 1.5 | Paramagnetic (1 |
| $O_2^{2-}$ (Peroxide) | 18 | $(\pi^* 2p)^4$ | 1.0 | Diamagr |
| $C_2$ | 12 | $(\pi 2p)^4$ | 2.0 | Diamagnetic (Double bond c |

Table B.3: Molecular Orbital Configurations, Bond Orders and Paramagnetism for Homonuclear Diatomics.

# B.3 Coordination Compounds (Unit 12)

## B.3.1 Werner's Theory, Ligands & Chelate Effect

**Primary vs Secondary Valency & Chelate Stability**

$$\text{Primary Valency} = \text{Oxidation state of metal; ionizable; non-directional} \tag{B.12}$$

$$\text{Secondary Valency} = \text{Coordination number of metal; non-ionizable; directional (g} \tag{B.13}$$

$$\text{Chelate Effect : Polydentate ligands form stable ring structures} \implies \Delta S^\circ > \tag{B.14}$$

## B.3.2 Crystal Field Theory (CFT) & Splitting Formulations

**Octahedral & Tetrahedral Crystal Field Splitting Parameters**

$$\Delta_t = \frac{4}{9}\Delta_o \implies \Delta_t < P \quad \text{(Tetrahedral complexes are ALWA} \tag{B.15}$$

$$\text{CFSE}_{\text{octahedral}} = \left(-0.4\, n_{t_{2g}} + 0.6\, n_{e_g}\right)\Delta_o + mP \tag{B.16}$$

$$\text{Spectrochemical Series : } \mathrm{I^- < Br^- < S^{2-} < SCN^- < Cl^- < N_3^- < F^- < OH^-} \tag{B.17}$$

$$< \mathrm{C_2O_4^{2-} \approx H_2O < NCS^- < EDTA^{4-} < NH_3 \approx py} \tag{B.18}$$

$$< \mathrm{en < bpy < phen < NO_2^- < PPh_3 < CN^- \approx CO} \tag{B.19}$$

**High-Spin vs Low-Spin Splitting Conditions for Octahedral Complexes:**

$$\Delta_o > P \implies \text{Strong field; } t_{2g} \text{ pairing} \implies \text{Low-Spin / Inner Orbital} \tag{B.20}$$

$$\Delta_o < P \implies \text{Weak field; no } t_{2g} \text{ pairing} \implies \text{High-Spin / Outer Orbital} \tag{B.21}$$

| Ion Con-fig-u-ra-tion | Weak Field ($\Delta_o < P$) | CFSE (Weak) | Strong Field ($\Delta_o > P$) | CFSE (Strong) |
|---|---|---|---|---|
| $d^4$ ($Cr^{2+}, Mn^{3+}$) | $t_{2g}^3 e_g^1$ ($n = 4$) | $-0.6\,\Delta_o$ | $t_{2g}^4 e_g^0$ ($n = 2$) | $-1.6\,\Delta_o + P$ |
| $d^5$ ($Fe^{3+}, Mn^{2+}$) | $t_{2g}^3 e_g^2$ ($n = 5$) | $0\,\Delta_o$ | $t_{2g}^5 e_g^0$ ($n = 1$) | $-2.0\,\Delta_o + 2P$ |
| $d^6$ ($Fe^{2+}, Co^{3+}$) | $t_{2g}^4 e_g^2$ ($n = 4$) | $-0.4\,\Delta_o$ | $t_{2g}^6 e_g^0$ ($n = 0$) | $-2.4\,\Delta_o + 2P$ |
| $d^7$ ($Co^{2+}$) | $t_{2g}^5 e_g^2$ ($n = 3$) | $-0.8\,\Delta_o$ | $t_{2g}^6 e_g^1$ ($n = 1$) | $-1.8\,\Delta_o + P$ |

Table B.4: Crystal Field Electronic Configurations and CFSE Formulations for $3d^4 - 3d^7$ Octahedral Ions.

### B.3.3 Stereoisomerism in Coordination Complexes

**Geometrical and Optical Isomerism Rules**

- **Square Planar Complexes ($CN = 4$):**
  - $[MA_2B_2]$: Exhibits 2 geometrical isomers (cis and trans).
  - $[MABCD]$: Exhibits 3 geometrical isomers (2 cis, 1 trans).
  - Square planar complexes **never exhibit optical isomerism** because the molecular plane is an inherent plane of symmetry ($\sigma$).
- **Octahedral Complexes ($CN = 6$):**
  - $[MA_4B_2]$: Exhibits 2 geometrical isomers (cis and trans). Both are optically inactive.
  - $[MA_3B_3]$: Exhibits 2 geometrical isomers (facial *fac* and meridional *mer*). Both are optically inactive.
  - $[M(AA)_3]$ (e.g. $[Co(en)_3]^{3+}$): Exhibits 0 geometrical, but exists as a pair of **optical enantiomers** ($d$ and $l$).
  - $[M(AA)_2B_2]$ (e.g. $[Co(en)_2Cl_2]^{+}$): *cis*-isomer is chiral and optically active ($d/l$ pair); *trans*-isomer has plane of symmetry and is **optically inactive**.

# B.4 The $d$- and $f$-Block Elements (Unit 11)

## B.4.1 Transition Metal Trends & Variable Oxidation States

**Magnetic Moment, Enthalpy of Atomization & Electrode Potentials**

$$\mu_{\text{spin-only}} = \sqrt{n(n+2)}\,\text{BM} \tag{B.22}$$

Max Oxidation States : Sc (+3), Ti (+4), V (+5), Cr (+6), Mn (+7), Fe (+6), Os (B.23)

## B.4.2 Potassium Dichromate ($K_2Cr_2O_7$) & Permanganate ($KMnO_4$)

| Reagent & Reaction Medium | Change in Oxidation State | $n$-Factor | Canonical Half-Reaction |
|---|---|---|---|
| $K_2Cr_2O_7$ in Acidic Medium | $Cr^{6+} \to Cr^{3+}$ | 6 | $Cr_2O_7^{2-} + 14H^+ + 6e^- \to 2Cr^{3+} + 7H_2O$ |
| $KMnO_4$ in Acidic Medium | $Mn^{7+} \to Mn^{2+}$ | 5 | $MnO_4^- + 8H^+ + 5e^- \to Mn^{2+} + 4H_2O$ |
| $KMnO_4$ in Neutral / Faintly Alkaline | $Mn^{7+} \to Mn^{4+}$ | 3 | $MnO_4^- + 2H_2O + 3e^- \to MnO_2 + 4OH^-$ |
| $KMnO_4$ in Strongly Alkaline Medium | $Mn^{7+} \to Mn^{6+}$ | 1 | $MnO_4^- + e^- \to MnO_4^{2-}$ |

Table B.5: Stoichiometric Redox $n$-Factors for Dichromate and Permanganate Titrations.

**High-Yield Oxidation Reactions of $KMnO_4$ and $K_2Cr_2O_7$**

- **Acidic $KMnO_4$ converts:**

  $Fe^{2+} \to Fe^{3+}$, $C_2O_4^{2-} \to CO_2$, $S^{2-} \to S$, $SO_3^{2-} \to SO_4^{2-}$, $NO_2^- \to NO_3^-$, $I^-$

- **Faintly Alkaline $KMnO_4$ (Baeyer's Reagent) converts:**

  $I^- \to IO_3^-$ (Iodate ion, NOT $I_2$!), $S_2O_3^{2-} \to SO_4^{2-}$ (Thiosulfate to Sulfate)

- **Chromyl Chloride Test for Chloride ($Cl^-$) Detection:**

  $K_2Cr_2O_7 + 4NaCl + 6H_2SO_4 \xrightarrow{\Delta} 2CrO_2Cl_2 \uparrow$ (Deep Red Vapour) $+ 2KHSO_4 + 4NaHSO$

## B.4.3 Lanthanoid Contraction & Oxidation States

**Lanthanoid Contraction Governing Principles**

$$\text{Radius Trend} : La^{3+}(106\,\text{pm}) \to Lu^{3+}(86\,\text{pm}) \quad [\text{Steady contraction of } 20\,\text{pm across} \tag{B.24}$$

$$\text{Radius Similarity} : Zr(160\,\text{pm}) \approx Hf(159\,\text{pm}), \quad Nb(146\,\text{pm}) \approx Ta(146\,\text{pm}), \quad Mo \approx W \tag{B.25}$$

$$\text{Basicity Order} : La(OH)_3 > Ce(OH)_3 > \cdots > Lu(OH)_3 \quad [\text{Covalent character increas} \tag{B.26}$$

**High-Yield Oxidation States of Lanthanoids:**

$$\text{Dominant: } +3; \quad Ce^{4+} \ ([Xe]\ 4f^0 \text{ powerful oxidizing agent}), \quad Eu^{2+} \ ([Xe]\ 4f^7 \text{ redu} \tag{B.27}$$

# B.5 $p$-Block Elements (Unit 10)

## B.5.1 Inert Pair Effect & Thermodynamic Stability Trends

| Group | Higher Oxidation State | Lower Oxidation State | Stability Order & Redox Conse-quence |
|---|---|---|---|
| Group 13 (B, Al, Ga, In, Tl) | +3 | +1 | $Tl^{+}$ > $Tl^{3+}$ $\Longrightarrow$ $Tl^{3+}$ is oxidizing; Al |
| Group 14 (C, Si, Ge, Sn, Pb) | +4 | +2 | $Pb^{2+}$ > $Pb^{4+}$ $\Longrightarrow$ $PbO_2$ is powerful ox |
| Group 15 (N, P, As, Sb, Bi) | +5 | +3 | $Bi^{3+}$ > $Bi^{5+}$ $\Longrightarrow$ $BiF_5$ only stable pe |

Table B.6: Inert Pair Effect Manifestation Across Heavy $p$-Block Elements.

**Group 15, 16 & 17 Hydrides High-Yield Periodic Trends**

- **Group 15 Hydrides ($\mathbf{NH_3, PH_3, AsH_3, SbH_3, BiH_3}$):**
  - Bond Angle: $NH_3(107.8°)$ > $PH_3(93.6°)$ > $AsH_3(91.8°)$ > $SbH_3(91.3°)$ (Drago's rule).
  - Thermal Stability: $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$.
  - Reducing Character: $BiH_3 > SbH_3 > AsH_3 > PH_3 > NH_3$ (strongest reducing agent is $BiH_3$).
  - Boiling Point: $BiH_3 > SbH_3 > \mathbf{NH_3} > AsH_3 > PH_3$ ($NH_3$ elevated due to H-bonding!).
- **Group 16 Hydrides ($\mathbf{H_2O, H_2S, H_2Se, H_2Te}$):**
  - Acidic Strength: $H_2O < H_2S < H_2Se < H_2Te$ ($E-$H bond enthalpy decreases).
  - Boiling Point: $\mathbf{H_2}O > H_2Te > H_2Se > H_2S$ (water is highest due to extensive intermolecular H-bonding).
- **Group 17 Hydrides ($\mathbf{HF, HCl, HBr, HI}$):**
  - Acidic Strength: HF < HCl < HBr < HI (HI is strongest acid).
  - Boiling Point: HF > HI > HBr > HCl (HF elevated due to Zig-Zag H-bonding).

– Reducing Power: HI > HBr > HCl > HF.

**$p$-Block Traps: Oxoacids of Phosphorus & Boric Acid Basicity**

- **Basicity and Reducing Nature of Phosphorus Oxoacids:**
  - $H_3PO_2$ (Hypophosphorous / Phosphinic acid): Contains 1 P – OH and 2 P – H bonds. **Monobasic**; powerful reducing agent.
  - $H_3PO_3$ (Phosphorous / Phosphonic acid): Contains 2 P – OH and 1 P – H bonds. **Dibasic**; moderate reducing agent.
  - $H_3PO_4$ (Orthophosphoric acid): Contains 3 P – OH and 0 P – H bonds. **Tribasic**; non-reducing.
- **Orthoboric Acid ($H_3BO_3$):** Is **not a protonic acid**! It is a weak, **monobasic Lewis acid** that accepts $OH^-$ from water:

$$B(OH)_3 + 2H_2O \rightleftharpoons [B(OH)_4]^- + H_3O^+$$

## B.6 General Organic Chemistry (GOC) & Electronic Effects

**Fundamental Electronic Effects in Organic Molecules**

Organic reaction mechanisms and intermediate stabilities are governed by four fundamental displacement effects:

1. **Inductive Effect ($I$-Effect):** Permanent polarization through $\sigma$-bonds due to electronegativity difference. Decreases rapidly with distance ($I \propto 1/d^3$, negligible beyond 3 carbons).
   - $-I$ **Order:** $-NF_3^+ > -NR_3^+ > -NO_2 > -CN > -COOH > -F > -Cl > -Br > -I > -OR > -OH > -C \equiv CH > -NH_2 > -C_6H_5 > -CH = CH_2 > -H$
   - $+I$ **Order:** $-O^- > -COO^- > -CR_3(3°) > -CHR_2(2°) > -CH_2R(1°) > -CH_3 > -T > -D > -H$
2. **Electromeric Effect ($E$-Effect):** Temporary complete transfer of a shared pair of $\pi$-electrons to one of the atoms in the presence of an attacking reagent.
   - $+E$ **Effect:** $\pi$-electrons transferred to the atom where the electrophile attacks (e.g., addition of $H^+$ to alkenes).
   - $-E$ **Effect:** $\pi$-electrons transferred to the atom from which the nucleophile moves away (e.g., addition of $CN^-$ to carbonyls).
3. **Resonance / Mesomeric Effect ($R/M$-Effect):** Permanent de-

localization of conjugated $\pi$-electrons or lone pairs.

- $+M$ **(Electron Donating via Lone Pair):** $-O^- > -NH_2 > -NHR > -OH > -OR > -NHCOR > -OCOR > -Ph > -X$ ($F > Cl > Br > I$).
- $-M$ **(Electron Withdrawing via Multiple Bonds):** $-NO_2 > -CN > -SO_3H > -CHO >> C=O > -COOH > -COOR > -CONH_2$.

4. **Hyperconjugation (Baker-Nathan Effect):** Delocalization of $\sigma_{C-H}$ electrons with an adjacent empty $p$-orbital or $\pi^*$ orbital.

$$\text{Number of hyperconjugative structures} = \text{Number of } \alpha\text{-H atoms} \tag{B.28}$$

**Stability Hierarchy of Reactive Intermediates**

**Carbocations** ($sp^2, 6e^-$): Tropylium > CPM > $Ph_3C^+$ > Allyl/Benzyl ≈ 3
Stabilized by: $+M$ > Hyperconjugation ($\alpha$-H) >
Destabilized by: $-M, -I$.

**Carbanions** ($sp^3$ or $sp, 8e^-$): $CH \equiv C^- > CH_2=CH^- > CH_3^- > 1^\circ > 2^\circ > 3^\circ$
Stabilized by: $-M, -I$, high % $s$-character, aro

**Free Radicals** ($sp^2, 7e^-$): $Ph_3C^\bullet$ > Allyl/Benzyl > $3^\circ > 2^\circ > 1^\circ > CH_3^\bullet$
Governed by hyperconjugation and resonance.

**Carbocation Rearrangements: Hydride, Alkyl Shift & Ring Expansion**

Carbocations spontaneously undergo 1,2-shifts to achieve higher thermodynamic stability:

- **Migratory Aptitude in Carbocations:** $H^-$ > Aryl (Ar) > Alkyl ($3^\circ > 2^\circ > 1^\circ > CH_3$).
- **Ring Expansion:** 4-membered ring carbocation → 5-membered ring (relieves angle strain from 90° → 108°); 5-membered ring carbocation → 6-membered ring (relieves strain to stable chair conformation).
- **Ring Contraction:** 7-membered ring carbocation undergoes contraction to stable 6-membered ring.

**Quantitative Acidity & Basicity Sequences**

**1. Acidity Orders (Stabilization of Conjugate Base $A^-$):**

General Class: $R-SO_3H > R-COOH > H_2CO_3 > Ph-OH > H_2O > R-OH > HC\equiv CH > NH_3 > RH$

Haloacetic Acids: $CF_3COOH > CCl_3COOH > CHCl_2COOH > CH_2FCOOH > CH$

Substituted Phenols: Picric acid > *p*-nitrophenol > *o*-nitrophenol > *m*-nitrophenol > I

Ortho-Effect: Any ortho-substituted benzoic acid is stronger than benzoic acid

**2. Basicity Orders of Amines (Availability of Lone Pair on N):**

**Gas Phase:** $3^\circ$ Amine > $2^\circ$ Amine > $1^\circ$ Amine > $NH_3$ (strictly $+I$ gover

**Aqueous (Methyl):** $(CH_3)_2NH(2^\circ) > CH_3NH_2(1^\circ) > (CH_3)_3N(3^\circ) > NH_3$ **[Trick:**

**Aqueous (Ethyl):** $(C_2H_5)_2NH(2^\circ) > (C_2H_5)_3N(3^\circ) > C_2H_5NH_2(1^\circ) > NH_3$ **[Tri**

**Aromatic Amines:** Aliphatic amines $\gg NH_3 >$ *p*-toluidine > Aniline > *p*-nitroanilin

**NEET Trap: Ortho-Effect, H-Bonding & Volatility**

- ***o*-Nitrophenol vs *p*-Nitrophenol:** *o*-nitrophenol possesses **intramolecular H-bonding** (chelation), making it steam volatile with lower boiling point and less water solubility. *p*-nitrophenol possesses **intermolecular H-bonding**, giving it a higher boiling point and higher acidity ($pK_a = 7.15$ vs 7.23 for ortho due to absence of steric repulsion).
- **Picric Acid ($pK_a = 0.38$):** Stronger than carbonic acid and even many carboxylic acids; readily gives effervescence with aqueous $NaHCO_3$.
- **Phenol with $NaHCO_3$:** Phenol does NOT evolve $CO_2$ with $NaHCO_3$ ($pK_a$ phenol $\approx 10 > pK_a$ $H_2CO_3 \approx 6.4$). Nitrophenols with $\geq 2$ nitro groups DO dissolve in $NaHCO_3$.

## B.7 Hydrocarbons: Alkanes, Alkenes, Alkynes & Arenes

## Alkane Synthesis & Free Radical Transformations

**1. Wurtz Reaction:**

$$2R - X + 2Na \xrightarrow{\text{dry ether}} R - R + 2NaX \quad \text{(Free radical mechanism; best for symmetr} \tag{B.29}$$

**2. Decarboxylation (Soda Lime):**

$$R - COONa + NaOH \xrightarrow[\Delta]{\text{CaO (3:1)}} R - H + Na_2CO_3 \quad \text{(Product contains 1 fewer carbo} \tag{B.30}$$

**3. Kolbe Electrolytic Synthesis:**

$$\text{At Anode: } 2R - COO^- \xrightarrow{-2e^-} 2R - COO^\bullet \xrightarrow{-2CO_2} 2R^\bullet \rightarrow R - R + 2CO_2$$

$$\text{At Cathode: } 2H_2O + 2e^- \rightarrow H_2 \uparrow + 2OH^- \quad \text{(pH of solution increases during elect}$$

**4. Free Radical Halogenation Reactivity & Selectivity:**

$$\text{Reactivity: } F_2 > Cl_2 > Br_2 > I_2 \qquad \text{Selectivity: } 3^\circ\text{-H} > 2^\circ\text{-H} > 1^\circ\text{-H} \tag{B.31}$$

Relative rates of abstraction at 300°C:

- Chlorination ($1^\circ : 2^\circ : 3^\circ = 1 : 3.8 : 5.0$) → Low selectivity, governed by probability factor.
- Bromination ($1^\circ : 2^\circ : 3^\circ = 1 : 82 : 1600$) → Highly selective for 3° carbon.

## Alkene Addition Reactions & Regiochemical Rules

**1. Electrophilic Addition of HX (Markownikoff's Rule):** The electrophile ($H^+$) adds to the double bond to form the more stable carbocation intermediate (allows rearrangement!).

$$CH_3 - CH = CH_2 + HCl \longrightarrow CH_3 - CH(Cl) - CH_3 \quad \text{(2-chloropropane)} \tag{B.32}$$

**2. Peroxide Effect / Kharasch Effect (Anti-Markownikoff Addition):** Occurs **only with** HBr in the presence of organic peroxides ($R - O - O - R$) via free radical mechanism.

$$CH_3 - CH = CH_2 + HBr \xrightarrow{\text{peroxide}} CH_3 - CH_2 - CH_2 - Br \quad \text{(1-bromopropane)} \tag{B.33}$$

*Why only* HBr*?* Addition of $Cl^\bullet$ to alkene is endothermic in step 2; abstraction of H from HI by radical is endothermic in step 1. Only with HBr are both propagation steps exothermic!

**3. Hydroboration-Oxidation (Syn-Addition, Anti-Markownikoff Hydration):**

$$6R - CH = CH_2 \xrightarrow{B_2H_6/THF} 2(R - CH_2 - CH_2)_3B \xrightarrow{H_2O_2/OH^-} 3R - CH_2 - CH_2 - OH + \tag{B.34}$$

Regioselectivity: Anti-Markownikoff alcohol; Stereochemistry: **Syn-addition**; No carbocation rearrangement!

**4. Oxymercuration-Demercuration (Markownikoff Hydration without Rearrangement):**

$$R - CH = CH_2 \xrightarrow[H_2O]{Hg(OAc)_2} R - CH(OH) - CH_2 - HgOAc \xrightarrow{NaBH_4/OH^-} R - CH(OH) - C \tag{B.35}$$

Proceeds via cyclic mercurinium ion $\implies$ strictly NO rearrangement.

**5. Ozonolysis (Reductive vs Oxidative Cleavage):**

$$R_2C = CHR' \xrightarrow{O_3/CH_2Cl_2} \text{Molar Ozonide} \xrightarrow[Zn/H_2O]{\text{Reductive}} R_2C = O + R'CHO + ZnO \tag{B.36}$$

If oxidative ($H_2O_2$ or $Ag_2O$), any aldehyde formed is oxidized to carboxylic acid ($R'COOH$).

## Alkynes: Acidity, Hydration & Stereoselective Reduction

**1. Terminal Alkyne Acidity:** *sp*-hybridized carbon has 50% *s*-character, stabilizing conjugate carbanion:

$$R - C \equiv CH + NaNH_2 \longrightarrow R - C \equiv C^-Na^+ + NH_3 \uparrow$$

$$R - C \equiv CH + [Ag(NH_3)_2]^+ \longrightarrow R - C \equiv C - Ag \downarrow \quad \text{(White precipitate, Tollens}$$

$$R - C \equiv CH + Cu_2Cl_2 + NH_4OH \longrightarrow R - C \equiv C - Cu \downarrow \quad \text{(Red precipitate)}$$

**2. Kucherov Reaction (Hydration of Alkynes):**

$$HC \equiv CH + H_2O \xrightarrow[333\,K]{HgSO_4/\text{dil. } H_2SO_4} [CH_2 = CH - OH] \xrightarrow{\text{tautomerism}} CH_3 - CHO \quad \text{(Acetal} \tag{B.37}$$

All other terminal and non-terminal alkynes yield ketones:

$$CH_3 - C \equiv CH + H_2O \xrightarrow{Hg^{2+}/H^+} CH_3 - CO - CH_3 \quad \text{(Acetone)} \tag{B.38}$$

**3. Stereospecific Reduction of Alkynes:**

- **Lindlar's Catalyst ($H_2/Pd - CaCO_3$/quinoline):** Syn-addition → ***cis*-Alkene**.
- **Birch Reduction (Na/liq. $NH_3$ with EtOH):** Anti-addition → ***trans*-Alkene**.

**Aromatic Hydrocarbons: EAS Directing Groups & Activating Power**

Electrophilic Aromatic Substitution (EAS) involves arenium ion ($\sigma$-complex / Wheland intermediate):

| Group Category | Examples | Directing |
|---|---|---|
| Strong Activators | $-OH, -O^-, -NH_2, -NHR, -NR_2$ | *ortho / para* |
| Moderate Activators | $-OR, -NHCOR, -OCOR$ | *ortho / para* |
| Weak Activators | $-CH_3, -C_2H_5, -Ar$ | *ortho / para* |
| **Deactivating $o/p$-directors** | $-F, -Cl, -Br, -I$ | ***ortho / para*** |
| Moderate Deactivators | $-CHO, -COR, -COOH, -COOR, -SO_3H$ | *meta* |
| Strong Deactivators | $-NO_2, -NR_3^+, -CF_3, -CCl_3, -CN$ | *meta* |

**NEET Trap: Friedel-Crafts Reactions Limitations**

1. **Aniline does NOT undergo Friedel-Crafts reaction:** The Lewis acid catalyst ($AlCl_3$) complexes with the lone pair on the $-NH_2$ group: $Ph-NH_2 + AlCl_3 \rightarrow Ph-\overset{+}{N}H_2-\bar{A}lCl_3$. The resulting positive charge strongly deactivates the aromatic ring.
2. **Strongly deactivated rings do not react:** Nitrobenzene, benzoic acid, and benzonitrile do not undergo alkylation or acylation (used as solvents).
3. **Polyalkylation vs Monoacylation:** Alkylation introduces an activating alkyl group, leading to polyalkylation. Acylation introduces a deactivating acyl group ($-COR$), so it strictly stops at monoacylation.
4. **Rearrangement in Alkylation:** 1-chloropropane $+ Ph-H \xrightarrow{AlCl_3}$ isopropylbenzene (Cumene), due to $1^\circ \rightarrow 2^\circ$ carbocation hydride shift!

## B.8 Haloalkanes & Haloarenes: $S_N1$, $S_N2$, $E1$, $E2$

## Nucleophilic Substitution Mechanisms: $S_N1$ vs $S_N2$ Core Matrix

| Parameter | $S_N1$ (Unimolecular) | $S_N2$ (Bimolecular) |
|---|---|---|
| Kinetics | First Order: Rate $= k[\mathrm{R-X}]$ | Second Order: Rate $= k[\mathrm{R-X}][\mathrm{Nu^-}]$ |
| Mechanism | Two steps (carbocation intermediate) | One step concerted (transition state) |
| Reactivity of RX | $3^\circ > 2^\circ > 1^\circ > \mathrm{CH_3X}$ (carbocation) | $\mathrm{CH_3X} > 1^\circ > 2^\circ > 3^\circ$ (sterics) |
| Stereochemistry | **Racemization** (partial inversion) | **100% Walden Inversion** |
| Nucleophile | Weak ($\mathrm{H_2O}, \mathrm{ROH}$); low conc. | Strong ($\mathrm{OH^-}, \mathrm{RO^-}, \mathrm{CN^-}$); high conc. |
| Solvent | **Polar Protic** ($\mathrm{H_2O}, \mathrm{EtOH}, \mathrm{AcOH}$) | **Polar Aprotic** (DMSO, DMF, Acetone) |
| Rearrangement | **Possible** (hydride/alkyl shift) | **Strictly Impossible** |
| Substrate Limits | Allylic & Benzylic react rapidly | Vinyl ($\mathrm{CH_2=CHX}$) and Aryl unreactive |

## Ambident Nucleophiles in Haloalkane Reactions

Ambident nucleophiles possess two nucleophilic centers with differing electronegativities:

$$\mathrm{R-X + KCN} \xrightarrow{\text{aq. EtOH}} \mathrm{R-CN} \quad \text{(Ionic; C-attack forms stronger C-C bond)}$$

$$\mathrm{R-X + AgCN} \xrightarrow{\text{aq. EtOH}} \mathrm{R-NC} \quad \text{(Covalent; lone pair on N attacks)}$$

$$\mathrm{R-X + KNO_2} \xrightarrow{\text{aq. EtOH}} \mathrm{R-O-N=O} \quad \text{(Ionic; negative charge on O attacks)}$$

$$\mathrm{R-X + AgNO_2} \xrightarrow{\text{aq. EtOH}} \mathrm{R-NO_2} \quad \text{(Covalent; lone pair on N attacks)}$$

## Elimination Regiochemistry: Saytzeff vs Hofmann Product

- **Saytzeff (Zaitsev) Rule:** Base removes $\beta$-hydrogen from the $\beta$-carbon having fewer hydrogens, yielding the more substituted, thermodynamically more stable alkene (major product with small bases: $\mathrm{OH^-}, \mathrm{EtO^-}, \mathrm{MeO^-}$).
- **Hofmann Rule:** Base removes $\beta$-hydrogen from the least hindered $\beta$-carbon, yielding the less substituted alkene as major product. Favored when:
  1. Attacking base is bulky (e.g., Potassium *tert*-butoxide: $t\text{-BuO}^-\mathrm{K}^+$).
  2. Leaving group is poor (e.g., Fluoride $-\mathrm{F}$, or quarternary ammonium $-\mathrm{NR_3^+}$).

**NEET Trap: Nucleophilic Inactivity of Haloarenes**

Chlorobenzene ($C_6H_5Cl$) does not undergo nucleophilic substitution under normal conditions due to:

1. **Resonance:** Delocalization of chlorine lone pair gives partial double bond character to the $C-Cl$ bond (169 pm vs 177 pm in $R-Cl$).
2. **Hybridization of Carbon:** Chlorine is attached to an $sp^2$ hybridized carbon (33.3% $s$-character), which is more electronegative and holds electrons tighter than $sp^3$ carbon.
3. **Instability of Phenyl Cation:** Self-ionization to phenyl cation is energetically forbidden.
4. **Electrostatic Repulsion:** Electron-rich $\pi$-cloud of benzene repels incoming nucleophile.

**Remedy ($S_NAr$ Mechanism):** Introduction of strong electron-withdrawing groups ($-NO_2$) at ***ortho* and *para* positions** stabilizes the Meisenheimer intermediate anionic complex, drastically decreasing required temperature:

- Chlorobenzene: NaOH, 623 K, 300 atm (Dow Process).
- $p$-Nitrochlorobenzene: NaOH, 443 K, $H^+$.
- 2,4-Dinitrochlorobenzene: NaOH, 368 K, $H^+$.
- 2,4,6-Trinitrochlorobenzene: Warm with $H_2O$ alone at 323 K!

## B.9 Oxygen Compounds: Alcohols, Phenols, Ethers, Carbonyls & Acids

**Alcohols & Phenols Preparation & Conversions**

**1. Grignard Reagent (RMgX) Carbonyl Additions:**

$$HCHO + RMgX \xrightarrow{H_3O^+} 1^\circ \text{ Alcohol} \qquad R'CHO + RMgX \xrightarrow{H_3O^+} 2^\circ \text{ Alcohol}$$

$$R'COR'' + RMgX \xrightarrow{H_3O^+} 3^\circ \text{ Alcohol} \qquad CO_2 + RMgX \xrightarrow{H_3O^+} RCOOH \quad \text{(Carbox}$$

**2. Cumene Hydroperoxide Process (Industrial Phenol):**

$$\text{Benzene} + CH_3 - CH = CH_2 \xrightarrow{H_3PO_4} \text{Cumene} \xrightarrow{O_2} \text{Cumene Hydroperoxide}$$
$$\xrightarrow{\text{dil. } H_2SO_4} \text{Phenol} + CH_3COCH_3 \quad \text{(Acetone by-prod}$$

**3. Kolbe-Schmitt Reaction (Synthesis of Salicylic Acid):**

$$\text{Sodium phenoxide} + CO_2 \xrightarrow[4-7\,\text{atm}]{400\,\text{K}} \text{Sodium salicylate} \xrightarrow{H^+} \text{Salicylic acid (2-hydroxybenzo} \quad \text{(B.39)}$$

**4. Reimer-Tiemann Reaction (Synthesis of Salicylaldehyde):**

$$\text{Phenol} + CHCl_3 + 3NaOH \xrightarrow{340\,\text{K}} \text{Salicylaldehyde} \quad \textbf{(Electrophile: Dichlorocarbene} \quad \text{(B.40)}$$

*Note:* If $CCl_4$ is used instead of $CHCl_3$, salicylic acid is the main product.

**5. Esterification (Fischer):**

$$RCOOH + R'OH \underset{}{\overset{\text{conc. } H_2SO_4}{\rightleftharpoons}} RCOOR' + H_2O \quad \text{(Reactivity of ROH: } CH_3OH > 1^\circ > 2^\circ > \quad \text{(B.41)}$$

## Ethers: Williamson Synthesis & Cleavage with HI

**1. Williamson Ether Synthesis:**

$$R - X + R' - O^-Na^+ \longrightarrow R - O - R' + NaX \quad \textbf{(Mechanism: Strictly } S_N2) \quad \text{(B.42)}$$

- Alkyl halide must be $1^\circ$ (or $CH_3X$).
- If $3^\circ$ alkyl halide is used with alkoxide, **elimination ($E2$) dominates exclusively**:

$$(CH_3)_3C - Br + CH_3O^-Na^+ \longrightarrow CH_3 - C(CH_3) = CH_2 + CH_3OH + NaBr \quad \text{(Isobutyle} \quad \text{(B.43)}$$

**2. Cleavage of Ethers by Halogen Acid (HI):**

- **Alkyl Alkyl Ethers ($1^\circ/2^\circ$ groups):** Follows $S_N2$. Iodine ion attacks the less sterically hindered alkyl group:

$$CH_3 - O - C_2H_5 + HI \xrightarrow{\text{cold}} CH_3 - I + C_2H_5 - OH \quad \text{(B.44)}$$

- **One group is $3^\circ$ (or Benzyl):** Follows $S_N1$ via stable carbocation; iodine attacks the $3^\circ$ carbon:

$$(CH_3)_3C - O - CH_3 + HI \xrightarrow{\text{cold}} (CH_3)_3C - I + CH_3 - OH \quad \text{(B.45)}$$

- **Alkyl Aryl Ethers (Anisole):** The $O - Aryl$ bond has partial double bond character and never cleaves:

$$C_6H_5 - O - CH_3 + HI \xrightarrow{\Delta} C_6H_5 - OH\ (\text{Phenol}) + CH_3 - I \quad \text{(B.46)}$$

## Carbonyl Compounds: Nucleophilic Addition & Carbonyl Condensations

**1. Relative Reactivity Towards Nucleophilic Addition:**

$$\text{HCHO} > \text{CH}_3\text{CHO} > \text{CH}_3\text{COCH}_3 > \text{PhCHO} > \text{PhCOCH}_3 > \text{PhCOPh} \tag{B.47}$$

*Governed by:* Steric hindrance at carbonyl carbon and $+I/+M$ electron donation reducing electrophilicity of carbonyl carbon.

**2. Aldol Condensation ($\alpha$-H Required):** Carbonyl compounds with at least one $\alpha$-hydrogen react in dilute alkali (10% NaOH or $\text{Ba(OH)}_2$):

$$2\text{CH}_3\text{CHO} \xrightarrow{\text{dil. NaOH}} \text{CH}_3 - \text{CH(OH)} - \text{CH}_2 - \text{CHO} \xrightarrow[\Delta]{-\text{H}_2\text{O}} \text{CH}_3 - \text{CH} = \text{CH} - \text{C} \tag{B.48}$$

**3. Cannizzaro Reaction (No $\alpha$-H Present):** Self-oxidation and self-reduction in concentrated alkali (50% NaOH):

$$2\text{HCHO} + \text{conc. NaOH} \longrightarrow \text{CH}_3\text{OH (Methanol)} + \text{HCOONa (Sodium formate)} \tag{B.49}$$

$$2\text{C}_6\text{H}_5\text{CHO} + \text{conc. NaOH} \longrightarrow \text{C}_6\text{H}_5\text{CH}_2\text{OH (Benzyl alcohol)} + \text{C}_6\text{H}_5\text{COONa (Sodi} \tag{B.50}$$

*Cross Cannizzaro with Formaldehyde:* Formaldehyde undergoes oxidation exclusively ($\text{HCOO}^-$) because hydride transfer occurs from the more nucleophilically attacked formaldehyde.

## Named Carbonyl Reductions, Oxidations & Acid Reactions

**1. Clemmensen Reduction:** $\text{Zn} - \text{Hg} + \text{conc. HCl}$ ($> \text{C} = \text{O} \rightarrow > \text{CH}_2$); *Avoid if acid-sensitive groups (e.g.* $-\text{OH}, -\text{C} = \text{C}-, -\text{OR}$*) present.*

**2. Wolff-Kishner Reduction:** $\text{NH}_2 - \text{NH}_2 + \text{KOH/glycol}, \ \Delta$ ($> \text{C} = \text{O} \rightarrow > \text{CH}_2$); *Avoid if base-sensitive groups present.*

**3. Rosenmund Reduction:** $\text{R} - \text{COCl} + \text{H}_2 \xrightarrow{\text{Pd/BaSO}_4\text{,quinoline}} \text{R} - \text{CHO} + \text{HCl}$.

**4. Stephen Reaction:** $\text{R} - \text{CN} + \text{SnCl}_2/\text{HCl} \longrightarrow \text{R} - \text{CH} = \text{NH} \cdot \text{HCl} \xrightarrow{\text{H}_3\text{O}^+} \text{R} - \text{CHO} + \text{NH}_4\text{Cl}$.

**5. Etard Reaction:** $\text{C}_6\text{H}_5 - \text{CH}_3 + \text{CrO}_2\text{Cl}_2/\text{CS}_2 \rightarrow$ Chromium complex $\xrightarrow{\text{H}_3\text{O}^+} \text{C}_6\text{H}_5 - \text{CHO}$.

**6. Gattermann-Koch Reaction:** $\text{C}_6\text{H}_6 + \text{CO} + \text{HCl} \xrightarrow{\text{anhyd. AlCl}_3/\text{CuCl}} \text{C}_6\text{H}_5 - \text{CHO} + \text{HCl}$.

**7. Hell-Volhard-Zelinsky (HVZ) Reaction:** Carboxylic acids

with $\alpha$-hydrogen react with halogen in presence of red phosphorus:

$$\mathrm{R-CH_2-COOH + X_2 \xrightarrow[(ii)\ H_2O]{(i)\ Red\ P} R-CH(X)-COOH} \quad (\alpha\text{-halocarboxylic acid; X = Cl} \tag{B.51}$$

## B.10 Nitrogen Compounds: Amines & Diazonium Salts

### Synthesis & Reactions of Amines

**1. Gabriel Phthalimide Synthesis (Pure 1° Aliphatic Amines):**

$$\text{Phthalimide} \xrightarrow{\text{KOH}} \text{K-phthalimide} \xrightarrow{\text{R-X}\,(S_N2)} N\text{-Alkylphthalimide} \xrightarrow{\text{NaOH(aq)}} \mathrm{R-NH_2} +$$

*Limitation:* Aromatic primary amines cannot be prepared because aryl halides do not undergo $S_N2$ displacement.

**2. Hoffmann Bromamide Degradation:** Conversion of an amide to a 1° amine containing **one carbon less**:

$$\mathrm{R-CONH_2 + Br_2 + 4NaOH \longrightarrow R-NH_2 + Na_2CO_3 + 2NaBr + 2H_2O} \tag{B.52}$$

Intermediate is alkyl isocyanate ($\mathrm{R-N=C=O}$); migration of alkyl group occurs with **retention of configuration**.

**3. Carbylamine Test (Isocyanide Test - Strictly 1° Amines):** Both aliphatic and aromatic 1° amines react with chloroform and alc. KOH to form foul-smelling isocyanides:

$$\mathrm{R-NH_2 + CHCl_3 + 3KOH \xrightarrow{\Delta} R-NC\ (Alkyl\ isocyanide) + 3KCl + 3H_2O} \tag{B.53}$$

2° and 3° amines do NOT show this test.

**4. Hinsberg Test (Distinction of 1°, 2°, 3° Amines):** Reagent: Benzenesulfonyl chloride ($\mathrm{C_6H_5SO_2Cl}$).

- 1° Amine → $N$-alkylbenzenesulfonamide → possesses acidic H on N → **Soluble in alkali** (KOH).
- 2° Amine → $N,N$-dialkylbenzenesulfonamide → no acidic H → **Insoluble in alkali**.
- 3° Amine → No replaceable H on N → **Does not react with Hinsberg reagent**.

**Diazonium Chemistry: Arenediazonium Salt Reactions**

Diazotization: $\mathrm{Ar-NH_2 + NaNO_2 + 2HCl \xrightarrow{273-278\,K} Ar-N_2^+Cl^- + NaCl + 2H_2O}$.

$$\text{Sandmeyer: } \mathrm{Ar-N_2^+Cl^- \xrightarrow{Cu_2X_2/HX} Ar-X \quad (X = Cl, Br, CN)}$$

$$\text{Gattermann: } \mathrm{Ar-N_2^+Cl^- \xrightarrow{Cu/HX} Ar-X \quad (X = Cl, Br)}$$

$$\text{Balz-Schiemann: } \mathrm{Ar-N_2^+Cl^- \xrightarrow{HBF_4} Ar-N_2^+BF_4^- \xrightarrow{\Delta} Ar-F + BF_3 + N_2\uparrow}$$

$$\text{Iodination: } \mathrm{Ar-N_2^+Cl^- + KI \xrightarrow{warm} Ar-I + KCl + N_2\uparrow}$$

$$\text{Reduction: } \mathrm{Ar-N_2^+Cl^- + H_3PO_2 + H_2O \xrightarrow{Cu^+} Ar-H + H_3PO_3 + HCl + N}$$

$$\mathrm{Ar-N_2^+Cl^- + CH_3CH_2OH \longrightarrow Ar-H + CH_3CHO + HCl + N_2}$$

$$\text{Hydrolysis: } \mathrm{Ar-N_2^+Cl^- + H_2O \xrightarrow{373\,K} Ar-OH + HCl + N_2\uparrow}$$

**Azo Coupling Reactions (Formation of Dyes):**

With Phenol: $\mathrm{Ph-N_2^+Cl^- + C_6H_5-OH \xrightarrow{pH\approx 9-10}}$ *p*-hydroxyazobenzene (Orang

With Aniline: $\mathrm{Ph-N_2^+Cl^- + C_6H_5-NH_2 \xrightarrow{pH\approx 4-5}}$ *p*-aminoazobenzene (Yellow c

## B.11 Biomolecules: Carbohydrates, Amino Acids & Nucleic Acids

**Carbohydrates Classification & Reactions of D-Glucose**

1. **Classification & Reducing Nature:**
   - **Reducing Sugars:** All monosaccharides (aldoses & ketoses) and disaccharides with a free hemiacetal/hemiketal -OH at anomeric carbon (Maltose, Lactose). Reduce Tollens and Fehling solutions.
   - **Non-Reducing Sugars:** Glycosidic linkage involves anomeric carbons of both units (e.g., **Sucrose** = $\alpha$-D-glucopyranosyl-(1→2)-$\beta$-D-fructofuranoside).
   - **Invert Sugar:** Sucrose ($[\alpha]_D = +66.5°$) on hydrolysis gives D-glucose ($+52.5°$) and D-fructose ($-92.4°$). The mixture is levorotatory ($-39.9°/2 = -19.9°$).

**2. Chemical Reactions of D-Glucose ($C_6H_{12}O_6$):**

$$\text{Glucose} + \text{HI},\ \Delta \longrightarrow n\text{-hexane} \quad \text{(Proves open chain of 6 carbons)}$$
$$\text{Glucose} + NH_2OH \longrightarrow \text{Glucose oxime} \quad \text{(Confirms carbonyl group } > C = O\text{)}$$
$$\text{Glucose} + HCN \longrightarrow \text{Glucose cyanohydrin}$$
$$\text{Glucose} + Br_2/H_2O \longrightarrow \text{Gluconic acid (mild oxidation of } -CHO \text{ to } -COOH\text{)}$$
$$\text{Glucose} + \text{conc. } HNO_3 \longrightarrow \text{Saccharic acid / Glucaric acid (oxidizes both } -CHO \text{ and}$$
$$\text{Glucose} + 5(CH_3CO)_2O \longrightarrow \text{Glucose pentaacetate} \quad \text{(Confirms presence of 5 independe}$$

**3. Why the Cyclic Structure of Glucose?** D-Glucose does not give Schiff's test, 2,4-DNP test, and does not form hydrogen sulphite addition product with $NaHSO_3$. The pentaacetate does not react with hydroxylamine (absence of free $-CHO$).

$$\alpha\text{-D-(+)-glucose } ([\alpha]_D = +112^\circ) \underset{\text{in solution}}{\overset{\text{Mutarotation}}{\rightleftharpoons}} \text{Equilibrium } (+52.7^\circ) \overset{\beta}{\rightleftharpoons} \text{-D-(+)-glucose (} \tag{B.54}$$

$\alpha$ and $\beta$ isomers differ only at $C-1$ (anomeric carbon) $\Longrightarrow$ **Anomers**.

## Amino Acids, Peptide Bonds & Protein Structure

**1. Zwitterion & Isoelectric Point (pI):** In aqueous solution, the carboxyl group loses a proton and amino group accepts it:

$$R - CH(NH_2) - COOH \rightleftharpoons R - CH(NH_3^+) - COO^- \quad \text{(Dipolar Zwitterion; net charge } = \tag{B.55}$$

$$\text{At pH} < \text{pI}: \text{ exists as cation } R - CH(NH_3^+) - COOH \quad \text{(migrates towards cathode).} \tag{B.56}$$

$$\text{At pH} > \text{pI}: \text{ exists as anion } R - CH(NH_2) - COO^- \quad \text{(migrates towards anode).} \tag{B.57}$$

For neutral amino acid: $\text{pI} = \frac{pK_{a1}+pK_{a2}}{2}$.

- **Chirality:** All naturally occurring amino acids are optically active and possess **L-configuration**, **except Glycine** ($H_2N - CH_2 - COOH$, achiral).
- **Essential Amino Acids (10):** Valine, Leucine, Isoleucine, Phenylalanine, Methionine, Tryptophan, Threonine, Lysine, Histidine, Arginine. [**Mnemonic: PVT TIM HALL**].

**2. Protein Structural Levels:**

- **Primary:** Specific sequence of amino acids linked by covalent peptide bonds ($-CO - NH-$).
- **Secondary:** Local conformation of polypeptide chain stabilized by **hydrogen bonds** between $> C = O$ and $-NH-$ groups: $\alpha$-helix

(intramolecular H-bonds) and $\beta$-pleated sheets (intermolecular H-bonds).

- **Tertiary:** Overall 3D folding of polypeptide chains stabilized by H-bonds, disulphide linkages ($-S-S-$), van der Waals forces, and electrostatic attractions.
- **Quaternary:** Spatial arrangement of two or more subunits (e.g., Hemoglobin with $2\alpha$ and $2\beta$ chains).
- **Denaturation:** Heat, pH change breaks $2°, 3°, 4°$ structures; **Primary structure remains completely intact!** (e.g., coagulation of egg white, curdling of milk).

**Nucleic Acids: Nucleotides, Chargaff's Rules & Double Helix**

1. **Structural Components:**
- **Pentose Sugar:** $\beta$-D-ribose (RNA) vs $\beta$-D-2-deoxyribose (DNA).
- **Nitrogenous Bases:**
  - **Purines (Bicyclic):** Adenine (A), Guanine (G).
  - **Pyrimidines (Monocyclic):** Cytosine (C), Thymine (T, DNA only), Uracil (U, RNA only).
- **Nucleoside:** Sugar + Base (attached at $C-1'$ of sugar via $\beta$-glycosidic linkage).
- **Nucleotide:** Nucleoside + Phosphate (esterified at $C-5'$ of sugar).
- **Phosphodiester Linkage:** Connects $C-3'$ of one nucleotide to $C-5'$ of adjacent nucleotide.

2. **Chargaff's Rules (for Double-Stranded DNA):**

$$\%A = \%T \quad \text{and} \quad \%G = \%C \implies \frac{A+G}{T+C} = 1 \qquad \text{(Purines = Pyrimidines)} \tag{B.58}$$

Base pairing: A pairs with T via **2 hydrogen bonds** ($A = T$); G pairs with C via **3 hydrogen bonds** ($G \equiv C$).

## B.12 Master Identification Tests & Distinction Matrix

## Comprehensive Tabular Matrix for Distinguishing Organic Functional Pairs

| Compound Pair | Reagent / Test | Observed Visual Distinction |
|---|---|---|
| $1^\circ, 2^\circ, 3^\circ$ Alcohols | Lucas Reagent (conc. $HCl$ + anhyd. $ZnCl_2$) | $3^\circ$: immediate cloudiness; $2^\circ$: turbidity in 5 min; $1^\circ$: no turbidity at room temp. |
| Aliphatic vs Aromatic Aldehydes | Fehling's Solution ($A + B$) | Aliphatic: Red ppt ($Cu_2O$); Benzaldehyde: No reaction (does not reduce Fehling). |
| Aldehyde vs Ketone | Tollens' Reagent ($[Ag(NH_3)_2]^+$) | Aldehydes: Bright silver mirror (Ag); Ketones: Negative. |
| $CH_3 - C = O$ or $CH_3 - CH(OH) -$ | Iodoform Test ($I_2 + NaOH$) | Yellow crystalline $CHI_3 \downarrow$ (m.p. $119^\circ C$) with antiseptic odor. Positive for ethanol, acetaldehyde, 2-propanol. |
| Carboxylic Acid vs Phenol | $NaHCO_3$ (aqueous) | Carboxylic acid: Brisk effervescence of $CO_2$; Phenol: No effervescence. |
| Phenol vs Alcohol | Neutral $FeCl_3$ solution | Phenol: Violet coloration ($[Fe(OC_6H_5)_6]^{3-}$); Alcohols: Negative. |
| $1^\circ$ Amine vs $2^\circ/3^\circ$ | Carbylamine Test ($CHCl_3$ + alc. KOH) | Only $1^\circ$ amine gives extremely foul smell of isocyanide ($R - NC$). |
| $1^\circ, 2^\circ, 3^\circ$ Amines | Hinsberg Reagent ($C_6H_5SO_2Cl$) | $1^\circ$: Soluble in KOH; $2^\circ$: Insoluble in KOH; $3^\circ$: No reaction. |
| Terminal vs Internal Alkyne | Ammoniacal $Cu_2Cl_2$ | Terminal ($R - C \equiv CH$): Red ppt of copper acetylide; Internal: Negative. |
| Alkene/Alkyne vs Alkane | Bromine water ($Br_2/CCl_4$) | Unsaturated: Decolorization of reddish-brown bromine color without HBr evolution. |

## High-Yield Reagent Conversion Master Reference

- PCC **(Pyridinium Chlorochromate) / Collins Reagent:** Oxidizes $1^\circ$ alcohols strictly to aldehydes (RCHO) without overoxidation to carboxylic acid. Oxidizes $2^\circ$ alcohols to ketones.
- **DIBAL-H (Diisobutylaluminium Hydride):** Selectively reduces esters (RCOOR′) and nitriles (RCN) at low temperature ($-78^\circ C$) to **aldehydes** (RCHO).
- $LiAlH_4$ **vs** $NaBH_4$**:** $LiAlH_4$ reduces acids, esters, amides, nitriles, aldehydes, and ketones. $NaBH_4$ is weaker and selectively reduces **only aldehydes, ketones, and acid halides** (does not touch esters, acids, or amides).
- $CrO_3$ **in glacial acetic acid vs Jones Reagent** ($CrO_3/H_2SO_4$)**:** Mild vs strong chromic acid oxidation.
- **Fenton's Reagent** ($FeSO_4 + H_2O_2$)**:** Generates hydroxyl free radicals ($OH^\bullet$) for selective oxidation of alcohols to carbonyls.

# About the Author & Errata Submissions

**Satya Prakash** is an educator, curriculum researcher, and author dedicated to creating rigorous, student-first learning resources for competitive examinations in India. His focus is on eliminating cognitive overload, dismantling obsolete legacy syllabi, and restoring conceptual rigor and mathematical transparency to competitive pre-medical preparation.

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